

Invent Cast

Deep Intelligence Report

<https://inventcast.in/>

Executive Summary

Invent Cast operates within the industrial manufacturing sector, specializing in B2B manufacturing solutions that focus on precision-driven investment casting and machining. The company offers a comprehensive range of high-tolerance metal parts and components, particularly emphasizing non-ferrous nickel alloys and copper alloys. Their product line is designed to meet the rigorous demands of various industries, including engineering and automotive, providing customizable solutions tailored to specific engineering needs. Invent Cast prides itself on its integrated manufacturing environment, which enhances efficiency and ensures high-quality outputs that comply with global standards.

Founded with a commitment to innovation and quality, Invent Cast has established itself as a leader in the manufacturing of complex geometries and high-quality materials. The company leverages advanced manufacturing technologies, including robotic shelling, to maintain consistency and precision in its products. Its robust quality assurance processes and comprehensive documentation ensure compliance with ASTM standards and facilitate export-ready documentation, making it a trusted partner for manufacturing and production managers across multiple geographies, including the United States, Europe, and Asia.

Invent Cast's market positioning is distinguished by its expertise in non-ferrous alloys and its ability to deliver high-quality, durable products that cater to a wide range of applications. The company's unique selling propositions include its rigorous testing protocols, commitment to precision, and a diverse product range that addresses the specific needs of various industries. By focusing on high-quality materials and customizable solutions, Invent Cast not only meets but exceeds the expectations of its clients, solidifying its reputation as a reliable and innovative manufacturing partner in the global market.

Business Identity

Official Name: Invent Cast

Industry: Manufacturing

Sub-Industry: Industrial manufacturing

Business Model: B2B manufacturing

Product Catalog

Valves

Category: Valves

Forged or Rolled Alloy Pipe Flanges, Forged Fittings, and Valves and Parts for Corrosive High-Temperature Service

Key Features:

- High precision and durability
- Leak-proof operation
- Meets international performance standards
- Forged or rolled alloy construction
- Designed for high-temperature applications
- Suitable for corrosive environments
- Intended for Valve, Pump, General Engineering, Automotive and other Industries
- High precision and accuracy
- Leak-proof sealing
- Durable and reliable performance
- High tensile strength
- Corrosion resistance
- Versatile applications in engineering
- High strength
- Suitable for high-temperature applications
- High corrosion resistance
- Durability under high pressure
- Suitable for various industrial applications
- Manufactured from high-quality materials
- Designed for durability and reliability
- Suitable for various industries
- Chemical Composition
- Heat Treatment
- Mechanical Properties
- Versatile applications
- Durable construction
- High pressure tolerance
- Corrosion-resistant materials
- Dimensional stability

Technical Specifications:

Material	Various steel alloys
Standards	ASTM B271, ASTM B275
Heat Treatment	As-Cast or Annealing or Age Hardening
Chemical Composition	Various alloys including C95410, C95400, C89320, etc.
Mechanical Properties	
Applications	Valve, Pump, General Engineering, Automotive
Aluminium	6.000 - 8.000
Copper	88.000 min.
Tensile Strength in Mpa	415 min.
Yield Strength in Mpa	195 min.

Elongation in %	10 min.
Material Standard	ASTM B505
Pressure Rating	Up to 1000 psi
Temperature Range	-20°C to 200°C
Materials	Stainless steel, duplex steel, super alloys, Hastelloy
Casting Weight	50g to 50kg
Surface Finish	Up to 3.2 Ra without machining
Tolerances	As per ISO 8062, up to ± 0.1 mm

Use Cases:

- Oil & gas
- Water treatment
- General engineering
- Valve applications in automotive
- Pumping systems
- General Engineering
- Automotive
- Oil & Gas
- Water Treatment
- Chemical Processing
- Power Generation
- Industrial applications
- Automotive components
- Oil and gas industry
- Chemical processing
- General engineering projects
- Other Industries
- Automotive applications
- Pump applications
- Medical
- Industrial Applications
- Pump Applications
- Oil and gas

Export Markets:

- Global
- USA
- International
- Germany
- UK
- Middle East
- Southeast Asia
- Europe
- Asia
- Australia
- Korea
- North America
- India

Sub-Products / Variations:

- ASTM B 462 N10276: Standard specification for forged or rolled alloy pipe flanges, forged fittings, and valves.
- ASTM B 462 R20033: Standard specification for forged or rolled alloy pipe flanges, forged fittings, and valves.

- ASTM B 463 N08020: Standard specification for nickel alloy plate, sheet, and strip.
- ASTM B 463 N08024: Standard specification for nickel alloy plate, sheet, and strip.
- ASTM B 463 N08026: Standard specification for nickel alloy bar and wire.
- ASTM B 464 N08020: Standard specification for nickel alloy pipe.
- ASTM B 464 N08026: Standard specification for nickel alloy pipe.
- ASTM B 468 N08020: Standard specification for nickel alloy tubes.
- ASTM B 472 N06002: Standard specification for nickel alloy billets and bars for reforging.
- ASTM B 472 N06030: Standard specification for nickel alloy billets and bars for reforging.

(+ 109 more specific variations recorded in internal database)

AI Layman Summary:

Valves are essential components used in various industries to control the flow of liquids and gases. They are made from strong materials designed to withstand high temperatures and corrosive environments, making them reliable for demanding applications such as oil and gas, automotive, and chemical processing. These valves ensure that systems operate smoothly and safely by preventing leaks and maintaining pressure.

Key benefits of using these valves include:

- ****High Precision and Durability****: Built to last, they can handle extreme conditions.
- ****Leak-Proof Operation****: Their design prevents leaks, enhancing safety and efficiency.
- ****Versatile Applications****: Suitable for a wide range of industries, including automotive, water treatment, and power generation.

Pumps

Category: Pumps

Forged or Rolled Alloy Pipe Flanges, Forged Fittings, and Valves and Parts for Corrosive High-Temperature Service

Key Features:

- Optimized fluid dynamics
- Long-term durability
- High tolerance metal parts
- Durable construction
- High efficiency
- Suitable for various industrial applications
- Intended for Valve, Pump, General Engineering, Automotive and other Industries
- High precision and accuracy
- Leak-proof operation
- Durable and reliable performance
- Durability under high pressure
- Resistance to wear and tear
- Suitable for various fluids
- Efficient performance
- Wide range of applications
- Efficient fluid transfer
- Robust construction
- Versatile for different fluids
- High efficiency and performance
- Versatile applications
- Chemical Composition
- Heat Treatment
- Mechanical Properties
- Robust design
- High strength and durability
- Corrosion-resistant materials

- Efficient fluid dynamics

Technical Specifications:

Material	Stainless steel, cast iron
Standards	ASTM B271, ASTM B275
Heat Treatment	As-Cast or Annealing or Age Hardening
Chemical Composition	Various alloys including C95410, C95400, C89320, etc.
Mechanical Properties	
Applications	Valve, Pump, General Engineering, Automotive
Aluminium	6.000 - 8.000
Copper	88.000 min.
Tensile Strength in Mpa	415 min.
Yield Strength in Mpa	195 min.
Elongation in %	10 min.
Material Standard	ASTM B505
Flow Rate	Up to 5000 L/min
Pressure Rating	Up to 1500 psi
Materials	Stainless steel, duplex steel, super alloys
Casting Weight	50g to 50kg
Surface Finish	Up to 3.2 Ra without machining
Tolerances	As per ISO 8062, up to ±0.1mm

Use Cases:

- Industrial applications
- Automotive
- General engineering
- Pumping systems in automotive
- General engineering applications
- General Engineering
- Fire Protection
- Food Processing
- Chemical Processing
- Fluid transport
- Chemical processing
- Water supply systems
- Mining operations
- Automotive components
- General engineering projects
- Other Industries
- Fluid transfer
- Hydraulic systems
- Medical
- Industrial Applications
- Valve Applications
- Agricultural irrigation
- HVAC systems
- Water supply
- Food processing

Standards	International performance standards
-----------	-------------------------------------

Use Cases:

- Industrial applications

Export Markets:

- Global

AI Layman Summary:

The Vacuum Pump Impeller is a crucial component used in vacuum pumps, which are devices that remove air and other gases from a sealed environment to create a vacuum. This particular impeller is made from high-quality alloys and is designed through precision investment casting, ensuring a perfect fit and optimal performance. It operates efficiently while preventing leaks, making it a reliable choice for various industrial applications around the world.

****Benefits of the Vacuum Pump Impeller:****

- ****High Efficiency:**** Engineered to maximize performance, helping to save energy and reduce operational costs.
- ****Leak-Proof Operation:**** Ensures that the vacuum created remains intact, which is essential for processes requiring a controlled environment.
- ****Long-Term Durability:**** Built to withstand harsh conditions, this impeller is a long-lasting investment for businesses needing reliable vacuum solutions.

Nickel Alloy Bar and Wire

Category: Bar and Wire

Nickel alloy bars and wires for various applications.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump manufacturing

AI Layman Summary:

Nickel Alloy Bar and Wire is a versatile material used in various industrial applications, particularly in the manufacturing of valves and pumps. Made primarily from nickel and other elements, it offers exceptional strength and durability, making it suitable for harsh environments. This material can withstand high temperatures and resist corrosion, which is essential for equipment that operates under challenging conditions.

****Key Benefits:****

- ****High Strength:**** With a minimum tensile strength of 551, it ensures that products made from this alloy can endure significant stress without breaking.
- ****Corrosion Resistance:**** The chemical composition, rich in nickel and chromium, helps prevent rust and degradation, extending the lifespan of the equipment.
- ****Versatile Applications:**** Ideal for use in critical components like valves and pumps, where reliability and performance are crucial.

Nickel Alloy Billets and Bars for Reforging

Category: Billets and Bars

Billets and bars made from nickel alloys for reforging.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- General Engineering
- Automotive

AI Layman Summary:

Nickel Alloy Billets and Bars for Reforging are specially designed metal products made from a strong mixture of nickel and other elements. These billets and bars are primarily used in the manufacturing and engineering industries, particularly in automotive applications. They provide a reliable foundation for creating various metal parts through a process called reforging, which reshapes the metal to meet specific needs.

Benefits include:

- **Strength and Durability**: With a minimum tensile strength of 690 and yield strength of 310, these products are built to withstand heavy use.
- **Versatile Use**: Ideal for general engineering and automotive sectors, making them suitable for a wide range of applications.
- **Customizable**: The reforging process allows for tailoring the metal's shape and properties to match specific requirements, ensuring optimal performance in final products.

Nickel Alloy Round Weaving Wire

Category: Wire

Round weaving wire made from nickel alloys.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- General Engineering
- Automotive

AI Layman Summary:

Nickel Alloy Round Weaving Wire is a specialized type of wire made from a mixture of metals, primarily nickel, designed for strength and durability. This wire is particularly useful in various industries, including engineering and automotive, due to its excellent resistance to corrosion and high tensile strength. Think of it as a strong and reliable material that can be used for making components that need to withstand tough conditions.

Key Benefits:

- **Durable and Strong**: With a minimum tensile strength of 690, this wire can handle significant stress without breaking.
- **Corrosion Resistant**: The nickel content provides resistance to rust and other forms of corrosion, making it ideal for outdoor or harsh environments.
- **Versatile Applications**: Suitable for general engineering and automotive uses, allowing for a wide range of practical applications.

Nickel-Chromium-Iron Alloy Tubes

Category: Tubes

Tubes made from nickel-chromium-iron alloys.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- General Engineering
- Automotive

AI Layman Summary:

Nickel-Chromium-Iron Alloy Tubes are specialized tubes made from a strong combination of nickel, chromium, and iron. These materials provide excellent resistance to heat and corrosion, making them suitable for demanding environments. They are primarily used in general engineering and automotive applications, where durability and reliability are crucial.

Benefits of Nickel-Chromium-Iron Alloy Tubes include:

- **High Strength**: With a minimum tensile strength of 680 MPa, these tubes can withstand significant pressure and stress.
- **Corrosion Resistance**: The unique alloy composition helps prevent rust and wear, prolonging the lifespan of products using these tubes.
- **Versatility**: Ideal for various applications, they can be used in different sectors, ensuring a wide range of usability.

ASTM A 240 S31277

Category: Plate, Sheet and Strip

Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 240 S31277 is a type of stainless steel plate, sheet, and strip specifically designed for use in pressure vessels. This stainless steel is made with a blend of chromium, nickel, and molybdenum, giving it excellent resistance to corrosion and high temperatures. It's commonly used in industries like automotive, general engineering, and for making valves and pumps where durability and safety are crucial.

Key Benefits:

- **Durability:** Its strong chemical composition ensures that it withstands harsh environments without degrading.
- **High Strength:** The material has a minimum tensile strength of 770 and yield strength of 360, making it reliable for heavy-duty applications.
- **Versatility:** Ideal for many applications, from automotive parts to industrial machinery, enhancing performance and safety.

ASTM A 249 N08904

Category: Tube

Standard Specification for Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 249 N08904 is a high-quality tube designed for various industrial applications, particularly in boilers, superheaters, and heat exchangers. It is made from a special alloy-steel that is both strong and resistant to corrosion, making it ideal for handling extreme temperatures and pressures. This seamless tube construction ensures reliability and longevity in demanding environments.

Key benefits include:

- **Durability:** The advanced alloy composition provides excellent strength and resistance to wear and tear.
- **Versatility:** Suitable for various uses including valves, pumps, and general engineering tasks, making it a valuable component in many industries, including automotive.
- **Performance:** With high tensile and yield strength, it can withstand significant stress, ensuring safety and efficiency in operations.

ASTM A 269 N08904

Category: Tube

Standard Specification for Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 269 N08904 is a type of tube made from a special blend of metals designed to withstand high temperatures and pressure, ideal for use in boilers, heat exchangers, and other engineering applications. This tube is seamless, meaning it has no joints that could weaken it, making it reliable for critical functions in industries like automotive and general engineering.

Key benefits include:

- **Durability**: It can handle extreme conditions without failing.
- **Versatility**: Suitable for a variety of applications, including valves and pumps.
- **Strong Composition**: Made with a mix of metals that enhance its strength and resistance to corrosion, ensuring it lasts longer in harsh environments.

ASTM A 249 S31277

Category: Tube

Standard Specification for Austenitic Steel Boiler, Superheater, Heat-Exchanger, and Condenser Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 249 S31277 is a type of tube made from a special kind of steel designed for high-performance applications like boilers, heat exchangers, and condensers. These tubes are engineered to withstand high temperatures and pressures, making them ideal for use in industries such as automotive, engineering, and valve and pump manufacturing.

Key benefits of ASTM A 249 S31277 tubes include:

- **Durability**: Made from strong austenitic steel, these tubes can handle extreme conditions without breaking down.
- **Corrosion Resistance**: With a carefully balanced chemical composition, they resist rust and degradation, ensuring a longer lifespan.
- **Versatility**: Suitable for various applications, they can be used in different machinery and systems, enhancing overall efficiency.

ASTM A 312 N08020

Category: Pipe

Standard Specification for Seamless and Heavily Cold Worked Austenitic Stainless Steel Pipes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 312 N08020 is a type of pipe made from high-quality stainless steel that is designed to be strong and durable. These pipes are seamless, meaning they have no joints or welds that can weaken them, and are especially tough due to a special cold-working process. This makes them ideal for use in demanding environments where reliability is crucial.

Here are some key benefits of ASTM A 312 N08020 pipes:

- **Strong and Reliable**: With a minimum tensile strength of 550, these pipes can withstand high pressure and stress.
- **Corrosion Resistant**: Made with a specific mixture of metals including nickel and chromium, they are highly resistant to rust and other forms of corrosion.
- **Versatile Use**: Suitable for a variety of applications such as valves, pumps, general engineering, and automotive parts, making them a great choice for many industries.

ASTM A 312 Alloy 20

Category: Pipe

Standard Specification for Seamless and Heavily Cold Worked Austenitic Stainless Steel Pipes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 312 Alloy 20 is a type of stainless steel pipe designed for demanding applications. It is created using a special process that makes it both seamless and heavily cold worked, which enhances its strength and durability. This pipe is particularly useful in industries where resistance to corrosion is critical, such as in valves, pumps, and various automotive and engineering applications.

Key benefits of ASTM A 312 Alloy 20 include:

- **High Strength**: With a minimum tensile strength of 550, it can withstand significant pressure and stress.
- **Corrosion Resistance**: Its unique chemical composition, which includes high levels of chromium and nickel, protects against rust and deterioration, making it perfect for harsh environments.
- **Versatile Applications**: Ideal for use in valves, pumps, and general engineering, making it suitable for a wide range of industrial needs.

ASTM A 297 Grade HU

Category: Casting

Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel Corrosion Resistant.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM A 297 Grade HU is a special type of casting material made from a mix of iron, chromium, and nickel that is designed to resist corrosion. This means it can withstand harsh environments without breaking down, making it ideal for use in various industries. It works by using specific chemical properties that give it strength and durability, ensuring that products made from this material last longer and perform better under pressure.

Key Benefits:

- **Corrosion Resistance**: Protects against rust and damage in tough conditions.
- **Strong and Durable**: With a minimum tensile strength of 450 and good elongation properties, it can handle significant stress without failing.
- **Versatile Applications**: Suitable for valves, pumps, automotive parts, and medical equipment, making it a go-to choice for engineers and manufacturers.

ASTM B 163 N06600

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N06600 is a high-quality tube made from seamless nickel and nickel alloy, specifically designed for use in condensers and heat exchangers. This product is ideal for various applications, including valves, pumps, general engineering, and automotive components. Its unique composition and properties make it suitable for environments that require durability and resistance to corrosion.

Key benefits of ASTM B 163 N06600 include:

- **High Strength**: With a tensile strength ranging from 552 to 758 MPa, it can withstand significant pressure and stress.
- **Corrosion Resistance**: The nickel content (minimum 72%) ensures that the tube resists rust and other forms of corrosion, extending its lifespan in harsh conditions.
- **Versatile Applications**: This tube can be used in a wide range of industries, making it a valuable addition to

any engineering project.

ASTM B 163 N06603

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N06603 is a type of seamless tube made from nickel and nickel alloy, specifically designed for use in condensers and heat exchangers. These tubes are used in various industries, including automotive and general engineering, to efficiently transfer heat and fluids. Their unique composition allows them to withstand high temperatures and corrosive environments, making them a reliable choice for demanding applications.

Key benefits include:

- ****Durability****: Made from a strong nickel alloy that provides excellent resistance to heat and corrosion.
- ****Versatility****: Suitable for use in various applications, from valves and pumps to general engineering projects.
- ****Reliable Performance****: With high tensile and yield strength, these tubes ensure longevity and efficiency in heat exchange processes.

ASTM B 163 N06686

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 163 N06686 is a type of tube made from nickel and nickel alloys, specifically designed for use in condensers and heat exchangers. These tubes are seamless, meaning they have no joints or welds, which makes them stronger and more reliable in handling high temperatures and pressures. They are particularly useful in industries that require efficient heat transfer, such as automotive and general engineering sectors.

Key benefits of using ASTM B 163 N06686 tubes include:

- **Durability**: Made from high-quality materials that resist corrosion and wear, ensuring a long lifespan.
- **High Performance**: Capable of withstanding extreme conditions, these tubes maintain their strength and flexibility under pressure.
- **Versatility**: Suitable for a variety of applications, including valves, pumps, and general engineering tasks, making them a practical choice for many industries.

ASTM B 163 N06690

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N06690 is a specialized type of tube made from nickel and nickel alloys, primarily used in heat exchangers and condensers. These tubes are seamless, which means they are manufactured without any joints, making them stronger and more reliable for transferring heat in various industrial applications. The unique chemical composition of this tube includes a high percentage of nickel and chromium, which enhances its resistance to corrosion and high temperatures, ensuring longevity and efficiency in demanding environments.

Key benefits of ASTM B 163 N06690 include:

- **Durability**: The strong material can withstand high pressures and temperatures, making it ideal for use in valves, pumps, and automotive applications.
- **Corrosion Resistance**: With its nickel-rich composition, this tube effectively resists rust and deterioration, increasing the lifespan of equipment.
- **Versatility**: It can be utilized in a range of engineering projects, providing reliable performance across different sectors.

ASTM B 163 N08800

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N08800 is a type of seamless tube made from nickel and nickel alloy, specifically designed for use in condensers and heat exchangers. These tubes are essential components in various industries where heat transfer is crucial, such as automotive, engineering, and pumping systems. Their unique composition allows them to withstand high temperatures and resist corrosion, making them ideal for demanding applications.

Key benefits of ASTM B 163 N08800 tubes include:

- **Durability**: Made with a strong blend of nickel, chromium, and other elements, these tubes can handle tough conditions without breaking down.
- **Versatility**: Suitable for a range of uses, including valves and pumps, making them a reliable choice for engineers and manufacturers.
- **Performance**: With good tensile and yield strength, these tubes ensure efficient operation in systems that require effective heat exchange.

ASTM B 163 N08120

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N08120 is a type of seamless tube made from nickel and nickel alloys, designed specifically for use in heat exchangers and condensers. These tubes are crucial in various industries where efficient heat transfer is necessary. They work by allowing fluids to flow through them, facilitating heat exchange without the risk of leaks often associated with welded tubes.

Benefits of ASTM B 163 N08120:

- **Durability**: Made from high-strength materials, these tubes can withstand high pressure and temperature, making them ideal for demanding applications.
- **Corrosion Resistance**: The nickel content provides excellent resistance to corrosion, ensuring a longer lifespan in harsh environments.
- **Versatility**: Suitable for a wide range of applications including valves, pumps, and automotive components, making it a valuable addition to various engineering projects.

ASTM B 163 N08801

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 163 N08801 is a special type of tube made from nickel and nickel alloy. It's designed for use in systems that need to transfer heat, like condensers and heat exchangers. This means it's very effective in cooling or heating liquids in various industries, ensuring that machinery operates efficiently without overheating.

****Key Benefits:****

- ****Durable Material:**** Made from a blend of nickel and other metals, it resists corrosion and withstands high temperatures, making it ideal for challenging environments.
- ****Versatile Use:**** This tube is useful in many applications, including valves, pumps, and automotive systems, ensuring it can meet various engineering needs efficiently.

ASTM B 163 N08811

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 163 N08811 is a type of seamless tube made from nickel and nickel alloy, specifically designed for use in condenser and heat-exchanger applications. This tube is essential in various industries where heat transfer is crucial, such as in valves, pumps, general engineering, and automotive systems. Its seamless design ensures durability and efficiency, making it a reliable choice for high-performance applications.

Key benefits of the ASTM B 163 N08811 tube include:

- ****High Resistance:**** The nickel alloy composition provides excellent resistance to corrosion and extreme temperatures, which helps maintain performance and longevity.
- ****Strong Mechanical Properties:**** With a minimum tensile strength of 448 and yield strength of 172, this tube can withstand significant stress, ensuring it performs well under pressure.
- ****Versatile Applications:**** Suitable for various uses across multiple industries, this tube is ideal for any application that requires reliable heat exchange and fluid transport.

ASTM B 163 N08810

Category: Tube

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 163 N08810 is a specialized tube made from seamless nickel and nickel alloy, primarily used in condensers and heat exchangers. These tubes are designed to efficiently transfer heat in various applications such as valves, pumps, and automotive systems. The material is known for its strong resistance to corrosion and high temperatures, making it ideal for demanding engineering environments.

****Key Benefits:****

- ****Durable and Strong:**** With a minimum tensile strength of 448 and yield strength of 172, it can withstand significant pressure and stress.
- ****Heat and Corrosion Resistant:**** The nickel and chromium content provides excellent protection against harsh environments, ensuring longevity and reliability in use.
- ****Versatile Applications:**** Suitable for a wide range of industries, enhancing efficiency in engineering, automotive, and general applications.

ASTM B 164 N04400

Category: Rod, Bar and Wire

Nickel-Copper Alloy Rod, Bar and Wire.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 164 N04400 is a type of rod, bar, and wire made from a special blend of nickel and copper, making it an ideal choice for various industrial applications. This material is particularly known for its strength and durability, which enables it to withstand tough conditions in places like valves and pumps, as well as in general engineering and automotive components.

Key benefits of ASTM B 164 N04400 include:

- ****High Strength**:** With a tensile strength ranging from 480 to 965 MPa, it can handle significant stress without breaking.
- ****Corrosion Resistance**:** The nickel content provides excellent resistance against corrosion, ensuring longevity in harsh environments.
- ****Versatile Use**:** Suitable for a wide range of applications, from automotive parts to general engineering tools.

ASTM B 167 N06025

Category: Pipe and Tube

Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Seamless Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 167 N06025 is a type of seamless pipe and tube made from special metal alloys, including nickel, chromium, and iron. These materials make the pipes strong and resistant to corrosion, which is essential for various industrial applications. This product is used in areas like valves, pumps, general engineering, and automotive industries, where durability and reliability are crucial.

Key Benefits:

- **Strength and Durability**: With a tensile strength of 680 to 860 Mpa, these pipes can withstand heavy loads and pressure.
- **Corrosion Resistance**: The unique chemical composition helps prevent rust and degradation, extending the life of the pipes in harsh environments.
- **Versatile Applications**: Ideal for use in valves and pumps, as well as in general engineering and automotive applications, making it a versatile choice for manufacturers.

ASTM B 626 N06007

Category: Tube

Nickel and Nickel-Cobalt Alloy Tube.

Key Features:

- High tensile strength
- Corrosion resistant
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump manufacturing

Export Markets:

- USA
- Global

AI Layman Summary:

The ASTM B 626 N06007 is a specialized tube made from a strong blend of nickel and nickel-cobalt alloy. This unique combination gives it impressive properties, making it ideal for demanding environments. It is particularly known for its high tensile strength, meaning it can withstand significant pressure without breaking, and is resistant to corrosion, ensuring durability even in harsh conditions. Additionally, it can operate efficiently at high temperatures, making it suitable for various industrial applications.

Key benefits include:

- **High Strength**: Able to handle tough conditions without damage.
- **Corrosion Resistance**: Stays intact and functional over time, even in corrosive environments.
- **Versatile Use**: Perfect for manufacturing valves and pumps, which are critical components in many industries.

This product is exported globally, including markets in the USA, providing businesses with reliable materials for their engineering needs.

ASTM B 626 N06035

Category: Tube

Nickel and Nickel-Cobalt Alloy Tube.

Key Features:

- High tensile strength
- Corrosion resistant
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump manufacturing

Export Markets:

- USA
- Global

AI Layman Summary:

ASTM B 626 N06035 is a specialized tube made from a strong combination of nickel and nickel-cobalt alloys. This tube is designed for demanding applications where durability and resistance to harsh conditions are crucial. It stands out due to its high tensile strength and ability to withstand corrosion, making it ideal for use in environments that involve extreme temperatures and exposure to chemicals.

Key benefits of ASTM B 626 N06035 include:

- **High Strength**: It can handle heavy loads without breaking, ensuring reliability in tough situations.
- **Corrosion Resistance**: It resists rust and damage from chemicals, leading to longer-lasting products.
- **High-Temperature Suitability**: Perfect for applications like valve and pump manufacturing, where heat can be a concern.

This tube is exported worldwide, including markets in the USA, and is a preferred choice for industries that require robust materials for critical components.

ASTM B 626 N06058

Category: Tube

Nickel and Nickel-Cobalt Alloy Tube.

Key Features:

- High tensile strength
- Corrosion resistant
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump manufacturing

Export Markets:

- USA
- Global

AI Layman Summary:

ASTM B 626 N06058 is a specialized tube made from a strong blend of nickel and nickel-cobalt alloy. This material is designed to withstand harsh conditions, making it ideal for use in high-temperature environments. The tube is known for its durability and resistance to corrosion, ensuring it lasts longer even in challenging applications.

Key benefits of ASTM B 626 N06058 include:

- **High strength**: It can handle significant stress without breaking, which is crucial for manufacturing robust machinery like valves and pumps.
- **Corrosion resistance**: Its unique composition prevents rust and degradation, making it suitable for use in various industries, including those that operate in extreme conditions.
- **Versatile applications**: It is widely used in industries across the USA and globally, proving its reliability and effectiveness in critical manufacturing processes.

ASTM B 626 N06210

Category: Tube

Nickel and Nickel-Cobalt Alloy Tube.

Key Features:

- High tensile strength
- Corrosion resistant
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing

- Pump manufacturing

Export Markets:

- USA
- Global

AI Layman Summary:

The ASTM B 626 N06210 is a specialized tube made from a strong nickel and nickel-cobalt alloy. This means it is designed to withstand extreme conditions, making it perfect for use in heavy-duty applications like valves and pumps. The unique combination of materials gives it impressive strength and resistance to corrosion, so it lasts longer even in harsh environments.

Key benefits include:

- ***High tensile strength***: It can handle significant stress without breaking.
- ***Corrosion resistant***: It won't easily rust or degrade, ensuring reliability over time.
- ***High-temperature suitability***: It performs well even in very hot conditions, making it versatile for different industries.

This product is available for export to the USA and globally, making it a great choice for manufacturers looking for durable components in their machinery.

ASTM B 564 N10624

Category: Nickel Alloy Forgings

Standard specification for nickel alloy forgings.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump components
- General engineering applications

Export Markets:

- USA
- Europe
- Asia

AI Layman Summary:

ASTM B 564 N10624 is a special type of nickel alloy used to create strong and durable metal parts. This material is designed to withstand tough conditions, making it perfect for various industrial applications. It contains a mix of metals that give it unique properties, such as high strength and resistance to rust and corrosion, especially in high-temperature environments.

Key benefits of ASTM B 564 N10624 include:

- **High Strength**: It can endure heavy loads without breaking.
- **Corrosion Resistance**: It doesn't easily rust or degrade, ensuring long-lasting performance.
- **High-Temperature Suitability**: It can be used in environments with extreme heat, making it ideal for industries like valve and pump manufacturing, as well as general engineering tasks.

This versatile alloy is used across various markets, including the USA, Europe, and Asia, making it a reliable choice for businesses looking for quality materials for their engineering needs.

ASTM B 564 N10629

Category: Nickel Alloy Forgings

Standard specification for nickel alloy forgings.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump components
- General engineering applications

Export Markets:

- USA
- Europe
- Asia

AI Layman Summary:

ASTM B 564 N10629 is a type of nickel alloy used to create strong and durable metal parts. This material is particularly known for its ability to resist rust and withstand high temperatures, making it ideal for various industrial applications. It is like a superhero for metal products, offering strength and protection in challenging environments.

Key Benefits:

- **High Strength**: Can handle heavy loads and stresses without breaking.
- **Corrosion Resistance**: Stays intact even in harsh, wet, or chemical-filled environments.
- **High-Temperature Suitability**: Performs well in very hot conditions, making it perfect for parts in engines and pumps.

Common uses for this material include making valves and pump components, as well as other engineering applications where strong and reliable metal is needed. It is popular in markets across the USA, Europe, and Asia.

ASTM B 564 N10665

Category: Nickel Alloy Forgings

Standard specification for nickel alloy forgings.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump components
- General engineering applications

Export Markets:

- USA
- Europe
- Asia

AI Layman Summary:

ASTM B 564 N10665 is a high-quality nickel alloy used primarily for making strong and durable forgings. This material is designed to withstand tough conditions, making it ideal for industries that require reliable components in challenging environments. Its unique properties allow it to perform well in high-temperature applications, offering excellent resistance to corrosion.

Key benefits of ASTM B 564 N10665 include:

- **High Strength**: Can handle heavy loads and stress without breaking.
- **Corrosion Resistance**: Withstands rust and other forms of deterioration, ensuring longevity.
- **Versatile Applications**: Suitable for various uses, such as valve manufacturing and pump components, making it a go-to choice for engineers in different fields across the USA, Europe, and Asia.

ASTM B 564 N10675

Category: Nickel Alloy Forgings

Standard specification for nickel alloy forgings.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump components
- General engineering applications

Export Markets:

- USA

- Europe
- Asia

AI Layman Summary:

ASTM B 564 N10675 is a type of nickel alloy that is specifically designed for making strong and durable forged products. These nickel alloys are known for their excellent resistance to corrosion and can withstand high temperatures, making them ideal for use in challenging environments. This material is often used in the manufacturing of components such as valves and pumps, which are essential in various engineering applications.

Key benefits of ASTM B 564 N10675 include:

- **High Strength:** It can handle significant stress without breaking.
- **Corrosion Resistance:** It does not easily rust or degrade when exposed to harsh chemicals or environments.
- **High-Temperature Suitability:** It remains stable and functional even in extreme heat conditions, making it reliable for various industrial applications.

ASTM B 564 N12160

Category: Nickel Alloy Forgings

Standard specification for nickel alloy forgings.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump components
- General engineering applications

Export Markets:

- USA
- Europe
- Asia

AI Layman Summary:

ASTM B 564 N12160 is a special type of metal known as a nickel alloy forging, designed to be incredibly strong and resistant to rust and corrosion. It's made with a blend of metals, including nickel and chromium, which gives it unique properties that make it useful in various demanding environments. This material can handle high temperatures and is often used in industries where reliability is crucial, such as making valves and pump components.

Key Benefits:

- **High Strength:** Can withstand significant pressure and stress, making it ideal for heavy-duty applications.
- **Corrosion Resistance:** Resists rust and wear, ensuring long-lasting performance even in harsh conditions.
- **Versatile Applications:** Suitable for various uses in engineering, particularly in the manufacturing of valves

and pumps.

This product is exported to markets in the USA, Europe, and Asia, showcasing its global demand and reliability in engineering solutions.

ASTM B 572 N06002

Category: Nickel Alloy Rod

Standard specification for nickel-iron-chromium-molybdenum alloy rod.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve manufacturing
- Pump components
- General engineering applications

Export Markets:

- USA
- Europe
- Asia

AI Layman Summary:

ASTM B 572 N06002 is a type of rod made from a special nickel alloy that consists of nickel, iron, chromium, and molybdenum. This material is known for its high strength and excellent ability to resist corrosion, making it ideal for use in tough environments. It can handle high temperatures, which means it's suitable for various industrial applications, particularly in manufacturing parts like valves and pumps.

****Key Benefits:****

- ****Durability:**** *With strong resistance to wear and tear, it lasts longer than many other materials.*
- ****Versatility:**** *Can be used in multiple industries across the USA, Europe, and Asia, making it a reliable choice for engineers.*
- ****High Performance:**** *Its mechanical properties, including high tensile and yield strength, ensure that it performs well under stress.*

ASTM B 575 N06022

Category: Nickel Alloy Plate

Standard specification for low-carbon nickel-chromium-molybdenum alloy plate.

Key Features:

- High strength
- Excellent corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Chemical Composition

Mechanical Properties

- Use Cases:
- Valve manufacturing
 - Pump components
 - General engineering applications

- Export Markets:
- USA
 - Europe
 - Asia

AI Layman Summary:

ASTM B 575 N06022 is a type of metal plate made from a special nickel alloy that includes chromium and molybdenum. This alloy is designed to be incredibly strong and resistant to rust and other forms of corrosion, making it ideal for use in tough environments. It can handle high temperatures, which is essential for industries that rely on machinery and equipment that operate under extreme conditions.

- Key benefits of ASTM B 575 N06022 include:
- **High Strength**: It can withstand significant stress and pressure, ensuring durability in demanding applications.
 - **Excellent Corrosion Resistance**: Protects against damage from chemicals and environmental factors, extending the lifespan of products made with it.
 - **Versatile Applications**: Commonly used in manufacturing valves, pump components, and various engineering tasks, making it a reliable choice for many industries around the world, including markets in the USA, Europe, and Asia.

ASTM A 753 Type 4

Category: Materials

Nickel-Iron Soft Magnetic Alloys.

- Key Features:
- Chemical Composition: Nickel 79.000 - 82.000%
 - Tensile Strength: 580 - 1100 MPa
 - Yield Strength: 220 - 1030 MPa

Technical Specifications:

Carbon	0.050 max.
Silicon	0.500 max.
Manganese	0.800 max.
Chromium	0.300 max.
Sulphur	0.010 max.
Molybdenum	3.500 - 6.000
Phosphorus	0.020 max.
Cobalt	0.500 max.
Copper	0.300 max.
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 753 Type 4 is a specialized type of material known as Nickel-Iron Soft Magnetic Alloys. These materials are designed to effectively conduct magnetic fields and are particularly useful in various engineering applications. They consist mainly of nickel, making them well-suited for tasks that require good magnetic performance and strength.

Key benefits of ASTM A 753 Type 4 include:

- **High Strength:** With a tensile strength ranging from 580 to 1100 MPa, this alloy can withstand significant stress without breaking.
- **Versatile Applications:** It is suitable for use in valves, pumps, general engineering projects, and the automotive industry, making it a flexible choice for manufacturers.
- **Controlled Composition:** The alloy has a carefully balanced chemical makeup, ensuring optimal performance in magnetic applications.

ASTM A 946 Grade S70003

Category: Materials

Chromium, Chromium-Nickel and Silicon Alloy Steel Plate, Sheet and Strip.

Key Features:

- Chemical Composition: Nickel 22.000 - 25.000%
- Tensile Strength: 540 min. MPa
- Yield Strength: 240 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	6.500 - 8.000
Manganese	2.000 max.
Chromium	8.000 - 11.000
Sulphur	0.010 max.
Molybdenum	0.500 max.
Phosphorus	0.025 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 946 Grade S70003 is a special type of steel that is made using a mix of chromium, nickel, and silicon. This unique combination gives it exceptional strength and durability, making it an ideal choice for various industrial applications. The material is available in plates, sheets, and strips, which can be easily shaped and used in different projects, from machinery to automotive parts.

Key benefits of ASTM A 946 Grade S70003 include:

- **High Strength**: With a minimum tensile strength of 540 MPa, it can withstand heavy loads and pressures.
- **Versatile Use**: Suitable for making valves, pumps, and parts in general engineering and automotive industries, it adapts to various needs.
- **Durability**: Its chemical composition helps prevent wear and tear, ensuring long-lasting performance in demanding environments.

ASTM A 954 S70003

Category: Materials

Austenitic Chromium-Nickel-Silicon Alloy Steel Seamless and Pipe.

Key Features:

- Chemical Composition: Nickel 22.000 - 25.000%
- Tensile Strength: 540 min. MPa
- Yield Strength: 240 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	6.500 - 8.000
Manganese	2.000 max.
Chromium	8.000 - 11.000
Sulphur	0.010 max.
Molybdenum	0.500 max.
Phosphorus	0.025 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 954 S70003 is a special type of steel that's made without seams, which means it has a smooth and strong structure. This steel is created using a mix of metals, including chromium, nickel, and silicon, making it very durable and resistant to wear and tear. It's known for its high tensile strength, meaning it can handle heavy loads without breaking, and has a good yield strength, so it can withstand pressure without deforming.

Some key benefits of using ASTM A 954 S70003 include:

- **Versatile Applications**: Ideal for valves, pumps, and general engineering projects, as well as use in the automotive industry.
- **Strong and Reliable**: With a minimum tensile strength of 540 MPa, this steel is built to last, providing safety and durability in demanding conditions.
- **Corrosion Resistance**: The alloy's chemical properties help it resist damage from environmental factors, making it a smart choice for long-term use.

ASTM A 953 S70003

Category: Materials

Austenitic Chromium-Nickel-Silicon Alloy Steel Seamless Tubing.

Key Features:

- Chemical Composition: Nickel 22.000 - 25.000%
- Tensile Strength: 540 min. MPa
- Yield Strength: 240 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	6.500 - 8.000
Manganese	2.000 max.
Chromium	8.000 - 11.000
Sulphur	0.010 max.
Molybdenum	0.500 max.
Phosphorus	0.025 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 953 S70003 is a specialized type of seamless tubing made from a unique steel alloy that includes chromium, nickel, and silicon. This material is designed for various industrial applications, providing excellent strength and durability. It is particularly useful in sectors such as automotive, valve manufacturing, and general engineering where robust materials are essential.

Key benefits of ASTM A 953 S70003 include:

- High tensile strength of at least 540 MPa, ensuring it can withstand heavy loads and pressures.
- Good yield strength starting at 240 MPa, which means it can maintain its shape under stress without deforming.
- Its chemical composition includes a significant amount of nickel (22-25%), contributing to its resistance to corrosion and heat, making it ideal for demanding environments.

ASTM A 959 S31277

Category: Materials

Seamless and Heavily Cold Worked Austenitic Stainless Steel Pipes.

Key Features:

- Chemical Composition: Nickel 26.000 - 28.000%
- Tensile Strength: 1620 min. MPa
- Yield Strength: 1345 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	0.500 max.
Manganese	3.000 max.
Chromium	20.500 - 23.000
Molybdenum	6.500 - 8.000

Copper	0.500 - 1.500
Nitrogen	0.300 - 0.400
Phosphorus	0.030 max.
Sulphur	0.010 max.
Nickel	26.000 - 28.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 959 S31277 is a type of high-quality stainless steel pipe known for its strength and durability. Made from a special blend of metals, including a significant amount of nickel, this pipe is designed to withstand tough conditions in various applications. It is seamless and heavily cold worked, which means it is produced without any joints and has been strengthened through a specific manufacturing process. This makes it ideal for use in industries such as automotive, general engineering, pumps, and valves.

****Key Benefits:****

- ****High Strength:**** With a tensile strength of at least 1620 MPa, it can handle heavy loads and pressure.
- ****Corrosion Resistance:**** The unique chemical composition ensures that it resists rust and other forms of corrosion, extending its lifespan.
- ****Versatile Usage:**** Suitable for a wide range of applications, making it a reliable choice for various engineering needs.

ASTM A 959 Type 660

Category: Materials

Seamless Nickel and Nickel Alloy Pipe, Tube, Sheet, Strip, Plate, Round Bar, Flat Bar, Forging Stock, Hexagon and Wire.

Key Features:

- Chemical Composition: Nickel 24.000 - 27.000%
- Tensile Strength: 600 - 1500 MPa
- Yield Strength: 737.4 min. MPa

Technical Specifications:

Carbon	0.080 max.
Silicon	1.000 max.
Manganese	2.000 max.
Chromium	13.500 - 16.000
Sulphur	0.030 max.
Molybdenum	1.000 - 1.500
Aluminium	0.350 max.
Boron	0.003 - 0.010
Titanium	1.900 - 2.300
Vanadium	0.100 - 0.500

Iron

Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 959 Type 660 is a type of pipe and metal product made from nickel and nickel alloys. These materials come in various forms such as seamless pipes, tubes, sheets, and bars, making them versatile for different applications. They are designed to withstand high levels of stress and corrosion, making them ideal for use in demanding environments like automotive and engineering sectors.

Key benefits of ASTM A 959 Type 660 include:

- **High Strength**: With a tensile strength ranging from 600 to 1500 MPa, these materials are strong and durable.
- **Corrosion Resistance**: The nickel composition helps in resisting rust and corrosion, extending the life of products made from this material.
- **Versatile Applications**: Suitable for various uses such as in valves, pumps, and automotive components, making it a reliable choice for manufacturers.

ASTM A 965 N08904

Category: Materials

Steel Forgings, Austenitic, for Pressure and High Temperature Parts.

Key Features:

- Chemical Composition: Nickel 23.000 - 28.000%
- Tensile Strength: 490 min. MPa
- Yield Strength: 215 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	1.000 max.
Manganese	2.000 max.
Chromium	19.000 - 23.000
Sulphur	0.030 max.
Molybdenum	4.000 - 5.000
Phosphorus	0.040 max.
Copper	1.000 - 2.000
Nitrogen	0.100 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 965 N08904 is a special type of steel designed for making strong and durable parts that need to withstand high pressure and temperature. This material is particularly useful in industries where equipment must operate safely under tough conditions, such as in valves, pumps, and automotive applications. Its austenitic structure helps it resist corrosion and maintain stability even in extreme environments.

Key benefits of ASTM A 965 N08904 include:

- **High Strength**: With a minimum tensile strength of 490 MPa, it can handle heavy loads without breaking.
- **Corrosion Resistance**: The chemical composition, including a high nickel content, ensures that it resists rust and other forms of deterioration.
- **Versatility**: Ideal for various applications, including general engineering, automotive, and more, making it a reliable choice for manufacturers.

ASTM A 968 S70003

Category: Materials

Chromium, Chromium-Nickel, and Silicon Alloy Steel Bars and Shapes for Corrosion and Heat-Resisting Service.

Key Features:

- Chemical Composition: Nickel 22.000 - 25.000%
- Tensile Strength: 540 min. MPa
- Yield Strength: 240 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	6.500 - 8.000
Manganese	2.000 max.
Chromium	8.000 - 11.000
Sulphur	0.010 max.
Molybdenum	0.500 max.
Phosphorus	0.025 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 968 S70003 is a specialized type of steel designed to withstand harsh conditions, including exposure to heat and corrosion. This steel is made from a blend of chromium, nickel, and silicon, which gives it unique properties that make it suitable for a variety of applications. It is primarily used in industries like automotive, engineering, and for manufacturing components like valves and pumps.

Key benefits of ASTM A 968 S70003 include:

- **High Strength**: It has a minimum tensile strength of 540 MPa, which means it can handle a lot of stress without breaking.
- **Corrosion Resistance**: The alloy composition helps protect it from rust and other types of damage due to moisture and chemicals.
- **Versatility**: Its properties make it suitable for a wide range of applications, ensuring durability and reliability

in various engineering tasks.

ASTM A 988 N08028

Category: Materials

Hot Isostatically-Pressed Stainless Steel Flanges, Fittings, Valves and Parts for High Temperature Service.

Key Features:

- Chemical Composition: Nickel 30.000 - 34.000%
- Tensile Strength: 500 min. MPa
- Yield Strength: 214 min. MPa

Technical Specifications:

Carbon	0.030 max.
Silicon	1.000 max.
Manganese	2.500 max.
Chromium	26.000 - 28.000
Sulphur	0.030 max.
Molybdenum	3.000 - 4.000
Copper	0.600 - 1.400
Phosphorus	0.030 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 988 N08028 is a type of high-quality stainless steel designed for making flanges, fittings, valves, and other components that work in high-temperature environments. This material is hot isostatically pressed, meaning it has been treated under high pressure and temperature to enhance its strength and durability. It is particularly useful in industries where metal parts are exposed to extreme conditions, such as valves in pumps and automotive applications.

****Key Benefits:****

- ****High Strength:**** It has a minimum tensile strength of 500 MPa, ensuring it can withstand heavy loads without deforming.
- ****Corrosion Resistance:**** The chemical composition, which includes a high nickel content, helps resist rust and corrosion, extending the lifespan of the components made from it.
- ****Versatile Applications:**** Ideal for various uses in industries like general engineering and automotive, making it a reliable choice for manufacturers.

ASTM A 312 N08028

Category: Materials

Seamless and Heavily Cold Worked Austenitic Stainless Steel Pipe.

Key Features:

- Chemical Composition: Nickel 30.000 - 34.000%
- Tensile Strength: 500 min. MPa

- Yield Strength: 214 min. MPa

Technical Specifications:

Carbon	0.030 max.
Silicon	1.000 max.
Manganese	2.500 max.
Chromium	26.000 - 28.000
Sulphur	0.030 max.
Molybdenum	3.000 - 4.000
Copper	0.600 - 1.400
Phosphorus	0.030 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 312 N08028 is a type of stainless steel pipe that is made without seams and is specially treated to enhance its strength. This material is particularly useful in various industries due to its durability and resistance to corrosion. It contains a high percentage of nickel, which contributes to its exceptional properties. The pipe is designed to withstand high pressure and is commonly used in applications such as valves, pumps, and general engineering tasks.

****Key Benefits:****

- ****High Strength:**** With a tensile strength of at least 500 MPa, it can handle significant stress without breaking.
- ****Corrosion Resistance:**** The chemical composition, including nickel and chromium, helps protect against rust and deterioration, making it ideal for long-lasting use in harsh environments.
- ****Versatile Applications:**** Suitable for various industries, including automotive and engineering, ensuring it can meet diverse needs.

ASTM A 213 Alloy 20

Category: Materials

Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater and Heat-Exchanger Tubes.

Key Features:

- Chemical Composition: Nickel 32.000 - 38.000%
- Tensile Strength: 550 min. MPa
- Yield Strength: 240 min. MPa

Technical Specifications:

Carbon	0.070 max.
Silicon	1.000 max.
Manganese	2.000 max.
Chromium	19.000 - 21.000
Sulphur	0.035 max.

Molybdenum	2.000 - 3.000
Copper	3.000 - 4.000
Nb + Ta	1.000 max.
Phosphorus	0.045 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A 213 Alloy 20 is a type of high-quality steel tube designed for use in boilers, superheaters, and heat exchangers. These tubes are seamless, meaning they are made without any joints, which enhances their strength and reliability. The alloy is composed mainly of nickel and features a unique blend of other elements like chromium and molybdenum to improve its performance in high-temperature and corrosive environments.

****Benefits of ASTM A 213 Alloy 20:****

- ****Strong and Durable:**** With a minimum tensile strength of 550 MPa, these tubes can withstand significant pressure and stress, making them ideal for demanding applications such as valves and pumps.
- ****Corrosion Resistance:**** The alloy's chemical composition enables it to resist corrosion, ensuring a longer lifespan and less maintenance.
- ****Versatile Applications:**** Suitable for general engineering and automotive uses, making it a valuable material across various industries.

ASTM B 124 C48500

Category: Materials

Naval Brass Rod, Bar and Shapes.

Key Features:

- Chemical Composition: Copper 59.000 - 62.000%
- Tensile Strength: 345 min. MPa
- Yield Strength: 140 min. MPa

Technical Specifications:

Iron	0.100 max.
Lead	1.300 - 2.200
Tin	0.500 - 1.000
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 124 C48500 is a type of naval brass that comes in rods, bars, and various shapes. It's made primarily of copper, with specific amounts of other metals like lead and tin. This material is designed to be strong and

durable, making it ideal for various applications, especially in marine environments. Its chemical composition ensures that it can withstand harsh conditions without corroding easily.

- Key benefits of ASTM B 124 C48500 include:*
- *Excellent strength: With a tensile strength of at least 345 MPa, it can handle heavy loads without breaking.*
 - *Versatility: This brass can be used in valves, pumps, automotive parts, and general engineering projects, making it suitable for a range of industries.*
 - *Corrosion resistance: The mixture of metals helps it resist rust and degradation, increasing its lifespan in applications exposed to moisture and saltwater.*

ASTM B 124 CW617N

Category: Materials

Copper and Copper Alloy Rod for Free Machining Purpose.

Key Features:

- Chemical Composition: Copper 57.000 - 59.000%
- Tensile Strength: 360 min. MPa
- Yield Strength: 150 min. MPa

Technical Specifications:

Aluminium	0.050 max.
Iron	0.300 max.
Nickel	0.300 max.
Lead	1.600 - 2.500
Tin	0.300 max.
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 124 CW617N is a type of copper rod designed specifically for easy machining. This means it's ideal for making parts that require precise shapes and sizes without causing excessive wear on tools. With a high copper content of 57% to 59%, this material is both durable and reliable, making it suitable for various applications in industries such as automotive, pumps, and valves.

- Key benefits of using ASTM B 124 CW617N include:*
- *High tensile strength of at least 360 MPa, ensuring it can withstand significant force without breaking.*
 - *Good yield strength starting at 150 MPa, which means it maintains its shape under pressure.*
 - *Versatile use in many engineering sectors, helping manufacturers produce high-quality components efficiently.*

ASTM B 124 CW612N

Category: Materials

Copper and Copper Alloy Rod for Free Machining Purpose.

Key Features:

- Chemical Composition: Copper 59.000 - 60.000%
- Tensile Strength: 360 min. MPa
- Yield Strength: 150 min. MPa

Technical Specifications:

Aluminium	0.050 max.
Iron	0.300 max.
Nickel	0.300 max.
Lead	1.600 - 2.500
Tin	0.300 max.
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 124 CW612N is a type of copper rod specifically designed for easy machining, making it an ideal material for various engineering applications. This product is composed mainly of copper, with a high percentage ensuring durability and strength. It is commonly used in manufacturing components for valves, pumps, and automotive parts due to its excellent mechanical properties and resistance to wear and tear.

Key Benefits:

- ****Strong and Durable****: With a tensile strength of at least 360 MPa, this copper alloy can withstand significant stress and pressure, ensuring reliable performance in demanding environments.
- ****Versatile Applications****: Suitable for a range of uses in general engineering, automotive, and more, making it a flexible choice for manufacturers.
- ****Easy to Machine****: Designed for free machining, it can be easily shaped and cut into various forms, saving time and resources during production.

ASTM B 127 N04400

Category: Materials

Nickel Plate, Sheet and Strip.

Key Features:

- Chemical Composition: Nickel 63.000 min.
- Tensile Strength: 485 - 585 MPa
- Yield Strength: 195 min. MPa

Technical Specifications:

Carbon	0.300 max.
Silicon	0.500 max.
Manganese	2.000 max.
Copper	28.000 - 34.000
Sulphur	0.024 max.
Iron	2.500 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 127 N04400 is a type of nickel plate, sheet, and strip that is used in various industries for its strength and resistance to corrosion. This material is primarily made of nickel, providing it with excellent durability and reliability for different applications. It is especially useful in environments where high strength and resistance to harsh conditions are required.

Key benefits of ASTM B 127 N04400 include:

- **High Strength**: With a tensile strength of 485 to 585 MPa, it can withstand heavy loads and stress.
- **Corrosion Resistance**: The nickel composition ensures that it resists rust and degradation over time, making it ideal for use in pumps, valves, and automotive parts.
- **Versatile Applications**: It is suitable for a wide range of uses in general engineering and automotive sectors, ensuring it meets various industrial needs.

ASTM B 160 N02200

Category: Materials

Nickel Rod and Bar.

Key Features:

- Chemical Composition: Nickel 99.000 min.
- Tensile Strength: 380 min. MPa
- Yield Strength: 105 min. MPa

Technical Specifications:

Carbon	0.150 max.
Silicon	0.350 max.
Manganese	0.350 max.
Copper	0.250 max.
Sulphur	0.010 max.
Iron	0.400 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 160 N02200 is a high-quality nickel rod and bar designed for various industrial applications. Made with a minimum of 99% nickel, this material offers excellent strength and durability, making it ideal for demanding environments. Its key features include a tensile strength of at least 380 MPa and a yield strength of at least 105 MPa, ensuring it can withstand significant pressure and stress without failing.

This nickel product is suitable for use in critical components such as valves and pumps, as well as in general engineering and automotive applications. The chemical composition is strictly controlled, with low levels of

impurities like carbon and sulfur, which contributes to its reliable performance.

- **Benefits:****
- *High strength and durability for long-lasting performance.*
 - *Versatile applications across various industries.*
 - *Low impurity levels enhance material integrity.*

ASTM B 160 N02201

Category: Materials

Nickel Rod and Bar.

- Key Features:**
- Chemical Composition: Nickel 99.000 min.
 - Tensile Strength: 345 min. MPa
 - Yield Strength: 70 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	0.350 max.
Manganese	0.350 max.
Copper	0.250 max.
Sulphur	0.010 max.
Iron	0.400 max.

- Use Cases:**
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM B 160 N02201 is a type of high-quality nickel rod and bar used in various industrial applications. It is made primarily of nickel, ensuring excellent resistance to corrosion and durability. This material is particularly useful in environments where strength and reliability are essential, making it a popular choice for manufacturing components in valves, pumps, general engineering, and automotive industries.

- Key benefits include:*
- ****High Strength**:** *With a minimum tensile strength of 345 MPa, it can withstand significant stress without breaking.*
 - ****Corrosion Resistance**:** *Made from 99% pure nickel, it offers superior resistance to rust and chemical damage.*
 - ****Versatile Applications**:** *Ideal for use in critical components like valves and pumps, ensuring longevity and performance in demanding conditions.*

ASTM B 161 N02200

Category: Materials

Nickel Seamless Pipe and Tube.

Key Features:

- Chemical Composition: Nickel 99.000 min.
- Tensile Strength: 380 - 515 MPa
- Yield Strength: 80 min. MPa

Technical Specifications:

Carbon	0.150 max.
Silicon	0.350 max.
Manganese	0.350 max.
Copper	0.250 max.
Sulphur	0.010 max.
Iron	0.400 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 161 N02200 is a high-quality nickel seamless pipe and tube that is designed for various industrial applications. Made primarily of nickel, this material is known for its strength and durability, making it an excellent choice for components that require resistance to corrosion and high temperatures. The pipe boasts impressive mechanical properties, including a tensile strength that ranges from 380 to 515 MPa, ensuring it can withstand significant stress and pressure.

****Key Benefits:****

- ****High Nickel Content:**** At least 99% nickel, which provides excellent resistance to corrosion and oxidation.
- ****Strong Performance:**** With a yield strength of 80 MPa or more, this pipe can handle demanding conditions in various applications.
- ****Versatile Use:**** Ideal for valves, pumps, and general engineering projects, as well as in the automotive industry.

ASTM B 161 N02201

Category: Materials

Nickel Seamless Pipe and Tube.

Key Features:

- Chemical Composition: Nickel 99.000 min.
- Tensile Strength: 345 - 480 MPa
- Yield Strength: 70 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	0.350 max.
Manganese	0.350 max.
Copper	0.250 max.
Sulphur	0.010 max.
Iron	0.400 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 161 N02201 is a high-quality nickel seamless pipe and tube designed for various industrial applications. Made with a minimum of 99% nickel, this material is known for its excellent strength and durability. It can withstand significant pressure, making it ideal for use in valves, pumps, and general engineering tasks, as well as in the automotive industry.

Key benefits of ASTM B 161 N02201 include:

- **Strength and Durability**: With a tensile strength ranging from 345 to 480 MPa, this pipe can handle high-stress environments without compromising its integrity.
- **Versatile Applications**: Its reliable performance makes it suitable for a wide range of applications, ensuring you can use it in various engineering projects.
- **Chemical Composition**: The low levels of elements like carbon and sulfur enhance its corrosion resistance, making it a long-lasting choice for demanding conditions.

ASTM B 162 N02201

Category: Materials

Nickel Plate, Sheet and Strip.

Key Features:

- Chemical Composition: Nickel 99.000 min.
- Tensile Strength: 345 min. MPa
- Yield Strength: 80 min. MPa

Technical Specifications:

Carbon	0.020 max.
Silicon	0.350 max.
Manganese	0.350 max.
Copper	0.250 max.
Sulphur	0.010 max.
Iron	0.400 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 162 N02201 is a high-quality nickel plate, sheet, and strip material used in various industries. This product is made of at least 99% nickel, making it highly resistant to corrosion and suitable for applications that require durability and strength. It has a minimum tensile strength of 345 MPa and a yield strength of 80 MPa, ensuring it can withstand significant stress without breaking.

Benefits of using ASTM B 162 N02201 include:

- **Corrosion Resistance**: Perfect for environments where other materials might fail.
- **High Strength**: Offers excellent tensile and yield strength for demanding applications such as valves, pumps, and automotive components.
- **Versatile Applications**: Suitable for general engineering and a wide range of industrial uses.

ASTM B 163 N02200

Category: Materials

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Key Features:

- Chemical Composition: Nickel 99.000 min.
- Tensile Strength: 379 - 552 MPa
- Yield Strength: 103 min. MPa

Technical Specifications:

Carbon	0.150 max.
Silicon	0.350 max.
Manganese	0.350 max.
Copper	0.250 max.
Sulphur	0.010 max.
Iron	0.400 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N02200 is a type of seamless tubing made from nickel and nickel alloys, which are specially designed for use in condensers and heat exchangers. These tubes are highly durable and can handle high temperatures and pressures, making them ideal for applications in various industries such as automotive, engineering, and more.

Key benefits include:

- **High Strength**: With a tensile strength ranging from 379 to 552 MPa, these tubes are strong and can withstand demanding conditions.
- **Corrosion Resistance**: Made from 99% pure nickel, they offer excellent resistance to corrosion, ensuring a longer lifespan in harsh environments.
- **Versatile Applications**: Suitable for use in valves, pumps, and general engineering projects, making them a valuable addition to many industrial processes.

ASTM B 163 N06025

Category: Materials

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Key Features:

- Chemical Composition: Nickel Balance
- Tensile Strength: 680 min. MPa

- Yield Strength: 270 min. MPa

Technical Specifications:

Carbon	0.150 - 0.250
Silicon	0.500 max.
Manganese	0.150 max.
Copper	0.100 max.
Sulphur	0.010 max.
Chromium	24.000 - 26.000
Aluminium	1.800 - 2.400
Titanium	0.100 - 0.2000
Phosphorus	0.020 max.
Iron	8.000 - 11.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N06025 is a type of tube made from seamless nickel and nickel alloy, specially designed for use in condensers and heat exchangers. These tubes are known for their strong and durable properties, making them ideal for high-performance applications in various industries. They work by allowing heat transfer efficiently while resisting corrosion, which is essential in environments where heat and moisture are present.

****Key Benefits:****

- ****Strong and Reliable:**** With a tensile strength of at least 680 MPa, these tubes can handle high pressure and temperature without failing.
- ****Corrosion Resistance:**** The unique chemical composition helps prevent rust and degradation, ensuring a longer lifespan in tough conditions.
- ****Versatile Applications:**** Suitable for use in valves, pumps, general engineering, and the automotive sector, making them a flexible choice for many engineering needs.

ASTM B 163 N04400

Category: Materials

Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes.

Key Features:

- Chemical Composition: Nickel 63.000 min.
- Tensile Strength: 483 - 621 MPa
- Yield Strength: 193 min. MPa

Technical Specifications:

Carbon	0.300 max.
Silicon	0.500 max.
Manganese	2.000 max.

Copper	28.000 - 34.000
Sulphur	0.024 max.
Iron	2.500 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 163 N04400 is a type of seamless tubes made from nickel and nickel alloys, designed specifically for use in condensers and heat exchangers. These tubes are essential components in various industries, ensuring efficient heat transfer in systems like valves, pumps, and automotive applications. Their construction allows them to withstand high pressures and temperatures, making them reliable for demanding environments.

****Key Benefits:****

- ****Durability:**** Made from high-quality nickel, these tubes offer excellent strength and resistance to corrosion.
- ****Versatility:**** Suitable for a wide range of applications including general engineering and automotive uses, ensuring broad usability across different sectors.
- ****Performance:**** With a tensile strength range of 483 to 621 MPa, they are designed to handle significant stress, ensuring long-lasting performance.

ASTM B 619 N06985

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Carbon	0.015 max.
Silicon	1.000 max.
Manganese	1.000 max.
Chromium	21.000 - 23.500
Sulphur	0.030 max.
Molybdenum	6.000 - 8.000
Phosphorus	0.040 max.
Nickel	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06985 is a type of pipe made from a special blend of nickel and nickel-cobalt alloys. This means it is designed to be durable and resistant to various environmental factors, making it ideal for demanding applications. The pipe is utilized in industries such as automotive, valve manufacturing, and pump production, where strength and reliability are crucial.

- Key benefits include:
- **Strong and Durable**: The chemical composition ensures that the pipe can withstand harsh conditions without degrading.
 - **Versatile Use**: Suitable for a wide range of applications, from general engineering to automotive parts.
 - **High-Quality Standards**: Adheres to strict specifications for mechanical properties, ensuring it performs well in various situations.

ASTM B 619 N08320

Category: Pipe
Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

- Key Features:
- Chemical Composition
 - Mechanical Properties

Technical Specifications:

Carbon	0.050 max.
Silicon	1.000 max.
Manganese	2.500 max.
Chromium	21.000 - 23.000
Sulphur	0.030 max.
Molybdenum	4.000 - 6.000
Nickel	25.000 - 27.000

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM B 619 N08320 is a type of pipe made from nickel and nickel-cobalt alloy, designed to withstand harsh environments and extreme conditions. This standard specification outlines the chemical composition and mechanical properties of the pipe, ensuring it is durable and reliable for various applications. It is commonly used in industries such as automotive, general engineering, and in components like valves and pumps.

- Key benefits include:
- **High Durability**: Its robust material composition makes it suitable for demanding applications.
 - **Versatile Use**: Ideal for many industries, including automotive and engineering.
 - **Standardized Quality**: Adherence to strict specifications guarantees consistent performance.

ASTM B 619 N08031

Category: Pipe
Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Carbon	0.015 max.
Silicon	0.300 max.
Manganese	2.000 max.
Chromium	26.000 - 28.000
Sulphur	0.010 max.
Molybdenum	6.000 - 7.000
Nickel	30.000 - 32.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N08031 is a type of pipe made from a special combination of nickel and cobalt alloys. This pipe is designed for various applications, including valves and pumps, making it essential in industries like automotive and general engineering. The unique chemical makeup gives it strong resistance to corrosion and wear, ensuring longevity and reliability in demanding environments.

Key benefits include:

- **Durability**: The high nickel content provides excellent resistance to harsh conditions.
- **Versatility**: Suitable for use in multiple industries, from automotive to general engineering.
- **High Performance**: Mechanical properties are tailored for optimal function in critical applications.

ASTM B 619 N10001

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Carbon	0.050 max.
Silicon	1.000 max.
Manganese	1.000 max.
Chromium	1.000 max.
Nickel	Balance

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N10001 is a type of pipe made from nickel and nickel-cobalt alloys, designed for various industrial applications. This pipe is known for its strong chemical composition and mechanical properties, making it suitable for environments where durability and resistance to corrosion are essential. It can be used in valves, pumps, general engineering projects, and even in the automotive industry.

- Key Benefits:
- **Durability**: The materials used in this pipe ensure it can withstand harsh conditions and last longer than standard pipes.
 - **Versatility**: Its applications range from automotive to general engineering, making it a reliable choice for many industries.
 - **High Quality**: With strict specifications on chemical composition, this pipe guarantees consistent performance and reliability in various settings.

ASTM B 619 N10242

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Carbon	0.030 max.
Silicon	0.800 max.
Manganese	0.800 max.
Chromium	7.000 - 9.000
Nickel	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 619 N10242 is a type of pipe made from a special blend of nickel and nickel-cobalt alloys. This pipe is designed to be strong and resistant to various environmental factors, making it ideal for use in demanding applications. It has defined chemical and mechanical properties that ensure it performs well under pressure and in different temperatures.

- Key benefits of using ASTM B 619 N10242 pipe include:
- **Durability**: The alloy composition provides excellent resistance to corrosion and wear.
 - **Versatility**: Suitable for various applications such as valves, pumps, general engineering, and automotive parts.
 - **Quality Assurance**: Manufactured to meet specific industry standards, ensuring reliable performance in critical situations.

ASTM B 622 N06002

Category: Pipe and Tube

Standard Specification for Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Carbon	0.050 - 0.150
Silicon	1.000 max.
Manganese	1.000 max.
Chromium	20.500 - 23.000
Nickel	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N06002 is a type of pipe and tube made from a special combination of nickel and nickel-cobalt alloy. This product is designed to be strong and resistant to wear, making it ideal for a variety of applications. It is particularly useful in industries like automotive, general engineering, and for parts like valves and pumps.

Key benefits include:

- **Durability**: The material composition ensures it can withstand harsh conditions and resist corrosion.
- **Versatility**: Suitable for multiple uses across different sectors, from engineering to automotive applications.
- **Reliable Performance**: The specific chemical makeup guarantees consistent quality and mechanical properties, ensuring safety and efficiency in operations.

ASTM B 649 N08031

Category: Bar and Wire

Ni-Fe-Cr-Mo-Cu Low-Carbon Alloy.

Technical Specifications:

Carbon	0.015 max.
Silicon	0.300 max.
Manganese	2.000 max.
Chromium	26.000 - 28.000
Sulphur	0.010 max.
Molybdenum	6.000 - 7.000
Phosphorus	0.020 max.
Nitrogen	0.150 - 0.250
Copper	1.000 - 1.400

Nickel	30.000 - 32.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 649 N08031 is a specialized type of low-carbon alloy made from nickel, iron, chromium, molybdenum, and copper. This material is designed to be strong and resistant to corrosion, making it ideal for various engineering applications. It contains specific amounts of elements like chromium (26-28%) and nickel (30-32%), which enhance its durability and performance under challenging conditions.

Key benefits of ASTM B 649 N08031 include:

- **Corrosion Resistance**: Its unique composition helps prevent rust and degradation, extending the lifespan of products made from it.
- **Versatile Applications**: This alloy is suitable for manufacturing valves, pumps, and components in the automotive industry, ensuring reliability in diverse environments.
- **Strong and Durable**: With its low carbon content and high nickel and chromium levels, it provides excellent strength and toughness, making it a preferred choice for general engineering needs.

ASTM B 649 N08925

Category: Bar and Wire
Ni-Fe-Cr-Mo-Cu Low-Carbon Alloy.

Technical Specifications:

Carbon	0.020 max.
Silicon	0.500 max.
Manganese	1.000 max.
Chromium	19.000 - 21.000
Sulphur	0.030 max.
Molybdenum	6.000 - 7.000
Phosphorus	0.045 max.
Copper	0.800 - 1.500
Nitrogen	0.100 - 0.200
Nickel	24.000 - 26.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 649 N08925 is a specialized type of metal alloy that combines nickel, iron, chromium, molybdenum, and copper. This low-carbon alloy is designed for use in various industrial applications, providing excellent resistance to corrosion and wear. It is particularly beneficial in environments where durability and longevity are essential, such as in valves, pumps, and automotive components.

****Key Benefits:****

- ****Corrosion Resistance:**** The alloy's composition helps it withstand harsh conditions, making it ideal for use in demanding environments.
- ****Versatility:**** Suitable for a wide range of applications in general engineering and automotive industries.
- ****Durability:**** The low-carbon content contributes to the strength and longevity of the components made from this alloy.

ASTM B 649 N08926

Category: Bar and Wire

Ni-Fe-Cr-Mo-Cu Low-Carbon Alloy.

Technical Specifications:

Carbon	0.020 max.
Silicon	0.500 max.
Manganese	2.000 max.
Chromium	19.000 - 21.000
Sulphur	0.010 max.
Molybdenum	6.000 - 7.000
Phosphorus	0.030 max.
Copper	0.500 - 1.500
Nitrogen	0.150 - 0.250
Nickel	24.000 - 26.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 649 N08926 is a specialized type of metal alloy made primarily from nickel, iron, chromium, molybdenum, and copper, designed for various industrial applications. This low-carbon alloy is known for its durability and resistance to corrosion, making it an excellent choice for items that need to withstand harsh environments. It is commonly used in the manufacturing of valves, pumps, and components in general engineering and automotive industries.

Key benefits of ASTM B 649 N08926 include:

- ****Corrosion Resistance**:** Its composition allows it to resist rust and degradation, ensuring a longer lifespan for products made with it.
- ****Versatile Applications**:** Suitable for a wide range of uses, from automotive parts to general engineering components, making it a flexible option for manufacturers.
- ****Durability**:** The alloy's strength ensures that it can handle high-stress conditions without failing, which is crucial in industrial environments.

ASTM B 673 N08925

Category: Pipe

Nickel-Iron-Chromium-Molybdenum Pipe.

Technical Specifications:

Carbon	0.020 max.
Silicon	0.500 max.
Manganese	1.000 max.
Chromium	19.000 - 21.000
Sulphur	0.030 max.
Molybdenum	6.000 - 7.000
Phosphorus	0.045 max.
Copper	0.800 - 1.500
Nitrogen	0.100 - 0.200
Nickel	24.000 - 26.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 673 N08925 is a type of pipe made from a special blend of nickel, iron, chromium, and molybdenum. This unique combination of metals gives the pipe excellent strength and resistance to corrosion, making it ideal for use in various applications like valves, pumps, and automotive parts. Essentially, it's a durable and reliable option for industries that require high-performance materials.

****Benefits:****

- ****Corrosion Resistance:**** The materials used prevent rust and degradation, ensuring the pipe lasts longer even in harsh environments.
- ****Strength and Durability:**** This pipe can handle high pressure and extreme conditions, making it suitable for demanding engineering tasks.
- ****Versatile Applications:**** It can be used in different sectors, enhancing its value for businesses looking for a multi-purpose piping solution.

ASTM B 677 N08925

Category: Pipe and Tube

Nickel-Iron-Chromium-Molybdenum Seamless Pipe.

Technical Specifications:

Carbon	0.020 max.
Silicon	0.500 max.
Manganese	1.000 max.

Chromium	19.000 - 21.000
Sulphur	0.030 max.
Molybdenum	6.000 - 7.000
Phosphorus	0.045 max.
Copper	0.800 - 1.500
Nitrogen	0.100 - 0.200
Nickel	24.000 - 26.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 677 N08925 is a type of seamless pipe made from a special blend of metals including nickel, iron, chromium, and molybdenum. This unique combination gives the pipe excellent resistance to corrosion and high temperatures, making it ideal for various industrial applications. It's commonly used in valves, pumps, general engineering projects, and automotive parts, where durability and strength are crucial.

Key benefits of ASTM B 677 N08925 include:

- **Corrosion Resistance**: The high nickel and chromium content helps prevent rust and degradation, especially in harsh environments.
- **High Strength**: This pipe can withstand high pressures and temperatures, ensuring reliable performance in demanding applications.
- **Versatility**: Suitable for a wide range of uses, from machinery to automotive components, making it a valuable addition to any industrial setting.

ASTM B 705 N06219

Category: Pipe

Nickel Alloy Pipe.

Technical Specifications:

Carbon	0.050 max.
Silicon	0.700 - 1.100
Manganese	0.500 max.
Copper	0.500 max.
Sulphur	0.010 max.
Chromium	18.000 - 22.000
Aluminium	0.500 max.
Molybdenum	7.000 - 9.000
Iron	2.000 - 4.000
Phosphorus	0.020 max.
Titanium	0.500 max.

hardening, allowing for tailored performance based on specific needs.

Nickel Alloy Pipe

Category: Pipe

ASTM B 834 N06690 Standard Specification for Nickel Alloy Pipe.

Key Features:

- Chemical Composition: Carbon 0.050 max, Silicon 0.500 max, Nickel 58.000 min
- Mechanical Properties: Tensile Strength 585 min MPa, Yield Strength 240 min MPa

Technical Specifications:

Heat Treatment As-Cast or Annealing or Age Hardening

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

Nickel Alloy Pipe is a specialized type of pipe made from a unique nickel-based material that is designed to be strong and durable. It meets specific industry standards (ASTM B 834 N06690) to ensure quality and reliability. This pipe is particularly known for its ability to withstand harsh conditions, making it ideal for various applications, including valves, pumps, general engineering, and automotive uses.

Key benefits include:

- **Strong and Reliable**: With a minimum tensile strength of 585 MPa and a yield strength of 240 MPa, this pipe can handle significant pressure and stress.
- **Versatile Applications**: Suitable for a wide range of industries, from automotive to general engineering, making it a go-to choice for many engineers and manufacturers.
- **Customizable Treatment**: The pipe can undergo different heat treatments, such as annealing or age hardening, to enhance its properties for specific requirements.

Precipitation Hardening Nickel-Copper-Aluminum Alloy

Category: Bar, Rod, Wire and Forging

ASTM B 865 N05500 Standard Specification for Precipitation Hardening Nickel-Copper-Aluminum Alloy.

Key Features:

- Chemical Composition: Carbon 0.180 max, Silicon 0.500 max, Nickel 63.000 min
- Mechanical Properties: Tensile Strength 760 - 1070 MPa, Yield Strength 585 min MPa

Technical Specifications:

Heat Treatment As-Cast or Annealing or Age Hardening

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The Precipitation Hardening Nickel-Copper-Aluminum Alloy is a specialized metal used in various industries for its strength and durability. This alloy is manufactured according to strict standards (ASTM B 865 N05500) and is designed to withstand high stress and corrosion, making it ideal for demanding applications. It can be processed in different ways, such as being cast or heat-treated to enhance its properties further.

Benefits of this alloy include:

- **High Strength**: With a tensile strength ranging from 760 to 1070 MPa, it can handle significant forces without breaking.
- **Versatile Applications**: Common uses include components for valves, pumps, general engineering projects, and automotive parts, showcasing its adaptability in various settings.

Pressure Consolidated Powder Nickel Alloy Pipe Flanges, Fittings, Valves, and Parts

Category: Pipe Flanges, Forged Fittings, and Valve

ASTM B 834 N08367 Standard Specification for Pressure Consolidated Powder Nickel Alloy.

Key Features:

- Chemical Composition: Carbon 0.030 max, Silicon 1.000 max, Nickel 23.500 - 25.500
- Mechanical Properties: Tensile Strength 655 min MPa, Yield Strength 310 min MPa

Technical Specifications:

Heat Treatment As-Cast or Annealing or Age Hardening

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

Pressure Consolidated Powder Nickel Alloy Pipe Flanges, Fittings, Valves, and Parts are specialized components used in various industries, including automotive and general engineering. These products are made from a unique nickel alloy that provides exceptional strength and durability, ensuring they can withstand high pressure and harsh conditions. The materials are processed in a way that enhances their mechanical properties, making them ideal for critical applications such as valves and pumps.

Key Benefits:

- **Strong and Reliable**: With a tensile strength of at least 655 MPa, these components are designed to handle extreme pressures without failing.
- **Versatile Applications**: Suitable for use in valves, pumps, and various engineering projects, they can adapt to different industrial needs.
- **Durable Material**: The use of nickel alloy ensures resistance to corrosion and wear, extending the lifespan of the components.

Precipitation-Hardening Nickel Alloys Plate, Sheet and Strip

Category: Plate, Sheet and Strip

ASTM B 872 N07725 Standard Specification for Precipitation-Hardening Nickel Alloys.

Key Features:

- Chemical Composition: Carbon 0.030 max, Silicon 0.200 max, Nickel 55.000 - 59.000
- Mechanical Properties: Tensile Strength 1034 min MPa, Yield Strength 827 min MPa

Technical Specifications:**Heat Treatment**

As-Cast or Annealing or Age Hardening

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

Precipitation-Hardening Nickel Alloys Plate, Sheet and Strip is a type of metal product designed for various industrial applications. It is made from a special nickel alloy that becomes stronger and more durable when treated with heat. This makes it suitable for use in demanding environments, such as in valves, pumps, and automotive parts.

Key features of this product include:

- **High Strength**: It has a minimum tensile strength of 1034 MPa, making it very strong and resistant to breaking under pressure.
- **Versatile Use**: Suitable for a range of applications in general engineering and automotive industries.

The alloy has a specific chemical composition, including a maximum carbon content of 0.030 and nickel content ranging from 55% to 59%. It can be processed through various heat treatments like annealing or age hardening, ensuring it meets the required strength and durability for critical applications.

Precipitation Hardened or Cold Worked, Seamless Nickel Alloy Pipe and Tube

Category: Pipe and Tube

ASTM B 983 N07725 Type 2 Standard Specification for Precipitation Hardened or Cold Worked, Seamless Nickel Alloy Pipe and Tube.

Key Features:

- Chemical Composition: Carbon 0.030 max, Silicon 0.200 max, Nickel 55.000 - 59.000
- Mechanical Properties: Tensile Strength 1035 min MPa, Yield Strength 827 min MPa

Technical Specifications:**Heat Treatment**

As-Cast or Annealing or Age Hardening

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The Precipitation Hardened or Cold Worked, Seamless Nickel Alloy Pipe and Tube is a durable and strong material used in various industries. It is designed according to specific standards to ensure it can withstand high pressures and temperatures. This type of pipe and tube is made from nickel alloy, which provides excellent resistance to corrosion and mechanical stress, making it suitable for critical applications.

Here are some key points about this product:

- **High Strength**: It has a minimum tensile strength of 1035 MPa and yield strength of 827 MPa, ensuring it can handle tough conditions.

- **Versatile Applications**: Commonly used in valves, pumps, automotive parts, and general engineering projects, allowing for wide-ranging use.
- **Customizable Heat Treatment**: It can be treated in different ways, such as annealing or age hardening, to enhance its properties further, tailoring it to specific needs.

Copper-Alloy Bearing and Expansion Plates and Sheets

Category: Plate and Sheet

ASTM B100 C61300 Standard Specification for Wrought Copper-Alloy Bearing and Expansion Plates and Sheets.

Key Features:

- **Chemical Composition:** Aluminium 6.000 - 7.500, Iron 2.000 - 3.000
- **Mechanical Properties:** Tensile Strength 415 min MPa, Yield Strength 170 min MPa

Technical Specifications:

Heat Treatment	As Drawn or Stress Relieving
-----------------------	------------------------------

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

Copper-Alloy Bearing and Expansion Plates and Sheets are specialized materials made from a durable copper alloy, designed primarily for use in various engineering applications. These plates and sheets meet the ASTM B100 C61300 standard, ensuring they are reliable and effective in supporting heavy machinery and components. They are particularly beneficial in industries like automotive, valve manufacturing, and general engineering due to their excellent mechanical properties and resistance to wear and tear.

Key benefits include:

- **Strong and Durable**: With a minimum tensile strength of 415 MPa, these materials can withstand significant stress without breaking.
- **Versatile Use**: Ideal for applications in valves, pumps, and automotive parts, making them a go-to choice for engineers.
- **Customizable Treatment**: Available in different heat treatment states, they can be tailored to meet specific performance needs.

ASTM B 335 N10665

Category: Material

Standard Specification for Nickel-Molybdenum Alloy Rod.

Key Features:

- Chemical Composition: C %0.020 max, Si %0.100 max, Mo %26.000 - 30.000, Ni %Balance
- Mechanical Properties: Tensile Strength in Mpa 760 min, Yield Strength in Mpa 350 min, Elongation in % 40 min

Technical Specifications:

Carbon	0.020 max
Silicon	0.100 max

Manganese	1.000 max
Phosphorus	0.040 max
Cobalt	1.000 max
Sulphur	0.030 max
Chromium	1.000 max
Iron	2.000 max
Molybdenum	26.000 - 30.000
Nickel	Balance

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 335 N10665 is a specialized metal rod made from a high-strength Nickel-Molybdenum alloy. This material is designed to withstand extreme conditions, making it ideal for various industrial applications. The alloy's unique chemical composition ensures durability and reliability in demanding environments such as valve and pump manufacturing, general engineering, and automotive industries.

****Key Features:****

- ****Strong and Durable:**** With a minimum tensile strength of 760 MPa and yield strength of 350 MPa, this alloy can handle heavy loads without breaking.
- ****Flexible:**** It allows for a minimum elongation of 40%, meaning it can bend without cracking, which is crucial for many engineering applications.
- ****Versatile Use:**** Perfect for creating parts in valves, pumps, and automotive components, making it a go-to choice for engineers.

Overall, ASTM B 335 N10665 offers a combination of strength, flexibility, and chemical resistance, ensuring that products made from this alloy perform reliably in tough conditions.

ASTM B 335 N10675

Category: Material

Standard Specification for Nickel-Molybdenum Alloy Rod.

Key Features:

- Chemical Composition: C %0.010 max, W %3.000 max, Ni %65.000 min
- Mechanical Properties: Tensile Strength in Mpa 760 min, Yield Strength in Mpa 350 min, Elongation in % 40 min

Technical Specifications:

Carbon	0.010 max
--------	-----------

Tungsten	3.000 max
Silicon	0.100 max
Manganese	3.000 max
Phosphorus	0.030 max
Nickel	65.000 min
Cobalt	3.000 max

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 335 N10675 is a specialized rod made from a strong nickel-molybdenum alloy, designed for various industrial applications. This material is known for its exceptional strength and resistance to harsh environments, making it ideal for use in critical components like valves and pumps. It can withstand high stress and has a significant ability to stretch without breaking, ensuring reliability and durability in demanding situations.

Key benefits include:

- **High Strength:** With a minimum tensile strength of 760 MPa, this alloy is robust enough for tough applications.
- **Corrosion Resistance:** The unique chemical composition helps it resist rust and deterioration, enhancing the lifespan of machinery.
- **Versatile Use:** It can be utilized in a variety of industries including automotive and general engineering, making it a valuable addition to any material inventory.

ASTM B 344 Ni60-Cr16

Category: Material

Nickel-Chromium Alloy for Electrical Heating Elements.

Key Features:

- Chemical Composition: C %0.150 max, Cr %14.000 - 18.000, Ni %57.000 min
- Mechanical Properties: Elongation in % 10 min

Technical Specifications:

Carbon	0.150 max
Silicon	0.750 - 1.750
Manganese	1.000 max
Chromium	14.000 - 18.000
Sulphur	0.010 max
Nickel	57.000 min

IronBalance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 344 Ni60-Cr16 is a specialized nickel-chromium alloy primarily used to create electrical heating elements. This material is designed to withstand high temperatures and provide excellent resistance to oxidation, making it ideal for applications in various industries such as automotive, valve and pump manufacturing, and general engineering. Its unique composition, with a minimum of 57% nickel and 14% to 18% chromium, ensures durability and efficiency in heating applications.

****Benefits of ASTM B 344 Ni60-Cr16:****

- ****High Performance:**** The alloy's ability to maintain strength and functionality at elevated temperatures ensures reliable operation in heating elements.
- ****Versatile Applications:**** Suitable for a range of industries, this material can be used in automotive components, valves, and pumps, making it a flexible choice for manufacturers.
- ****Enhanced Durability:**** With low carbon and sulfur content, the material offers superior resistance to wear and corrosion, prolonging the lifespan of products made from it.

ASTM B 344 Ni35-Cr20

Category: Material

Nickel-Chromium Alloy for Electrical Heating Elements.

Key Features:

- Chemical Composition: C %0.150 max, Cr %18.000 - 21.000, Ni %34.000 - 37.000
- Mechanical Properties: Elongation in % 10 min

Technical Specifications:

Carbon	0.150 max
Silicon	1.000 - 3.000
Manganese	1.000 max
Chromium	18.000 - 21.000
Sulphur	0.010 max
Nickel	34.000 - 37.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 344 Ni35-Cr20 is a specialized material made from a blend of nickel and chromium, designed primarily for use in electrical heating elements. This alloy is known for its excellent durability and high resistance to heat, making it ideal for applications where reliable and efficient heating is crucial. With a specific chemical makeup that includes up to 37% nickel and between 18% to 21% chromium, this material can withstand high temperatures while maintaining its structural integrity.

- **Benefits of ASTM B 344 Ni35-Cr20:****
- ****Durability:**** This alloy is built to last, making it suitable for demanding environments like valves and pumps.
 - ****Versatility:**** It is used in various industries, including general engineering and automotive, ensuring it meets diverse heating requirements.
 - ****High Performance:**** The mechanical properties, such as a minimum elongation of 10%, ensure flexibility and strength under stress.

ASTM B 344 Ni80-Cr20

Category: Material

Nickel-Chromium Alloy for Electrical Heating Elements.

Key Features:

- Chemical Composition: C %0.150 max, Cr %19.000 - 21.000, Ni %Balance
- Mechanical Properties: Elongation in % 10 min

Technical Specifications:

Carbon	0.150 max
Silicon	0.750 - 1.750
Manganese	1.000 max
Chromium	19.000 - 21.000
Sulphur	0.010 max
Nickel	Balance
Iron	1.000 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 344 Ni80-Cr20 is a special alloy made primarily of nickel and chromium, designed specifically for use in electrical heating elements. This material is highly valued in various industries, including automotive and general engineering, for its ability to withstand high temperatures while maintaining durability. Its unique chemical composition ensures that it performs effectively in demanding environments, making it a reliable choice for applications like valves and pumps.

Key benefits of ASTM B 344 Ni80-Cr20 include:

- **High Resistance to Heat**: Ideal for applications that require stable performance at elevated temperatures.
- **Durability**: The alloy's strong mechanical properties ensure longevity and reliability in various engineering tasks, reducing maintenance needs.
- **Versatile Applications**: Useful in a wide range of sectors, from automotive to general industrial use, making it a flexible option for manufacturers.

ASTM B 39

Category: Material

Nickel Alloy.

Key Features:

- Chemical Composition: C %0.030 max, Ni %99.800 min
- Mechanical Properties: Tensile Strength in Mpa 300 - 450

Technical Specifications:

Carbon	0.030 max
Silicon	0.005 max
Manganese	0.005 max
Phosphorus	0.005 max
Sulphur	0.010 max
Cobalt	0.150 max
Copper	0.020 max
Iron	0.020 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 39 is a high-quality nickel alloy widely used in various engineering applications. It is known for its excellent strength and resistance to corrosion, making it ideal for tough environments. This material primarily consists of nickel, with very low amounts of other elements, ensuring its durability and reliability.

Key Benefits:

- **Strong and Durable**: With a tensile strength ranging from 300 to 450 MPa, ASTM B 39 can withstand significant stress, making it suitable for demanding applications like valves and pumps.
- **Versatile Applications**: This alloy is used in general engineering and automotive industries, proving its versatility in various manufacturing processes.
- **Export Quality**: Primarily available for export to the USA, it meets stringent international standards for quality and performance.

ASTM B 407 N08120

Category: Material

Nickel-Iron-Chromium Alloy Seamless Pipe and Tube.

Key Features:

- Chemical Composition: C %0.020 - 0.100, Ni %35.000 - 39.000
- Mechanical Properties: Tensile Strength in Mpa 621 min, Yield Strength in Mpa 276 min

Technical Specifications:

Carbon	0.020 - 0.100
Nickel	35.000 - 39.000
Silicon	1.000 max
Iron	Balance
Manganese	1.500 max
Chromium	23.000 - 27.000
Sulphur	0.030 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

The ASTM B 407 N08120 is a special type of seamless pipe and tube made from a Nickel-Iron-Chromium alloy. This material is designed to be strong and durable, making it ideal for various industrial applications. It works by combining specific elements like nickel and chromium, which enhance its resistance to corrosion and improve its overall strength. This means it can handle tough conditions without breaking down easily.

Benefits of ASTM B 407 N08120 include:

- **High Strength**: With a minimum tensile strength of 621 MPa, this material can withstand significant force, making it reliable for heavy-duty uses.
- **Versatile Applications**: It is suitable for various uses, including valves, pumps, general engineering, and automotive parts, making it a flexible choice for manufacturers.
- **Corrosion Resistance**: The combination of materials helps protect against rust and wear, extending the lifespan of products made with it.

ASTM B 366 N06600

Category: Material

Nickel Alloy Forgings.

Key Features:

- Chemical Composition: C %0.150 max, Ni %72.000 min
- Mechanical Properties: Tensile Strength in Mpa 551.6 min, Yield Strength in Mpa 241 min

Technical Specifications:

Carbon	0.150 max
Silicon	0.500 max

Manganese	1.000 max
Chromium	14.000 - 17.000
Sulphur	0.015 max
Copper	0.500 max
Iron	6.000 - 10.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 366 N06600 is a type of nickel alloy material that is primarily used for creating strong and durable components like valves and pumps. This material is known for its excellent resistance to corrosion and high strength, making it ideal for various engineering applications, including automotive parts. The alloy consists mainly of nickel, with specific limits on other elements to enhance its performance.

Key benefits of using ASTM B 366 N06600 include:

- **High Strength**: It has a minimum tensile strength of 551.6 MPa, ensuring that it can withstand significant stress and pressure.
- **Corrosion Resistance**: The unique chemical composition helps it resist wear and tear, especially in harsh environments, prolonging the life of the parts made from it.
- **Versatility**: Suitable for a wide range of applications, including valves, pumps, and general engineering projects.

ASTM B 407 N08800

Category: Material

Nickel-Iron-Chromium Alloy Seamless Pipe and Tube.

Key Features:

- Chemical Composition: C %0.100 max, Ni %30.000 - 35.000
- Mechanical Properties: Tensile Strength in Mpa 450 - 690

Technical Specifications:

Carbon	0.100 max
Chromium	19.000 - 23.000
Silicon	1.000 max
Manganese	1.500 max
Aluminium	0.150 - 0.600
Sulphur	0.015 max
Copper	0.750 max
Iron	39.500 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 407 N08800 is a type of seamless pipe and tube made from a special alloy that includes nickel, iron, and chromium. This material is designed to be strong and resistant to corrosion, making it ideal for various applications in industries like automotive, engineering, and manufacturing. Its seamless design means there are no joints or welds, which enhances its durability and reliability in high-pressure systems.

Key benefits of ASTM B 407 N08800 include:

- **High Strength**: It has a tensile strength ranging from 450 to 690 MPa, providing excellent structural integrity under stress.
- **Corrosion Resistance**: The alloy composition helps it resist rust and deterioration, prolonging its lifespan in challenging environments.
- **Versatile Applications**: Suitable for use in valves, pumps, and general engineering tasks, making it a valuable material across multiple sectors.

ASTM B 407 N08801

Category: Material

Nickel-Iron-Chromium Alloy Seamless Pipe and Tube.

Key Features:

- Chemical Composition: C %0.100 max, Ni %30.000 - 35.000
- Mechanical Properties: Tensile Strength in Mpa 450 min

Technical Specifications:

Carbon	0.100 max
Chromium	19.000 - 22.000
Silicon	1.000 max
Manganese	1.500 max
Copper	0.500 max
Iron	39.500 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

The ASTM B 407 N08801 is a special type of pipe and tube made from a durable nickel-iron-chromium alloy. This material is designed to be seamless, meaning it has no joints or welds that can weaken it, making it strong

and reliable for various applications. It is particularly useful in industries like automotive, engineering, and for manufacturing valves and pumps.

****Key Benefits:****

- ****Strong and Durable:**** With a minimum tensile strength of 450 MPa, this alloy can withstand high pressure and stress, ensuring long-lasting performance.
- ****Versatile Applications:**** It is suitable for a wide range of uses, from automotive parts to general engineering projects, making it a flexible choice for many industries.
- ****High Corrosion Resistance:**** The chemical composition allows it to resist rust and corrosion, which is crucial for equipment that operates in harsh environments.

ASTM B 407 N08811

Category: Material

Nickel-Iron-Chromium Alloy Seamless Pipe and Tube.

Key Features:

- Chemical Composition: C %0.060 - 0.100, Ni %30.000 - 35.000
- Mechanical Properties: Tensile Strength in Mpa 450 min

Technical Specifications:

Carbon	0.060 - 0.100
Chromium	19.000 - 23.000
Silicon	1.000 max
Manganese	1.500 max
Copper	0.750 max
Iron	39.500 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 407 N08811 is a type of seamless pipe and tube made from a special blend of nickel, iron, and chromium. This unique material is designed to withstand tough conditions, making it ideal for various applications. It has a high tensile strength, ensuring that it can handle significant pressure and stress without breaking. This product is particularly useful in industries like automotive, engineering, and for components such as valves and pumps.

Benefits of ASTM B 407 N08811 include:

- ****Durability**:** The strong material composition allows it to resist wear and tear over time.
- ****Versatility**:** Suitable for a wide range of applications, from automotive to general engineering.
- ****Reliability**:** Ensures long-lasting performance in demanding environments, minimizing the need for frequent replacements.

ASTM B 409 N08120

Category: Material

Nickel-Iron-Chromium Alloy Plate, Sheet and Strip.

Key Features:

- Chemical Composition: C %0.020 - 0.100, Ni %35.000 - 39.000
- Mechanical Properties: Tensile Strength in Mpa 621 min, Yield Strength in Mpa 276 min

Technical Specifications:

Carbon	0.020 - 0.100
Nickel	35.000 - 39.000
Silicon	1.000 max
Iron	Balance
Manganese	1.500 max
Chromium	23.000 - 27.000
Sulphur	0.030 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 409 N08120 is a specialized type of metal made from a blend of nickel, iron, and chromium, designed to be used in various industrial applications. This material is available in forms such as plates, sheets, and strips, making it versatile for different manufacturing needs. It is particularly valued for its strength and resistance to corrosion, which makes it ideal for use in environments that require durable materials.

****Key Benefits:****

- ****Strong and Durable:**** With a minimum tensile strength of 621 MPa, this alloy can withstand significant stress, making it reliable for heavy-duty applications.
- ****Corrosion Resistance:**** The combination of nickel and chromium provides excellent resistance to rust and other forms of degradation, extending the lifespan of components made from this material.
- ****Wide Applications:**** Commonly used in valves, pumps, and automotive parts, it is suitable for various engineering projects, ensuring that businesses have the right materials for their specific needs.

ASTM B 409 N08800

Category: Material

Nickel-Iron-Chromium Alloy Plate, Sheet and Strip.

Key Features:

- Chemical Composition: C %0.100 max, Ni %30.000 - 35.000
- Mechanical Properties: Tensile Strength in Mpa 520 min, Yield Strength in Mpa 205 min

Technical Specifications:

Carbon	0.100 max
--------	-----------

Chromium	19.000 - 23.000
Silicon	1.000 max
Manganese	1.500 max
Aluminium	0.150 - 0.600
Sulphur	0.015 max
Copper	0.750 max
Iron	39.500 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 409 N08800 is a specialized type of metal known as a Nickel-Iron-Chromium Alloy. This material is available in different forms like plates, sheets, and strips, making it versatile for various applications. It's designed to withstand harsh environments, making it an ideal choice for industries such as automotive, general engineering, as well as in manufacturing valves and pumps.

****Key Benefits:****

- ****Strong and Durable:**** With a minimum tensile strength of 520 MPa, this alloy provides excellent strength and durability for demanding applications.
- ****Corrosion Resistant:**** The unique composition of nickel, chromium, and iron ensures that the material can resist corrosion, extending the lifespan of products made from it.
- ****Versatile Applications:**** Suitable for a range of uses including automotive parts and engineering components, making it a flexible choice for manufacturers.

ASTM B 409 N08810

Category: Material

Nickel-Iron-Chromium Alloy Plate, Sheet and Strip.

Key Features:

- Chemical Composition: C %0.050 - 0.100, Ni %30.000 - 35.000
- Mechanical Properties: Tensile Strength in Mpa 450 min, Yield Strength in Mpa 170 min

Technical Specifications:

Carbon	0.050 - 0.100
Chromium	19.000 - 23.000
Silicon	1.000 max
Manganese	1.500 max
Aluminium	0.150 - 0.600
Sulphur	0.015 max
Copper	0.750 max

Iron39.500 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 409 N08810 is a specialized metal alloy made primarily of nickel, iron, and chromium, designed for various industrial applications. This alloy is available in the form of plates, sheets, and strips, making it versatile for different manufacturing needs. Its unique composition gives it excellent strength and resistance to corrosion, making it suitable for use in challenging environments.

****Key Benefits:****

- ****Durable and Strong:**** With a minimum tensile strength of 450 MPa and yield strength of 170 MPa, it can withstand high pressures and stress, ideal for components like valves and pumps.
- ****Corrosion Resistance:**** The combination of nickel and chromium provides excellent protection against rust and corrosion, enhancing the longevity of the products made from it.
- ****Versatile Applications:**** Commonly used in industries such as automotive and general engineering, this alloy can be tailored to meet specific needs in various sectors.

ASTM B 409 N08890

Category: Material

Nickel-Iron-Chromium Alloy Plate, Sheet and Strip.

Key Features:

- Chemical Composition: C %0.060 - 0.140, Ni %40.000 - 45.000
- Mechanical Properties: Tensile Strength in Mpa 520 min, Yield Strength in Mpa 205 min

Technical Specifications:

Carbon	0.060 - 0.140
Chromium	23.500 - 28.500
Silicon	1.000 - 2.000
Manganese	1.500 max
Aluminium	0.050 - 0.600
Sulphur	0.015 max
Copper	0.750 max
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 409 N08890 is a specialized material made of a strong alloy that consists primarily of nickel, iron, and chromium. This alloy is available in various forms such as plates, sheets, and strips, making it versatile for different applications. It is known for its impressive strength and resistance to corrosion, which makes it ideal for use in demanding environments.

Key features of ASTM B 409 N08890 include:

- **Strong Mechanical Properties**: It has a minimum tensile strength of 520 MPa and a minimum yield strength of 205 MPa, ensuring that it can withstand significant stress and pressure.
- **Chemical Composition**: The alloy contains specific percentages of carbon, chromium, silicon, and other elements that contribute to its durability and performance.

This material is commonly used in applications such as valves and pumps, as well as in general engineering and automotive industries. Its ability to maintain integrity in harsh conditions provides significant benefits, making it a preferred choice for manufacturers looking for reliable and long-lasting materials.

ASTM B 423 N08825

Category: Material

Nickel-Iron-Chromium-Molybdenum-Copper Alloy Seamless Pipe and Tube.

Key Features:

- Chemical Composition: C %0.050 max, Ni %38.000 - 46.000
- Mechanical Properties: Tensile Strength in Mpa 517 - 793, Yield Strength in Mpa 172 min

Technical Specifications:

Carbon	0.050 max
Silicon	0.500 max
Manganese	1.000 max
Aluminium	0.200 max
Sulphur	0.030 max
Copper	1.500 - 3.000
Chromium	19.500 - 23.500
Molybdenum	2.500 - 3.500
Titanium	0.600 - 1.200
Iron	22.000 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B 423 N08825 is a type of high-quality alloy pipe and tube made from a mixture of nickel, iron, chromium, molybdenum, and copper. This special combination of metals makes it strong and resistant to corrosion, which is essential for various industrial applications. The pipes are seamless, meaning they don't have any joints, which enhances their durability and reliability in tough environments.

- Key Benefits:**
- **Durability:** Strong mechanical properties ensure that the pipes can handle high pressure and resist wear and tear over time.
 - **Versatility:** Suitable for various applications such as valves, pumps, and automotive parts, making it a great choice for engineers and manufacturers.
 - **Corrosion Resistance:** The alloy composition provides excellent protection against rust and chemical damage, extending the lifespan of the products made from it.

ASTM B 434 N10003

Category: Material

Nickel-Molybdenum-Chromium-Iron Alloys Plate, Sheet and Strip.

- Key Features:**
- Chemical Composition: C %0.040 - 0.080, Ni %Balance
 - Mechanical Properties: Tensile Strength in Mpa 690 min, Yield Strength in Mpa 280 min

Technical Specifications:

Carbon	0.040 - 0.080
Silicon	1.000 max
Manganese	1.000 max
Chromium	6.000 - 8.000
Sulphur	0.020 max
Copper	0.350 max
Iron	5.000 max
Molybdenum	15.000 - 18.000

- Use Cases:**
- Valve
 - Pump
 - General Engineering
 - Automotive

- Export Markets:**
- USA

AI Layman Summary:

ASTM B 434 N10003 is a special type of metal alloy made primarily of nickel, molybdenum, chromium, and iron. This alloy is available in forms such as plates, sheets, and strips, making it versatile for various applications. It is known for its strong resistance to corrosion and high temperatures, making it ideal for environments where durability is crucial.

- Key benefits of using ASTM B 434 N10003 include:
- **High Strength:** It has a minimum tensile strength of 690 MPa, meaning it can withstand heavy loads without breaking.

- **Versatile Applications**: This alloy is suitable for use in valves, pumps, and automotive parts, making it valuable across different industries.
- **Chemical Stability**: Its balanced chemical composition ensures reliability and longevity in engineering applications, particularly in harsh conditions.

ASTM B147-7A

Category: Material

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Aluminium	0.500 - 1.500%
Iron	0.400 - 2.000%
Manganese	0.100 - 1.000%
Copper	56.000 - 62.000%

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B147-7A is a standard specification that outlines the requirements for a type of copper alloy used in sand casting, which is a method of shaping metal. This specification ensures that the copper alloy meets certain chemical and mechanical properties, making it suitable for a variety of applications. In simpler terms, it tells manufacturers what the copper alloy should contain and how strong it should be, so they can create reliable products.

Key benefits of ASTM B147-7A include:

- **Versatile Use**: It's perfect for making parts like valves and pumps, as well as for general engineering and automotive applications.
- **Quality Assurance**: The specification ensures that the materials used are consistent in their composition and mechanical strength, leading to safer and more durable products.

ASTM B148 Aluminum Bronze UNS C95200

Category: Material

Standard Specification for Aluminum-Bronze Sand Castings.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Iron	2.500 - 4.000%
Aluminium	8.500 - 9.500%

Copper 86.000 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 Aluminum Bronze UNS C95200 is a special type of metal made from a mix of aluminum and copper, designed specifically for making strong and durable castings. This material is particularly useful in industries that require high resistance to wear and corrosion, such as valves, pumps, and automotive parts. It is known for its excellent mechanical properties and unique chemical composition that enhance its performance in various applications.

Key Benefits:

- ****Durability****: This aluminum bronze alloy is designed to withstand tough conditions, making it ideal for heavy-duty machinery and components.
- ****Versatility****: It can be used in a variety of applications, including general engineering and automotive parts, ensuring reliability across different sectors.

ASTM B148 Aluminum Bronze UNS C95300

Category: Material

Standard Specification for Aluminum-Bronze Sand Castings.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Iron	0.800 - 1.500%
Aluminium	9.000 - 11.000%
Copper	86.000 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 Aluminum Bronze UNS C95300 is a type of metal that combines aluminum and copper, specifically designed for making sand castings. This material is known for its strength and resistance to corrosion, making it ideal for various applications. It has a specific blend of elements, including 9% to 11% aluminum and at least 86% copper, along with small amounts of iron.

Key benefits of using this material include:

- ****Durability****: It is strong and can withstand harsh conditions, making it suitable for use in valves, pumps, and automotive components.
- ****Corrosion Resistance****: The aluminum content helps protect the metal from rust and other forms of

deterioration, extending the life of the products made from it.

- **Versatility**: Its properties make it useful in general engineering applications, ensuring it can be used in a wide range of industries.

ASTM B148 Aluminum Bronze UNS C95400

Category: Material

Standard Specification for Aluminum-Bronze Sand Castings.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Iron	3.000 - 5.000%
Aluminium	10.000 - 11.500%
Copper	83.000 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 Aluminum Bronze UNS C95400 is a type of metal alloy made primarily from copper, aluminum, and iron. It is specifically designed for sand casting, which is a process used to create metal parts by pouring molten metal into molds. This particular alloy is known for its strong mechanical properties, making it reliable for various applications.

Benefits of using ASTM B148 Aluminum Bronze UNS C95400 include:

- **Durability**: The combination of metals provides strength and resistance to wear, making it ideal for demanding environments.
- **Versatility**: It can be used in a variety of industries, including automotive, valve manufacturing, and general engineering, allowing for a wide range of applications.
- **Consistent Quality**: The standard specifications ensure that the material meets specific chemical and mechanical property requirements, providing assurance of performance in critical applications.

ASTM B148 Aluminum Bronze UNS C95900

Category: Material

Standard Specification for Aluminum-Bronze Sand Castings.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Iron	3.000 - 5.000%
Aluminium	12.000 - 13.500%
Copper	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 Aluminum Bronze UNS C95900 is a specialized metal alloy that combines aluminum and copper, primarily used for creating strong and durable sand castings. This material is known for its excellent mechanical properties, making it suitable for various industrial applications. It has a specific chemical makeup, with 12% to 13.5% aluminum and 3% to 5% iron, which contributes to its strength and resistance to corrosion.

Key benefits of ASTM B148 Aluminum Bronze UNS C95900 include:

- **Durability**: With its robust mechanical properties, it can withstand high-stress conditions, making it ideal for applications in valves, pumps, automotive parts, and general engineering.
- **Corrosion Resistance**: This alloy's composition allows it to resist oxidation and other forms of corrosion, ensuring a longer lifespan for products made from it.
- **Versatile Applications**: It's widely used in various industries, providing solutions for both everyday and specialized engineering needs.

ASTM B150 C62300

Category: Material

Standard Specification for Aluminum Bronze Rod, Bar and Shapes.

Key Features:

- Chemical Composition
- Mechanical Properties

Technical Specifications:

Aluminium	8.500 - 10.000%
Iron	2.000 - 4.000%
Manganese	0.500 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B150 C62300 is a specific type of material known as aluminum bronze, which comes in the form of rod, bar, and various shapes. This material is designed to meet set standards for its chemical makeup and mechanical strength, making it reliable for various applications. Essentially, it combines aluminum and copper with small amounts of iron and manganese to create a strong and durable metal suitable for different industries.

Key benefits of using ASTM B150 C62300 include:

- **Durability**: Its strong composition makes it ideal for high-stress environments like valves and pumps.
- **Versatility**: It's used in general engineering and automotive applications, showcasing its adaptability to different uses.
- **Standardization**: The material adheres to established specifications, ensuring consistent quality and performance across different projects.

Copper, Bus Bar, Rod and Shapes

Category: Materials

ASTM B187 C11000 standard specification for copper materials.

Key Features:

- Chemical Composition: Cu + AgCu% + Ag% 99.900 min.
- Heat Treatment: Normalizing or Annealing or Tempering

Technical Specifications:

Tensile Strength in Mpa	154.500 - 698.200
Yield Strength in Mpa	55 - 367
Hardness in HRB	30 - 97

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

Copper bus bars, rods, and shapes are high-quality copper materials that conform to the ASTM B187 C11000 standard. They are designed for various applications including valves, pumps, and automotive components. These products are known for their excellent conductivity, strength, and durability, making them ideal for general engineering projects.

Key benefits of using copper materials include:

- **High Purity**: With a minimum copper content of 99.900%, you can trust that the materials will perform reliably in demanding environments.
- **Versatile Applications**: Suitable for a range of uses in industries such as automotive and engineering, ensuring that you can find the right solution for your needs.
- **Strong and Durable**: With tensile strengths ranging from 154.500 to 698.200 MPa, these materials offer both flexibility and resistance to deformation, making them a dependable choice for various applications.

Copper Alloy Sand Casting

Category: Materials

ASTM B198-12A standard specification for copper alloy sand casting.

Key Features:

- Chemical Composition: Iron Fe %0.200 max., Manganese Mn %0.800 - 1.500, Lead Pb %0.090 max.
- Heat Treatment: As-Cast

Technical Specifications:

Tensile Strength in Mpa	310 min.
Yield Strength in Mpa	124 min.
Elongation in %	20 min.

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

Copper Alloy Sand Casting is a specialized manufacturing process that creates components from a mixture of copper and other metals using sand molds. This method follows strict standards (ASTM B198-12A) to ensure high quality and durability in the final products. The casting is particularly strong and can withstand significant stress, making it suitable for various applications, including valves, pumps, and automotive parts.

Key benefits include:

- **Strong and Durable**: With a minimum tensile strength of 310 MPa, these castings can handle heavy loads and resist wear.
- **Versatile Applications**: Ideal for use in industries like automotive and general engineering, providing reliable parts for critical machinery.
- **Standardized Quality**: Adheres to established specifications, ensuring consistent performance across all products.

Copper-Nickel-Zinc Wire

Category: Materials

ASTM B206 C75700 standard specification for copper-nickel-zinc wire.

Key Features:

- Chemical Composition: Iron Fe %0.250 max., Manganese Mn %0.500 max.
- Heat Treatment: Normalizing or Annealing or Tempering

Technical Specifications:

Tensile Strength in Mpa 505 - 895

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- India
- USA

AI Layman Summary:

Copper-Nickel-Zinc Wire is a highly durable and versatile material used in various industries, including automotive, valves, and pumps. It adheres to the ASTM B206 C75700 standard, ensuring high quality and

reliability. This wire is made from a specific blend of metals, which gives it excellent strength and resistance to corrosion.

The wire can be heat-treated through processes like normalizing, annealing, or tempering to enhance its properties further. Its tensile strength ranges from 505 to 895 MPa, making it suitable for demanding applications. Here are some key benefits:

- **Durability**: Strong resistance to wear and tear, ideal for heavy-duty applications.
- **Versatility**: Suitable for various uses, from automotive parts to general engineering projects.
- **Quality Assurance**: Meets stringent industry standards, ensuring reliability and performance.

Magnesium-Alloy Permanent Mold Castings

Category: Materials

ASTM B199 AZ92A standard specification for magnesium-alloy castings.

Key Features:

- Chemical Composition: Aluminium Al %8.300 - 9.700, Copper Cu %0.250 max.
- Heat Treatment: As-Cast or Solution Treated or Fully Treated

Technical Specifications:

Tensile Strength in Mpa	158 - 234
Yield Strength in Mpa	76 - 124
Elongation in %	6 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

Magnesium-Alloy Permanent Mold Castings are specialized materials made from a unique magnesium-aluminum alloy, designed to be strong yet lightweight. They are produced according to strict industry standards (ASTM B199 AZ92A), ensuring high quality and reliability. These castings can be heat-treated in various ways to enhance their strength and performance, making them versatile for different applications.

- Benefits of Magnesium-Alloy Castings:**
- **Lightweight and Strong**: Their high tensile strength (158 - 234 MPa) allows for durable products without the added weight, making them ideal for industries like automotive and engineering.
 - **Versatile Applications**: Suitable for making components like valves and pumps, they can be used in a variety of fields, ensuring that businesses can find uses in many areas.
 - **Customizable Heat Treatments**: Options for heat treatment allow manufacturers to tailor the properties of the castings to meet specific requirements, enhancing their usability in different environments.

Zinc and Zinc-Aluminum Alloys

Category: Materials

ASTM B240 standard specification for zinc and zinc-aluminum alloys.

Key Features:

- Chemical Composition: Aluminium Al %3.900 - 4.300, Cadmium Cd %0.003 max.
- Heat Treatment: As-Cast or Aging

Technical Specifications:

Tensile Strength in Mpa	328 min.
Yield Strength in Mpa	228 min.
Elongation in %	7 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

Zinc and Zinc-Aluminum Alloys are specialized materials used primarily in manufacturing and engineering applications. These alloys are designed to meet specific standards, ensuring they are reliable and effective for various uses. They are particularly useful in making parts for valves, pumps, and automotive components due to their strength and durability.

Key benefits include:

- **Strong Performance**: With a minimum tensile strength of 328 MPa and yield strength of 228 MPa, these materials can withstand significant stress, making them ideal for heavy-duty applications.
- **Versatile Applications**: Suitable for a wide range of industries including general engineering and automotive, allowing for diverse use cases.
- **Chemical Stability**: The alloy's composition ensures resistance to corrosion and wear, enhancing the longevity of the products made from them.

ASTM B 167 N06690

Category: Pipe and Tube

Standard Specification for Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Seamless Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 167 N06690 is a type of seamless pipe and tube made from a special nickel-chromium-iron alloy. This material is designed to be strong and resistant to corrosion, making it ideal for various industrial applications.

The seamless design means that the pipes are made without any joints or welds, which enhances their strength and reliability.

How it works: The composition of the alloy includes a high percentage of nickel (at least 58%), which provides excellent resistance to heat and chemicals. The pipes can withstand high pressure and maintain their integrity, which is crucial for applications like valves, pumps, and automotive parts.

- Benefits:
- **Durability**: High resistance to corrosion and heat, ensuring a longer lifespan.
 - **Strength**: Capable of handling extreme conditions, making it suitable for demanding environments.
 - **Versatility**: Useful in various sectors, including engineering and automotive, allowing for widespread applications.

ASTM B 168 N06600

Category: Plate, Sheet and Strip

Standard Specification for Nickel Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 168 N06600 is a type of nickel plate, sheet, and strip that is designed for various industrial applications. This material is known for its high nickel content, making it extremely resistant to corrosion and heat. It is primarily used in manufacturing components such as valves and pumps, as well as in general engineering and automotive industries.

- Key Benefits:**
- **Durability**: With a tensile strength of at least 550, this nickel product can withstand significant stress and pressure.
 - **Corrosion Resistance**: The high nickel content (minimum 72%) ensures that it performs well in harsh environments, extending the lifespan of the components made from it.
 - **Versatility**: Suitable for a range of applications, from industrial machinery to automotive parts, making it a valuable choice for manufacturers.

ASTM B 168 N06601

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 168 N06601 is a type of specialized metal that is primarily made of nickel, chromium, and iron, along with some other elements. It's commonly used in manufacturing plates, sheets, and strips for various industrial applications. This material is known for its excellent resistance to corrosion and high temperatures, making it ideal for environments that are harsh or require durability. It's especially useful in products like valves, pumps, and in general engineering and automotive applications.

Here are some key points about ASTM B 168 N06601:

- **Durable Composition**: Contains high levels of nickel (58-63%) and chromium (21-25%), providing strong resistance to rust and wear.
- **Strong Performance**: With a minimum tensile strength of 550 and yield strength of 205, it can handle significant stress without breaking.
- **Versatile Use**: Suitable for various applications, including automotive parts and engineering components, ensuring reliability in critical systems.

ASTM B 167 N06693

Category: Pipe and Tube

Standard Specification for Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Seamless Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 167 N06693 is a type of seamless pipe and tube made from a special blend of nickel, chromium, and iron, designed to withstand tough conditions. These pipes are used in various industries, including automotive and general engineering, where durability and resistance to corrosion are essential. They are particularly useful for manufacturing valves and pumps that need to perform reliably over time.

Key benefits include:

- **Strength and Durability**: With a minimum tensile strength of 690 MPa and yield strength of 345 MPa, these pipes can handle high pressure and stress.
- **Corrosion Resistance**: The unique chemical composition helps resist rust and degradation, making them ideal for harsh environments.
- **Versatile Applications**: Suitable for a range of uses in automotive and engineering sectors, ensuring broad applicability in various projects.

ASTM B 168 N06603

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 168 N06603 is a specialized material made from a mix of nickel, chromium, and iron, which is particularly strong and resistant to corrosion. This product comes in the form of plates, sheets, and strips, making it versatile for various manufacturing and engineering applications. Essentially, it can be used to create components in industries such as valves, pumps, and even automotive parts, where durability and reliability are essential.

Here are some key benefits of ASTM B 168 N06603:

- **High Strength**: With a minimum tensile strength of 650 and a yield strength of 300, it can withstand significant stress.
- **Corrosion Resistance**: The unique chemical composition ensures it resists corrosion, making it ideal for harsh environments.
- **Versatile Applications**: Its use in various industries like general engineering and automotive shows its adaptability and importance in manufacturing durable products.

ASTM B 168 N06617

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 168 N06617 is a type of high-performance metal alloy primarily used in various industrial applications. It is made from a combination of nickel, chromium, iron, and other elements, which makes it strong and resistant to corrosion. This alloy is commonly found in products like valves and pumps, contributing to their durability and

reliability in demanding environments.

Key benefits of ASTM B 168 N06617 include:

- **High strength**: With a minimum tensile strength of 655, this alloy can withstand significant stress and strain without breaking.
- **Corrosion resistance**: The unique chemical composition allows it to resist rust and deterioration, making it ideal for use in challenging conditions.
- **Versatile applications**: Suitable for general engineering, automotive parts, and various other industrial uses, this alloy is a reliable choice for many manufacturing needs.

ASTM B 168 N06690

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 168 N06690 is a type of strong metal material used to make plates, sheets, and strips. This specific alloy is made from nickel, chromium, iron, and other elements, which gives it excellent resistance to corrosion and high temperatures. It is commonly used in various industries, including automotive, general engineering, and in making valves and pumps, where durability and reliability are essential.

Key Benefits:

- **Durability**: The alloy's composition allows it to withstand harsh environments and extreme conditions, making it ideal for industrial use.
- **Versatility**: Suitable for a range of applications, from automotive parts to engineering components, enhancing its usability in different sectors.
- **Strength**: With a minimum tensile strength of 514 and yield strength of 206, it provides robust support for various mechanical applications.

ASTM B 240 AC41A

Category: Ingot and Casting

Standard Specification For Zinc and Zinc Alloy Castings.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 240 AC41A is a type of metal casting made primarily of zinc and zinc alloys. It is designed to meet specific standards for quality and performance, making it suitable for various applications in industries like automotive and engineering. This product is especially important for creating parts that need to be durable and reliable, such as valves and pumps.

****Key Benefits:****

- ****Strong and Durable:**** With a minimum tensile strength of 330, this material can withstand significant stress and pressure, making it ideal for heavy-duty applications.
- ****Corrosion Resistance:**** The chemical composition ensures that the castings can resist corrosion, which is crucial for parts exposed to harsh environments.
- ****Versatile Applications:**** It can be used in diverse areas such as general engineering, automotive parts, and machinery, highlighting its flexibility and utility.

ASTM B 240 AG40A

Category: Casting

Standard Specification For Zinc and Zinc Alloy Castings.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 240 AG40A is a standard specification for zinc and zinc alloy castings, which are materials commonly used in various industries for making durable and lightweight components. This specification outlines the chemical and mechanical properties of these castings, ensuring they meet specific quality standards for performance and safety. Essentially, it helps manufacturers create reliable parts that can withstand pressure and stress while remaining resistant to corrosion.

Key benefits of ASTM B 240 AG40A include:

- ****Durability**:** High tensile and yield strength ensure that parts made from this alloy can handle significant stress without breaking.
- ****Versatility**:** This material is suitable for a wide range of applications, including valves, pumps, and automotive components.
- ****Quality Assurance**:** The specification provides guidelines that help manufacturers produce consistent and high-quality castings, which is crucial for safety and efficiency in engineering projects.

ASTM B 240 ZA27

Category: Ingot and Casting

Standard Specification For Zinc and Zinc Alloy Ingot and Die Castings.

Technical Specifications:**Chemical Composition****Mechanical Properties**

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 240 ZA27 is a type of metal ingot and casting material made primarily from zinc and zinc alloys. It's designed to be used in various applications like valves, pumps, and automotive parts due to its strong and durable properties. Essentially, it serves as a raw material that can be melted down and shaped into parts that need to withstand pressure and stress.

Key benefits of ASTM B 240 ZA27 include:

- **High Strength**: With a minimum tensile strength of 425, it can handle heavy loads and resist breaking.
- **Versatile Applications**: Suitable for a wide range of industries, including engineering and automotive, making it adaptable for many manufacturing processes.
- **Controlled Composition**: The precise chemical makeup ensures consistency and reliability in performance, contributing to the longevity of the products made from it.

ASTM B 240 ZA12

Category: Ingot and Casting

Standard Specification for Zinc and Zinc Alloys Ingot and Die Castings.

Technical Specifications:**Chemical Composition****Mechanical Properties**

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 240 ZA12 is a type of metal ingot and die casting material made primarily from zinc and its alloys. It is engineered to meet specific standards for quality and performance, making it suitable for various applications. This material is particularly useful in industries such as automotive, general engineering, and manufacturing of valves and pumps.

Key benefits of ASTM B 240 ZA12 include:

- **Durability**: It offers strong mechanical properties, including a minimum tensile strength of 400, ensuring that products made from it are robust and long-lasting.
- **Versatility**: Its composition allows for use in a wide range of products from automotive parts to engineering components, making it a flexible choice for manufacturers.
- **High Quality**: The strict chemical composition standards ensure that the material is free from harmful levels of elements like lead and cadmium, promoting safety and reliability in its applications.

ASTM B 240 ZA8

Category: Casting

Standard Specification For Zinc and Zinc Alloy Ingot and Die Castings.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 240 ZA8 is a standard specification for zinc and zinc alloy materials used in casting processes. This means it outlines the guidelines for creating high-quality zinc ingots and die castings that can be used in various applications. Essentially, it helps manufacturers ensure that they are using the right materials to create durable and reliable products.

****Key Benefits:****

- ****Versatility:**** Suitable for a range of applications, including valves, pumps, general engineering, and automotive parts.
- ****Strength:**** Offers good mechanical properties with a minimum tensile strength of 370 and yield strength of 220, ensuring that products made from this material are strong and durable.
- ****Controlled Composition:**** The chemical makeup includes a balance of zinc along with other elements like aluminum and copper, ensuring high performance and longevity in different environments.

ASTM B 275 GS10A

Category: Bar, Rod, Shapes, Tube and Wire, Extrusion

Standard Specification For Aluminium Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 275 GS10A is a type of aluminum alloy that is used to create various shapes like bars, rods, tubes, and wires. This material is designed to meet specific standards, ensuring it has the right balance of strength and flexibility for different applications. The alloy is made up of several elements, each contributing to its overall performance, such as silicon, magnesium, and manganese.

Key benefits of ASTM B 275 GS10A include:

- **Versatility**: It can be used in a wide range of industries, including automotive, general engineering, and for making components like valves and pumps.
- **Strength and Durability**: The alloy offers good tensile and yield strength, making it suitable for demanding applications where reliability is essential.
- **Standardized Quality**: Being a standard specification means it adheres to specific quality guidelines, ensuring consistent performance across different batches.

ASTM B 283 C48500

Category: Forging

Standard Specification for Copper and Copper-Alloy Die Forgings.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 283 C48500 is a type of copper and copper-alloy material specifically designed for creating strong and durable forged products. This standard specification ensures that the material meets certain quality and performance criteria, making it suitable for various industrial applications. It contains a mix of copper, lead, tin, and zinc, which gives it unique properties that enhance its strength and resistance to wear and corrosion.

Key benefits of ASTM B 283 C48500 include:

- **High Strength**: With a minimum tensile strength of 358 MPa, it can withstand heavy loads and stress.
- **Versatile Use**: Ideal for manufacturing components in valves, pumps, and automotive parts, making it useful across different engineering fields.
- **Durability**: The chemical composition ensures long-lasting performance in demanding environments.

ASTM B 333 N10001

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 333 N10001 is a specialized type of metal plate, sheet, and strip made from a strong nickel-molybdenum alloy. This material is designed to be used in various applications where durability and resistance to harsh conditions are essential. It features a unique chemical composition that includes nickel, molybdenum, and other elements to enhance its performance and longevity.

Here are some key benefits of using ASTM B 333 N10001:

- **High Strength**: It has a minimum tensile strength of 690, making it suitable for demanding applications.
- **Versatile Applications**: Ideal for use in valves, pumps, and automotive parts, as well as general engineering needs.
- **Resistant to Corrosion**: The alloy's composition allows it to withstand tough environments, ensuring reliability and reducing maintenance costs.

ASTM B 333 N10624

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 333 N10624 is a type of high-quality metal material known as a nickel-molybdenum alloy, specifically available in the form of plates, sheets, and strips. This material is designed to be strong and durable, making it ideal for various applications such as valves, pumps, and general engineering projects, including automotive uses. Its unique composition of nickel and molybdenum provides excellent resistance to corrosion and wear, ensuring longevity and reliability in demanding environments.

Key Benefits:

- **Strong and Durable**: With a minimum tensile strength of 720 MPa, it can withstand significant stress and pressure.
- **Corrosion Resistant**: The nickel-molybdenum composition allows it to resist corrosion, making it suitable for harsh operating conditions.
- **Versatile Applications**: Perfect for use in critical components across industries, including automotive and general engineering.

ASTM B 333 N10629

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 333 N10629 is a type of strong and durable metal made from a mix of nickel and molybdenum, specifically designed for making plates, sheets, and strips. This material is used in various industries, including automotive and general engineering, where reliable performance is crucial. It stands out for its resistance to corrosion and its ability to withstand high temperatures, making it a preferred choice for components like valves and pumps.

Key benefits include:

- **Strong and Reliable**: With a tensile strength of at least 760 MPa, it can handle heavy loads without breaking.
- **Versatile Use**: Suitable for different applications, including automotive and engineering, which require materials that can endure harsh conditions.
- **Corrosion Resistance**: Its chemical composition helps it resist wear and tear, prolonging the life of the products made from it.

ASTM B 333 N10665

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 333 N10665 is a type of metal known as a nickel-molybdenum alloy, primarily available in plates, sheets, and strips. This material is designed for durability and strength, making it ideal for various applications, including valves, pumps, and general engineering tasks. The unique composition of nickel and molybdenum gives it excellent resistance to corrosion and heat, allowing it to perform well in challenging environments, such as in automotive industries.

Key Benefits:

- **Strong and Durable**: With a tensile strength of at least 760, this material can withstand significant stress.
- **Corrosion Resistance**: The alloy's composition helps it resist rust and degradation, ensuring longevity in various applications.
- **Versatile Use**: Suitable for a wide range of applications, from automotive to engineering, making it a practical choice for many industries.

ASTM B 333 N10675

Category: Plate, Sheet and Strip

Standard Specification for Nickel-Molybdenum Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 333 N10675 is a specialized type of metal alloy made from nickel and molybdenum, available in the form of plates, sheets, and strips. This alloy is designed to withstand high levels of stress and corrosion, making it ideal for various demanding applications. Essentially, it combines the strength and durability of nickel with the toughness of molybdenum, resulting in a material that can perform well in tough environments.

****Key Benefits:****

- ****High Strength:**** With a minimum tensile strength of 760, this alloy can handle significant pressure and force without breaking.
- ****Corrosion Resistance:**** Its unique chemical makeup allows it to resist rust and wear, which is crucial for parts used in valves, pumps, and automotive components.
- ****Versatile Uses:**** Suitable for a wide range of industries, including general engineering and automotive, making it a go-to choice for engineers and manufacturers.

ASTM B 335 N10001

Category: Rod

Standard Specification for Nickel-Molybdenum Alloy Rod.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 335 N10001 is a special type of rod made from a nickel-molybdenum alloy. This rod is known for its high strength and durability, making it an excellent choice for various industrial applications. It's designed to withstand tough conditions, which is why it's commonly used in valves, pumps, and automotive parts, as well as in general engineering projects.

Key benefits of the ASTM B 335 N10001 include:

- ****High Strength**:** With a minimum tensile strength of 690 MPa, this rod can handle heavy loads without breaking.
- ****Corrosion Resistance**:** The unique alloy composition helps resist rust and wear, ensuring longevity in

various environments.

- **Versatile Applications**: Ideal for use in critical manufacturing sectors, it supports a range of functions from automotive components to industrial machinery.

ASTM B 335 N10629

Category: Rod

Standard Specification for Nickel-Molybdenum Alloy Rod.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 335 N10629 is a type of rod made from a special nickel-molybdenum alloy. This material is designed to be strong and resistant to wear, making it ideal for various industrial applications. It is commonly used in valves, pumps, and in the automotive industry, among other engineering needs. The rod's unique composition allows it to withstand high stress and harsh environments, ensuring reliability and durability in critical components.

Key benefits of the ASTM B 335 N10629 include:

- **High Strength**: With a minimum tensile strength of 760 MPa, it can handle significant forces without breaking.

- **Corrosion Resistance**: The alloy composition makes it resistant to rust and corrosion, extending the lifespan of products made from it.

- **Versatile Applications**: Suitable for a range of uses, from automotive parts to general engineering, making it a valuable material in many industries.

ASTM B 335 N10624

Category: Rod

Standard Specification for Nickel-Molybdenum Alloy Rod.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 335 N10624 is a type of rod made from a special alloy that includes nickel and molybdenum, designed for various industrial applications. This rod is particularly useful in creating strong and reliable components for machinery and vehicles. It is known for its durability and ability to withstand harsh conditions, making it ideal for use in valves, pumps, and general engineering projects.

- Key benefits of ASTM B 335 N10624 include:
- **High Strength**: With a minimum tensile strength of 720 MPa, it can withstand significant force without breaking.
 - **Resistance to Corrosion**: The chemical composition helps prevent rust and degradation, extending the lifespan of products made from this rod.
 - **Versatile Applications**: Suitable for a wide range of uses in automotive and engineering industries, ensuring flexibility in manufacturing and design.

ASTM B 622 N06255

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 622 N06255 is a type of high-quality pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed to be seamless, meaning it has no joints or welds, which makes it stronger and more reliable for various applications. It is commonly used in industries such as automotive, engineering, and for components like valves and pumps.

- Key benefits of ASTM B 622 N06255 include:
- **Durability**: The alloy's strong chemical composition, including elements like chromium and molybdenum, helps it resist corrosion and wear, making it ideal for tough environments.
 - **Strength**: With a high tensile strength and yield strength, this pipe can withstand significant pressure, ensuring safety and performance in demanding applications.
 - **Versatility**: Its use in a variety of industries means it can adapt to different engineering challenges, providing reliable solutions across multiple sectors.

ASTM B 622 N06455

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N06455 is a high-quality pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed to be seamless, which means it is strong and reliable for a variety of applications. The unique composition of this alloy helps it withstand harsh conditions, making it suitable for industries like automotive, general engineering, and in components like valves and pumps.

Key benefits include:

- **Durability**: Resistant to corrosion and wear, ensuring a longer lifespan in demanding environments.
- **Strength**: Offers high tensile and yield strength, making it capable of handling significant pressure and stress.
- **Versatility**: Can be used in various applications, from automotive parts to engineering projects, which makes it a valuable material for many businesses.

ASTM B 622 N06686

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N06686 is a type of pipe and tube made from a special combination of nickel and cobalt alloys. This means it can withstand harsh conditions and is perfect for industries that require strong, durable materials. The seamless design enhances its strength and reliability, making it ideal for various applications such as valves, pumps, and automotive components.

Key benefits include:

- **High Resistance**: Its composition allows it to resist corrosion and high temperatures, making it suitable for challenging environments.
- **Strong Performance**: With a minimum tensile strength of 690 and a yield strength of 310, the material is tough and can handle heavy loads.
- **Versatile Usage**: This alloy pipe can be used in different sectors, including general engineering and automotive, providing flexibility for manufacturers.

ASTM B 622 N06985

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N06985 is a type of seamless pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed to handle demanding environments, making it ideal for various applications such as valves, pumps, and general engineering tasks. The pipe's unique chemical composition ensures it is strong and resistant to corrosion, which helps extend the life of the products made from it.

Key Benefits:

- **Durability:** With a tensile strength of at least 621, this pipe can withstand significant pressure, making it reliable for tough jobs.
- **Corrosion Resistance:** The alloy is engineered to resist rust and wear, which is crucial for applications in automotive and industrial settings.
- **Versatile Applications:** Suitable for use in valves, pumps, and various engineering projects, ensuring it meets a wide range of industry needs.

ASTM B 622 N06975

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 622 N06975 is a type of seamless pipe and tube made from a special nickel and nickel-cobalt alloy. This material is designed to be used in various applications, including valves, pumps, general engineering, and automotive parts. The seamless design means there are no welds or joints, making it stronger and more reliable for transporting fluids and gases under pressure.

Key benefits of using ASTM B 622 N06975 include:

- **High Strength:** With a minimum tensile strength of 586, this alloy can withstand significant force without breaking.
- **Corrosion Resistance:** The chemical composition includes chromium and molybdenum, which help prevent rust and degradation, ensuring a longer lifespan in harsh environments.
- **Versatile Applications:** Ideal for various industries, this alloy can be utilized in critical components like

valves and pumps, where durability and performance are essential.

ASTM B 622 N08135

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N08135 is a type of seamless pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed to be strong and resistant to various environmental factors, making it perfect for a wide range of applications. Whether you're working on valves, pumps, or general engineering tasks, this pipe can handle the demands of your project with ease.

Key Benefits:

- **Durable**: High tensile and yield strength ensure it can withstand heavy use.
- **Versatile**: Suitable for various applications, including automotive and engineering.
- **Corrosion Resistant**: The alloy composition provides excellent resistance against harsh conditions, extending the life of your components.

ASTM B 622 N08031

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 622 N08031 is a high-quality seamless pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed to handle tough environments, making it perfect for various applications such as valves, pumps, and automotive parts. Its unique composition helps it resist corrosion and wear, ensuring durability and longevity in use.

Key benefits include:

- **High Strength**: With a tensile strength of at least 650 units, this material can withstand significant pressure and stress.
- **Corrosion Resistance**: The alloy's chemical makeup includes chromium and molybdenum, which protect against rust and deterioration, making it ideal for harsh conditions.
- **Versatile Applications**: Suitable for various industries, it can be used in general engineering and automotive applications, providing flexibility for manufacturers.

ASTM B 622 N08320

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 622 N08320 is a type of seamless pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed for various industrial applications, particularly where resistance to corrosion and high strength is essential. It is commonly used in valves, pumps, general engineering tasks, and in the automotive industry.

The benefits of using ASTM B 622 N08320 include:

- **Durability**: Its strong chemical composition, which includes elements like chromium and molybdenum, provides excellent resistance to wear and tear, ensuring a longer lifespan in tough environments.
- **High Strength**: With a minimum tensile strength of 517 and yield strength of 193, this alloy can withstand significant stress, making it ideal for heavy-duty applications.
- **Versatility**: This material can be utilized in a variety of settings, from manufacturing pumps to automotive components, making it a reliable choice for engineers and manufacturers.

ASTM B 622 N08535

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 622 N08535 is a type of pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This seamless design makes it strong and resistant to corrosion, which is essential for various industrial applications. It is particularly valuable in environments where durability and reliability are crucial, such as in valves, pumps, general engineering, and automotive sectors.

- Key Benefits:
- **High Strength**: It has a minimum tensile strength of 503, ensuring it can handle heavy pressure and stress.
 - **Corrosion Resistance**: The unique chemical composition, including a high nickel content, protects it from rust and degradation, extending its lifespan and reducing maintenance costs.
 - **Versatile Use**: Suitable for a variety of applications, making it a go-to choice for engineers and manufacturers looking for dependable materials.

ASTM B 622 N10001

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N10001 is a type of seamless pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This material is designed to handle various engineering applications and is especially useful in environments where strength and resistance to corrosion are essential. The seamless design means it has no joints, making it more durable and reliable for transporting fluids and gases.

- Key features and benefits include:
- **High Strength**: With a minimum tensile strength of 690, these pipes can withstand significant pressure without breaking.
 - **Corrosion Resistance**: The unique alloy composition offers excellent resistance to chemicals and harsh environments, making it ideal for use in valves, pumps, and automotive applications.
 - **Versatile Applications**: Suitable for general engineering tasks, this pipe can be used in a wide range of industries, ensuring you have the right solution for your needs.

ASTM B 622 N10242

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N10242 is a type of seamless pipe and tube made primarily from nickel and nickel-cobalt alloy. This material is specially designed for various industrial applications, making it strong and resistant to corrosion. It is commonly used in environments where durability and reliability are critical, such as in valves, pumps, and automotive components.

Key benefits of the ASTM B 622 N10242 include:

- **High Strength**: With a minimum tensile strength of 725, it can withstand significant pressure and stress.
- **Corrosion Resistance**: The unique composition helps it resist rust and wear, ensuring a longer lifespan in tough conditions.
- **Versatile Applications**: Suitable for a variety of uses in engineering and automotive sectors, making it a flexible choice for manufacturers.

ASTM B 622 N10624

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 622 N10624 is a type of pipe and tube made from a special blend of nickel and nickel-cobalt alloy. This unique material is designed to be seamless, meaning it has no joints or welds, which makes it highly durable and reliable for various applications. It is primarily used in industries that require high-performance materials, such as automotive, pumps, valves, and general engineering tasks.

Key Benefits:

- **Strength and Durability**: With a tensile strength of at least 720 units, this pipe can withstand significant pressure and stress, making it ideal for demanding environments.
- **Corrosion Resistance**: The chemical composition, which includes nickel and molybdenum, provides excellent resistance to corrosion, ensuring longevity even in harsh conditions.
- **Versatile Applications**: Suitable for a range of uses from automotive parts to pumps and valves, it meets the needs of various engineering projects.

ASTM B 622 N10276

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N10276 is a type of pipe and tube made from a special combination of nickel and nickel-cobalt alloy. This material is designed to be strong and durable, making it ideal for various applications, especially in environments that require resistance to corrosion and high temperatures. The seamless design means there are no joints or welds, which adds to its strength and reliability.

Key benefits include:

- **High Strength**: With a tensile strength of at least 690, this pipe can withstand significant pressure and stress.
- **Corrosion Resistance**: The unique chemical composition helps it resist damage from harsh chemicals and environments, making it perfect for industries like automotive and general engineering.
- **Versatile Use**: Suitable for valves, pumps, and other engineering applications, providing flexibility in design and manufacturing.

ASTM B 622 N10629

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N10629 is a type of seamless pipe and tube made from a special combination of nickel and nickel-cobalt alloy. This product is designed to be strong and durable, making it ideal for various applications where reliability is crucial. It is particularly useful in industries like automotive, general engineering, and for components such as valves and pumps.

Key benefits of the ASTM B 622 N10629 include:

- **High Strength**: With a minimum tensile strength of 760, this pipe can withstand significant pressure and stress, ensuring it performs well in demanding environments.
- **Corrosion Resistance**: The unique chemical composition helps the pipe resist corrosion, extending its lifespan in harsh conditions.
- **Versatility**: Suitable for a wide range of applications, from automotive parts to engineering projects, making

it a flexible choice for manufacturers.

ASTM B 622 N10665

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 N10665 is a type of high-quality pipe and tube made from a special blend of nickel and nickel-cobalt alloy. These pipes are seamless, meaning they have no welds, which makes them stronger and more reliable for various applications. They are designed to withstand harsh conditions and are commonly used in industries like automotive, engineering, and for components like valves and pumps.

Key benefits of the ASTM B 622 N10665 include:

- ****Durability****: The alloy composition gives it high resistance to corrosion and wear, ensuring a long lifespan.
- ****Strength****: With impressive tensile and yield strength, these pipes can handle heavy loads and high pressure without failing.
- ****Versatility****: Suitable for a range of applications, making it a go-to choice for engineers and manufacturers looking for reliable piping solutions.

ASTM B 622 R20033

Category: Pipe and Tube

Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 622 R20033 is a special type of pipe and tube made from seamless Nickel and Nickel-Cobalt alloys. These materials are chosen for their strength and resistance to corrosion, making them ideal for use in challenging environments. This product is commonly used in applications such as valves, pumps, general engineering, and automotive parts, where durability and reliability are critical.

Key benefits of the ASTM B 622 R20033 include:

- **High Strength**: With a minimum tensile strength of 750 and yield strength of 380, these pipes can handle significant stress and pressure.
- **Corrosion Resistance**: The chemical makeup, including a high percentage of Nickel and Chromium, helps the pipes resist rust and degradation, extending their lifespan.
- **Versatile Applications**: Suitable for various industries, this product can be utilized in many different systems, ensuring it meets diverse engineering needs.

ASTM B 625 N08925

Category: Plate, Sheet and Strip

Ni-Fe-Cr-Mo-Cu-N Low-Carbon Alloy, Ni-Fe-Cr-Si Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 625 N08925 is a special type of metal alloy used in various industrial applications. This alloy is made primarily of nickel, iron, chromium, and molybdenum, making it highly resistant to corrosion and suitable for harsh environments. Its low carbon content and high strength ensure durability, making it an excellent choice for industries like automotive, general engineering, and for components such as valves and pumps.

Key benefits of ASTM B 625 N08925 include:

- **Corrosion Resistance**: It can withstand harsh chemicals and environments, which prolongs the lifespan of tools and machinery.
- **High Strength**: With a minimum tensile strength of 600 and yield strength of 295, this alloy can handle heavy loads without breaking.
- **Versatile Applications**: Ideal for various uses such as automotive parts, pumps, and valves, making it a go-to material for engineers and manufacturers.

ASTM B 625 N08926

Category: Plate, Sheet and Strip

Ni-Fe-Cr-Mo-Cu-N Low-Carbon Alloy, Ni-Fe-Cr-Si Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 625 N08926 is a specialized type of metal alloy designed for use in a variety of industrial applications. This alloy is made up of nickel, iron, chromium, molybdenum, and copper, making it highly resistant to corrosion and suitable for harsh environments. It is primarily used in the creation of components such as valves and pumps, as well as in general engineering and automotive applications.

The key benefits of this alloy include:

- **Durability**: With a high tensile strength of 650 min and yield strength of 295 min, it can withstand significant stress and strain without breaking.
- **Corrosion Resistance**: Its unique chemical composition helps it resist degradation, making it ideal for use in environments that may expose it to corrosive substances.
- **Versatility**: Suitable for a wide range of applications, this alloy can be used in various fields including automotive and general engineering, ensuring it meets diverse industry needs.

ASTM B 625 R20033

Category: Plate, Sheet and Strip

Ni-Fe-Cr-Mo-Cu-N Low-Carbon Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 625 R20033 is a special type of low-carbon alloy made primarily of nickel, iron, chromium, and molybdenum. It's designed for use in demanding applications where resistance to corrosion and strength are crucial. This material is often used in valves, pumps, and automotive components due to its durability and ability to withstand harsh environments.

Key benefits of ASTM B 625 R20033 include:

- **High Strength**: With a tensile strength of at least 750, it can handle significant stress without breaking.
- **Corrosion Resistance**: The unique chemical composition helps prevent rust and degradation, extending the lifespan of products made from it.
- **Versatile Applications**: Suitable for various industries, including engineering and automotive, making it a reliable choice for manufacturers.

ASTM B150 UNS C63200

Category: Materials

Aluminum Bronze Rod, Bar and Shapes.

Key Features:

- Chemical Composition: Al % 8.700 - 9.500, Cu % Balance
- Mechanical Properties: Tensile Strength in Mpa 620 min, Yield Strength in Mpa 275 min

Technical Specifications:

Lead Pb %	0.020 max.
Silicon Si %	0.100 max.
Manganese Mn %	1.200 - 2.000
Nickel Ni %	4.000 - 4.800

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B150 UNS C63200 is a type of aluminum bronze material that comes in various shapes like rods and bars. This metal is primarily made of aluminum and copper, which gives it unique properties that make it ideal for a range of applications. It is known for its strength and durability, making it suitable for use in valves, pumps, and general engineering projects, as well as in the automotive industry.

Key benefits of ASTM B150 UNS C63200 include:

- High tensile strength and yield strength, ensuring that it can withstand significant pressure and stress.
- Excellent resistance to corrosion, making it a reliable choice for components that are exposed to harsh environments.
- Versatile applications in various industries, allowing manufacturers to use it in a wide range of products.

ASTM B150 UNS C64200

Category: Materials

Aluminum Bronze Rod, Bar and Shapes.

Key Features:

- Chemical Composition: Al % 6.300 - 7.600, Cu % Balance
- Mechanical Properties: Tensile Strength in Mpa 485 min, Yield Strength in Mpa 170 min

Technical Specifications:

Lead Pb %	0.050 max.
Silicon Si %	1.500 - 2.200
Manganese Mn %	0.100 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B150 UNS C64200 is a type of Aluminum Bronze material that comes in the form of rods, bars, and shapes. This specialized metal is known for its strong combination of aluminum and copper, which gives it excellent resistance to corrosion and wear. It is commonly used in various industries, such as automotive and general engineering, to create durable components like valves and pumps.

Key Benefits:

- **Strong and Durable:** With a minimum tensile strength of 485 MPa, it can withstand significant stress, making it ideal for heavy-duty applications.
- **Corrosion Resistant:** Its unique chemical composition helps it resist rust and degradation, ensuring a longer lifespan of parts made from it.
- **Versatile Use:** Suitable for a range of applications, including valves and automotive components, making it a reliable choice for engineers and manufacturers.

ASTM B151 C70620

Category: Materials

Copper-Nickel-Zinc Alloy and Copper-Nickel Rod and Bar.

Key Features:

- Chemical Composition: Cu + Ag % 86.500 min.
- Mechanical Properties: Tensile Strength in Mpa 260 - 415, Yield Strength in Mpa 105 - 260

Technical Specifications:

Lead Pb %	0.020 max.
Iron Fe %	1.000 - 1.800
Manganese Mn %	1.000 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B151 C70620 is a special type of metal made primarily from copper, nickel, and zinc, designed for a variety of engineering applications. This alloy is known for its strong mechanical properties, making it suitable for use in tough environments where durability is essential. It is commonly used in valves and pumps, as well as in general engineering and automotive parts, ensuring reliability in performance.

Key benefits of ASTM B151 C70620 include:

- **High Strength:** With a tensile strength that ranges from 260 to 415 MPa, this metal can withstand significant stress without breaking.
- **Versatile Applications:** Its properties make it ideal for use in various industries, including automotive and engineering, where strong, corrosion-resistant materials are required.

ASTM B151 C75200

Category: Materials

Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Rod and Bar.

Key Features:

- Chemical Composition: Cu + Ag % 63.000 - 66.500
- Mechanical Properties: Tensile Strength in Mpa 415 - 690

Technical Specifications:

Lead Pb %	0.050 max.
Iron Fe %	0.250 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B151 C75200 is a special type of metal alloy made from copper, nickel, and zinc, commonly known as nickel silver. This durable material is commonly used in a variety of applications such as valves, pumps, general engineering tasks, and the automotive industry. It's known for its strength and resistance to corrosion, making it a reliable choice for many industrial needs.

Key benefits of ASTM B151 C75200 include:

- **Strong and Durable**: With a tensile strength ranging from 415 to 690 MPa, it can withstand significant stress without breaking.
- **Corrosion Resistance**: This alloy is designed to resist rust and wear, ensuring longevity in various environments.
- **Versatile Applications**: Ideal for use in automotive parts, pumps, and valves, it is suitable for multiple engineering applications.

ASTM B151 C71520

Category: Materials

Copper-Nickel-Zinc Alloy and Copper-Nickel Rod and Bar.

Key Features:

- Chemical Composition: Cu + Ag % 65.000 min.
- Mechanical Properties: Tensile Strength in Mpa 310 - 550

Technical Specifications:

Lead Pb %	0.020 max.
Iron Fe %	0.400 - 1.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B151 C71520 is a special type of metal made from a mixture of copper, nickel, and zinc. This alloy is designed to be strong and durable, making it ideal for various applications, including valves, pumps, and automotive parts. It has a minimum copper content of 65% and is engineered to withstand significant stress, with a tensile strength ranging from 310 to 550 megapascals.

Key benefits of using this material include:

- **Strength and Durability**: Capable of handling tough conditions without breaking.
- **Versatile Applications**: Suitable for a wide range of industries, from engineering to automotive, ensuring reliability in critical components.
- **Low Impurities**: Contains minimal lead and iron, enhancing its quality and performance.

ASTM B152 C11000

Category: Materials

Copper Sheet, Strip, Plate and Rolled Bar.

Key Features:

- Chemical Composition: Cu + Ag % 99.900 min.
- Mechanical Properties: Tensile Strength in Mpa 154.500 - 698.200

Technical Specifications:

Lead Pb % 0.010 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B152 C11000 is a high-quality copper material available in various forms such as sheets, strips, plates, and rolled bars. It is primarily made of copper with a minimum purity of 99.900%, which means it is very effective for conductive applications. This copper material is known for its excellent tensile strength, ranging from 154.500 to 698.200 MPa, making it durable and reliable for various engineering needs.

Key benefits of ASTM B152 C11000 include:

- **Versatile Use**: Ideal for applications in valves, pumps, general engineering, and the automotive industry.
- **High Purity**: The high copper content ensures excellent conductivity and performance in electrical applications.
- **Strong and Durable**: With impressive tensile strength, this material can withstand significant stress, making it suitable for demanding environments.

ASTM B171 C36500

Category: Materials

Copper-Alloy Plate and Sheet for Pressure Vessels.

Key Features:

- Chemical Composition: Cu % 58.000 - 61.000
- Mechanical Properties: Tensile Strength in Mpa 275 - 345

Technical Specifications:

Lead Pb % 0.250 - 0.700

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B171 C36500 is a specialized type of copper-alloy plate and sheet designed for use in pressure vessels, which are containers that hold gases or liquids at high pressure. This material is made primarily of copper, with

a specific chemical composition that ensures durability and strength. It has mechanical properties that provide a good balance between flexibility and hardness, making it suitable for various engineering applications.

Key benefits of ASTM B171 C36500 include:

- **Versatile Applications**: Ideal for use in valves, pumps, and automotive parts, making it a go-to choice for many industries.
- **Strong and Reliable**: With a tensile strength ranging from 275 to 345 MPa, it can withstand significant stress and pressure, ensuring safety and longevity in demanding environments.

ASTM B171 C61300

Category: Materials

Copper-Alloy Plate and Sheet for Pressure Vessels.

Key Features:

- Chemical Composition: Al % 6.000 - 7.500
- Mechanical Properties: Tensile Strength in Mpa 450 - 515

Technical Specifications:

Lead Pb % 0.010 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B171 C61300 is a specialized type of copper-alloy plate and sheet designed for use in pressure vessels, which are containers that hold gases or liquids at high pressure. This material is made from a specific blend of copper with aluminum, ensuring it has the right balance of strength and durability. With a tensile strength of 450 to 515 MPa, it can withstand significant stress without breaking, making it ideal for demanding applications.

Key benefits of using ASTM B171 C61300 include:

- **High Strength**: It offers excellent tensile strength, which means it can handle heavy loads and pressure.
- **Versatile Applications**: Suitable for various uses, including valves, pumps, and general engineering tasks, as well as in the automotive industry.
- **Low Lead Content**: The lead content is kept to a minimum (0.010% max), promoting safety and environmental compliance in manufacturing processes.

ASTM B171 C61400

Category: Materials

Copper-Alloy Plate and Sheet for Pressure Vessels.

Key Features:

- Chemical Composition: Al % 6.000 - 8.000
- Mechanical Properties: Tensile Strength in Mpa 450 - 485

Technical Specifications:

Lead Pb % 0.010 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B171 C61400 is a type of copper-alloy plate and sheet specifically designed for use in pressure vessels, which are containers used to hold gases or liquids at high pressure. This material is made from a special blend of copper and aluminum, making it strong and durable while also being resistant to corrosion. Its key features include a chemical composition that contains 6% to 8% aluminum and impressive mechanical properties with a tensile strength ranging from 450 to 485 megapascals, ensuring it can withstand tough conditions without breaking.

The benefits of using ASTM B171 C61400 include:

- Excellent strength and durability for various applications, including valves, pumps, and general engineering tasks.
- Resistance to corrosion, which extends the lifespan of equipment and reduces maintenance costs.
- Versatility, making it suitable for a wide range of industries, including automotive and engineering.

ASTM B171 C63000

Category: Materials

Copper-Alloy Plate and Sheet for Pressure Vessels.

Key Features:

- Chemical Composition: Al % 9.000 - 11.000
- Mechanical Properties: Tensile Strength in Mpa 550 - 620

Technical Specifications:

Lead Pb % 0.010 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B171 C63000 is a special type of metal made primarily from copper, designed specifically for use in pressure vessels. These vessels can be found in various applications such as valves and pumps, as well as in general engineering and automotive industries. The alloy contains a specific range of aluminum, which enhances its strength and durability, making it suitable for demanding environments where safety and reliability are crucial.

Key features of ASTM B171 C63000 include:

- **Chemical Composition**: Contains 9% to 11% aluminum, which improves its properties.
- **Mechanical Strength**: Offers high tensile strength between 550 to 620 MPa, ensuring it can withstand significant pressure without failing.

Overall, this material is beneficial for manufacturers looking for a strong, reliable metal that performs well under pressure, ensuring the longevity and safety of their products.

ASTM B171 C63200

Category: Materials

Aluminum Bronze Rod, Bar and Shapes.

Key Features:

- Chemical Composition: Al % 8.700 - 9.500
- Mechanical Properties: Tensile Strength in Mpa 620 min

Technical Specifications:

Lead Pb % 0.020 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B171 C63200 is a type of aluminum bronze material, specifically available in rod, bar, and various shapes. This material is known for its durability and strength, making it a popular choice in different industries. It contains a specific percentage of aluminum (between 8.7% and 9.5%), which contributes to its excellent mechanical properties, including a minimum tensile strength of 620 MPa, meaning it can withstand a lot of stress without breaking.

Key benefits of ASTM B171 C63200 include:

- **Versatility**: It can be used in a variety of applications such as valves, pumps, general engineering projects, and automotive parts.
- **Durability**: The aluminum bronze composition provides resistance to corrosion and wear, extending the lifespan of products made from it.
- **Strength**: With high tensile strength, it ensures reliable performance in demanding environments.

ASTM B171 C70620

Category: Materials

Copper-Alloy Plate and Sheet for Pressure Vessels.

Key Features:

- Chemical Composition: Cu + Ag % 86.500 min.
- Mechanical Properties: Tensile Strength in Mpa 275 min

Technical Specifications:

Lead Pb % 0.020 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B171 C70620 is a special type of copper-alloy plate and sheet designed for use in pressure vessels,

which are containers that hold gases or liquids under pressure. This material is highly durable and has a specific chemical composition that includes at least 86.5% copper and silver. Its strong mechanical properties mean it can withstand significant stress, making it ideal for various industrial applications.

Key benefits of ASTM B171 C70620 include:

- **High Strength**: With a minimum tensile strength of 275 MPa, it can handle tough conditions without failing.
- **Versatile Applications**: It is suitable for use in valves, pumps, general engineering, and automotive components, ensuring it can meet diverse industry needs.

ASTM B 575 N06455

Category: Plate, Sheet and Strip

Standard Specification for Low-Carbon Nickel-Chromium-Molybdenum Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 575 N06455 is a specialized metal alloy used primarily in the production of plates, sheets, and strips. This low-carbon nickel-chromium-molybdenum alloy is designed to withstand various industrial applications, particularly in environments that require high strength and resistance to corrosion. It is composed mainly of nickel, chromium, and molybdenum, making it highly durable and versatile.

Key Benefits:

- **Strength and Durability**: With a minimum tensile strength of 690 and yield strength of 276, this alloy can endure significant stress, making it ideal for heavy-duty applications like valves and pumps.
- **Corrosion Resistance**: The unique chemical composition provides excellent resistance to corrosive environments, ensuring a longer lifespan for products made from this alloy.
- **Versatile Applications**: Suitable for various sectors, including general engineering and automotive, this alloy can be used in a wide range of manufacturing processes.

ASTM B 575 N06210

Category: Plate, Sheet and Strip

Standard Specification for Low-Carbon Nickel-Chromium-Molybdenum Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 575 N06210 is a specialized metal alloy designed for high-performance applications. This alloy is primarily made of low-carbon nickel, chromium, and molybdenum, making it strong and resistant to corrosion. It's commonly used in industries such as automotive, engineering, and for manufacturing valves and pumps. The unique combination of elements gives it exceptional mechanical properties, ensuring it can withstand significant stress and elongation without breaking.

Key benefits include:

- **Strength and Durability**: With a minimum tensile strength of 690, this alloy can handle heavy loads and stress.
- **Corrosion Resistance**: Its composition helps prevent rust and damage from harsh environments, making it ideal for long-lasting applications.
- **Versatility**: Suitable for various uses, including valves, pumps, and general engineering tasks, enhancing its appeal across multiple industries.

ASTM B 575 N06686

Category: Plate, Sheet and Strip

Standard Specification for Low-Carbon Nickel-Chromium-Molybdenum Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 575 N06686 is a special type of metal alloy known for its low carbon content and high durability. This alloy primarily consists of nickel, chromium, and molybdenum, making it strong and resistant to corrosion. It is often used in harsh environments where equipment is exposed to chemicals and extreme temperatures, such as in valves, pumps, and automotive applications.

Key Benefits:

- **Strong Performance**: With a minimum tensile strength of 690, this alloy can handle heavy loads without breaking.
- **Corrosion Resistance**: Its unique composition helps prevent rust and degradation, extending the life of products made from it.
- **Versatile Uses**: Suitable for various industries including general engineering and automotive, making it a reliable choice for manufacturers.

ASTM B 575 N10276

Category: Plate, Sheet and Strip

Standard Specification for Low-Carbon Nickel-Chromium-Molybdenum Alloy.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 575 N10276 is a type of specialty metal that is designed for various industrial applications, particularly where strength and resistance to corrosion are important. This metal is made from a low-carbon alloy that includes nickel, chromium, and molybdenum, which gives it enhanced durability and performance in challenging environments. It is commonly used in valves, pumps, and automotive components, making it ideal for engineering projects that require reliable materials.

Key benefits of ASTM B 575 N10276 include:

- **High Strength**: With a tensile strength of at least 690 MPa, this material can withstand heavy loads without breaking.
- **Corrosion Resistance**: The chemical composition helps it resist rust and wear, extending the lifespan of components made from it.
- **Versatile Use**: Suitable for a range of applications, from general engineering to automotive, making it a flexible choice for manufacturers.

ASTM B 581 N06007

Category: Rod

Standard Specification for Nickel-Chromium-Iron-Molybdenum-Copper Alloy Rod.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 581 N06007 is a specialized rod made from a unique blend of metals, including nickel, chromium, iron, molybdenum, and copper. This rod is designed to be strong and durable, making it ideal for various applications such as valves, pumps, general engineering projects, and automotive components. Its specific chemical composition ensures that it can withstand harsh environments and high-stress situations, providing reliability in critical applications.

Here are some key points about the ASTM B 581 N06007:

- **Strong and Durable**: With a minimum tensile strength of 586 and yield strength of 207, this rod is built to handle significant stress and pressure.
- **Versatile Applications**: Suitable for use in valves, pumps, and automotive parts, making it a valuable choice for engineers and manufacturers in different industries.
- **Corrosion Resistance**: The nickel and chromium content helps protect against rust and corrosion, extending the lifespan of products made from this material.

ASTM B 581 N06975

Category: Rod

Standard Specification for Nickel-Chromium-Iron-Molybdenum-Copper Alloy Rod.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 581 N06975 is a type of metal rod made from a special blend of nickel, chromium, iron, molybdenum, and copper. This unique combination gives the rod excellent strength and durability, making it suitable for a variety of industrial applications. It is commonly used in industries like automotive, engineering, and in the manufacturing of valves and pumps.

Key Benefits:

- **High Strength:** With a minimum tensile strength of 586, this rod can withstand significant force without breaking.
- **Corrosion Resistance:** The alloy's composition helps it resist rust and corrosion, which is essential for long-lasting performance in harsh environments.
- **Versatile Applications:** Ideal for use in valves, pumps, and general engineering, making it a versatile choice for many projects.

ASTM B 619 N06002

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06002 is a high-quality pipe made from a special combination of nickel and nickel-cobalt alloys. This type of pipe is designed to be strong and resistant to corrosion, making it ideal for various demanding applications. It's particularly useful in industries like automotive, general engineering, and for components like valves and pumps.

Key benefits include:

- **Durability**: With a tensile strength of 690 MPa, this pipe can withstand significant stress and pressure.
- **Corrosion Resistance**: The chemical composition, including chromium and molybdenum, helps protect the pipe from rust and damage in harsh environments.
- **Versatility**: Suitable for a range of uses, whether in automotive parts or industrial machinery, making it a smart choice for engineers and manufacturers.

ASTM B 619 N06022

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06022 is a type of pipe made from a special blend of nickel and nickel-cobalt alloy. This material is designed to be strong and resistant to harsh environments, making it ideal for a variety of applications including valves, pumps, and automotive parts. The unique chemical composition of this alloy ensures that it can withstand high temperatures and corrosive conditions, which is crucial in many engineering fields.

Key benefits of ASTM B 619 N06022 include:

- **Durability**: With a tensile strength of at least 690 MPa, it can handle heavy loads without breaking.
- **Corrosion Resistance**: The alloy's composition, particularly the high chromium and molybdenum content, helps prevent rust and deterioration.
- **Versatility**: It's suitable for several industries, allowing for broad applications in general engineering, automotive, and more.

ASTM B 619 N06030

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06030 is a special type of pipe made from a nickel and nickel-cobalt alloy. This pipe is designed to be strong and resistant to corrosion, making it ideal for use in harsh environments. It is particularly useful in applications like valves, pumps, and automotive parts, where reliability and durability are crucial.

- Key benefits of ASTM B 619 N06030 include:
- **High Strength**: With a minimum tensile strength of 586, this pipe can withstand heavy pressure and stress.
 - **Corrosion Resistance**: The unique chemical composition helps prevent rust and damage over time, ensuring longer-lasting performance.
 - **Versatile Applications**: Suitable for various industries, it's used in general engineering and automotive sectors, ensuring it meets diverse needs.

ASTM B 619 N06035

Category: Pipe
Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 619 N06035 is a specialized type of pipe made from a combination of nickel and cobalt alloys. This material is designed to withstand harsh conditions and is often used in applications where durability and resistance to corrosion are crucial. It is commonly found in industries such as automotive, engineering, and in components like valves and pumps.

- Key benefits of using ASTM B 619 N06035 pipes include:
- **High Strength**: With a minimum tensile strength of 586, these pipes are strong and can handle significant pressure.
 - **Corrosion Resistance**: The unique chemical composition helps to prevent rust and degradation, making it reliable for long-term use in challenging environments.
 - **Versatile Applications**: Ideal for various uses, from automotive parts to general engineering, providing flexibility in manufacturing and construction.

ASTM B 619 N06059

Category: Pipe
Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06059 is a type of pipe made from a special blend of nickel and nickel-cobalt alloy. This pipe is designed to be strong and resistant to various environmental factors, making it ideal for use in demanding industries. It can be used in applications such as valves, pumps, and automotive components, ensuring reliable performance in critical situations.

Key benefits of ASTM B 619 N06059 include:

- **High Strength**: With a tensile strength of at least 690, this pipe can withstand heavy stress without breaking.
- **Corrosion Resistance**: The unique chemical composition, including elements like chromium and molybdenum, helps resist rust and damage from harsh chemicals.
- **Versatile Applications**: Suitable for various applications in general engineering, automotive, and more, making it a flexible choice for many projects.

ASTM B 619 N06058

Category: Pipe

Standard Specification for Welded Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 619 N06058 is a type of pipe made from a special blend of nickel and nickel-cobalt alloys. This means it's designed to be very strong and resistant to various kinds of wear and tear. It's especially useful in industries where high temperatures and corrosive environments are common, such as in valves, pumps, and automotive applications. The pipe is made according to strict standards to ensure that it meets specific quality and performance requirements.

Benefits of ASTM B 619 N06058 Pipe:

- **Durability**: It can withstand harsh conditions, making it ideal for critical applications.
- **High Strength**: With a tensile strength of at least 760, it can support heavy loads without breaking.
- **Versatile Use**: Suitable for various applications, including engineering and automotive, ensuring broad usability.

ASTM B 619 N06200

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06200 is a type of pipe made from a special combination of nickel and cobalt alloys. This material is designed to withstand harsh environments and is commonly used in various industries such as automotive, engineering, and for components like valves and pumps. The unique chemical composition helps it resist corrosion and maintain strength under pressure, making it a reliable choice for many applications.

Key benefits of ASTM B 619 N06200 include:

- **Durability**: With high tensile and yield strength, it can handle demanding conditions without failing.
- **Corrosion Resistance**: The nickel and cobalt alloys protect against rust and degradation, extending the life of the pipes.
- **Versatility**: Ideal for a wide range of uses, from automotive components to engineering projects, ensuring it meets diverse industry needs.

ASTM B 619 N06210

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06210 is a type of pipe made from a special alloy consisting mainly of nickel and cobalt. This pipe is designed to withstand harsh conditions and is often used in various industries such as automotive, engineering, and for valves and pumps. It is recognized for its durability and resistance to corrosion, making it a reliable choice for demanding applications.

Key benefits of ASTM B 619 N06210 include:

- **High Strength**: With a minimum tensile strength of 690, this pipe can handle significant pressure without breaking.
- **Corrosion Resistance**: The alloy composition helps it resist rust and wear, ensuring a longer lifespan.
- **Versatile Use**: Suitable for multiple applications, including valves, pumps, and general engineering, making it a practical option for various projects.

ASTM B 619 N06455

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06455 is a type of pipe made from a special nickel and nickel-cobalt alloy. This material is designed to withstand harsh conditions and is commonly used in various industries such as automotive, engineering, and in parts like valves and pumps. The alloy's unique chemical composition makes it resistant to corrosion and wear, ensuring long-lasting performance even in demanding environments.

****Key Benefits:****

- ****Durability****: The pipe is built to last, with a high tensile strength and resistance to corrosion.
- ****Versatility****: It can be used in a variety of applications across different sectors.
- ****High Performance****: With excellent mechanical properties, it can handle significant stress and strain, making it reliable for critical applications.

ASTM B 619 N06230

Category: Pipe

Standard Specification for Nickel and Nickel-Cobalt Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 619 N06230 is a specialized type of pipe made from a unique combination of nickel and nickel-cobalt alloys. These materials give the pipe exceptional resistance to high temperatures and corrosive environments, making it ideal for use in challenging industries such as automotive, general engineering, valves, and pumps. Essentially, this pipe is designed to perform reliably where standard materials might fail, ensuring long-lasting durability and safety in various applications.

Key benefits of ASTM B 619 N06230 include:

- ****High Strength****: With a minimum tensile strength of 760, this pipe can withstand significant stress without

breaking.

- **Corrosion Resistance**: The chemical composition allows it to resist rust and corrosion, extending its lifespan and reducing maintenance costs.
- **Versatile Applications**: Suitable for a wide range of uses, from automotive parts to industrial machinery, making it a valuable choice for manufacturers.

ASTM B 710 N08330

Category: Pipe
Nickel-Iron-Chromium-Silicon Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 710 N08330 is a specialized type of pipe made from a unique alloy containing nickel, iron, chromium, and silicon. This combination gives the pipe excellent strength and resistance to various environmental factors, making it suitable for demanding applications. It is particularly useful in industries such as automotive, general engineering, and for components like valves and pumps.

Key Benefits:

- **Durability**: The alloy's composition ensures that the pipe can withstand high stress and maintains its shape under pressure.
- **Versatility**: It can be used in a variety of applications, from automotive parts to general engineering projects.
- **Strength**: With a minimum tensile strength of 483 and a yield strength of 207, it provides reliable performance in tough conditions.

ASTM B 729 N08024

Category: Pipe and Tube
Seamless Nickel-Iron-Chromium-Molybdenum-Copper Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 729 N08024 is a special type of pipe and tube made from a strong and durable nickel-iron-chromium-molybdenum-copper alloy. This means it is designed to resist corrosion and withstand high temperatures, making it suitable for various industrial applications. It is seamless, which means there are no joints or welds that can weaken it, ensuring a consistent and reliable performance.

This alloy pipe is perfect for use in valves, pumps, and general engineering tasks, as well as in the automotive industry. Some key benefits include:

- High strength and durability to resist wear and tear.
- Excellent resistance to corrosion in harsh environments.
- Versatile applications across different industries.

ASTM B 729 N08020

Category: Pipe and Tube

Seamless Nickel-Iron-Chromium-Molybdenum-Copper Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 729 N08020 is a special type of pipe and tube made from a strong and durable nickel-iron-chromium-molybdenum-copper alloy. This seamless material is designed to be used in various industries, such as automotive, engineering, and in components like valves and pumps. Its unique composition allows it to withstand harsh environments and high temperatures, making it a reliable choice for demanding applications.

****Benefits of ASTM B 729 N08020:****

- ****High Strength:**** With a minimum tensile strength of 550, it can handle significant pressure without breaking.
- ****Corrosion Resistant:**** The alloy's composition helps it resist rust and corrosion, extending its lifespan.
- ****Versatile Use:**** Suitable for various applications, including pumps, valves, and general engineering tasks, making it a flexible option for different projects.

ASTM B 729 N08026

Category: Pipe and Tube

Seamless Nickel-Iron-Chromium-Molybdenum-Copper Alloy Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 729 N08026 is a type of seamless pipe and tube made from a special blend of metals, including nickel, iron, chromium, molybdenum, and copper. This unique combination gives it excellent resistance to corrosion and high temperatures, making it suitable for various demanding applications. The product is particularly useful in industries such as automotive, engineering, and for components like valves and pumps.

****Key Benefits:****

- ****Durability:**** The strong chemical composition ensures that the pipe can withstand harsh environments.
- ****Versatility:**** Ideal for multiple applications, from automotive parts to engineering projects.
- ****Strong Performance:**** With high tensile and yield strength, it maintains integrity under stress.

ASTM B 730 N02200

Category: Tube

Nickel and Nickel Copper Alloy Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 730 N02200 is a specialized tube made from a unique blend of nickel and nickel-copper alloy, designed for various industrial applications. This tube is particularly known for its strength and resistance to corrosion, making it a reliable choice for use in challenging environments such as valves, pumps, and automotive systems.

How it works: The alloy's composition ensures that it can withstand high temperatures and pressures, while its mechanical properties provide excellent durability. With a minimum tensile strength of 380 MPa and an elongation of 40%, the tube can handle significant stress without breaking.

Benefits include:

- High resistance to corrosion and wear
- Suitable for a wide range of applications including automotive and engineering
- Strong mechanical properties ensure longevity and reliability in demanding conditions

ASTM B 725 N04400

Category: Pipe

Nickel and Nickel Copper Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 725 N04400 is a type of pipe made from a special blend of nickel and nickel copper alloy. This material is known for its excellent resistance to corrosion and high temperatures, making it ideal for various applications. The pipe is commonly used in industries like automotive, general engineering, and for components such as valves and pumps.

Key features of ASTM B 725 N04400 include:

- **Durability**: Its strong composition allows it to withstand harsh conditions without degrading.
- **Versatility**: Suitable for a wide range of applications, from automotive parts to industrial machinery.
- **High Strength**: With a minimum tensile strength of 480 and yield strength of 170, it can handle significant stress and strain.

ASTM B 730 N02201

Category: Tube

Nickel and Nickel Copper Alloy Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 730 N02201 is a specialized type of tube made from a blend of nickel and nickel copper alloy. This material is known for its strength and durability, making it suitable for various applications in industries such as automotive, general engineering, and for use in valves and pumps. The tube is designed to withstand demanding environments, offering both corrosion resistance and good mechanical properties.

Key Benefits:

- **Durability**: With a minimum tensile strength of 345, this tube can handle high pressures and stresses without failing.
- **Versatility**: Ideal for a range of applications, including automotive parts and engineering components.
- **Corrosion Resistance**: Made from nickel and copper, it resists rust and degradation, ensuring a longer lifespan in various conditions.

ASTM B 739 N08332

Category: Tube

Nickel-Iron-Chromium-Silicon Alloy Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 739 N08332 is a specialized tube made from a unique blend of nickel, iron, chromium, and silicon. This alloy is designed to provide excellent strength and durability for various applications. It is particularly useful in industries like automotive, engineering, and for components such as valves and pumps. The tube is engineered to withstand tough conditions, making it reliable for long-term use.

****Key Benefits:****

- ****High Strength****: With a minimum tensile strength of 462, it's built to handle demanding tasks.
- ****Versatile Applications****: Ideal for use in automotive systems, valves, and pumps, catering to a wide range of engineering needs.
- ****Durability****: The alloy composition ensures resistance to corrosion and wear, extending the product's lifespan.

ASTM B 739 N08330

Category: Tube

Nickel-Iron-Chromium-Silicon Alloy Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 739 N08330 is a specialized tube made from a unique blend of nickel, iron, chromium, and silicon. This alloy tube is designed to be strong and durable, making it an ideal choice for various applications in industries like automotive, general engineering, and pump and valve manufacturing. Its composition allows it to withstand harsh environments, ensuring that equipment operates reliably over time.

Key benefits of the ASTM B 739 N08330 tube include:

- ****High Strength****: With a minimum tensile strength of 483, this tube can handle significant pressure and stress.
- ****Corrosion Resistance****: The nickel and chromium content helps protect against rust and degradation, extending the lifespan of the equipment.
- ****Versatile Applications****: Suitable for making valves and pumps, as well as for general engineering purposes, this tube meets a wide range of industry needs.

ASTM B 755 N06110

Category: *Plate, Sheet and Strip*

Nickel-Chromium-Molybdenum-Tungsten Alloys Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 755 N06110 is a specialized type of metal plate, sheet, and strip made from a strong combination of nickel, chromium, molybdenum, and tungsten. This alloy is designed to withstand extreme environments, making it ideal for use in various industries such as automotive, valves, pumps, and general engineering. Its unique chemical composition allows it to resist corrosion and maintain strength under high stress, ensuring durability and reliability in tough applications.

Key benefits include:

- **High Strength**: With a minimum tensile strength of 655, this material provides excellent support for heavy-duty applications.
- **Corrosion Resistance**: The alloy's composition makes it resistant to rust and degradation, extending the lifespan of products made from it.
- **Versatility**: Suitable for various applications, including automotive parts and industrial machinery, it meets the demands of different sectors efficiently.

ASTM B 751 N06600

Category: *Pipe*

Nickel-Chromium-Iron-Alloy Pipe.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 751 N06600 is a specialized type of pipe made from a strong and durable Nickel-Chromium-Iron alloy. This material is designed to withstand harsh environments, making it perfect for various applications such as valves, pumps, and general engineering tasks in industries, including automotive. The unique composition of this alloy provides excellent resistance to corrosion and heat, ensuring longevity and reliability in tough conditions.

Key benefits of using ASTM B 751 N06600 pipes include:

- **High Strength**: With a tensile strength of at least 655 units, these pipes can handle significant pressure and stress.
- **Corrosion Resistance**: The nickel and chromium content helps prevent rust and degradation, extending the lifespan of the pipes.
- **Versatility**: Suitable for a wide range of applications in different industries, making it a go-to choice for engineers and manufacturers.

ASTM B 751 N06601

Category: Rod, Bar and Wire

Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Rod, Bar and Wire.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 751 N06601 is a type of strong and durable metal rod, bar, and wire made from a special blend of nickel, chromium, and iron. This alloy is designed to withstand harsh conditions, making it suitable for various industrial applications. It is particularly useful in making components for valves and pumps, as well as in general engineering and automotive industries.

Key benefits of ASTM B 751 N06601 include:

- **High Strength**: With a minimum tensile strength of 550, it can handle heavy loads and pressure without breaking.
- **Corrosion Resistance**: The nickel and chromium content helps protect against rust and deterioration, ensuring a longer lifespan in tough environments.
- **Versatile Use**: Ideal for a range of applications, from automotive parts to general engineering projects.

ASTM B 758 N06110

Category: Tube

Nickel-Chromium-Molybdenum-Tungsten-Alloys Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 758 N06110 is a specialized tube made from a mix of nickel, chromium, molybdenum, and tungsten alloys. These materials give the tube exceptional strength and resistance to corrosion, making it ideal for demanding environments. This tube is commonly used in industries such as automotive, valve manufacturing, and pumps, where durability and performance are critical.

****Key Benefits:****

- ****High Strength:**** It has a minimum tensile strength of 655, ensuring it can withstand heavy loads and pressure.
- ****Corrosion Resistance:**** The unique alloy composition makes it resistant to rust and wear, prolonging its lifespan.
- ****Versatile Applications:**** Suitable for various uses including valves, pumps, and general engineering applications.

ASTM B 759 N06110

Category: Pipe and Tube

Nickel-Chromium-Molybdenum-Tungsten Alloys Pipe and Tube.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 759 N06110 is a specialized type of pipe and tube made from a strong blend of nickel, chromium, molybdenum, and tungsten. This unique combination of materials gives it excellent durability and resistance to harsh environments, making it ideal for a variety of applications. It's commonly used in valves, pumps, and automotive engineering, providing reliable performance in critical situations.

Key benefits include:

- ****High Strength**:** With a tensile strength of at least 655, it can withstand significant stress and pressure.
- ****Corrosion Resistance**:** The chemical composition helps it resist rust and degradation, extending its lifespan in challenging conditions.
- ****Versatility**:** Suitable for various industries, including general engineering and automotive, making it a valuable asset for manufacturers.

ASTM B 805 N07716

Category: Bar and Wire

Precipitation Hardening Nickel Alloys Bar and Wire.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 805 N07716 is a type of strong and durable metal bar and wire made from a special nickel alloy that hardens when treated. This material is particularly useful in industries like automotive, engineering, and for making valves and pumps. Its unique composition gives it high strength and resistance to wear and tear, making it ideal for demanding applications.

Key benefits include:

- **High Strength**: It has excellent tensile and yield strength, meaning it can withstand heavy loads and pressure.
- **Durability**: The alloy is resistant to corrosion and other environmental factors, ensuring longevity in use.
- **Versatility**: Suitable for a range of applications, making it a go-to choice for various engineering projects.

ASTM B 805 N07725

Category: Bar and Wire

Precipitation Hardening Nickel Alloys Bar and Wire.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 805 N07725 is a type of strong and durable metal bar and wire made from a special blend of nickel and other metals. This material is known as precipitation hardening nickel alloy, which means it gets tougher when treated with heat. It's primarily used in various industries such as automotive, general engineering, and for making valves and pumps, where strength and resistance to corrosion are crucial.

Here are some of the benefits of using ASTM B 805 N07725:

- **High Strength**: It has a minimum tensile strength of 758 MPa, making it very strong and able to withstand heavy loads.
- **Corrosion Resistance**: The nickel content helps protect it from rust and deterioration, ensuring long-lasting performance even in harsh environments.
- **Versatile Applications**: Suitable for various uses including valves, pumps, and other engineering tasks, making it a go-to choice for many manufacturers.

ASTM B 805 N07776

Category: Bar and Wire

Precipitation Hardening Nickel Alloys Bar and Wire.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 805 N07776 is a type of strong and durable metal bar and wire made from nickel alloys. It is specifically designed to resist wear and tear, making it ideal for tough environments. This material is created through a process called precipitation hardening, which means it becomes stronger when it undergoes certain heat treatments.

Key benefits of ASTM B 805 N07776 include:

- **High Strength**: With a minimum tensile strength of 621, it can withstand significant pressure and stress.
- **Versatile Use**: Suitable for various applications such as valves, pumps, and automotive components.
- **Chemical Resistance**: Its unique composition helps it resist corrosion and degradation, extending the lifespan of products made from it.

ASTM B 805 N09777

Category: Bar and Wire

Precipitation Hardening Nickel Alloys Bar and Wire.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B 805 N09777 is a high-strength bar and wire made from a special type of nickel alloy that hardens when treated. This means it can withstand heavy use and stress, making it ideal for important industries like automotive, engineering, and manufacturing. The alloy is designed to resist wear and tear, ensuring that parts made from it last longer and perform better under tough conditions.

Here's a quick overview of its benefits:

- **Durability**: The material is engineered to endure significant pressure and stress, which is crucial for machinery and components in various applications.
- **Versatility**: It can be used in multiple applications, including valves and pumps, making it a reliable choice for engineers and manufacturers looking for high-performance materials.
- **Strong Composition**: With a robust chemical makeup that includes nickel, chromium, and molybdenum, this alloy offers excellent resistance to corrosion and fatigue, enhancing the longevity of the products made from it.

ASTM B 814 N06920

Category: Plate, Sheet and Strip

Nickel-Chromium-Iron-Molybdenum-Tungsten Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 814 N06920 is a special type of metal plate, sheet, and strip made from a durable alloy that includes nickel, chromium, iron, molybdenum, and tungsten. This combination of metals gives it unique properties that make it strong and resistant to corrosion. It's commonly used in various industries such as automotive, general engineering, and for making valves and pumps.

Here are some of the benefits of using ASTM B 814 N06920:

- **High Strength**: It has a minimum tensile strength of 655, making it strong enough for demanding applications.
- **Corrosion Resistance**: The alloy's composition helps it withstand harsh environments, ensuring long-lasting performance.
- **Versatile Applications**: Suitable for various uses in industries, providing flexibility for different engineering needs.

ASTM B 818 R31233

Category: Plate, Sheet and Strip

Cobalt-Chromium-Nickel-Molybdenum-Tungsten Alloy Plate, Sheet and Strip.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

The ASTM B 818 R31233 is a specially designed alloy made from cobalt, chromium, nickel, molybdenum, and tungsten. This product comes in the form of plates, sheets, and strips, making it versatile for various applications. It is known for its strength and durability, making it an excellent choice for industries that require reliable materials, such as valves, pumps, automotive components, and even medical devices.

****Key Benefits:****

- ****High Strength:**** With a tensile strength of at least 896, this alloy can withstand significant stress, making it ideal for heavy-duty applications.
- ****Corrosion Resistance:**** The unique combination of metals ensures that it can resist corrosion and wear, extending the lifespan of the products made from it.
- ****Versatile Applications:**** Suitable for a wide range of industries, including automotive and medical, ensuring that it meets various operational needs.

ASTM B86 Alloy 3

Category: Materials

Zinc and Zinc-Aluminum alloy for foundry and die castings.

Key Features:

- Tensile Strength: 283 MPa min
- Yield Strength: 221 MPa min

Technical Specifications:

Aluminium	3.700 - 4.300%
Cadmium	0.004 max%
Chromium	0.020 max%
Copper	0.100 max%
Iron	0.050 max%
Magnesium	0.020 - 0.060%
Manganese	0.060 max%
Nickel	0.020 max%
Lead	0.005 max%
Silicon	0.035 max%
Tin	0.002 max%
Titanium	0.001 max%
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 Alloy 3 is a specialized material made primarily of zinc and zinc-aluminum, designed for use in foundry and die casting applications. This alloy is known for its strength, making it ideal for producing parts that need to withstand considerable stress. With a minimum tensile strength of 283 MPa and a yield strength of 221 MPa, it ensures durability and reliability in a variety of engineering applications.

Key benefits of ASTM B86 Alloy 3 include:

- ****Versatility**:** Suitable for various applications such as valves, pumps, and automotive components.
- ****Enhanced Strength**:** Provides robust performance with high tensile and yield strength, making it a dependable choice for demanding environments.
- ****Low Alloy Content**:** Contains minimal amounts of elements like cadmium and lead, ensuring safer usage in

industrial processes.

ASTM B86 Alloy 5

Category: Materials

Zinc and Zinc-Aluminum alloy for foundry and die castings.

Key Features:

- Tensile Strength: 328 MPa min
- Yield Strength: 228 MPa min

Technical Specifications:

Aluminium	3.700 - 4.300%
Cadmium	0.004 max%
Chromium	0.020 max%
Copper	0.700 - 1.200%
Iron	0.050 max%
Magnesium	0.020 - 0.060%
Manganese	0.060 max%
Nickel	0.020 max%
Lead	0.005 max%
Silicon	0.035 max%
Tin	0.002 max%
Titanium	0.001 max%
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 Alloy 5 is a special type of metal made primarily of zinc and zinc-aluminum, designed for use in casting processes, which means it's molded into specific shapes for various products. This alloy is known for its strong performance, making it suitable for creating durable components in different industries. It has impressive strength characteristics, with a minimum tensile strength of 328 MPa and a yield strength of 228 MPa, ensuring that the parts made from it can withstand significant pressure and stress.

Key benefits of using ASTM B86 Alloy 5 include:

- **High Strength**: It maintains its shape and integrity even under heavy use.
- **Versatile Applications**: Ideal for manufacturing valves, pumps, and automotive parts, making it a go-to choice for engineers and manufacturers.
- **Reliable Performance**: With low levels of impurities, this alloy offers consistent quality, which is critical for high-performance applications.

ASTM B86 Alloy 7

Category: Materials

Zinc and Zinc-Aluminum alloy for foundry and die castings.

Key Features:

- Tensile Strength: 283 MPa min
- Yield Strength: 221 MPa min

Technical Specifications:

Aluminium	3.700 - 4.300%
Cadmium	0.002 max%
Chromium	0.020 max%
Copper	0.100 max%
Iron	0.050 max%
Magnesium	0.005 - 0.020%
Manganese	0.060 max%
Nickel	0.005 - 0.020%
Lead	0.003 max%
Silicon	0.035 max%
Tin	0.001 max%
Titanium	0.001 max%
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 Alloy 7 is a special type of metal made primarily from zinc and zinc-aluminum, designed for use in manufacturing processes like foundry work and die casting. This alloy is known for its strength and durability, making it an excellent choice for creating various components in industries such as automotive and engineering. Its properties ensure that parts made from this alloy can withstand significant stress without breaking, which is crucial for parts like valves and pumps that need to perform reliably.

Key Benefits:

- ****High Strength****: With a minimum tensile strength of 283 MPa and yield strength of 221 MPa, this alloy is tough and capable of handling demanding applications.
- ****Versatile Use****: Ideal for a range of applications including automotive parts, pumps, and general engineering products, making it a versatile material choice for manufacturers.

ASTM B86 ZA12

Category: Materials

Zinc and Zinc-Aluminum alloy for foundry and die castings.

Key Features:

- Tensile Strength: 276 - 404 MPa
- Yield Strength: 211 - 320 MPa

Technical Specifications:

Aluminium	10.500 - 11.500%
Cadmium	0.006 max%
Chromium	0.010 max%
Copper	0.500 - 1.200%
Iron	0.075 min%
Magnesium	0.010 - 0.030%
Manganese	0.010 max%
Nickel	0.010 max%
Lead	0.006 max%
Tin	0.003 max%
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 ZA12 is a special type of metal made from a combination of zinc and aluminum, designed specifically for use in foundries and die casting. This alloy is known for its strong physical properties, which means it can withstand a lot of pressure and stress without breaking. With a tensile strength ranging from 276 to 404 MPa and a yield strength between 211 to 320 MPa, it provides durability for various applications.

Key benefits of ASTM B86 ZA12 include:

- ****High Strength****: With impressive tensile and yield strength, it is suitable for demanding environments.
- ****Versatile Applications****: It can be used in many industries, including automotive, general engineering, and for making components like valves and pumps.
- ****Corrosion Resistance****: The composition helps in resisting corrosion, extending the lifespan of the parts made from it.

ASTM B86 Zamak 3

Category: Materials

Zinc and Zinc-Aluminum alloy for foundry and die castings.

Key Features:

- Tensile Strength: 283 MPa min
- Yield Strength: 221 MPa min

Technical Specifications:

Aluminium	3.700 - 4.300%
Cadmium	0.004 max%
Chromium	0.020 max%
Copper	0.100 max%
Iron	0.050 max%

Magnesium	0.020 - 0.060%
Manganese	0.060 max%
Nickel	0.020 max%
Lead	0.005 max%
Silicon	0.035 max%
Tin	0.002 max%
Titanium	0.001 max%
Zinc	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 Zamak 3 is a type of metal alloy primarily made from zinc, with a mix of aluminum and other elements. It's specifically designed for use in foundries and die casting, which means it's melted down and shaped into various products. This alloy is known for its strength and durability, making it a popular choice for manufacturing important parts in various industries.

Key benefits of ASTM B86 Zamak 3 include:

- **High Strength**: With a minimum tensile strength of 283 MPa, it can withstand significant force without breaking.
- **Versatile Applications**: It's suitable for making parts like valves and pumps, as well as components used in automotive and general engineering.
- **Durability**: The combination of zinc and aluminum gives it excellent resistance to corrosion, extending the lifespan of the products made from it.

ASTM B124 UNS C36500

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.250 - 0.700, Tin Sn %0.250 max., Iron Fe %0.150 max., Zinc Zn %Balance, Copper Cu %58.000 - 61.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C36500 is a type of copper alloy used primarily for forging into rods, bars, and other shapes. This material is known for its strong and durable properties, making it ideal for various industrial applications. It contains a specific mix of metals, including copper, lead, tin, iron, and zinc, which contribute to its performance and reliability in tough environments.

Key Benefits:

- **Strong and Durable**: With a minimum tensile strength of 344 MPa, this alloy can withstand significant stress and pressure.
- **Versatile Heat Treatment Options**: It can undergo processes like normalizing, annealing, or tempering to

enhance its properties depending on the specific needs of your project.

- **Consistent Quality**: The precise chemical composition ensures that the material performs reliably, making it suitable for a wide range of applications in various industries.

ASTM B124 UNS C37000

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.800 - 1.500, Iron Fe %0.150 max., Zinc Zn %Balance, Copper Cu %59.000 - 62.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

The ASTM B124 UNS C37000 is a type of copper and copper alloy that is specifically designed for creating rods, bars, and various shapes used in different industries. This material is made from a combination of copper, zinc, lead, and iron, which helps it achieve strong mechanical properties. It can be heat-treated in different ways, such as normalizing or annealing, to enhance its performance in various applications.

Benefits of ASTM B124 UNS C37000 include:

- **Durability**: With a minimum tensile strength of 344 Mpa, it offers excellent strength and resilience.
- **Versatile Applications**: Ideal for manufacturing components that require good strength and corrosion resistance.
- **Customizable Properties**: Heat treatment options allow for adjustments to suit specific needs in different environments.

ASTM B124 UNS C37700

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %1.500 - 2.500, Iron Fe %0.300 max., Zinc Zn %Balance, Copper Cu %58.000 - 61.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C37700 is a type of copper and copper alloy used primarily for making various forged products like rods, bars, and shapes. This material is designed to meet specific standards for quality and performance, ensuring that it is reliable for various industrial applications. It contains a mix of copper, zinc, lead, and iron, which gives it unique properties that make it suitable for different uses.

Key benefits of ASTM B124 UNS C37700 include:

- **Durability**: With a minimum tensile strength of 344 MPa, it can withstand significant stress and pressure.
- **Versatility**: It can be heat-treated through processes like normalizing, annealing, or tempering to enhance its performance characteristics, making it adaptable for different manufacturing needs.
- **Quality Assurance**: The precise chemical composition ensures that it meets industry standards, providing confidence in its use for critical applications.

ASTM B124 UNS C46400

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.200 max., Tin Sn %0.500 - 1.000, Iron Fe %0.100 max., Copper Cu %59.000 - 62.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa -

AI Layman Summary:

ASTM B124 UNS C46400 is a type of copper and copper alloy material specifically designed for making rods, bars, and various shapes. It is known for its strong and durable properties, making it suitable for a range of industrial applications. This material is made with a careful blend of metals, ensuring that it meets high standards for quality and performance.

Key features of ASTM B124 UNS C46400 include:

- Chemical Composition: Contains a maximum of 0.200% lead, 0.500% to 1.000% tin, and 0.100% maximum iron, with copper content ranging from 59.000% to 62.000%.
- Heat Treatment: Can be processed through normalizing, annealing, or tempering to enhance its properties.
- Mechanical Strength: Offers good tensile strength, making it reliable for various applications.

This copper alloy is beneficial because it combines strength with excellent corrosion resistance, making it ideal for use in electrical and plumbing components, as well as in construction and manufacturing industries.

ASTM B124 UNS C48200

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.400 - 1.000, Tin Sn %0.500 - 1.000, Iron Fe %0.100 max., Copper Cu %59.000 - 62.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa -

AI Layman Summary:

ASTM B124 UNS C48200 is a type of copper and copper alloy used primarily for forging, which is the process of shaping metal using compressive forces. This standard specification outlines the quality and characteristics of copper and its alloys when made into rods, bars, and other shapes. It's designed to ensure that the materials meet certain performance criteria for various industrial applications.

Key features of ASTM B124 UNS C48200 include:

- A specific chemical composition that includes copper, lead, tin, and iron, ensuring that the material has the right properties for strength and durability.
- Versatile heat treatment options like normalizing, annealing, or tempering, which enhance the material's mechanical properties.
- High tensile strength, making it suitable for demanding applications where reliability and longevity are critical.

Overall, this product is beneficial for industries that require robust and reliable copper materials, providing flexibility in manufacturing and ensuring high performance in end-use applications.

ASTM B124 UNS C61900

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.020 max., Tin Sn %0.600 max., Iron Fe %3.000 - 4.500, Aluminium Al %8.500 - 10.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C61900 is a type of copper alloy material specifically designed for making rods, bars, and various shapes used in different industries. This material adheres to strict standards that ensure its quality and performance. It consists of a mixture of copper and other elements, which gives it unique properties suitable for forging and manufacturing.

Key benefits include:

- ****Durability****: With a minimum tensile strength of 344 MPa, this alloy can withstand significant stress and wear, making it ideal for heavy-duty applications.
- ****Versatility****: It can be shaped through processes like normalizing, annealing, or tempering, allowing for a wide range of uses in manufacturing.
- ****Quality Assurance****: The specific chemical composition ensures that it is free from excessive lead and tin, making it safer and more reliable for various applications.

ASTM B124 UNS C48500

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %1.300 - 2.200, Tin Sn %0.500 - 1.000, Iron Fe %0.100 max., Copper Cu %59.000 - 62.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa -

AI Layman Summary:

ASTM B124 UNS C48500 is a type of copper alloy that is specifically designed for making various forged products like rods, bars, and other shapes. This material is known for its unique combination of metals, including copper, lead, and tin, which gives it excellent strength and durability. It goes through specific heat treatment processes like normalizing or annealing to enhance its mechanical properties, making it suitable for demanding applications.

Benefits of ASTM B124 UNS C48500:

- ****Strong and Durable****: The alloy's composition allows it to withstand high stress and pressure, making it reliable for various industrial uses.
- ****Versatile Use****: It can be formed into different shapes and sizes, catering to a wide range of manufacturing needs.
- ****Improved Workability****: The heat treatment process provides flexibility in production, allowing for easy shaping and forging of the material.

ASTM B124 UNS C62300

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Tin Sn %0.600 max., Iron Fe %2.000 - 4.000, Ni + Cu Ni % + Cu %1.000 max., Silicon Si %0.250 max.
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C62300 is a type of copper and copper alloy material that is used to create various shapes like rods and bars. This specification ensures that the material meets certain quality standards, particularly in its chemical makeup and mechanical strength. Essentially, it defines how the metal should be made and what it should be capable of, ensuring it performs well in its intended applications.

Key benefits include:

- **Durability**: With a minimum tensile strength of 344 MPa, this material is strong and can withstand considerable stress without breaking.
- **Versatile Applications**: It can be treated through processes like normalizing, annealing, or tempering, making it adaptable for different uses in industries requiring reliable metal products.
- **Quality Assurance**: The specific chemical composition limits ensure that the material maintains high performance and reliability, making it a trusted choice for various engineering and manufacturing needs.

ASTM B124 UNS C63000

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Tin Sn %0.200 max., Iron Fe %2.000 - 4.000, Ni + Cu Ni % + Cu % 4.000 - 5.500
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C63000 is a standard type of copper alloy used primarily for making rods, bars, and various shapes through a forging process. This material is notable for its specific chemical composition, which includes a maximum of 0.200% tin, between 2.000% and 4.000% iron, and 4.000% to 5.500% nickel combined with copper. These elements contribute to its overall strength and durability, making it suitable for a range of industrial applications.

The benefits of using ASTM B124 UNS C63000 include:

- **Strong and Durable**: With a minimum tensile strength of 344 MPa, this alloy can withstand significant stress and strain.
- **Versatile Treatment Options**: It can be heat treated through normalizing, annealing, or tempering, which enhances its mechanical properties and adaptability for various uses.
- **Widely Applicable**: Suitable for a variety of industries that require robust materials for their products, ensuring reliability and longevity.

ASTM B124 UNS C63200

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.020 max., Iron Fe %3.500 - 4.300, Ni + Cu Ni % + Cu % 4.000 - 4.800
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C63200 is a standard guideline for a type of copper alloy used in forging various products like rods, bars, and shapes. This material is made of copper mixed with small amounts of lead, iron, and nickel, which helps enhance its properties. It can be heat-treated through processes like normalizing or annealing to improve its strength and performance.

Key benefits of ASTM B124 UNS C63200 include:

- **Durability**: With a minimum tensile strength of 344 MPa, it offers resistance to wear and tear.
- **Versatility**: Suitable for a variety of applications due to its chemical composition and ability to undergo different heat treatments.
- **Reliability**: Conforming to industry standards ensures consistent quality and performance in manufacturing processes.

ASTM B124 UNS C64200

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.050 max., Tin Sn %0.200 max., Iron Fe %0.300 max., Ni + Cu Ni % + Cu %0.250 max.
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C64200 is a type of copper alloy that is specifically designed for making various forged products such as rods, bars, and other shapes. This material is known for its excellent strength and durability, making it suitable for different industrial applications. The alloy includes controlled amounts of elements like lead, tin, and iron to enhance its performance while ensuring it meets safety and quality standards.

Key benefits include:

- **Strength and Durability**: With a minimum tensile strength of 344 MPa, this alloy is robust enough for demanding environments.
- **Versatile Applications**: It can be heat-treated in various ways (normalizing, annealing, or tempering) to suit different manufacturing needs, allowing for flexibility in production.
- **Controlled Composition**: The specific chemical composition ensures that the alloy performs reliably while minimizing risks associated with impurities.

ASTM B124 UNS C64210

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.050 max., Tin Sn %0.200 max., Iron Fe %0.300 max., Ni + Cu Ni % + Cu %0.250 max.
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C64210 is a type of copper and copper alloy material specifically designed for forging into rods, bars, and other shapes. This standard specification ensures that the material meets certain chemical and mechanical properties, making it suitable for various applications. It contains a small amount of lead, tin, and iron, which helps enhance its performance while maintaining a high level of strength.

Here are some key features of this product:

- **Chemical Composition**: It has controlled levels of lead, tin, and iron, ensuring durability and reliability.
- **Heat Treatment Options**: The material can undergo processes like normalizing, annealing, or tempering to improve its properties.
- **Strength**: It boasts a minimum tensile strength of 344 MPa, making it strong enough for demanding applications.

Overall, ASTM B124 UNS C64210 is an excellent choice for industries that require strong, durable, and well-defined copper alloy components.

ASTM B124 UNS C67500

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.020 max., Tin Sn %0.500 - 1.500, Iron Fe %0.800 - 2.000, Manganese Mn %0.050 - 0.500
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C67500 is a type of copper and copper alloy material primarily used for forging into rods, bars, and various shapes. This specification ensures that manufacturers produce high-quality copper products that meet specific standards for composition and mechanical properties. The main features include a controlled chemical makeup, with minimal lead and specific amounts of tin, iron, and manganese, ensuring the material's durability and reliability.

Key Benefits:

- **Durability**: With a minimum tensile strength of 344 MPa, this material is strong and can withstand significant stress.
- **Versatile Processing**: It can undergo various heat treatments like normalizing, annealing, or tempering, enhancing its performance for different applications.
- **Quality Assurance**: The strict chemical composition guidelines ensure consistent quality, making it suitable for various industrial uses.

ASTM B124 UNS C65500

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.050 max., Iron Fe %0.800 max., Ni + Cu Ni % + Cu %0.600 max., Silicon Si %2.800 - 3.800
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C65500 is a standard specification for high-quality copper and copper alloy products, particularly used in forging. This means it is designed to create various shapes like rods and bars from copper and its alloys, which are essential materials in many industries. The specification ensures that these products meet specific chemical and mechanical standards, making them reliable for various applications.

Key benefits of ASTM B124 UNS C65500 include:

- **Durability**: With a minimum tensile strength of 344 MPa, these materials are strong and can withstand significant stress.
- **Versatility**: Suitable for heat treatment processes like normalizing, annealing, or tempering, allowing for improved performance in different conditions.
- **Quality Assurance**: The strict chemical composition ensures that the product maintains consistent quality, making it ideal for professional use in manufacturing and construction.

ASTM B124 UNS C67600

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.500 - 1.000, Tin Sn %0.500 - 1.500, Iron Fe %0.400 - 1.300, Manganese Mn %0.050 - 0.500
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C67600 is a standard specification for a type of copper and copper alloy material that is commonly used in manufacturing various products. It is available in forms such as rods, bars, and other shapes, making it versatile for different applications. This material is designed to have specific chemical compositions, ensuring it meets certain quality and performance standards.

Key benefits of ASTM B124 UNS C67600 include:

- **Durability**: With a minimum tensile strength of 344 MPa, it can withstand significant stress and pressure, making it suitable for demanding environments.
- **Versatility**: The material can be heat-treated through processes like normalizing or annealing, allowing manufacturers to tailor its properties for specific uses.
- **Chemical Composition**: It contains controlled amounts of elements like lead, tin, iron, and manganese, which enhance its performance and reliability in various applications.

ASTM B124 UNS C70620

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.020 max., Iron Fe %1.000 - 1.800, Ni + Cu Ni % + Cu % 9.000 - 11.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C70620 is a type of copper and copper alloy material used for making various forged products like rods and bars. This standard specification ensures that the material meets specific chemical and mechanical properties, making it reliable for various applications. The key feature of this alloy includes a very

low maximum lead content, specific levels of iron, and a defined nickel and copper percentage, which contribute to its strength and durability.

How it works: This material undergoes processes such as normalizing, annealing, or tempering to enhance its performance and flexibility under different conditions. With a minimum tensile strength of 344 MPa, it is designed to withstand significant stress without breaking, which is crucial for industries that require robust components.

Benefits include:

- *High strength and durability, ideal for demanding applications.*
- *Low lead content for safety and environmental compliance.*
- *Versatile processing options to meet specific project needs.*

ASTM B124 UNS C71520

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.020 max., Iron Fe %0.400 - 1.000, Ni + Cu Ni % + Cu % 29.000 - 33.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C71520 is a type of copper and copper alloy material that is specifically designed for forging into various shapes such as rods and bars. This material is known for its strength and durability, making it ideal for a variety of industrial applications. It is made with a precise chemical composition, ensuring it meets quality standards for performance and reliability.

Key benefits of ASTM B124 UNS C71520 include:

- *****Strength****: With a minimum tensile strength of 344 MPa, it provides excellent resistance to wear and tear.*
- *****Versatility****: It can be processed through different heat treatments like normalizing, annealing, or tempering, allowing for customization based on specific needs.*
- *****Quality Assurance****: The controlled chemical composition ensures that it is free from excess lead and iron, contributing to its overall integrity and performance.*

ASTM B124 UNS C77400

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Lead Pb %0.020 max., Ni + Cu Ni % + Cu % 9.000 - 11.000, Copper Cu %43.000 - 47.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 344 min.

AI Layman Summary:

ASTM B124 UNS C77400 is a type of copper and copper alloy that is specifically designed for creating rods, bars, and various shapes through a forging process. This material is useful in industries that require strong and reliable metal components. It has a specific chemical makeup, ensuring it meets certain standards for strength and durability.

Key benefits of ASTM B124 UNS C77400 include:

- **Strong Structure**: With a minimum tensile strength of 344 MPa, it provides excellent resistance to stress and strain.
- **Versatile Treatment Options**: It can be heat-treated in various ways (normalizing, annealing, or tempering) to enhance its properties according to specific needs.
- **Controlled Composition**: The precise chemical composition, including limited lead content, ensures safety and compliance with industry standards.

ASTM B132-B

Category: Copper Alloy Sand Casting

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium Al %1.000 - 3.000, Iron Fe %1.000 - 3.000, Manganese Mn %1.000 - 3.500
- Heat Treatment: As-Cast
- Mechanical Properties: Tensile Strength in Mpa 552 min.

AI Layman Summary:

ASTM B132-B is a standard specification that outlines the requirements for copper alloy sand casting, which is a method used to create metal parts for various applications. This process involves pouring molten metal into a sand mold to form the desired shape, making it an efficient way to produce complex components. The specification ensures that the castings meet certain standards in terms of their composition and mechanical properties, guaranteeing their reliability and performance in use.

Key benefits of ASTM B132-B include:

- **Durable Composition**: The alloy typically contains aluminum, iron, and manganese, ensuring strength and resistance to wear.
- **Strong Performance**: With a minimum tensile strength of 552 MPa, the castings are robust enough for general applications, making them suitable for a variety of industrial uses.
- **Simple Heat Treatment**: These castings are used as-cast, meaning they do not require additional heat treatment, which simplifies the production process and reduces costs.

ASTM B132-A

Category: Copper Alloy Sand Casting

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium Al %0.500 - 1.500, Iron Fe %0.400 - 2.000, Manganese Mn %0.100 - 1.000
- Heat Treatment: As-Cast
- Mechanical Properties: Tensile Strength in Mpa 414 min.

AI Layman Summary:

ASTM B132-A is a standard specification for copper alloy sand casting, which is a method used to create metal parts by pouring molten metal into a sand mold. This process is particularly useful for producing various components that require a combination of strength and durability. The main advantage of using copper alloy in sand casting is its excellent resistance to corrosion and wear, making it ideal for general applications in various industries.

Key features of ASTM B132-A include:

- **Chemical Composition**: It includes specific percentages of aluminum, iron, and manganese, ensuring the

metal has the right properties for strength and durability.

- **Heat Treatment:** The parts are used in their as-cast state, which simplifies the manufacturing process.
- **Mechanical Properties:** The specification guarantees a minimum tensile strength of 414 MPa, ensuring that the finished products can withstand significant stress and pressure.

ASTM B133 C11000

Category: Copper and Copper Alloy Forging

Standard Specification for Copper and Copper Alloy Forging Rod, Bar and Shapes

Key Features:

- Chemical Composition: Cu + Ag Cu % + Ag % 99.900 min.
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 154.500 - 698.200

AI Layman Summary:

ASTM B133 C11000 is a high-quality material made from copper and its alloys, specifically designed for forging into rods, bars, and various shapes. This standard specification ensures that the product has a minimum copper purity of 99.900%, making it exceptionally conductive and durable. The material can undergo different heat treatments such as normalizing, annealing, or tempering, which enhances its mechanical properties, including a tensile strength ranging from 154.500 to 698.200 MPa.

Key benefits of using ASTM B133 C11000 include:

- **High Purity:** With a copper content of at least 99.900%, it offers excellent electrical and thermal conductivity.
- **Versatile Heat Treatments:** The ability to undergo various heat treatments allows for customization of the material's strength and flexibility, making it suitable for a wide range of applications.
- **Strong Mechanical Properties:** Its high tensile strength ensures that it can withstand significant stress and strain, making it ideal for demanding industrial environments.

ASTM B138 C67600

Category: Manganese Bronze Rod, Bar and Shapes

Standard Specification for Manganese Bronze Rod, Bar and Shapes

Key Features:

- Chemical Composition: Iron Fe %0.400 - 1.300, Manganese Mn %0.050 - 0.500, Lead Pb %0.500 - 1.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 380 - 550

AI Layman Summary:

ASTM B138 C67600 is a type of manganese bronze rod, bar, and shapes used in various industrial applications. It is made from a specific mixture of metals, including iron, manganese, and lead, which gives it unique properties that make it strong and durable. This material is often used in manufacturing parts that require resistance to wear and corrosion, making it suitable for marine, automotive, and machinery components.

Key benefits of ASTM B138 C67600 include:

- **High Strength:** With a tensile strength ranging from 380 to 550 MPa, it can withstand heavy loads and stress without breaking.
- **Versatile Processing:** It can be heat treated through normalizing, annealing, or tempering, allowing for adjustments in its hardness and flexibility to meet specific needs.
- **Durability:** The chemical composition ensures good resistance to wear and oxidation, extending the lifespan of the products made from it.

ASTM B135 C37000

Category: Seamless Brass Tube

Standard Specification for Seamless Brass Tube

Key Features:

- Chemical Composition: Iron Fe %0.150 max., Lead Pb %0.800 - 1.500, Copper Cu %59.000 - 62.000
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 372 min.

AI Layman Summary:

ASTM B135 C37000 is a type of seamless brass tube that is used in various applications due to its strong and durable nature. It is made from a specific mixture of metals, primarily copper, along with small amounts of iron and lead, which give it unique properties. This tube is processed through methods like normalizing, annealing, or tempering to enhance its strength and performance. With a minimum tensile strength of 372 MPa, it can withstand significant stress without breaking, making it reliable for different uses.

****Benefits of ASTM B135 C37000:****

- ****Durability:**** The seamless design and strong materials ensure that the tube is long-lasting and can handle tough conditions.
- ****Versatility:**** Its chemical composition and mechanical properties make it suitable for various industrial applications, from plumbing to electrical systems.
- ****Quality Assurance:**** Being a standard specification means that it meets certain quality and safety standards, giving users peace of mind about its performance.

ASTM B138 C67500

Category: Manganese Bronze Rod, Bar and Shapes

Standard Specification for Manganese Bronze Rod, Bar and Shapes

Key Features:

- Chemical Composition: Aluminium Al %0.250 max., Iron Fe %0.800 - 2.000, Manganese Mn %0.050 - 0.500
- Heat Treatment: Normalizing or Annealing or Tempering
- Mechanical Properties: Tensile Strength in Mpa 380 - 550

AI Layman Summary:

ASTM B138 C67500 is a type of manganese bronze rod, bar, and shapes used in various industrial applications. It is made from a specific blend of metals, primarily manganese, aluminum, and iron, which gives it unique properties. This material can be heat-treated in different ways, such as normalizing or annealing, to enhance its strength and durability. With a tensile strength of 380 to 550 MPa, it is capable of withstanding significant stress, making it ideal for heavy-duty uses.

****Key Benefits:****

- ****Strong and Durable:**** The special composition and heat treatment options provide high tensile strength, ensuring the material can handle demanding applications.
- ****Versatile Applications:**** Its unique properties make it suitable for a variety of uses in industries that require robust materials, such as manufacturing and construction.

ASTM B146-6A

Category: Copper Alloy Sand Casting

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium Al %0.005 max., Iron Fe %0.600 max., Ni + Co Ni % + Co %1.000 max.
- Heat Treatment: As-Cast
- Mechanical Properties: Tensile Strength in Mpa 241 min.

AI Layman Summary:

ASTM B146-6A is a standard specification for making sand castings from copper alloys, which are metals made primarily of copper mixed with other elements like aluminum and iron. This specification ensures that the castings meet certain quality and performance criteria for general use in various applications. The casting process involves pouring molten metal into a mold to create parts or products that can be used in different industries.

Key benefits of ASTM B146-6A include:

- **Consistency and Quality**: The specification defines the chemical makeup and mechanical properties required, ensuring that each casting meets strict standards for strength and durability.
- **Versatility**: The copper alloy sand castings produced under this standard can be used for a wide range of applications, making them suitable for many different sectors of the industry.

ASTM B981 C37700

Category: Materials

Low Leaded Brass Rod, Bar, Wire and Shapes.

Key Features:

- Chemical Composition: Fe 0.300 max, Pb 1.500 - 2.500, Cu 58.000 - 61.000, Zn Balance

Technical Specifications:

Heat Treatment	Normalizing or Annealing or Tempering
Tensile Strength (MPa)	310 - 430
Yield Strength (MPa)	135 - 190
Elongation (%)	10 - 25

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B981 C37700 is a type of low leaded brass that comes in various forms such as rods, bars, wires, and shapes. It is designed to be used in applications where corrosion resistance and durability are important, like in valves, pumps, and automotive components. Its unique chemical makeup includes a specific amount of copper, zinc, and lead, which gives it the right balance of strength and workability.

Some key benefits of ASTM B981 C37700 include:

- **Versatility**: Suitable for a variety of applications across industries, from general engineering to automotive.
- **Strength**: It has a tensile strength ranging from 310 to 430 MPa, making it durable and reliable in tough conditions.
- **Formability**: The material can be easily shaped and processed, allowing for efficient manufacturing of parts.

ASTM B98 C65500

Category: Materials

Copper-Silicon Alloy Rod, Bar and Shapes.

Key Features:

- Chemical Composition: Fe 0.800 max, Mn 0.500 - 1.300, Ni + Co 0.600 max, Pb 0.050 max, Si 2.800 - 3.800, Zn 1.500 max

Technical Specifications:

Heat Treatment	Normalizing or Annealing or Tempering
Tensile Strength (MPa)	360 - 690
Yield Strength (MPa)	105 - 380
Elongation (%)	7 - 35

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B98 C65500 is a type of copper-silicon alloy that is crafted into rods, bars, and various shapes. This material is known for its strength and versatility, making it suitable for a range of applications. The alloy is designed to withstand high pressure and stress, which is why it's commonly used in industries such as automotive, valves, and pumps.

Key benefits of ASTM B98 C65500 include:

- **Strong and Durable**: With a tensile strength ranging from 360 to 690 MPa, it can handle tough conditions without easily breaking.
- **Versatile Use**: Ideal for various engineering applications, including automotive and general machinery.
- **Adaptable**: Can be heat-treated in different ways to enhance its properties, ensuring it meets specific requirements for different projects.

ASTM B983 N07718

Category: Materials

Seamless Nickel Alloy Pipe and Tube.

Key Features:

- Chemical Composition: C 0.080 max, Si 0.350 max, Mn 0.350 max, Cr 17.000 - 21.000, Mo 2.800 - 3.300, Ni 50.000 - 55.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	1034 min
Yield Strength (MPa)	862 min
Elongation (%)	20 min

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B983 N07718 is a type of seamless pipe and tube made from nickel alloy, which is a special metal known for its strength and resistance to corrosion. It is designed to be used in various industrial applications, including valves, pumps, and automotive components. The manufacturing process involves heat treatment methods that enhance its durability and performance under pressure.

Key benefits include:

- **High Strength**: With a minimum tensile strength of 1034 MPa, it can withstand heavy loads and stress.
- **Corrosion Resistance**: The unique chemical composition allows it to resist rust and degradation in harsh environments.
- **Versatile Applications**: Ideal for use in general engineering, automotive, and various mechanical systems.

ASTM B983 N07740

Category: Materials

Precipitation-Hardening Nickel Alloys Plate, Sheet and Strip.

Key Features:

- Chemical Composition: C 0.005 - 0.080, Si 1.000 max, Cr 23.500 - 25.500, Mo 2.000 max, Ni Balance

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	1035 min
Yield Strength (MPa)	620 min
Elongation (%)	20 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B983 N07740 is a special type of material known as a precipitation-hardening nickel alloy, which comes in the form of plates, sheets, and strips. This material is designed to be incredibly strong and durable, making it an excellent choice for a variety of demanding applications. It is composed of nickel and other elements, which give it unique properties that enhance its performance in tough environments.

Key benefits of ASTM B983 N07740 include:

- **High Strength**: With a minimum tensile strength of 1035 MPa and yield strength of 620 MPa, it can withstand significant forces and pressures.
- **Versatile Applications**: It is ideal for use in valves, pumps, automotive parts, and general engineering projects, making it suitable for various industries.
- **Customizable Properties**: The material can be heat-treated in different ways, allowing for customization of its hardness and strength based on specific needs.

ASTM F 1537 Alloy 1

Category: Materials

Cobalt - 28 Chromium - 6 Molybdenum Alloys for Surgical Implants.

Key Features:

- Chemical Composition: C 0.140 max, Cr 26.000 - 30.000, Mo 5.000 - 7.000, Ni 1.000 max

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	897 min
Yield Strength (MPa)	517 min
Elongation (%)	12 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F 1537 Alloy 1 is a specialized metal alloy primarily used for making surgical implants. It is made up of cobalt, chromium, and molybdenum, which gives it excellent strength and durability. This alloy is designed to withstand the harsh conditions of the human body, ensuring that implants remain safe and effective over time.

****Key Benefits:****

- ****High Strength:**** With a minimum tensile strength of 897 MPa, it can endure significant stress without breaking.
- ****Corrosion Resistance:**** The unique composition helps prevent rust and degradation, making it ideal for medical applications.
- ****Versatile Uses:**** Besides surgical implants, it can also be used in valves, pumps, automotive parts, and other general engineering applications.

ASTM F 2281 601

Category: Materials

Stainless Steel and Nickel Alloy Bolts, Screw and Stud.

Key Features:

- Chemical Composition: C 0.100 max, Si 0.500 max, Cr 21.000 - 25.000, Ni 58.000 - 63.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	571 min
Yield Strength (MPa)	172 min
Elongation (%)	35 min

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

AI Layman Summary:

The ASTM F 2281 601 is a type of high-quality fastener made from stainless steel and nickel alloy, specifically designed for demanding applications. These bolts, screws, and studs are engineered to be strong and durable, making them ideal for use in various industries such as automotive, general engineering, and in equipment like valves and pumps. They are built to withstand harsh conditions, ensuring reliability in every use.

- Key benefits include:
- **Strength and Durability**: With a minimum tensile strength of 571 MPa and a yield strength of 172 MPa, these fasteners can handle heavy loads and stress without failing.
 - **Versatile Applications**: Suitable for a wide range of applications, from automotive parts to industrial machinery, making them a go-to choice for engineers and manufacturers.
 - **Resistant to Corrosion**: The chemical composition, featuring high nickel content, provides excellent resistance to corrosion, prolonging the life of the components they secure.

ASTM F 2527 R30003

Category: Materials

Seamless Drawn Cobalt Alloy Small Diameter Tubing for Surgical Implants.

Key Features:

- Chemical Composition: C 0.150 max, Cr 19.000 - 21.000, Mo 6.000 - 8.000, Ni 14.000 - 18.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	850 min
Yield Strength (MPa)	450 min
Elongation (%)	1 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F 2527 R30003 is a specialized type of tubing made from a strong cobalt alloy, designed specifically for use in surgical implants. This seamless tubing has a small diameter, making it ideal for intricate medical applications where durability and precision are essential. The alloy is composed of several key elements that enhance its performance, ensuring that it can withstand the demanding conditions often found in medical environments.

- Key Benefits:
- **High Strength**: With a minimum tensile strength of 850 MPa and yield strength of 450 MPa, it provides excellent resistance to deformation and breakage.
 - **Versatile Applications**: Besides medical uses, this tubing can also be utilized in valves, pumps, general engineering, and automotive industries, showcasing its adaptability.
 - **Heat Treatment Options**: It can be processed through heat treatment methods like annealing, allowing for customization based on specific operational requirements.

ASTM F 2527 R30008

Category: Materials

Alloy Wire and Strip for Surgical Implant Applications.

Key Features:

- Chemical Composition: C 0.150 max, Cr 18.500 - 21.500, Mo 6.500 - 7.500, Ni 15.000 - 18.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	850 min
Yield Strength (MPa)	450 min
Elongation (%)	1 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F 2527 R30008 is a special type of alloy wire and strip designed specifically for surgical implant applications. This material is made from a carefully balanced mix of metals, including chromium, molybdenum, and nickel, which helps it withstand the demanding conditions of medical environments. It is known for its strength and durability, making it ideal for use in various critical applications, including valves, pumps, and automotive parts.

Key benefits of ASTM F 2527 R30008 include:

- **High Strength**: With a minimum tensile strength of 850 MPa, it can handle significant stress without breaking.
- **Versatility**: Suitable for a range of industries from medical to automotive, providing flexibility in usage.
- **Durability**: Resistant to corrosion and wear, ensuring longevity in surgical implants and other applications.

ASTM F 2527 R30035

Category: Materials

Seamless Cobalt Alloy Small Diameter Tubing for Surgical Implants.

Key Features:

- Chemical Composition: C 0.025 max, Cr 19.000 - 21.000, Mo 9.000 - 10.500, Ni 33.000 - 37.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	793 - 2482
Yield Strength (MPa)	241 - 586
Elongation (%)	1.800 min

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

The ASTM F 2527 R30035 is a specialized type of tubing made from a strong cobalt alloy, designed specifically for use in surgical implants. This tubing is seamless, which means it has no joints or seams, making it more reliable and less likely to fail in critical medical applications. The unique chemical composition of the alloy helps it resist corrosion and maintain its strength even in harsh environments, ensuring the safety and longevity of medical devices that use it.

- Benefits of ASTM F 2527 R30035 include:
- **High Strength**: With a tensile strength ranging from 793 to 2482 MPa, it can withstand significant stress without breaking.
 - **Versatile Applications**: This tubing can be used in various industries, including medical, automotive, and general engineering, making it a flexible choice for different projects.
 - **Durability**: Its resistance to corrosion and ability to maintain strength over time means it is ideal for long-term use in surgical implants and other critical components.

ASTM F 2527 R31537

Category: Materials

Seamless Cobalt Alloy Small Diameter Tubing for Surgical Implants.

Key Features:

- Chemical Composition: C 0.140 max, Cr 26.000 - 30.000, Mo 5.000 - 7.000, Ni 1.000 max

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	897 min
Yield Strength (MPa)	517 min
Elongation (%)	12 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F 2527 R31537 is a specialized type of tubing made from a strong cobalt alloy, designed for use in surgical implants and other critical applications. This seamless tubing is particularly beneficial in medical and engineering fields due to its excellent strength and durability. It can be used in various products like valves, pumps, and automotive parts, making it a versatile choice for many industries.

- Key benefits include:
- **High Strength**: With a minimum tensile strength of 897 MPa, it can withstand significant pressure and stress.
 - **Corrosion Resistance**: The chemical composition ensures that it can resist corrosion, essential for medical implants that come into contact with bodily fluids.

- **Versatile Applications**: Suitable for multiple uses in both medical and general engineering contexts, providing flexibility for manufacturers.

ASTM F 563 R30563

Category: Materials

Cobalt-20 Nickel-20 Chromium-3.5 Molybdenum Alloy for Surgical Implant Applications.

Key Features:

- Chemical Composition: C 0.050 max, Cr 18.000 - 22.000, Mo 3.000 - 4.000, Ni 15.000 - 25.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	600 min
Yield Strength (MPa)	275 min
Elongation (%)	12 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

The ASTM F 563 R30563 is a specialized metal alloy designed for use in surgical implants. It is made up of a combination of cobalt, nickel, chromium, and molybdenum, which gives it unique properties that make it ideal for medical applications. This alloy is strong and durable, ensuring that implants made from it can withstand the stresses of the human body. Additionally, it can be treated in various ways to enhance its properties further, making it highly versatile for different uses.

Key Benefits:

- **Strength and Durability**: With a minimum tensile strength of 600 MPa, this alloy is robust enough for demanding applications.
- **Versatile Applications**: It is suitable for various fields including medical devices, automotive parts, and general engineering, making it a practical choice for manufacturers.
- **Customizable Properties**: The alloy can undergo different heat treatments to tailor its characteristics for specific needs, providing flexibility in its use.

ASTM F 75 R30075

Category: Materials

Cobalt-28 Chromium-6 Molybdenum Casting Alloy for Surgical Implants.

Key Features:

- Chemical Composition: C 0.350 max, Cr 27.000 - 30.000, Mo 5.000 - 7.000, Ni 0.500 max

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	655 min

Yield Strength (MPa)	450 min
Elongation (%)	8 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F 75 R30075 is a special type of metal alloy made primarily for surgical implants. It is composed of cobalt, chromium, and molybdenum, which gives it unique properties that make it strong and durable. This alloy can withstand high stress and is designed to be biocompatible, meaning it is safe for use in the human body. It can be used in various applications such as valves, pumps, and even in the automotive industry, but its primary purpose is in medical devices.

****Key Benefits:****

- ****Strength:**** With a minimum tensile strength of 655 MPa, it can handle significant force without breaking.
- ****Durability:**** The alloy is resistant to wear and corrosion, making it ideal for long-lasting medical implants.
- ****Versatility:**** Suitable for multiple industries, including medical and engineering, enhancing its usability across different sectors.

ASTM F 90 R30605

Category: Materials

Cobalt-20 Chromium-15 Tungsten-10 Cobalt Alloy for Surgical Implant Applications.

Key Features:

- Chemical Composition: C 0.050 - 0.150, Cr 19.000 - 21.000, Ni 9.000 - 11.000, W 14.000 - 16.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	860 min
Yield Strength (MPa)	310 min
Elongation (%)	15 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

The ASTM F 90 R30605 is a specialized metal alloy made primarily from cobalt, chromium, and tungsten, designed specifically for surgical implants. This material is known for its excellent strength and durability, making it a reliable choice for medical applications that require long-lasting performance under stress. The alloy can be treated in different ways, either as-cast or through annealing, to enhance its properties further.

Benefits of ASTM F 90 R30605 include:

- **Strong and Reliable**: With a minimum tensile strength of 860 MPa, it can withstand significant forces, making it suitable for critical applications.
- **Versatile Applications**: Beyond medical uses, it's also applicable in valves, pumps, general engineering, and automotive industries, showcasing its broad utility.
- **Durable Performance**: The alloy's composition ensures it remains resilient in challenging environments, providing longevity in surgical or mechanical applications.

ASTM F1058 Grade 1

Category: Materials

40 Cobalt-20 Chromium-16 Iron-15 Nickel-7 Molybdenum Alloy Wire and Strip for Surgical Implant Applications.

Key Features:

- Chemical Composition: C 0.150 max, Cr 19.000 - 21.000, Mo 6.000 - 8.000, Ni 14.000 - 16.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	850 min
Yield Strength (MPa)	450 min
Elongation (%)	1 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F1058 Grade 1 is a specialized alloy wire and strip made from a mix of metals like cobalt, chromium, iron, nickel, and molybdenum. This unique combination makes it particularly suitable for use in surgical implants, ensuring that medical devices are both strong and durable. The material can undergo heat treatment, allowing it to be customized for various applications, from valves and pumps to automotive parts and general engineering.

ASTM F1058 Grade 2

Category: Materials

Alloy Wire and Strip for Surgical Implant Applications.

Key Features:

- Chemical Composition: C 0.150 max, Cr 18.500 - 21.500, Mo 6.500 - 7.500, Ni 15.000 - 18.000

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	850 min
Yield Strength (MPa)	450 min
Elongation (%)	1 min

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F1058 Grade 2 is a specialized type of alloy wire and strip designed specifically for use in surgical implants. This material is known for its strong and durable properties, making it an ideal choice for medical applications where reliability is essential. It is made from a carefully controlled mix of metals that provide excellent resistance to corrosion and wear, which is crucial for items that will be used inside the human body.

Key benefits of ASTM F1058 Grade 2 include:

- **High Strength:** With a minimum tensile strength of 850 MPa, it can withstand significant stress without breaking.
- **Versatile Applications:** This alloy is suitable for various uses, including valves, pumps, and automotive parts, in addition to its primary function in medical devices.
- **Durability:** The material maintains its strength and integrity over time, even in challenging environments, making it a reliable choice for long-term applications.

ASTM F1537 Alloy 2

Category: Materials

Cobalt-Based alloy for Surgical Implants.

Key Features:

- Chemical Composition: C 0.150 - 0.350, Cr 26.000 - 30.000, Mo 5.000 - 7.000, Ni 1.000 max

Technical Specifications:

Heat Treatment	As-Cast or Annealing
Tensile Strength (MPa)	897 min
Yield Strength (MPa)	517 min
Elongation (%)	12 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive
- Medical

AI Layman Summary:

ASTM F1537 Alloy 2 is a special type of metal made primarily from cobalt, designed specifically for use in surgical implants. This material is engineered to be strong and durable, making it ideal for medical applications where reliability is crucial. It can withstand significant stress and has the ability to elongate under pressure without breaking, which is essential for implants that need to function effectively within the human body.

Key Benefits:

- **Strength and Durability:** With a minimum tensile strength of 897 MPa, this alloy can handle heavy loads, making it perfect for surgical use.
- **Versatile Applications:** Beyond medical implants, it can be used in valves, pumps, and even automotive engineering, showcasing its wide range of uses.
- **Chemical Resistance:** The specific chemical composition helps it resist corrosion, ensuring the implants

maintain integrity over time.

ASTM F3 Grade 4

Category: Materials

Nickel Strip for Electron Tubes.

Key Features:

- Chemical Composition: C 0.020 max, Ni 99.970 min

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	345 min
Yield Strength (MPa)	110 min
Elongation (%)	50 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F3 Grade 4 is a specialized nickel strip designed for use in electron tubes, which are often found in various electronic devices. This nickel strip is made with high purity nickel (99.970% minimum) and has strict chemical composition standards, ensuring reliability and performance in demanding applications. The material can be treated in different ways, such as being cast, annealed, or hardened, to enhance its strength and flexibility depending on the needs of the project.

Key benefits of ASTM F3 Grade 4 include:

- ****High Strength****: It has a minimum tensile strength of 345 MPa, making it durable for tough environments.
- ****Versatile Applications****: Ideal for use in valves, pumps, general engineering, and automotive industries, allowing for a wide range of uses.
- ****Good Flexibility****: With a minimum elongation of 50%, it can bend without breaking, which is crucial in many engineering applications.

ASTM F3056 N06625 CLASS B

Category: Materials

Additive Manufacturing Nickel Alloy.

Key Features:

- Chemical Composition: C 0.100 max, Cr 20.000 - 23.000, Mo 8.000 - 10.000, Ni Balance

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	485 min
Yield Strength (MPa)	275 min
Elongation (%)	30 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F3056 N06625 CLASS B is a specialized nickel alloy designed for additive manufacturing, which is a modern method of creating parts layer by layer using a computer-controlled process. This alloy is particularly beneficial in industries that require strong and durable components, such as valves, pumps, and automotive applications. Its unique chemical composition ensures that it can withstand high temperatures and pressures, making it ideal for various engineering needs.

Key benefits include:

- **Strength and Durability**: With a minimum tensile strength of 485 MPa and yield strength of 275 MPa, this alloy can handle demanding environments.
- **Versatility**: Suitable for various applications, including general engineering and automotive parts, allowing for flexibility in design and production.
- **Adaptable Heat Treatment**: Can undergo different heat treatments, enhancing its properties based on specific requirements.

ASTM F3056 N06625 CLASS C

Category: Materials
Additive Manufacturing Nickel Alloy.

Key Features:

- Chemical Composition: C 0.100 max, Cr 20.000 - 23.000, Mo 8.000 - 10.000, Ni Balance

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	485 min
Yield Strength (MPa)	275 min
Elongation (%)	30 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F3056 N06625 CLASS C is a special type of metal known as a nickel alloy, designed specifically for advanced manufacturing techniques, like 3D printing. This material is particularly valuable in industries that require strong and durable components, such as automotive and engineering. Its unique chemical makeup helps it resist wear and tear while maintaining its structural integrity even under high stress.

Key benefits of ASTM F3056 N06625 CLASS C include:

- **Strong Performance**: With a minimum tensile strength of 485 MPa, this material can withstand significant forces without breaking.
- **Versatile Applications**: It can be used in various parts such as valves and pumps, making it suitable for many engineering projects.

- **Flexible Processing**: It can be treated in different ways, such as annealing or age hardening, to enhance its properties further.

ASTM F3056 N06625 CLASS A

Category: Materials

Additive Manufacturing Nickel Alloy.

Key Features:

- Chemical Composition: C 0.100 max, Cr 20.000 - 23.000, Mo 8.000 - 10.000, Ni Balance

Technical Specifications:

Heat Treatment	As-Cast or Annealing or Age Hardening
Tensile Strength (MPa)	485 min
Yield Strength (MPa)	275 min
Elongation (%)	30 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F3056 N06625 CLASS A is a specialized nickel alloy designed for additive manufacturing, which is a process that creates objects layer by layer, often used in advanced engineering applications. This material is particularly strong and durable, making it ideal for producing critical components like valves and pumps in various industries such as automotive and general engineering.

Key benefits of using ASTM F3056 N06625 CLASS A include:

- **High Strength**: With a minimum tensile strength of 485 MPa, it can withstand significant stress without breaking.
- **Versatility**: It can be used in demanding applications, including valve and pump manufacturing, as well as in the automotive sector.
- **Flexible Processing**: The material can undergo different heat treatments to enhance its properties, allowing for customization based on specific needs.

ASTM F467 C46400

Category: Materials

Nonferrous Nuts for General Use.

Key Features:

- Chemical Composition: Fe 0.100 max, Pb 0.200 max, Cu 59.000 - 62.000

Technical Specifications:

Heat Treatment	Normalizing or Annealing or Tempering
Yield Strength (MPa)	344.7 - 372.3
Hardness (HRB)	55 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F467 C46400 is a type of nonferrous nut designed for general use in various industries. These nuts are primarily made of copper and are known for their excellent corrosion resistance and durability. They can be used in a variety of applications such as valves, pumps, and automotive components, making them a versatile choice for engineers and manufacturers alike.

Key benefits of ASTM F467 C46400 nuts include:

- **Strong and Reliable**: With a yield strength of up to 372.3 MPa, these nuts can withstand significant stress without deforming.
- **Heat Treatment Flexibility**: They can be heat treated through normalizing, annealing, or tempering processes, enhancing their performance in diverse environments.
- **Wide Application Range**: Ideal for use in general engineering, automotive, and other mechanical applications, ensuring they meet various industry demands.

ASTM B271 C87300

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Casting.

Key Features:

- Chemical Composition: Cu %94.000 min.
- Tensile Strength in Mpa: 310 min.
- Yield Strength in Mpa: 124 min.

Technical Specifications:

Iron	Fe %0.200 max.
Manganese	Mn %0.800 - 1.500
Lead	Pb %0.090 max.
Silicon	Si %3.500 - 4.500
Zinc	Zn %0.250 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C87300 is a type of copper-based metal that is created using a process called centrifugal casting. This means that the metal is poured into a mold that spins around, which helps to ensure that the metal is dense and strong. This specific alloy is known for its high copper content and impressive strength, making it ideal for various industrial applications.

Key Benefits:

- **Strong and Durable**: With a minimum tensile strength of 310 MPa, this alloy can withstand heavy use

without breaking.

- **Versatile Applications**: It's perfect for making parts in valves, pumps, and in the automotive industry, ensuring reliable performance in critical components.
- **Controlled Composition**: The alloy has a carefully balanced mix of other metals, such as manganese and silicon, which enhances its properties for specific uses.

ASTM B271 C87400

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Casting.

Key Features:

- Chemical Composition: Cu %79.000 min.
- Tensile Strength in Mpa: 345 min.
- Yield Strength in Mpa: 145 min.

Technical Specifications:

Aluminium	Al %0.800 max.
Lead	Pb %1.000 max.
Silicon	Si %2.500 - 4.000
Zinc	Zn %12.000 - 16.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C87400 is a specialized metal alloy used in manufacturing durable and strong components, especially in industries such as automotive and engineering. This copper-based alloy is created through a process called centrifugal casting, which ensures a high-quality finish and strength. With a minimum copper content of 79%, it provides excellent resistance to wear and tear, making it ideal for demanding applications like valves and pumps.

Key benefits include:

- **Strong and Durable**: With a minimum tensile strength of 345 MPa, it can withstand heavy stress and pressure.
- **Versatile Use**: Suitable for various applications including automotive parts, valves, and general engineering components, ensuring broad usability in different industries.
- **Specific Composition**: Contains controlled amounts of other elements like aluminum, lead, silicon, and zinc, enhancing its performance and longevity.

ASTM B271 C87500

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Casting.

Key Features:

- Chemical Composition: Cu %79.000 min.
- Tensile Strength in Mpa: 414 min.
- Yield Strength in Mpa: 165 min.

Technical Specifications:

Aluminium	Al %0.500 max.
Lead	Pb %0.090 max.
Silicon	Si %3.000 - 5.000
Zinc	Zn %12.000 - 16.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C87500 is a type of copper alloy that is created using centrifugal casting, which is a process where molten metal is spun in a mold to form strong, durable parts. This specific alloy is known for its high copper content, giving it excellent strength and resistance to wear. It's often used in industries requiring reliable components, such as valves and pumps, as well as in general engineering and automotive applications.

****Key Features:****

- Contains a minimum of 79% copper for strength and durability.
- Has a minimum tensile strength of 414 MPa and yield strength of 165 MPa, ensuring it can withstand significant stress.
- Compliant with strict chemical specifications to ensure consistent quality and performance.

ASTM B271 C87600

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Casting.

Key Features:

- Chemical Composition: Cu %88.000 min.
- Tensile Strength in Mpa: 414 min.
- Yield Strength in Mpa: 207 min.

Technical Specifications:

Iron	Fe %0.200 max.
Manganese	Mn %0.250 max.
Lead	Pb %0.090 max.
Silicon	Si %3.500 - 5.500
Zinc	Zn %4.000 - 7.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C87600 is a high-quality copper-based alloy that is specifically designed for centrifugal casting.

This means that it is produced using a spinning method that ensures a uniform and dense structure, making it ideal for creating durable components. The alloy contains at least 88% copper, which gives it excellent conductivity and corrosion resistance. It is particularly strong, with a minimum tensile strength of 414 MPa and a yield strength of 207 MPa, which means it can withstand significant stress without breaking.

This alloy is versatile and can be used in various applications, including:

- *Valves: Ensuring reliable operation in fluid control systems.*
- *Pumps: Providing strength and durability for various pumping solutions.*
- *General engineering: Suitable for a wide range of engineering projects.*
- *Automotive: Ideal for parts that require high performance and resistance to wear.*

Overall, ASTM B271 C87600 is a reliable choice for industries that need robust and long-lasting materials for their products.

ASTM B271 High-Lead Tin Bronze UNS C93500

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %83.000 - 86.000
- Tensile Strength in Mpa: 207 min.
- Yield Strength in Mpa: 97 min.

Technical Specifications:

Tin	Sn %4.300 - 6.000
Lead	Pb %8.000 - 10.000
Zinc	Zn %2.000 max.
Iron	Fe %0.200 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 High-Lead Tin Bronze UNS C93500 is a special type of metal alloy made primarily from copper, tin, and lead. It is designed for use in casting processes, which means it can be poured into molds to create various shapes and parts. This alloy is known for its strength and durability, making it a reliable choice for many industrial applications.

Key benefits of this alloy include:

- ***High Strength***: With a minimum tensile strength of 207 MPa, it can withstand heavy loads and pressures.
- ***Versatile Use***: It is commonly used in manufacturing valves, pumps, and automotive components, making it suitable for a wide range of engineering projects.
- ***Durability***: The chemical composition ensures that it is resistant to wear and tear, extending the life of the products made from it.

ASTM B271 High-Lead Tin Bronze UNS C94300

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %67.000 - 72.000
- Tensile Strength in Mpa: 179 min.
- Yield Strength in Mpa: 97 min.

Technical Specifications:

Tin	Sn %4.500 - 6.000
Lead	Pb %23.000 - 27.000
Zinc	Zn %0.800 max.
Iron	Fe %0.150 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 High-Lead Tin Bronze UNS C94300 is a specialized metal alloy designed for making strong and durable castings, particularly through a process called centrifugal casting, which helps produce parts with excellent strength and resistance. This alloy primarily consists of copper, tin, and lead, making it ideal for use in various industries. It is commonly utilized in components like valves and pumps, as well as in general engineering and automotive applications.

****Key Benefits:****

- ****High Strength:**** Offers a minimum tensile strength of 179 MPa and a yield strength of 97 MPa, ensuring that the parts can withstand significant pressure and stress.
- ****Versatile Applications:**** Suitable for a variety of uses in different sectors, including automotive, enhancing the reliability and performance of critical components.
- ****Durability:**** The chemical composition, with lead and tin, contributes to the alloy's resistance to wear and corrosion, extending the life of the products made from it.

ASTM B271 High-Strength Yellow Brass UNS C86200

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %60.000 - 66.000
- Tensile Strength in Mpa: 621 min.
- Yield Strength in Mpa: 310 min.

Technical Specifications:

Tin	Sn %0.200 max.
Lead	Pb %0.200 max.
Zinc	Zn %22.000 - 28.000
Iron	Fe %2.000 - 4.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 High-Strength Yellow Brass UNS C86200 is a type of metal that is specially made for creating strong and durable cast parts. This brass alloy is primarily composed of copper, alongside small amounts of tin, lead, zinc, and iron, which enhance its properties. Because of its robust tensile strength and yield strength, it is ideal for heavy-duty applications.

The benefits of using this material include:

- **High Strength**: It has a minimum tensile strength of 621 megapascals, making it suitable for demanding environments.
- **Versatile Uses**: Commonly used in valves, pumps, general engineering, and automotive components.
- **Durability**: Its unique composition allows it to withstand wear and tear, ensuring a long lifespan for the products made from it.

ASTM B271 Leaded Nickel Bronze UNS C97600

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %63.000 - 67.000
- Tensile Strength in Mpa: 276 min.
- Yield Strength in Mpa: 117 min.

Technical Specifications:

Tin	Sn %3.500 - 4.500
Lead	Pb %3.000 - 5.000
Zinc	Zn %3.000 - 9.000
Iron	Fe %1.500 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Nickel Bronze UNS C97600 is a type of metal alloy primarily made of copper and used in various industrial applications. It is known for its strength and durability, making it ideal for creating parts in machinery, such as valves and pumps. The alloy is created through a process called centrifugal casting, which ensures a high-quality finish and consistency in the material's properties.

Key benefits of this alloy include:

- **Strong and Reliable**: With a minimum tensile strength of 276 MPa, it can withstand significant stress and pressure, making it suitable for tough environments.
- **Versatile Applications**: It is commonly used in engineering, automotive, and other industries where reliable metal components are essential.
- **Good Corrosion Resistance**: The alloy's composition includes lead, tin, and zinc, which enhance its

resistance to wear and corrosion, extending the life of the parts made from it.

ASTM B271 Leaded Nickel Bronze UNS C97800

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %64.000 - 67.000
- Tensile Strength in Mpa: 345 min.
- Yield Strength in Mpa: 152 min.

Technical Specifications:

Tin	Sn %4.000 - 5.500
Lead	Pb %1.000 - 2.500
Zinc	Zn %1.000 - 4.000
Iron	Fe %1.500 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Nickel Bronze UNS C97800 is a durable metal alloy designed for making various industrial parts through a process called centrifugal casting. This means the alloy is poured into a mold that spins, ensuring a strong and uniform structure. It's primarily made of copper, tin, lead, and zinc, making it resistant to wear and capable of handling heavy loads.

Key benefits include:

- ****Strength and Durability****: With a minimum tensile strength of 345 MPa, this alloy is built to last, making it ideal for high-stress applications.
- ****Versatile Applications****: It can be used in valves, pumps, and automotive parts, showcasing its adaptability in different engineering fields.
- ****Easy to Work With****: The alloy's composition allows for effective casting, which means manufacturers can create complex shapes and components efficiently.

ASTM B271 Leaded Red Brass UNS C83600

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %84.000 - 86.000
- Tensile Strength in Mpa: 207 min.
- Yield Strength in Mpa: 97 min.

Technical Specifications:

Tin	Sn %4.000 - 6.000
Lead	Pb %4.000 - 6.000

Zinc	Zn %4.000 - 6.000
Iron	Fe %0.300 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Red Brass UNS C83600 is a type of metal alloy that is commonly used in manufacturing various products through a process called centrifugal casting. This alloy is made primarily of copper, with specific amounts of tin, lead, and zinc, making it strong and durable. It is especially valued in industries that require reliable materials for creating components like valves and pumps, as well as in general engineering and automotive applications.

Key benefits of this alloy include:

- **Strength**: It has a minimum tensile strength of 207 MPa, ensuring that it can withstand significant stress and pressure.
- **Versatility**: Suitable for a wide range of uses, from valves and pumps to automotive parts, making it a practical choice for manufacturers.
- **Durability**: The carefully controlled chemical composition provides excellent resistance to wear and corrosion, extending the lifespan of products made from this material.

ASTM B271 Leaded Red Brass UNS C83800

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %82.000 - 83.800
- Tensile Strength in Mpa: 207 min.
- Yield Strength in Mpa: 90 min.

Technical Specifications:

Tin	Sn %3.300 - 4.200
Lead	Pb %5.000 - 7.000
Zinc	Zn %5.000 - 8.000
Iron	Fe %0.300 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Red Brass UNS C83800 is a special type of metal alloy made primarily from copper, designed for making strong and durable cast parts. It is particularly useful in industries that require reliable components for machinery and automotive applications. This alloy is created using a process called centrifugal casting, which ensures that the metal is evenly distributed and solidifies in a uniform shape, resulting in high-

quality components.

Key benefits of using ASTM B271 Lead Red Brass include:

- **Strength and Durability**: With a minimum tensile strength of 207 MPa, this alloy can withstand significant stress, making it ideal for heavy-duty applications.
- **Versatility**: Its unique composition allows it to be used in various products such as valves, pumps, and automotive parts, making it a go-to choice for general engineering purposes.
- **Corrosion Resistance**: The addition of elements like tin and lead provides enhanced resistance against corrosion, ensuring a longer lifespan for the components made from this alloy.

ASTM B271 Lead Semi-Red Brass UNS C84800

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

- Key Features:
- Chemical Composition: Cu %75.000 - 77.000
 - Tensile Strength in Mpa: 193 min.
 - Yield Strength in Mpa: 83 min.

Technical Specifications:

Tin	Sn %2.000 - 3.000
Lead	Pb %5.500 - 7.000
Zinc	Zn %13.000 - 17.000
Iron	Fe %0.400 max.

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM B271 Lead Semi-Red Brass UNS C84800 is a type of metal alloy primarily made of copper, designed for uses in various industrial applications. This material is known for its durability and strength, making it ideal for creating components like valves and pumps. It is produced using a centrifugal casting technique, which helps in forming intricate shapes that are strong and reliable.

Key benefits of this brass alloy include:

- **High Strength**: With a minimum tensile strength of 193 MPa, it can withstand significant stress without breaking.
- **Versatile Use**: Suitable for many applications, including automotive parts, general engineering, and more, ensuring it meets diverse industry needs.
- **Durable Composition**: The alloy contains lead, tin, and zinc, which enhance its machinability and resistance to wear, making it a long-lasting choice for critical components.

ASTM B271 Lead Semi-Red Brass UNS C84400

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %78.000 - 82.000
- Tensile Strength in Mpa: 200 min.
- Yield Strength in Mpa: 90 min.

Technical Specifications:

Tin	Sn %2.300 - 3.500
Lead	Pb %6.000 - 8.000
Zinc	Zn %7.000 - 10.000
Iron	Fe %0.400 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Semi-Red Brass UNS C84400 is a type of metal alloy primarily made from copper, with added elements like lead, tin, and zinc. It is known for its durability and strength, making it suitable for various industrial applications. This alloy is created using a process called centrifugal casting, which ensures a uniform structure and high-quality finish.

Key benefits of this brass alloy include:

- **High Strength**: With a minimum tensile strength of 200 MPa, it can withstand significant pressure, making it ideal for heavy-duty applications.
- **Versatile Uses**: Commonly used in valves, pumps, and automotive parts, it is a go-to choice for engineers in multiple industries.
- **Reliable Performance**: The specific chemical composition ensures that it maintains its strength and integrity even under challenging conditions.

ASTM B271 Leaded Yellow Brass UNS C85200

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %70.000 - 74.000
- Tensile Strength in Mpa: 241 min.
- Yield Strength in Mpa: 83 min.

Technical Specifications:

Tin	Sn %0.700 - 2.000
Lead	Pb %1.500 - 3.800
Zinc	Zn %20.000 - 27.000
Iron	Fe %0.600 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Yellow Brass UNS C85200 is a type of metal alloy that is primarily used in making various mechanical parts. It is created through a process called centrifugal casting, which involves pouring molten metal into a rotating mold to create strong and durable components. This specific alloy consists mainly of copper, with additional elements like lead, zinc, and tin that enhance its properties.

Key benefits of this brass alloy include:

- **Strength and Durability**: With a minimum tensile strength of 241 MPa, it is designed to withstand high pressures and stresses, making it suitable for tough applications.
- **Versatile Applications**: It is commonly used in manufacturing valves, pumps, and automotive parts, making it a reliable choice for engineers and manufacturers looking for quality materials.
- **Corrosion Resistance**: The chemical composition provides good resistance to corrosion, extending the lifespan of components made from this alloy.

ASTM B271 Leaded Yellow Brass UNS C85700

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %58.000 - 64.000
- Tensile Strength in Mpa: 276 min.
- Yield Strength in Mpa: 97 min.

Technical Specifications:

Tin	Sn %0.500 - 1.500
Lead	Pb %0.800 - 1.500
Zinc	Zn %32.000 - 40.000
Iron	Fe %0.700 max.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Leaded Yellow Brass UNS C85700 is a special type of metal used for making precision parts through a process called centrifugal casting. This alloy is primarily made of copper, along with small amounts of tin, lead, zinc, and iron, which give it specific properties that make it suitable for various applications. It is known for its strength and durability, making it an ideal choice for producing components that need to withstand high pressure and wear.

Key Benefits:

- **Strong and Durable**: With a minimum tensile strength of 276 MPa, this brass can handle tough conditions without easily breaking.
- **Versatile Applications**: It is commonly used in making valves, pumps, and automotive parts, as well as in general engineering projects.
- **Quality Composition**: The controlled chemical composition ensures that the material has the right balance of elements for optimal performance.

ASTM B271 Nickel Alumium Bronze UNS C95500

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %78.000 min.
- Tensile Strength in Mpa: 620 min.
- Yield Strength in Mpa: 275 min.

Technical Specifications:

Iron	Fe %3.000 - 5.500
Ni + Co	Ni% + Co% 3.000 - 5.500
Aluminium	Al %10.000 - 11.500
Copper	Cu %78.000 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Nickel Aluminum Bronze UNS C95500 is a type of metal alloy primarily made of copper, aluminum, and nickel, designed for use in various industrial applications. This copper-base alloy is created using a centrifugal casting process, which ensures a strong and durable material ideal for making components like valves and pumps. Its unique chemical composition gives it excellent resistance to corrosion and wear, making it a reliable choice for both general engineering and automotive sectors.

Key benefits of using ASTM B271 Nickel Aluminum Bronze UNS C95500 include:

- High tensile strength (620 MPa minimum) and yield strength (275 MPa minimum), ensuring that it can withstand heavy loads and stress without breaking.
- Corrosion resistance due to its aluminum content, which helps it perform well even in harsh environments.
- Versatile applications, suitable for making critical components in machinery and vehicles, enhancing their longevity and performance.

ASTM B271 Nickel Alumium Bronze UNS C95520

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %74.500 min.
- Tensile Strength in Mpa: N/A
- Yield Strength in Mpa: N/A

Technical Specifications:

Tin	Sn %0.250 max.
Lead	Pb %0.030 max.
Zinc	Zn %0.300 max.
Iron	Fe %4.000 - 5.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Nickel Aluminum Bronze UNS C95520 is a specialized metal alloy used in various industrial applications. It is primarily made from copper, with a minimum copper content of 74.5%. This alloy is designed for creating strong and durable components through a process called centrifugal casting, which helps in producing high-quality parts. Its unique composition makes it suitable for harsh environments, making it a reliable choice for manufacturers.

****Key Benefits:****

- ****Versatile Applications:**** It can be used in valves, pumps, and general engineering projects, as well as in the automotive industry.
- ****Durability:**** The alloy's strength and resistance to corrosion make it ideal for long-lasting parts in demanding conditions.
- ****Quality Casting:**** The centrifugal casting process ensures high precision and uniformity in the final products, enhancing their performance.

ASTM B271 Nickel Alumium Bronze UNS C95800

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %79.000 min.
- Tensile Strength in Mpa: 585 min.
- Yield Strength in Mpa: 240 min.

Technical Specifications:

Lead	Pb %0.030 max.
Iron	Fe %3.500 - 4.500
Ni + Co	Ni% + Co% 4.000 - 5.000
Aluminium	Al %8.500 - 9.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Nickel Aluminum Bronze UNS C95800 is a special kind of metal alloy made primarily from copper, aluminum, and nickel. It is used in industries that require strong and durable materials, like automotive and engineering. This alloy is created using a process called centrifugal casting, which helps ensure it is strong and has consistent quality.

Key benefits of this alloy include:

- ****High Strength**:** It has a minimum tensile strength of 585 MPa, making it very strong for various applications.
- ****Corrosion Resistance**:** Its composition includes aluminum, which helps resist corrosion, ideal for use in

harsh environments.

- ****Versatile Applications****: This alloy is suitable for making valves, pumps, and other engineering components, ensuring reliability and longevity in their performance.

ASTM B271 Silicon Brass UNS C87400

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %79.000 min.
- Tensile Strength in Mpa: 345 min.
- Yield Strength in Mpa: 145 min.

Technical Specifications:

Lead	Pb %1.000 max.
Zinc	Zn %12.000 - 16.000
Aluminium	Al %0.800 max.
Silicon	Si %2.500 - 4.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Silicon Brass UNS C87400 is a type of metal alloy specifically designed for creating strong and durable castings. It's primarily made of copper, with a minimum copper content of 79%, which gives it excellent strength and resistance to corrosion. This alloy is particularly useful in manufacturing components that require high durability and reliability, such as valves and pumps in various engineering and automotive applications.

Key benefits of ASTM B271 Silicon Brass include:

- High tensile strength, ensuring that the parts can withstand significant pressure and forces (minimum 345 MPa).
- Good yield strength, allowing it to deform under stress without breaking (minimum 145 MPa).
- Versatile use in industries like automotive and general engineering, making it a valuable material for many manufacturing needs.

ASTM B271 Silicon Bronze UNS C87300

Category: Copper-Base Alloy Centrifugal Casting

Standard Specification for Copper-Base Alloy Centrifugal Castings.

Key Features:

- Chemical Composition: Cu %94.000 min.
- Tensile Strength in Mpa: 310 min.
- Yield Strength in Mpa: 124 min.

Technical Specifications:

Lead	Pb %0.200 max.
------	----------------

Zinc	Zn %0.250 max.
Iron	Fe %0.200 max.
Manganese	Mn %0.800 - 1.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 Silicon Bronze UNS C87300 is a specialized metal alloy known for its strength and durability, making it ideal for various industrial applications. This alloy primarily consists of copper, with at least 94% copper content, ensuring excellent resistance to corrosion and wear. It is commonly used in the manufacturing of valves, pumps, and other components in general engineering and automotive industries.

Key benefits include:

- **High Tensile Strength**: With a minimum tensile strength of 310 MPa, this alloy can withstand significant stress without breaking.
- **Versatile Applications**: Suitable for critical components in valves and pumps, ensuring reliability in demanding environments.
- **Durable Composition**: Low levels of lead, zinc, and iron enhance its performance while maintaining safety standards.

Aluminium Alloys for Casting

Category: Pumps

Aluminium alloys for casting applications.

Technical Specifications:

Chemical Composition

Mechanical Properties

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

Aluminium alloys for casting are specially formulated materials used to create various components through a casting process. These alloys are primarily made of aluminium, with small amounts of copper and iron added to enhance their properties. The result is a lightweight yet strong material that can be molded into complex shapes, making it ideal for various applications.

Key benefits include:

- **Strength and Durability**: With a minimum tensile strength of 305 MPa, these alloys are robust enough for demanding uses.
- **Versatile Applications**: Suitable for a range of uses including valves, pumps, general engineering projects, and automotive parts.
- **Ease of Casting**: The alloys can be easily cast into intricate designs, providing flexibility in manufacturing.

In summary, aluminium alloys for casting offer a reliable and efficient solution for producing high-quality components across various industries.

ASTM F467 C67500

Category: Nuts

Standard Specification for Nonferrous Nuts for General Use.

Key Features:

- Aluminium %0.250 max.
- Iron %0.800 - 2.000
- Copper %57.000 - 60.000

Technical Specifications:

Tensile Strength in Mpa	379.2 - 406.8
Hardness in HRB	60 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM F467 C67500 is a type of nut made from special nonferrous materials, designed to be used in various applications like valves, pumps, and automotive parts. These nuts are crafted to meet specific standards, ensuring they are reliable and durable in general engineering tasks. They are particularly notable for their composition, which includes a maximum of 0.25% aluminum, between 0.8% to 2% iron, and a significant 57% to 60% copper content.

****Key Benefits:****

- ****Strong and Durable:**** With a tensile strength ranging from 379.2 to 406.8 MPa and a minimum hardness of 60 HRB, these nuts can withstand significant stress and wear.
- ****Versatile Use:**** Ideal for various industries, including automotive and general engineering, making them a great choice for many different projects.

ASTM F467 C61300

Category: Nuts

Standard Specification for Nonferrous Nuts for General Use.

Key Features:

- Aluminium %6.000 - 7.500
- Copper %88.500 - 91.500

Technical Specifications:

Tensile Strength in Mpa	450 - 515
Yield Strength in Mpa	180 - 250

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F467 C61300 is a type of nut made from a special blend of nonferrous metals, mainly aluminum and copper. This product is designed for general use in various applications where strong and durable connections are needed. The material's composition ensures that it can withstand high levels of stress and pressure, making it reliable for important machinery and equipment.

- Key features of ASTM F467 C61300 include:
- Made with 6% to 7.5% aluminum for lightweight strength.
 - Contains 88.5% to 91.5% copper, providing excellent conductivity and corrosion resistance.

This nut is particularly useful in industries such as automotive, valve systems, pumps, and general engineering. Its high tensile strength (450 - 515 MPa) and yield strength (180 - 250 MPa) mean it can handle significant loads, ensuring safety and performance in demanding environments.

ASTM F467 Ni400

Category: Nickel-Copper Alloy

Standard Specification for Nickel-Copper Alloy, Corrosion Resistant Sheet, Strip and Plate.

- Key Features:**
- Nickel %63.000 - 70.000
 - Copper %Balance

Technical Specifications:

Yield Strength in Mpa	551.6 min.
Hardness in HRC	75 min.

- Use Cases:**
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM F467 Ni400 is a special type of metal known as a Nickel-Copper Alloy, designed to resist corrosion and wear. This material is made up of about 63% to 70% nickel, with the remaining portion being copper. It is available in various forms such as sheets, strips, and plates, making it versatile for different manufacturing needs.

- Key benefits of using ASTM F467 Ni400 include:
- ****Durability****: With a minimum yield strength of 551.6 MPa and a hardness rating of 75 HRC, this alloy is tough and can withstand harsh conditions.
 - ****Versatile Applications****: It is commonly used in valves, pumps, automotive parts, and general engineering projects, making it a go-to choice for many industries looking for reliable materials.

ASTM F562 R30035

Category: Cobalt Alloys

Standard Specification for Cobalt-35 Nickel-20 Chromium-10 Molybdenum Alloy for Surgical Implant Applications.

Key Features:

- Cobalt %Balance
- Nickel %33.000 - 37.000

Technical Specifications:

Tensile Strength in Mpa	793 - 2482
Yield Strength in Mpa	241 - 586

Use Cases:

- Valve
- Pump
- General Engineering
- Medical

AI Layman Summary:

ASTM F562 R30035 is a special type of metal alloy made primarily of cobalt, with additional elements like nickel, chromium, and molybdenum. This alloy is specifically designed for surgical implants, making it a critical material in the medical field. Its unique composition ensures that it is strong and durable, which is essential for devices that need to withstand the stresses of the human body.

Key benefits of ASTM F562 R30035 include:

- **High Strength**: It has impressive tensile strength, meaning it can handle a lot of force without breaking.
- **Versatile Applications**: Besides medical implants, it can also be used in valves, pumps, and general engineering projects, highlighting its adaptability across various industries.

ASTM F799 Alloy 2

Category: Cobalt Alloys

Standard Specification for Cobalt-Based alloy.

Technical Specifications:

Tensile Strength in Mpa	1172 min.
Yield Strength in Mpa	827 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F799 Alloy 2 is a high-performance cobalt-based alloy designed for demanding applications. It is known for its impressive strength and durability, making it suitable for various industries including automotive, engineering, and more. This alloy has a minimum tensile strength of 1172 MPa and a yield strength of 827 MPa, ensuring it can withstand heavy loads and stress without failing.

Key benefits of ASTM F799 Alloy 2 include:

- **Exceptional Strength**: It can handle high pressures and extreme conditions, which is crucial for components like valves and pumps.
- **Versatile Applications**: Ideal for use in automotive parts, general engineering projects, and other demanding environments, ensuring reliability and longevity.

ASTM F799 Alloy 3

Category: Cobalt Alloys

Standard Specification for Cobalt-Based alloy.

Technical Specifications:

Tensile Strength in Mpa	897 min.
Yield Strength in Mpa	517 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM F799 Alloy 3 is a special type of metal known as a cobalt alloy, designed specifically for various industrial applications. It is recognized for its strength and durability, making it an excellent choice for parts that need to withstand high levels of stress and pressure. This alloy is particularly useful in the production of valves and pumps, as well as in general engineering and automotive industries.

Key benefits of ASTM F799 Alloy 3 include:

- **High Strength**: With a minimum tensile strength of 897 MPa, it can handle tough environments without breaking.
- **Versatile Use**: Ideal for a range of applications, from automotive components to engineering solutions, ensuring reliability in critical systems.

ASTM F901 6061

Category: Aluminium Alloys

Standard Specification For Aluminium Alloy Transmission Tower Bolt and Nut.

Technical Specifications:

Tensile Strength in Mpa	121.2 - 328.5
Yield Strength in Mpa	49 - 305.1

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM F901 6061 is a type of aluminum alloy specifically designed for making strong and reliable bolts and nuts used in transmission towers. This product is made to meet industry standards, ensuring that it can

withstand significant weight and stress. It is particularly useful in various applications, such as valves, pumps, general engineering, and automotive sectors.

- Key benefits of ASTM F901 6061 include:
- **High Strength**: With a tensile strength range of 121.2 to 328.5 MPa, these bolts and nuts are capable of handling heavy loads.
 - **Durability**: The yield strength ranges from 49 to 305.1 MPa, allowing them to maintain their shape and function under pressure.
 - **Versatility**: Ideal for a wide range of applications, making it a go-to choice for engineers and manufacturers.

ASTM F961 R30035

Category: Cobalt Alloys

Standard Specification for 35Cobalt-35Nickel-20Chromium-10Molybdenum Alloy Forgings for Surgical Implants.

Technical Specifications:

Tensile Strength in Mpa	793 - 2482
Yield Strength in Mpa	241 - 586

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Medical

AI Layman Summary:

ASTM F961 R30035 is a special type of metal alloy that contains cobalt, nickel, chromium, and molybdenum. This alloy is primarily used in making surgical implants, which are medical devices that are placed inside the body to support or replace damaged structures. Its unique composition provides excellent strength and durability, making it ideal for high-stress applications in both medical and industrial settings.

How it works: The alloy is forged into various shapes and sizes to meet specific requirements for different applications. Its impressive tensile strength (which ranges from 793 to 2482 MPa) ensures that it can withstand significant forces without breaking, while its yield strength (from 241 to 586 MPa) indicates how much stress it can handle before deforming.

- Benefits include:
- High durability and strength, making it suitable for critical applications.
 - Versatile use in medical implants, valves, pumps, and general engineering projects.
 - Enhanced performance in harsh environments due to its robust composition.

ASTM SB-407 N08801

Category: Nickel Alloys

Standard Specification for Nickel-Iron-Chromium Alloy Seamless Pipe and Tube.

Technical Specifications:

Tensile Strength in Mpa	450 min.
Yield Strength in Mpa	170 min.

- Use Cases:
- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM SB-407 N08801 is a type of seamless pipe and tube made from a special nickel-iron-chromium alloy. This material is designed to be strong and durable, making it suitable for various applications, including valves, pumps, and automotive components. It's particularly valuable in industries that require reliable performance under stress and extreme conditions.

Key benefits of ASTM SB-407 N08801 include:

- **High Strength**: With a minimum tensile strength of 450 MPa and yield strength of 170 MPa, it can withstand significant pressure and force.
- **Versatile Applications**: Ideal for use in general engineering, automotive applications, and any situation that requires robust piping solutions.

ASTM A275 SC84B

Category: Aluminium Alloys

Standard Practice for Codification of Certain Nonferrous Metals and Alloys, Cast and Wrought.

Technical Specifications:

Tensile Strength in Mpa	315 min.
Yield Strength in Mpa	160 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM A275 SC84B is a type of aluminium alloy that follows a standard practice for categorizing different nonferrous metals and alloys, both cast and wrought. This alloy is designed to be strong and reliable, making it suitable for a variety of uses across multiple industries. It has a minimum tensile strength of 315 MPa and a minimum yield strength of 160 MPa, ensuring that it can withstand significant stress without deforming.

Here are some key points about ASTM A275 SC84B:

- **Versatile Applications**: It is commonly used in valves, pumps, and general engineering projects, as well as in the automotive sector.
- **Strong and Durable**: The alloy's high strength ratings mean it can be relied upon for demanding applications where performance is critical.
- **Standardized Quality**: Its codification according to established standards ensures consistency and quality across different manufacturing processes.

ASTM B 26 712.0

Category: Aluminium Alloys

Standard Specification for Aluminium Alloy Sand Casting.

Technical Specifications:

Tensile Strength in Mpa	235 min.
Yield Strength in Mpa	170 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 26 712.0 is a standard specification that outlines the requirements for aluminium alloy sand casting. This means it provides guidelines on how to create cast parts using a specific type of aluminium alloy, ensuring that these parts meet certain quality and performance standards. The aluminium used in these casts is strong and durable, making it suitable for a variety of applications.

Some benefits of using ASTM B 26 712.0 include:

- **Strength and Durability**: The aluminium alloy has a minimum tensile strength of 235 MPa and a yield strength of 170 MPa, which means it can withstand significant stress without breaking.
- **Versatile Applications**: This standard is used in various industries, including automotive, general engineering, and for making parts like valves and pumps, ensuring that the products are reliable and efficient.

ASTM B 145 - 5B

Category: Copper Alloys

Standard Specification for Copper Alloy Sand Casting for General Applications.

Technical Specifications:

Tensile Strength in Mpa	193 min.
Yield Strength in Mpa	83 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 145 - 5B is a type of copper alloy that is specifically designed for sand casting, which is a method used to create metal parts by pouring liquid metal into a mold. This particular alloy is used in various general applications, making it versatile for different industries. It is especially valued for its strength and durability, which means products made from this alloy are reliable and long-lasting.

Key Benefits:

- **High Strength**: With a minimum tensile strength of 193 MPa and a yield strength of 83 MPa, this alloy can withstand significant stress without breaking.
- **Versatile Applications**: It is suitable for use in valves, pumps, general engineering projects, and automotive components, making it a practical choice for manufacturers looking for a dependable material.

ASTM B 505 C93400

Category: Copper Alloys

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tensile Strength in Mpa	234 min.
Yield Strength in Mpa	138 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 505 C93400 is a type of copper alloy specifically designed for continuous casting processes. This means it can be manufactured in a steady stream, ensuring consistent quality and performance. The alloy is particularly strong, with a minimum tensile strength of 234 MPa and a yield strength of 138 MPa, making it suitable for various demanding applications.

Key benefits of ASTM B 505 C93400 include:

- **Versatility**: It's used in a variety of industries, such as valves, pumps, general engineering, and automotive, making it a reliable choice for many different products.
- **Strength and Durability**: The high tensile and yield strength ensure that products made from this alloy can withstand significant stress and wear, extending their lifespan and reducing maintenance costs.

ASTM B 108 308.0

Category: Aluminium Alloys

Standard Specification for Aluminum - Alloy Permanent Mold Castings.

Technical Specifications:

Tensile Strength in Mpa	165 min.
Hardness in HB	70 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 108 308.0 refers to a specific type of aluminum alloy that is used for making permanent mold castings. This means that it is designed to be poured into molds that can be reused multiple times, making it a cost-effective option for producing parts. The aluminum alloy has strong qualities, with a minimum tensile strength of 165 megapascals and a hardness rating of at least 70 on the Brinell scale. These properties ensure that the parts made from this alloy are durable and can withstand significant stress and wear.

This aluminum alloy is versatile and can be used in various applications, including:

- Manufacturing valves and pumps that require reliable performance.
- General engineering projects where strength and durability are crucial.
- The automotive industry for producing lightweight yet strong components.

By using ASTM B 108 308.0, businesses can benefit from a material that combines strength, durability, and the efficiency of reusable molds, ultimately leading to better quality products and reduced manufacturing costs.

ASTM B 145 - 5A

Category: Copper Alloys

Standard Specification for Copper Alloy Sand Casting for General Applications.

Technical Specifications:

Tensile Strength in Mpa	200 min.
Yield Strength in Mpa	90 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 145 - 5A is a standard specification for a type of copper alloy used in sand casting, which is a method for creating metal parts. This alloy is designed for general applications, making it versatile and useful in various industries. It is particularly known for its strength, with a minimum tensile strength of 200 MPa and a yield strength of at least 90 MPa, ensuring that it can withstand significant stress without breaking.

ASTM B 147 - 8A

Category: Copper Alloys

Standard Specification for Copper Alloy Sand Casting for General Applications.

Technical Specifications:

Tensile Strength in Mpa	448 min.
Yield Strength in Mpa	172 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 147 - 8A is a standard specification for a type of copper alloy that is used in sand casting processes for various general applications. Essentially, this means it is a guideline that ensures the quality and performance of copper alloy parts created through casting, which is a method of shaping metal by pouring it into a mold. This specific copper alloy is known for its strength and durability, making it suitable for a range of industries.

Key benefits of using ASTM B 147 - 8A include:

- **High Strength**: With a minimum tensile strength of 448 MPa and a yield strength of 172 MPa, this alloy is robust and capable of withstanding significant stress.
- **Versatile Applications**: It is ideal for manufacturing components in various sectors such as valves, pumps, and automotive parts, as well as in general engineering projects.

ASTM B 147 - 8B

Category: Copper Alloys

Standard Specification for Copper Alloy Sand Casting for General Applications.

Technical Specifications:

Tensile Strength in Mpa	621 min.
Yield Strength in Mpa	310 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 147 - 8B is a standard specification for a type of copper alloy that is used in sand casting, which is a method of shaping metal by pouring molten metal into a mold made of sand. This specific copper alloy is designed for general applications and is known for its strong physical properties, making it suitable for a variety of uses.

Key benefits of ASTM B 147 - 8B include:

- **High Strength**: It has a minimum tensile strength of 621 MPa and a yield strength of 310 MPa, meaning it can withstand significant stress before deforming or failing.
- **Versatile Applications**: This alloy is ideal for making components in valves, pumps, general engineering, and automotive parts, ensuring durability and reliability in demanding environments.

ASTM B 148 C95200

Category: Copper Alloys

Standard Specification for Aluminum - Bronze Sand Casting.

Technical Specifications:

Tensile Strength in Mpa	450 min.
Yield Strength in Mpa	170 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 148 C95200 is a type of copper alloy known as aluminum-bronze, specifically designed for sand casting. This material is renowned for its strength and durability, making it an excellent choice for a variety of industrial applications. It has a minimum tensile strength of 450 MPa and a yield strength of at least 170 MPa, ensuring that products made from this alloy can withstand significant stress and pressure without deforming.

Key benefits of ASTM B 148 C95200 include:

- **Versatility**: Ideal for manufacturing components like valves, pumps, and parts used in general engineering and automotive industries.

- **Strength and Durability**: Its high tensile and yield strength make it suitable for demanding applications, ensuring long-lasting performance in challenging environments.

ASTM B 148 C95300

Category: Copper Alloys

Standard Specification for Aluminum - Bronze Sand Casting.

Technical Specifications:

Tensile Strength in Mpa	450 - 550
Yield Strength in Mpa	170 - 275

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 148 C95300 is a type of copper alloy specifically designed for aluminum-bronze sand casting. This material is known for its durability and strength, making it an excellent choice for various applications. It has a tensile strength of 450 - 550 MPa and a yield strength of 170 - 275 MPa, which means it can withstand significant stress and pressure without breaking or deforming.

Key Benefits:

- **Versatile Use**: Ideal for manufacturing valves, pumps, and components in the automotive and general engineering sectors.
- **Strong and Durable**: High tensile and yield strength ensure longevity and reliability in demanding environments.

ASTM B 148 C95410

Category: Copper Alloys

Standard Specification for Aluminum - Bronze Sand Casting.

Technical Specifications:

Tensile Strength in Mpa	515 - 620
Yield Strength in Mpa	205 - 310

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 148 C95410 is a type of copper alloy known for its strength and durability, specifically designed for aluminum-bronze sand casting. This material is widely used in various industries due to its excellent mechanical properties, making it ideal for applications where reliable performance is essential. It can withstand high stress and has great resistance to wear, making it a popular choice for demanding environments.

Key benefits include:

- **High Strength**: With a tensile strength ranging from 515 to 620 MPa, it can handle significant loads without breaking.
- **Versatile Applications**: Suitable for manufacturing parts like valves and pumps, as well as in general engineering and automotive industries, ensuring it can meet various needs.

ASTM B 148 C95400

Category: Copper Alloys

Standard Specification for Aluminum - Bronze Sand Casting.

Technical Specifications:

Tensile Strength in Mpa	515 - 620
Yield Strength in Mpa	205 - 310

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B 148 C95400 is a type of copper alloy known as aluminum bronze, specifically designed for sand casting. This material is strong and durable, making it suitable for various industrial applications. It has impressive tensile strength, which means it can withstand significant pulling forces without breaking, and good yield strength, allowing it to maintain its shape under pressure.

Key benefits of ASTM B 148 C95400 include:

- **Versatility**: Ideal for use in valves, pumps, automotive parts, and general engineering projects.
- **High Strength**: Offers high tensile and yield strength, ensuring reliability in demanding environments.
- **Durability**: Resistant to wear and corrosion, making it a long-lasting choice for both commercial and industrial uses.

ASTM B30 C92300

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.200 max.
Copper	85.000 - 89.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C92300 is a high-quality copper alloy commonly used in various industries. It is designed for producing ingots and castings, which are essential materials in manufacturing components like valves, pumps, and automotive parts. This alloy is made primarily of copper, with a small amount of aluminum and iron to enhance its properties, making it suitable for engineering applications where durability and reliability are crucial.

- Key benefits of ASTM B30 C92300 include:
- **Versatility**: Ideal for use in different sectors such as general engineering and automotive.
 - **Durability**: The specific composition ensures that products made from this alloy can withstand wear and tear, enhancing their lifespan.
 - **Reliability**: Its standardized specification guarantees consistent quality, which is essential for critical components in machinery and vehicles.

ASTM B30 C92700

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.150 max.
Copper	86.000 - 89.000

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM B30 C92700 is a type of copper alloy that is specifically designed for creating ingots and castings. This material is made up of a high percentage of copper, which gives it excellent strength and durability. Additionally, it contains very small amounts of aluminum and iron, which help to enhance its overall performance in various applications.

- This copper alloy is particularly useful in industries such as automotive, general engineering, and machinery. Its versatility allows it to be used in a range of products including valves and pumps. Benefits of using ASTM B30 C92700 include:
- High strength and durability due to its copper content.
 - Good resistance to corrosion, making it suitable for various environments.
 - Versatile applications across multiple industries.

ASTM B30 C92800

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.150 max.
Copper	78.000 - 82.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C92800 is a type of copper alloy used mainly for making high-quality metal parts. This material is specifically designed for casting and comes in the form of ingots, which are blocks of metal that can be melted down and shaped into various products. Its composition includes a high percentage of copper, along with very small amounts of aluminum and iron, making it strong and durable for various applications.

This copper alloy is beneficial for industries such as automotive, general engineering, and pump manufacturing. The key advantages of using ASTM B30 C92800 include:

- **Strength and Durability**: The high copper content ensures that the final products are robust and long-lasting.
- **Versatility**: It can be used in a variety of applications, from valves to pumps, making it a flexible choice for manufacturers.

ASTM B30 C93200

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.200 max.
Copper	82.000 - 84.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93200 is a type of copper alloy that is primarily used for making ingots and castings. This means it's a metal material that can be shaped and molded into various parts that are essential in many industries. The alloy contains a high percentage of copper (between 82% and 84%) along with very small amounts of aluminum and iron, making it strong and durable for various applications.

This copper alloy is commonly used in products like valves and pumps, as well as in general engineering and automotive applications. Some benefits of using ASTM B30 C93200 include:

- **Durability**: The high copper content provides excellent resistance to wear and tear.
- **Versatility**: It can be used in different industries, making it suitable for various engineering needs.
- **Reliability**: The precise specifications ensure consistent quality in the products made from this alloy.

ASTM B30 C92900

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.150 max.
Copper	82.000 - 86.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C92900 is a type of copper alloy that is specifically designed for making ingots and castings. This material is composed mainly of copper, with small amounts of aluminum and iron. Its composition allows it to be strong and durable, making it suitable for various applications. The key benefit of using this alloy is its excellent corrosion resistance and high mechanical strength, which makes it ideal for demanding environments.

This copper alloy is commonly used in industries like automotive, general engineering, and for manufacturing components such as valves and pumps. Here are some benefits of ASTM B30 C92900:

- High strength and durability for long-lasting use
- Good resistance to corrosion, enhancing product lifespan
- Versatile applications in various industries, making it a reliable choice for engineers.

ASTM B30 C93400

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.200 max.
Copper	82.000 - 85.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93400 is a high-quality copper alloy that is mainly used for making ingots and castings. This material is known for its excellent strength and durability, making it ideal for various industrial applications. The alloy contains a specific blend of copper, aluminum, and iron, ensuring its reliability and performance in demanding environments.

Key benefits of ASTM B30 C93400 include:

- ****Versatility****: Suitable for a wide range of applications such as valves, pumps, and automotive components.
- ****Durability****: Offers high resistance to wear and corrosion, extending the lifespan of the products made from it.
- ****Consistent Quality****: Adheres to strict specifications, ensuring that every batch maintains the same high

standards for strength and performance.

ASTM B30 C93600

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.150 max.
Copper	79.000 - 82.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93600 is a type of copper alloy used mainly in manufacturing ingots and castings. This material is specifically designed to meet industry standards and is known for its quality and reliability. It contains a specific composition of copper, aluminum, and iron, ensuring that it performs well in various applications.

This copper alloy is particularly beneficial for use in critical components such as valves and pumps, and it is also widely utilized in general engineering and automotive industries. Key benefits include:

- High durability and resistance to wear and corrosion.
- Versatility for a range of engineering applications.
- Compliance with industry standards for quality and performance.

ASTM B30 C93800

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.100 max.
Copper	76.000 - 79.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93800 is a specific type of copper alloy used primarily for making metal products like valves and pumps. This standard specification ensures that the alloy meets certain quality criteria for ingots and castings, making it reliable for various engineering applications. Essentially, it combines copper with small amounts of aluminum and iron to create a strong and durable material that can withstand different environments.

****Benefits of ASTM B30 C93800:****

- ****Versatile Use**:** Suitable for a wide range of applications including automotive and general engineering.
- ****Quality Assurance**:** Adheres to strict standards, ensuring high performance and longevity of the products made from it.
- ****Durability**:** The specific composition of copper, aluminum, and iron enhances the strength and resistance to wear and corrosion.

ASTM B30 C93500

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.100 max.
Copper	83.000 - 85.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93500 is a type of copper alloy that is commonly used for making ingots and castings, which are essential materials in various industries. This specification ensures that the copper alloy meets certain quality standards, enabling it to perform effectively in different applications. The composition of this alloy includes a high percentage of copper, specifically between 83% and 85%, along with very low amounts of aluminium and iron, ensuring strength and durability.

Key benefits of ASTM B30 C93500 include:

- **Versatility:** It is suitable for a variety of applications such as valves, pumps, general engineering projects, and automotive parts.
- **Reliability:** The stringent specifications help in maintaining consistent quality, making it a trusted choice for engineers and manufacturers.
- **Performance:** Its high copper content provides excellent conductivity and corrosion resistance, which is vital for many mechanical and electrical applications.

ASTM B30 C93700

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.100 max.
Copper	78.000 - 81.000

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93700 is a high-quality copper alloy used primarily in various engineering applications. This material is designed for making ingots and castings, which are essential for creating parts like valves and pumps. It contains a specific range of copper, along with minimal amounts of aluminum and iron, ensuring durability and performance in demanding environments.

- **Key Benefits:****
- ****Versatile Use:**** Ideal for a range of applications including automotive and general engineering.
 - ****Durable Composition:**** The precise balance of copper and other metals leads to strong, reliable components that can withstand wear and tear.
 - ****Standard Specification:**** Meets industry standards, ensuring quality and consistency in manufacturing processes.

ASTM B30 C93900

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.350 max.
Copper	76.500 - 79.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C93900 is a type of copper alloy used primarily for manufacturing ingots and castings. This specification ensures that the copper used has specific qualities, making it suitable for various applications. The alloy contains a high percentage of copper (between 76.5% and 79.5%) with very low amounts of aluminum and iron.

- This copper alloy is highly versatile and is commonly used in industries such as automotive, pump manufacturing, and general engineering. Its benefits include:
- ****Durability**:** The specific composition of the alloy provides strength and longevity.
 - ****Versatility**:** Suitable for a range of applications including valves and pumps, making it a go-to material for engineers.
 - ****Quality Assurance**:** Compliance with ASTM standards ensures reliable and consistent performance.

ASTM B30 C94000

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.250 max.
Copper	69.000 - 72.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C94000 is a type of copper alloy that is specifically designed for making ingots and castings. This material is known for its durability and excellent performance in various applications. It contains a high percentage of copper, which gives it strong conductive properties, making it suitable for many industrial uses. The composition of this alloy is carefully controlled, with minimal amounts of aluminum and iron, ensuring that it meets specific standards for quality and reliability.

Key benefits include:

- **Versatile Applications**: Ideal for use in valves, pumps, general engineering projects, and automotive components.
- **High Performance**: Offers excellent conductivity and mechanical strength, making it a preferred choice in various manufacturing processes.

ASTM B30 C94300

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.100 max.
Copper	69.000 - 72.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C94300 is a type of copper alloy specifically designed for use in manufacturing ingots and castings. It is composed mainly of copper, with very small amounts of aluminum and iron. This alloy is known for its strength and durability, making it suitable for various industrial applications.

Key benefits include:

- **Versatility**: Used in a variety of industries such as valves, pumps, general engineering, and automotive, making it a reliable choice for many manufacturing needs.
- **Quality Assurance**: The alloy composition ensures high performance and longevity in products, which can lead to reduced maintenance costs over time.

ASTM B30 C94100

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.100 max.
Copper	74.000 - 79.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C94100 is a type of copper alloy that is used to create ingots and castings, which are essential materials for various industries. This alloy primarily consists of copper, along with very small amounts of aluminum and iron, making it strong and durable for different applications. The copper content ranges between 74% to 79%, which is key to its performance in various products.

This material is particularly beneficial for manufacturing components like valves and pumps, as well as in general engineering and automotive sectors. Its resistance to corrosion and excellent conductivity make it a reliable choice for engineers and manufacturers looking for high-quality materials.

Key Benefits:

- Strong and durable for long-lasting applications.
- Excellent corrosion resistance.
- Versatile use in multiple industries including automotive and engineering.

ASTM B30 C94700

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.200 max.
Copper	86.000 - 89.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C94700 is a type of copper alloy that is primarily used to create ingots and castings. This material is valuable in various industries due to its robust properties and versatility. It consists mainly of copper (between 86% and 89%) and has very low limits on aluminum and iron content, ensuring high quality and performance.

Key benefits of ASTM B30 C94700 include:

- **Durability**: Its composition makes it strong and resistant to wear, making it ideal for long-lasting applications.
- **Versatility**: This copper alloy can be used in a wide range of products, including valves, pumps, and automotive components, as well as in general engineering projects.
- **Quality Assurance**: The specifications ensure that the material meets industry standards, providing reliability in performance for various engineering needs.

ASTM B30 C94800

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.200 max.
Copper	85.000 - 89.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C94800 is a type of copper alloy that is specifically designed for making ingots and castings. This means it can be melted down and molded into various shapes, making it versatile for different industrial applications. The alloy consists mainly of copper, with small amounts of aluminum and iron to enhance its properties.

Key benefits of ASTM B30 C94800 include:

- **Durability**: Its composition gives it strength, making it suitable for demanding environments.
- **Versatility**: This alloy can be used in a variety of applications such as valves, pumps, and general engineering projects, as well as in the automotive industry.
- **Standardized Quality**: Being a standard specification ensures consistent quality in production, which is crucial for manufacturers looking for reliable materials.

ASTM B30 C95410

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	10.000 - 11.500
Iron	3.000 - 5.000
Copper	83.000 min.

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C95410 is a type of copper alloy that is specifically designed for making high-quality metal parts. This alloy includes a mix of copper, aluminum, and iron, which gives it unique properties that make it suitable for various applications. It has a minimum copper content of 83%, with aluminum and iron added to enhance its strength and durability.

This material is commonly used in industries such as automotive, general engineering, and for manufacturing valves and pumps. The benefits of using ASTM B30 C95410 include:

- High strength and durability, making it suitable for demanding applications.
- Excellent corrosion resistance, which helps extend the lifespan of components made from this alloy.
- Versatility, as it can be used in a wide range of industries and products.

ASTM B30 C95500

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	10.000 - 11.500
Iron	3.000 - 5.000
Copper	78.000 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C95500 is a type of copper alloy used primarily in manufacturing ingots and castings. This material is known for its strong properties and is mainly composed of copper, aluminum, and iron, making it suitable for various industrial applications. It contains at least 78% copper, with aluminum making up about 10% to 11.5% and iron ranging from 3% to 5%.

The benefits of using ASTM B30 C95500 include:

- ****Durability****: Its robust composition allows it to withstand significant wear and tear, making it ideal for products like valves and pumps.
- ****Versatility****: This alloy is used in a variety of sectors, including general engineering and automotive industries, ensuring it meets diverse manufacturing needs.

ASTM B30 C95520

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Aluminium	10.500 - 11.500
-----------	-----------------

Iron	4.000 - 5.500
Copper	74.500 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C95520 is a specific type of copper alloy that is used to make ingots and castings, which are essential materials in various industries. This alloy contains a significant amount of copper, along with aluminium and iron, which gives it strength and durability. It's designed to meet industry standards, ensuring that it performs well in demanding applications.

Key benefits of using ASTM B30 C95520 include:

- **High Strength**: With a minimum copper content of 74.5%, this alloy is robust and reliable.
- **Versatile Applications**: It's suitable for various uses, including valves, pumps, general engineering projects, and automotive parts, making it a versatile choice for manufacturers.
- **Standardized Quality**: Being a standard specification, it guarantees a consistent quality that meets industry requirements.

ASTM B30 C96400

Category: Copper Alloys

Standard Specification for Copper Alloys Ingot and Casting.

Technical Specifications:

Carbon	0.050 max.
Iron	0.250 - 1.000
Copper	65.000 - 67.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B30 C96400 is a type of copper alloy that is specifically designed for making ingots and castings. This means it's a material that can be melted down and poured into molds to create various shapes and components. It is primarily composed of copper, along with small amounts of carbon and iron, which enhance its properties. This alloy is known for its durability and resistance to wear, making it suitable for a variety of applications.

Key Benefits:

- **Versatile Use**: It is commonly used in industries such as automotive, valve manufacturing, and general engineering.
- **Strong and Reliable**: The specific composition of copper, carbon, and iron ensures that the materials produced are strong and can withstand significant stress and usage.
- **Easy to Work With**: Being a casting alloy, it is easy to mold and shape, making it ideal for creating customized components.

ASTM B369 C96400

Category: Copper Alloys

Standard Specification for Copper - Nickel Alloy Casting.

Technical Specifications:

Carbon	0.150 max.
Iron	0.250 - 1.500
Copper	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B369 C96400 is a type of copper alloy specifically designed for casting, which means it's used to create various metal parts by pouring molten metal into molds. This alloy is a mix mainly made of copper and nickel, which gives it unique properties suitable for different applications. It's known for its strength and resistance to corrosion, making it an excellent choice for industries that require durable materials.

Some key benefits of ASTM B369 C96400 include:

- **Versatility**: It's used in a range of applications such as valves, pumps, and automotive parts, making it a go-to material in general engineering.
- **Durability**: The alloy's composition helps it withstand harsh conditions, ensuring long-lasting performance in various environments.

ASTM B427 C90700

Category: Copper Alloys

Standard Specification for Gear Bronze Alloy Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.150 max.
Copper	88.000 - 90.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B427 C90700 is a type of gear bronze alloy specifically designed for casting. This material is primarily made of copper, with a composition of 88% to 90%, making it strong and durable. It also contains very low amounts of aluminum and iron, which helps enhance its performance in various applications. The key advantage of using this alloy is its excellent resistance to wear and corrosion, which makes it ideal for parts that experience

high friction and pressure.

This alloy is commonly used in several industries, including:

- **Valves**: Ensuring reliable performance in fluid control systems.
- **Pumps**: Providing durability in handling liquids.
- **General Engineering**: Suitable for various mechanical applications requiring robust materials.
- **Automotive**: Used in components that need to withstand tough conditions while maintaining efficiency.

ASTM B427 C92900

Category: Copper Alloys

Standard Specification for Gear Bronze Alloy Casting.

Technical Specifications:

Aluminium	0.005 max.
Iron	0.200 max.
Copper	82.000 - 86.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B427 C92900 is a type of copper alloy specifically designed for making gear bronze castings. This material is primarily used in various industries such as automotive and engineering due to its excellent strength and durability. It is composed mainly of copper, with small amounts of aluminum and iron, which help enhance its performance in demanding applications.

Key Benefits:

- **Versatile Applications**: Ideal for use in valves, pumps, and general engineering projects.
- **Durability**: Offers strong resistance to wear and tear, making it suitable for high-performance environments.
- **Standardized Quality**: Follows a standard specification, ensuring consistent quality and reliability in manufacturing.

Seamless Steel Tubes for Pressure Purposes

Category: Valves

High-strength seamless tubes designed for pressure applications.

Key Features:

- Excellent tensile strength
- Corrosion resistance
- High durability

Technical Specifications:

Material	Ferrous Mild Steel Alloys
Tensile Strength	510 - 850 MPa
Yield Strength	295 MPa min

Use Cases:

- Valve manufacturing
- Pump components
- General engineering

Export Markets:

- European Union
- USA
- Japan

AI Layman Summary:

Seamless Steel Tubes for Pressure Purposes are strong and durable tubes made specifically for use in high-pressure environments. These tubes are crafted from ferrous mild steel alloys, offering excellent resistance to corrosion and the ability to withstand significant stress without breaking. They are ideal for various applications, including valve manufacturing, pump components, and general engineering projects.

****Benefits:****

- ****High Strength:**** With a tensile strength ranging from 510 to 850 MPa, these tubes can handle heavy pressures.
- ****Corrosion Resistance:**** They are built to resist rust and wear, ensuring a longer lifespan in harsh environments.
- ****Versatile Applications:**** Suitable for multiple industries, making them a reliable choice for manufacturers in the European Union, USA, and Japan.

Flat Products Made of Steels for Pressure Purpose

Category: Valves

Flat steel products designed for pressure applications.

Key Features:

- High strength
- Versatile applications
- Cost-effective

Technical Specifications:

Material	Ferrous Mild Steel Alloys
Tensile Strength	450 - 630 MPa
Yield Strength	250 MPa min

Use Cases:

- Valve and pump manufacturing
- Automotive industry
- General engineering

Export Markets:

- Germany
- France
- Italy

AI Layman Summary:

Flat Products Made of Steels for Pressure Purpose are specially designed flat steel materials that can withstand high pressure. These products are crucial in manufacturing valves and pumps, commonly used in various

industries, including automotive and general engineering. Their robust construction ensures they can handle demanding applications while remaining cost-effective for businesses.

- **Key Benefits:****
- ****High Strength:**** These products are made from strong ferrous mild steel alloys, providing durability and reliability.
 - ****Versatile Applications:**** Suitable for a wide range of uses in different sectors, making them a flexible choice for manufacturers.
 - ****Cost-Effective:**** They offer a high performance-to-cost ratio, helping businesses save money while maintaining quality.

Seamless Steel Tubes for Pressure Purposes (BS 11 CrMo 9-10)

Category: Valves

Seamless tubes specifically designed for high-pressure applications.

Key Features:

- High corrosion resistance
- Excellent mechanical properties
- Suitable for high-temperature applications

Technical Specifications:

Material	Ferrous Mild Steel Alloys
Tensile Strength	450 - 680 MPa
Yield Strength	265 MPa min

Use Cases:

- Oil and gas pipelines
- Power generation
- Chemical processing

Export Markets:

- USA
- Canada
- Australia

AI Layman Summary:

Seamless Steel Tubes for Pressure Purposes (BS 11 CrMo 9-10) are specially crafted tubes designed to handle high-pressure situations, making them ideal for various industries. These tubes are made from ferrous mild steel alloys, providing them with exceptional strength and durability. They can withstand high temperatures and are resistant to corrosion, ensuring long-lasting performance even in tough environments.

- **Key Benefits:****
- ****High Corrosion Resistance:**** Protects the tubes from rust and degradation, extending their lifespan.
 - ****Excellent Mechanical Properties:**** Offers strong support in demanding applications, ensuring safety and reliability.
 - ****Versatile Applications:**** Suitable for use in oil and gas pipelines, power generation facilities, and chemical processing plants, making them a valuable asset across different sectors.

These tubes are exported to markets like the USA, Canada, and Australia, meeting the needs of global industries requiring robust and reliable piping solutions.

Castings for Pressure Purposes

Category: Valves

Castings designed for high-pressure applications.

Key Features:

- High strength
- Durability
- Versatile applications

Technical Specifications:

Material	Ferrous Mild Steel Alloys
Tensile Strength	480 MPa min
Yield Strength	241 MPa min

Use Cases:

- Valve and pump manufacturing
- General engineering
- Automotive industry

Export Markets:

- Europe
- Asia
- North America

AI Layman Summary:

Castings for Pressure Purposes are specialized parts made from strong steel materials, designed to handle high-pressure situations, making them ideal for various industrial uses. These castings are crucial in the manufacturing of valves and pumps, which are essential components in many machines and systems. With their high tensile strength and durability, they can withstand significant stress and strain, ensuring reliable performance over time.

Key benefits include:

- **High Strength**: Capable of withstanding high pressure without failure.
- **Durability**: Built to last, reducing the need for frequent replacements.
- **Versatile Applications**: Suitable for various industries, including automotive and general engineering, making them a flexible choice for manufacturers in Europe, Asia, and North America.

ASTM B505 Leaded Nickel Bronze UNS C97300

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Sn % 1.500 - 3.000, Pb % 8.000 - 11.000, Zn % 17.000 - 25.000
- Mechanical Properties: Tensile Strength in Mpa 207 min, Yield Strength in Mpa 103 min

Technical Specifications:

Tin	1.500 - 3.000
Lead	8.000 - 11.000
Zinc	17.000 - 25.000

Copper	53.000 - 58.000
--------	-----------------

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Nickel Bronze UNS C97300 is a specialized metal alloy primarily used in various engineering applications. This material is created by combining copper with other elements like tin, lead, and zinc, which enhances its performance in demanding environments. It is particularly valued for its strength and durability, making it suitable for parts that need to withstand pressure and wear, such as valves and pumps.

Key benefits of using this alloy include:

- **High Strength:** With a minimum tensile strength of 207 MPa, it can handle significant stress without breaking.
- **Corrosion Resistance:** The mixture of metals provides excellent resistance to corrosion, ensuring longevity in harsh conditions.
- **Versatile Applications:** Ideal for a wide range of uses in industries like automotive and general engineering, ensuring reliability in critical components.

ASTM B505 Alumium Bronze UNS C95900

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Ni + Cu % 0.500 max, Fe % 3.000 - 5.000
- Mechanical Properties: Hardness in HB 241 min

Technical Specifications:

Aluminium	12.000 - 13.500
Copper	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Aluminum Bronze UNS C95900 is a special type of metal made primarily from copper and aluminum, designed for various industrial applications. It is known for its strong and durable properties, making it ideal for use in valves, pumps, and automotive components. This metal is produced using a continuous casting process, which ensures consistency and high quality in its structure.

Key benefits of ASTM B505 Aluminum Bronze UNS C95900 include:

- **Strength and Hardness:** It has a minimum hardness rating of 241 HB, ensuring it can withstand significant wear and tear.
- **Corrosion Resistance:** The unique alloy composition, including copper and aluminum, provides excellent resistance to corrosion, making it suitable for harsh environments.

- **Versatile Applications**: Its reliability makes it perfect for various industries, including general engineering and automotive manufacturing.

ASTM B472 N06230

Category: Materials

Standard Specification for Nickel and Nickel-Related Alloys Billets and Bars for Reforging.

Key Features:

- Chemical Composition: C % 0.005 - 0.150, Cr % 20.000 - 24.000
- Mechanical Properties: Tensile Strength in Mpa 760 min, Yield Strength in Mpa 310 min

Technical Specifications:

Nickel	Balance
Iron	3.000 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B472 N06230 is a high-quality material made primarily from nickel and nickel-related alloys. It is specifically designed for creating billets and bars that can be reformed into various shapes and sizes. This material is ideal for industries such as automotive, general engineering, and manufacturing components like valves and pumps.

Key benefits include:

- **Durability**: With a minimum tensile strength of 760 MPa and a yield strength of 310 MPa, it can withstand heavy loads and stresses.
- **Versatility**: Its chemical composition allows it to be used in a wide range of applications, making it a go-to choice for engineers and manufacturers looking for reliable materials.

ASTM B505 Manganese Nickel Alumium Bronze UNS C95700

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Pb % 0.030 max, Cu % 71.000 min
- Mechanical Properties: Tensile Strength in Mpa 620 min

Technical Specifications:

Manganese	11.000 - 14.000
Aluminium	7.000 - 8.000

Use Cases:

- Valve
- Pump
- General Engineering

- Automotive

AI Layman Summary:

ASTM B505 Manganese Nickel Aluminum Bronze UNS C95700 is a type of metal alloy known for its strength and durability. It is primarily made of copper, manganese, and aluminum, making it suitable for a variety of applications. This material is designed for continuous casting processes, which means it can be formed into shapes and parts efficiently and consistently. Some key benefits include its resistance to corrosion, making it ideal for use in harsh environments, and its ability to withstand high levels of stress, which is important for components that must endure heavy loads.

Key Features:

- Strong and durable, with a minimum tensile strength of 620 Mpa.
- Composed of specific percentages of manganese (11-14%) and aluminum (7-8%), providing enhanced performance.

Common Uses:

- Ideal for manufacturing valves and pumps, as well as in general engineering and automotive applications.

ASTM B505 Leaded Nickel Bronze UNS C97600

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Sn % 3.500 - 4.500, Pb % 3.000 - 5.000
- Mechanical Properties: Tensile Strength in Mpa 276 min

Technical Specifications:

Copper	63.000 - 67.000
Yield Strength	138 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Nickel Bronze UNS C97600 is a special type of metal used in various industrial applications. This material is primarily made from copper, with added elements like tin and lead that enhance its properties. It's designed for continuous casting, meaning it can be produced in long, uniform shapes that are ideal for manufacturing. This bronze alloy is known for its strength and durability, making it a reliable choice for many products.

****Key Benefits:****

- ****Strong and Durable:**** With a minimum tensile strength of 276 MPa, it can withstand heavy loads and stress, making it suitable for demanding applications.
- ****Versatile Applications:**** Commonly used in valves, pumps, and automotive parts, it supports diverse engineering needs.
- ****Easy to Work With:**** The continuous casting process allows for efficient manufacturing, saving time and resources in production.

ASTM B505 Leaded Nickel Bronze UNS C97800

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Sn % 4.000 - 5.500, Pb % 1.000 - 2.500
- Mechanical Properties: Tensile Strength in Mpa 310 min

Technical Specifications:

Copper	64.000 - 67.000
Yield Strength	152 min

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Nickel Bronze UNS C97800 is a specialized metal alloy primarily used in various engineering applications. This alloy is composed of copper, tin, and lead, which gives it unique properties that make it strong and durable. It is specifically designed for continuous casting, meaning it can be produced in long sections, which is ideal for creating parts like valves and pumps.

Key benefits of this material include:

- **High Strength**: It has a minimum tensile strength of 310 MPa, making it suitable for demanding applications.
- **Versatility**: It is used in a wide range of industries, including automotive and general engineering, thanks to its reliable performance under pressure.
- **Durability**: The alloy's composition enhances resistance to wear and corrosion, ensuring longevity in various environments.

ASTM B505 Nickel-Alumium Bronze UNS C95800

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Pb % 0.030 max, Ni + Cu % 4.000 - 5.000
- Mechanical Properties: Tensile Strength in Mpa 586 min

Technical Specifications:

Copper	79.000 min
Iron	3.500 - 4.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Nickel-Aluminum Bronze UNS C95800 is a special type of metal that is mainly used in making various components like valves and pumps. It is made from a mixture of copper, nickel, and aluminum, which gives it strong properties. This metal is particularly known for its durability and resistance to corrosion, making it a reliable choice for engineers and manufacturers in different industries, including automotive and general engineering.

- **Key Benefits:****
- ****Strong and Durable:**** With a minimum tensile strength of 586 MPa, this material can withstand high pressure and stress without breaking.
 - ****Corrosion Resistant:**** The unique chemical composition helps it resist damage from water and other elements, prolonging the life of the products made from it.
 - ****Versatile Applications:**** Whether you're working on pumps, valves, or other engineering projects, this metal is a great option to consider.

ASTM B505 Special Alloy UNS C99500

Category: Materials

Standard Specification for Copper Alloy Continuous Castings.

Key Features:

- Chemical Composition: Pb % 0.250 max, Zn % 0.500 - 2.000
- Mechanical Properties: Tensile Strength in Mpa 483 min

Technical Specifications:

Copper	Balance
Iron	3.000 - 5.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Special Alloy UNS C99500 is a type of high-quality copper alloy that is produced through a continuous casting process. This means that it is made in a way that allows for a consistent shape and size, making it ideal for various industrial applications. The alloy contains a specific mix of metals, including a maximum of 0.25% lead and between 0.5% to 2% zinc, which enhances its durability and strength. It has a minimum tensile strength of 483 MPa, ensuring it can withstand significant stress without breaking.

This material is commonly used in a range of applications such as valves, pumps, and automotive parts. Its unique properties make it suitable for general engineering purposes as well. Benefits of using ASTM B505 include:

- High resistance to wear and corrosion, extending the lifespan of components.
- Excellent mechanical strength, allowing for safe and reliable performance in demanding environments.
- Versatile usage across multiple industries, simplifying procurement for manufacturers.

ASTM B516 N06601

Category: Materials

Standard Specification for Nickel-Chromium-Aluminum Alloy and Nickel-Chromium-Iron Alloy and Tubes.

Key Features:

- Chemical Composition: C % 0.100 max, Cr % 21.000 - 25.000
- Mechanical Properties: Tensile Strength in Mpa 550 min

Technical Specifications:

Nickel	58.000 - 63.000
Iron	Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B516 N06601 is a special type of metal alloy made primarily from nickel, chromium, and aluminum, designed to be used in various engineering applications. This alloy is known for its strength and resistance to heat and corrosion, making it ideal for demanding environments. It is commonly used in valves, pumps, and automotive parts, ensuring reliability and longevity in critical components.

Key Benefits:

- ****Strong and Durable****: With a minimum tensile strength of 550 MPa, it can withstand high pressure and stress.
- ****Corrosion-Resistant****: The unique chemical composition helps protect against rust and degradation, which is essential for machinery and automotive uses.
- ****Versatile Applications****: Suitable for use in different industries, including general engineering and automotive, making it a flexible choice for manufacturers.

ASTM B564 N06230

Category: Materials

Standard Specification for Nickel and Nickel-Related Alloys Forging.

Key Features:

- Chemical Composition: C % 0.005 - 0.150, Cr % 20.000 - 24.000
- Mechanical Properties: Tensile Strength in Mpa 758 min

Technical Specifications:

Nickel	Balance
Iron	3.000 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B564 N06230 is a type of specialized metal used primarily for making strong and durable parts in various industries. This material is a nickel alloy, which means it has a high nickel content along with other elements that enhance its performance. It's designed to be forged, making it ideal for creating components that need to withstand high stress and pressure, like valves and pumps in industrial machinery and automotive applications.

Here are some of the main benefits of using ASTM B564 N06230:

- **Strong and Reliable**: With a minimum tensile strength of 758 MPa, this alloy can handle heavy loads without breaking.
- **Versatile Applications**: It's perfect for a range of uses, including valves, pumps, and general engineering projects, making it a go-to choice for many manufacturers.
- **Durable Composition**: The specific chemical makeup ensures that the material resists wear and tear, which is crucial for long-lasting performance in demanding environments.

ASTM B584 C86400

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Al % 0.500 - 1.500, Cu % 56.000 - 62.000
- Mechanical Properties: Tensile Strength in Mpa 414 min

Technical Specifications:

Iron	0.400 - 2.000
Manganese	0.100 - 1.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C86400 is a type of copper alloy that is specifically designed for sand casting, which is a common method used to create metal parts. This material is used in various applications, including valves, pumps, and automotive components. Its unique composition ensures it is strong and durable, making it suitable for general engineering tasks.

Key Benefits:

- **Strong and Reliable**: With a minimum tensile strength of 414 MPa, this alloy can withstand significant stress and pressure in various applications.
- **Versatile Use**: Ideal for manufacturing parts in industries like automotive and engineering, this alloy can be used in a range of products, ensuring flexibility in design and application.
- **Controlled Composition**: The specific percentages of aluminum, copper, iron, and manganese in the alloy provide a balance of strength and performance, ensuring high-quality outcomes in production.

ASTM B584 C85400

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Al % 0.350 max, Cu % 65.000 - 70.000
- Mechanical Properties: Tensile Strength in Mpa 207 min

Technical Specifications:

Iron	0.700 max
Lead	1.500 - 3.800

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C85400 is a type of copper alloy specifically designed for sand casting, which is a common method used to create metal parts. This material is made primarily of copper and aluminum, making it strong and durable for various applications. With a tensile strength of at least 207 MPa, it can withstand significant stress without breaking.

****Key Benefits:****

- ****Versatile Applications:**** Ideal for manufacturing components in valves, pumps, and automotive parts, as well as other general engineering uses.
- ****Strong and Reliable:**** Its impressive mechanical properties ensure that it can handle demanding conditions while maintaining integrity.
- ****Standardized Quality:**** Follows a recognized standard specification, ensuring consistent quality and performance across products.

ASTM B584 C86700

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Al % 1.000 - 3.000, Cu % 55.000 - 60.000
- Mechanical Properties: Tensile Strength in Mpa 552 min

Technical Specifications:

Iron	1.000 - 3.000
Manganese	1.000 - 3.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C86700 is a specialized type of copper alloy that is primarily used for creating sand castings in various industries. This material is designed to be strong and durable, making it suitable for a wide range of applications such as valves, pumps, and automotive components. The alloy consists mainly of copper, along with specific amounts of aluminum, iron, and manganese, which enhance its properties and performance.

Key benefits of ASTM B584 C86700 include:

- ****High Strength**:** With a minimum tensile strength of 552 MPa, it can withstand significant stress and pressure, making it ideal for heavy-duty applications.

- **Versatility**: It can be used in various fields, including general engineering and automotive, allowing manufacturers to utilize it for different products and components.
- **Durability**: The combination of its chemical composition ensures that the material is resistant to wear and corrosion, extending the lifespan of the products made from it.

ASTM B584 C87300

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Fe % 0.200 max, Cu % 94.000 min
- Mechanical Properties: Tensile Strength in Mpa 310 min

Technical Specifications:

Manganese 0.800 - 1.500

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C87300 is a type of copper alloy specifically designed for creating sand castings used in various industries. This material is made primarily of copper, with a small amount of iron and manganese, which gives it strength and durability. It's commonly utilized in making items like valves and pumps, as well as in general engineering and automotive applications.

Key benefits of ASTM B584 C87300 include:

- **Strong and Reliable**: With a minimum tensile strength of 310 MPa, it can withstand significant stress and pressure.
- **Versatile Use**: Ideal for various applications, from automotive parts to general engineering projects, making it a flexible choice for manufacturers.

ASTM B584 C87600

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Fe % 0.200 max, Cu % 88.000 min
- Mechanical Properties: Tensile Strength in Mpa 414 min

Technical Specifications:

Manganese 0.250 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C87600 is a type of copper alloy that is primarily used for sand casting, which is a method of shaping metal by pouring it into molds. This alloy is designed for a variety of general applications, making it versatile for different industries. It has specific chemical and mechanical properties that ensure its strength and durability in various conditions.

Key features of ASTM B584 C87600 include:

- ****Strong Composition****: Contains at least 88% copper, ensuring high conductivity and corrosion resistance.
- ****Robust Strength****: It has a minimum tensile strength of 414 MPa, making it suitable for demanding applications.

This material is commonly used in the manufacturing of valves and pumps, as well as in general engineering and automotive components. Its reliable performance in various environments makes it a preferred choice for engineers and manufacturers looking for quality and resilience in their products.

ASTM B584 C87400

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Al % 0.800 max, Cu % 79.000 min
- Mechanical Properties: Tensile Strength in Mpa 345 min

Technical Specifications:

Lead 1.000 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C87400 is a type of copper alloy that is specifically designed for making sand castings, which are often used in various industries. It is made from a mixture of metals, with at least 79% copper and a maximum of 0.8% aluminum. This alloy is known for its strength and durability, making it suitable for applications where reliability is crucial.

Key benefits of ASTM B584 C87400 include:

- ****Strong Performance****: It has a minimum tensile strength of 345 MPa, meaning it can withstand significant stress without breaking.
- ****Versatile Uses****: This material is ideal for manufacturing items like valves and pumps, as well as in general engineering and automotive applications. Its robust properties ensure that products made from this alloy can perform well in demanding environments.

ASTM B584 C87500

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications.

Key Features:

- Chemical Composition: Al % 0.500 max, Cu % 79.000 min
- Mechanical Properties: Tensile Strength in Mpa 414 min

Technical Specifications:

Lead 0.090 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B584 C87500 is a type of copper alloy specifically designed for sand casting, which is a common manufacturing process where molten metal is poured into molds to create different shapes. This alloy is particularly suitable for general applications, making it versatile and useful in a variety of industries. It contains a maximum of 0.5% aluminum and at least 79% copper, ensuring strong and durable products.

Key benefits of ASTM B584 C87500 include:

- **High Strength**: With a minimum tensile strength of 414 MPa, products made from this alloy can withstand significant stress and pressure.
- **Versatile Applications**: It can be used in various components such as valves and pumps, as well as in general engineering and automotive sectors, making it a reliable choice for manufacturers.

ASTM B622 N06250

Category: Materials

Standard Specification for Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube.

Key Features:

- Chemical Composition: C % 0.020 max, Ni % 50.000 - 54.000
- Mechanical Properties: Tensile Strength in Mpa 621 min

Technical Specifications:

Iron 5.000 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B622 N06250 is a type of high-quality pipe and tube made from a special combination of nickel and nickel-cobalt alloy. It's designed to withstand tough conditions, making it ideal for various applications in industries like automotive, engineering, and pump manufacturing. This material is known for its strength and durability, ensuring that it can handle high stress and pressure without breaking down.

Key Benefits:

- **Strong and Reliable**: With a minimum tensile strength of 621 MPa, it provides the strength needed for demanding applications.
- **Corrosion Resistance**: The unique chemical composition, including a high nickel content, helps protect

against corrosion, extending the life of the pipes and tubes.

- **Versatile Use:** Suitable for various uses such as valves and pumps, making it a go-to choice for engineers and manufacturers.

ASTM B637 N07718

Category: Materials

Standard Specification for Nickel Alloy Bars, Forgings, and Forging Stock for Moderate or High Temperature Service.

Key Features:

- Chemical Composition: C % 0.080 max, Ni % 50.000 - 55.000
- Mechanical Properties: Tensile Strength in Mpa 1275 min

Technical Specifications:

Iron Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B637 N07718 is a special type of nickel alloy material used primarily in manufacturing bars and forgings that can withstand high temperatures. This means it's ideal for industries that require durable and reliable components, such as valves, pumps, automotive parts, and general engineering applications. The material is designed to perform well even in tough conditions, making it a preferred choice for many engineers and manufacturers.

Key benefits of using ASTM B637 N07718 include:

- **High Strength:** It has a minimum tensile strength of 1275 MPa, meaning it can endure significant stress without breaking.
- **Temperature Resilience:** It maintains its properties and performance in moderate to high-temperature environments, ensuring longevity and reliability in demanding applications.

ASTM B670 N07718

Category: Materials

Standard Specification for Nickel Alloy Plate, Sheet and Strip for High-Temperature Service.

Key Features:

- Chemical Composition: C % 0.080 max, Ni % 50.000 - 55.000
- Mechanical Properties: Tensile Strength in Mpa 1241 min

Technical Specifications:

Iron Balance

Use Cases:

- Valve
- Pump
- General Engineering

- Automotive

AI Layman Summary:

ASTM B670 N07718 is a specialized material made from a nickel alloy, designed to withstand high temperatures. This type of metal is used to create plates, sheets, and strips that are essential in various industries where heat resistance is crucial. It is particularly valuable in applications such as valves, pumps, general engineering, and automotive manufacturing.

Key benefits of ASTM B670 N07718 include:

- **High Strength:** With a minimum tensile strength of 1241 MPa, it can handle significant stress without breaking.
- **Temperature Resilience:** Its composition allows it to perform well in extreme heat, making it reliable for high-temperature applications.
- **Versatile Applications:** This material is suitable for a wide range of industries, ensuring that it meets various engineering needs.

ASTM B763 C 99500

Category: Materials

Standard Specification for Copper Alloy Sand Castings for Valve Applications.

Key Features:

- Chemical Composition: Al % 0.500 - 2.000, Ni % 3.500 - 5.500
- Mechanical Properties: Tensile Strength in Mpa 483 min

Technical Specifications:

Iron 3.000 - 5.000

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B763 C 99500 is a type of metal made from a special blend of copper and other materials, designed specifically for creating sand castings used in valves. This standard specification ensures that the metal has the right chemical and mechanical properties, making it suitable for various applications including pumps and automotive parts. Essentially, it provides manufacturers with a reliable material that meets industry standards for strength and durability.

Key benefits of using ASTM B763 C 99500 include:

- **Strong and Durable:** With a minimum tensile strength of 483 MPa, it can withstand significant stress.
- **Versatile Applications:** Ideal for valves, pumps, and general engineering projects, making it a go-to choice for various industries.
- **Quality Assurance:** Adhering to established standards ensures the material's reliability in critical applications.

ASTM B80 AZ92A

Category: Materials

Standard Specification for Magnesium Alloy Sand Castings.

Key Features:

- Chemical Composition: Al % 8.300 - 9.700, Mg % Balance
- Mechanical Properties: Tensile Strength in Mpa 158 - 234

Technical Specifications:

Copper 0.250 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B80 AZ92A is a special type of magnesium alloy specifically designed for sand casting. This means it can be shaped into various parts and components, making it useful in many industries. The alloy is composed mainly of aluminum and magnesium, offering a balance of strength and lightweight properties. It is particularly known for its tensile strength, which ranges from 158 to 234 MPa, ensuring that it can withstand significant stress without breaking.

Key benefits of ASTM B80 AZ92A include:

- ***Lightweight**: Being a magnesium alloy, it is lighter than many other metals, making it ideal for applications where weight is a concern, such as in automotive and engineering parts.*
- ***Versatile Applications**: This alloy is suitable for various uses, including valves, pumps, and general engineering projects, demonstrating its flexibility across different sectors.*

ASTM B86 AG40A

Category: Materials

Standard Specification for Zinc and Zinc-Aluminum (ZA) Alloy Foundry and Die Castings.

Key Features:

- Chemical Composition: Al % 3.700 - 4.300, Zn % Balance
- Mechanical Properties: Tensile Strength in Mpa 283 min

Technical Specifications:

Iron 0.050 max

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 AG40A is a special type of metal alloy made primarily of zinc and aluminum, designed specifically for use in manufacturing processes like casting. This alloy is known for its strong mechanical properties, making it a reliable choice for various applications. Essentially, it provides the strength and durability needed for products that require high performance, such as valves and pumps in automotive and general engineering industries.

Key points about ASTM B86 AG40A:

- **Composition**: It contains a specific percentage of aluminum (3.7% to 4.3%) while the rest is primarily zinc, ensuring a robust structure.
- **Strength**: It offers a minimum tensile strength of 283 MPa, which means it can withstand significant force without breaking.
- **Applications**: Widely used in manufacturing components for valves, pumps, and other engineering products, making it a versatile material choice.

ASTM B834 N07718

Category: Materials

Standard Specification for Nickel-Chromium-Molybdenum Alloy Pipe Flanges, Fittings, Valves and Parts.

Key Features:

- Chemical Composition: C % 0.080 max, Ni % 50.000 - 55.000
- Mechanical Properties: Tensile Strength in Mpa 1275 - 1365

Technical Specifications:

Iron Balance

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B834 N07718 is a specialized type of metal used to create pipe flanges, fittings, valves, and other parts. This material is made from a combination of nickel, chromium, and molybdenum, making it highly durable and resistant to corrosion. It is commonly used in industries such as automotive, engineering, and various applications involving valves and pumps.

Key benefits of using ASTM B834 N07718 include:

- **Strength and Durability**: With a high tensile strength of 1275 to 1365 MPa, it can withstand significant pressure and wear over time.
- **Corrosion Resistance**: The unique chemical composition ensures that the parts made from this alloy can resist damage from harsh environments, prolonging their lifespan and reducing maintenance costs.

ASTM B86 AG40B

Category: Materials

Standard Specification for Zinc and Zinc-Aluminum (ZA) Alloy Foundry and Die Castings.

Key Features:

- Chemical Composition: Al % 3.700 - 4.300, Zn % Balance
- Mechanical Properties: Tensile Strength in Mpa 283 min

Technical Specifications:

Iron 0.050 max

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

AI Layman Summary:

ASTM B86 AG40B is a specialized material made from zinc and zinc-aluminum alloys, designed for use in manufacturing cast parts. This standard specification outlines the quality and composition of the material, ensuring that it meets specific mechanical and chemical properties for reliable performance. It is particularly useful in creating durable components for various industries, including automotive and general engineering.

Key benefits of ASTM B86 AG40B include:

- **Strong and Durable**: With a minimum tensile strength of 283 MPa, it can withstand significant stress and strain, making it ideal for heavy-duty applications.
- **Versatile Applications**: This material is suitable for a range of uses, such as valves and pumps, providing flexibility for manufacturers in different sectors.

ASTM B148 C95500

Category: Valves

Standard Specification for Aluminum - Bronze Sand Casting.

Key Features:

- Aluminium Al % 10.000 - 11.500
- Iron Fe % 3.000 - 5.000
- Copper Cu % 78.000 min.

Technical Specifications:

Tensile Strength in Mpa	620 - 760
Yield Strength in Mpa	275 - 415
Elongation in %	5 - 6
Hardness in HB	190 - 200

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 C95500 is a type of valve made from a special aluminum-bronze material known for its durability and strength. This product is created using a sand casting process, which means the aluminum and bronze are poured into molds to form the final shape. The unique combination of metals makes it resistant to wear and corrosion, making it a great choice for various applications.

Key Benefits:

- **Strong and Durable**: With a tensile strength ranging from 620 to 760 MPa, these valves can withstand high pressure and stress.
- **Versatile Use**: Ideal for valves, pumps, and general engineering tasks, it's commonly used in automotive and industrial settings.
- **Corrosion Resistant**: The aluminum and copper content helps prevent rust and degradation, ensuring a longer lifespan and reliability in tough environments.

ASTM B148 C95800

Category: Valves

Standard Specification for Aluminum - Bronze Sand Casting.

Key Features:

- Aluminium Al % 8.500 - 9.500
- Iron Fe % 3.500 - 4.500
- Copper Cu % 79.000 min.

Technical Specifications:

Tensile Strength in Mpa	585 min.
Yield Strength in Mpa	240 min.
Elongation in %	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 C95800 is a high-quality aluminum-bronze alloy primarily used for making valves and other engineering components. This material is recognized for its excellent strength and durability, making it suitable for demanding applications in various industries, including automotive and general engineering. The alloy consists of at least 79% copper, along with precise amounts of aluminum and iron, which contribute to its robust properties.

Key benefits of ASTM B148 C95800 include:

- Exceptional tensile strength (minimum 585 MPa) ensures it can withstand heavy loads and stress.
- Good yield strength (minimum 240 MPa) improves its ability to maintain shape under pressure.
- A minimum elongation of 15% allows for flexibility, reducing the risk of breaking under strain.

Overall, this aluminum-bronze alloy is a reliable choice for manufacturers looking for durable and efficient materials for valves, pumps, and other engineering applications.

ASTM B148 Grade A

Category: Valves

Standard Specification for Aluminum - Bronze Sand Casting.

Key Features:

- Aluminium Al % 8.500 - 9.500
- Iron Fe % 2.500 - 4.000
- Copper Cu % 86.000 min.

Technical Specifications:

Tensile Strength in Mpa	450 min.
Yield Strength in Mpa	170 min.
Elongation in %	20 min.
Hardness in HB	110 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 Grade A is a type of valve made from a specific alloy known as aluminum-bronze. This material is created by mixing aluminum, copper, and iron, which gives it unique properties that make it strong and durable. The valve can withstand high pressure and is resistant to wear, making it ideal for various applications in industries like automotive, pumps, and general engineering.

****Key Benefits:****

- ****Durable and Strong:**** With a minimum tensile strength of 450 MPa, these valves can handle tough conditions without breaking.
- ****Versatile Use:**** Suitable for different applications including valves, pumps, and more, making it a reliable choice for various engineering needs.
- ****Corrosion Resistance:**** The aluminum-bronze material provides excellent resistance to corrosion, ensuring a longer lifespan for the product.

ASTM B176 C85700

Category: Valves

Standard Specification for Copper Alloy Casting.

Key Features:

- Copper Cu % 58.000 - 64.000
- Zinc Zn % 32.000 - 40.000

Technical Specifications:

Tensile Strength in Mpa	276 min.
Yield Strength in Mpa	97 min.
Elongation in %	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B176 C85700 is a type of copper alloy casting specifically designed for use in valves and other engineering applications. This material contains a high percentage of copper (58-64%) and zinc (32-40%), making it strong and durable. It is especially useful in environments where resistance to corrosion and wear is important, which is why it is commonly found in pumps and automotive parts.

Key benefits include:

- ****Strength and Durability:**** With a minimum tensile strength of 276 MPa and yield strength of 97 MPa, this alloy can withstand significant stress and pressure.
- ****Versatile Applications:**** Ideal for valves, pumps, and various engineering projects, making it a go-to choice for manufacturers in multiple industries.
- ****Ease of Fabrication:**** The properties of this alloy allow for straightforward casting and machining, simplifying

the manufacturing process.

ASTM B22 C91300

Category: Valves

Standard Specification for Bronze Casting.

Key Features:

- Copper Cu % 79.000 - 82.000
- Tin Sn % 18.000 - 20.000

Technical Specifications:

Hardness in HB170 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B22 C91300 is a type of bronze used primarily for making valves and other components in various engineering applications. It is known for its strong durability and resistance to wear and corrosion, making it an ideal choice for industries like automotive and general engineering. This bronze is made up of a specific mix of copper and tin, which gives it its unique properties.

Key benefits of ASTM B22 C91300 include:

- **High Strength**: With a minimum hardness of 170 HB, it can withstand significant pressure and stress.
- **Corrosion Resistance**: The copper and tin composition helps prevent rust and degradation, ensuring long-lasting performance.
- **Versatile Use**: Suitable for a range of applications, including valves and pumps, making it a reliable option for many engineering projects.

ASTM B148 Grade B

Category: Valves

Standard Specification for Aluminum - Bronze Sand Casting.

Key Features:

- Aluminium Al % 9.000 - 11.000
- Iron Fe % 0.800 - 1.500
- Copper Cu % 86.000 min.

Technical Specifications:

Tensile Strength in Mpa	450 - 550
Yield Strength in Mpa	170 - 275
Elongation in %	12 - 20
Hardness in HB	110 - 160

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B148 Grade B is a type of valve made from a special aluminum-bronze material. This standard specification indicates that the valve is manufactured using a sand casting process, which is a method to shape the metal by pouring it into a mold. The key components of this alloy include 9-11% aluminum, 0.8-1.5% iron, and at least 86% copper, making it strong and durable for various applications.

This valve is designed to withstand significant pressure and stress, with a tensile strength of 450-550 MPa and a yield strength of 170-275 MPa. It can elongate up to 20% before breaking, showing its flexibility. Here are some benefits of using ASTM B148 Grade B valves:

- **Durability**: Built to last, reducing the need for frequent replacements.
- **Versatility**: Suitable for various applications including valves, pumps, and automotive uses.
- **Strength**: High tensile strength ensures reliable performance in demanding environments.

ASTM B179 443.1

Category: Valves

Standard Specification for Aluminum Alloys in Ingot and Molten Forms for Casting.

Key Features:

- Copper Cu % 0.600 max.
- Iron Fe % 0.600 max.

Technical Specifications:

Tensile Strength in Mpa	145 min.
Yield Strength in Mpa	49 min.
Elongation in %	2 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B179 443.1 is a standard specification for aluminum alloys used in making valves and other engineering components. This specification ensures that the aluminum used in these products meets certain quality and performance standards, particularly in its ingot and molten forms. In simple terms, it lays out the rules for producing durable and reliable aluminum parts for various applications.

Key benefits of ASTM B179 443.1 include:

- **High Durability**: The aluminum alloys meet strict tensile and yield strength requirements, ensuring they can withstand pressure and use over time.
- **Versatile Applications**: These alloys are suitable for a wide range of uses, from valves and pumps to general engineering projects and automotive parts.
- **Quality Assurance**: By adhering to this specification, manufacturers can guarantee that their products have low levels of impurities, like copper and iron, which can affect performance.

ASTM B22 C93700

Category: Valves

Standard Specification for Bronze Casting.

Key Features:

- Copper Cu % 78.000 - 82.000
- Tin Sn % 9.000 - 11.000

Technical Specifications:

Tensile Strength in Mpa	205 - 240
Yield Strength in Mpa	85 - 140

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B22 C93700 is a type of bronze material specifically designed for making valves and other engineering components. It is made from a copper-tin alloy, which means it contains a high percentage of copper (78-82%) and a good amount of tin (9-11%). This combination gives it strength and durability, making it ideal for various applications in industries such as automotive and general engineering.

Key Benefits:

- **Strong and Durable**: With a tensile strength of 205-240 MPa, it can withstand significant pressure and wear, ensuring long-lasting performance.
- **Versatile Applications**: Suitable for use in valves, pumps, and other engineering components, allowing for broad usage across different sectors.
- **Reliable Performance**: The specific composition of copper and tin provides excellent resistance to corrosion, enhancing the lifespan of products made from this material.

ASTM B271 C84800

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 75.000 - 77.000
- Lead Pb % 5.500 - 7.000

Technical Specifications:

Tensile Strength in Mpa	193 min.
Yield Strength in Mpa	83 min.
Elongation in %	16 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C84800 is a type of valve made from a copper-based alloy that is designed for various engineering applications. This material is particularly strong and durable, making it suitable for use in valves, pumps, and even in the automotive industry. The valve is made with a high percentage of copper, which ensures excellent conductivity and resistance to corrosion, while also containing a small amount of lead to enhance its workability.

Key benefits of ASTM B271 C84800 include:

- **Durability**: With a minimum tensile strength of 193 MPa, this valve can withstand significant stress and pressure, making it reliable for industrial use.
- **Versatility**: It can be used in different applications, from general engineering tasks to specific needs in the automotive sector.
- **Workability**: The material's composition allows for easy machining and shaping, which is crucial in manufacturing processes.

ASTM B271 C83600

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 84.000 - 86.000
- Lead Pb % 4.000 - 6.000

Technical Specifications:

Tensile Strength in Mpa	207 min.
Yield Strength in Mpa	97 min.
Elongation in %	20 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C83600 is a type of high-quality valve made from a special copper-based alloy. This alloy is designed to withstand high stress and is used in various applications, such as valves, pumps, and automotive components. The primary materials in this alloy include 84-86% copper and 4-6% lead, which give it durability and strength.

Here are some key benefits of ASTM B271 C83600:

- **Strong and Reliable**: With a minimum tensile strength of 207 MPa and yield strength of 97 MPa, these valves are built to last and perform under pressure.
- **Versatile Use**: Suitable for a wide range of applications, making it ideal for industries like engineering and automotive.
- **Excellent Elongation**: This alloy can stretch up to 20% before breaking, allowing for flexibility in design and use.

ASTM B271 C84400

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 78.000 - 82.000
- Lead Pb % 6.000 - 8.000

Technical Specifications:

Tensile Strength in Mpa	200 min.
Yield Strength in Mpa	90 min.
Elongation in %	18 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C84400 is a type of valve made from a special copper-based alloy. This alloy is created through a process called centrifugal casting, which ensures that the final product is strong and durable. The key ingredients in this alloy are copper (78-82%) and lead (6-8%), which contribute to its excellent performance in various applications.

Key Benefits:

- **Strong and Reliable**: It has a minimum tensile strength of 200 MPa and yield strength of 90 MPa, making it suitable for high-pressure environments.
- **Versatile Uses**: This valve is ideal for various applications such as in pumps, general engineering projects, and automotive systems, ensuring flexibility in its use across different industries.

ASTM B271 C86200

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 60.000 - 66.000
- Lead Pb % 0.200 max.

Technical Specifications:

Tensile Strength in Mpa	621 min.
Yield Strength in Mpa	310 min.
Elongation in %	18 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C86200 is a high-quality copper-based alloy used primarily in the manufacturing of valves and pumps. This material is specially designed for centrifugal casting, ensuring durability and reliability in various applications. With a copper content of 60% to 66%, it offers excellent strength and corrosion resistance, making it an ideal choice for general engineering and automotive uses.

- **Key Benefits:****
- ****Strong and Durable:**** With a minimum tensile strength of 621 MPa, this alloy can withstand significant stress without breaking.
 - ****Versatile Applications:**** Suitable for valves, pumps, and other engineering needs, making it a flexible option for various industries.
 - ****Resistant to Wear:**** The alloy's properties ensure longevity, reducing the need for frequent replacements and maintenance.

ASTM B271 C86300

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 60.000 - 66.000
- Lead Pb % 0.200 max.

Technical Specifications:

Tensile Strength in Mpa	758 min.
Yield Strength in Mpa	414 min.
Elongation in %	12 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C86300 is a type of valve made from a special copper-based alloy known for its strength and durability. This alloy is created through centrifugal casting, which means it's formed by spinning molten metal, ensuring a strong and reliable product. The composition includes a high percentage of copper, making it a great choice for various industrial applications, including valves, pumps, and automotive parts.

- Key benefits of ASTM B271 C86300 include:
- ****High Strength**:** It has a minimum tensile strength of 758 MPa, ensuring it can withstand significant pressure and stress.
 - ****Durability**:** With a yield strength of 414 MPa and a minimum elongation of 12%, it is designed to last and maintain its shape under tough conditions.
 - ****Versatile Use**:** Ideal for various applications in engineering and automotive industries, making it a reliable choice for manufacturers.

ASTM B271 C90500

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 86.000 - 89.000
- Lead Pb % 0.300 max.

Technical Specifications:

Tensile Strength in Mpa	276 min.
Yield Strength in Mpa	124 min.
Elongation in %	20 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C90500 is a type of valve made from a special copper alloy that is designed for durability and strength. This material is created using a centrifugal casting process, which ensures that it is both robust and reliable for various applications. Essentially, it is a high-quality valve suitable for different industries, including automotive and general engineering, ensuring that operations run smoothly.

Key features of ASTM B271 C90500 include:

- High copper content (86% - 89%) for excellent conductivity and corrosion resistance.
- Low lead content (maximum 0.3%) to meet safety and environmental standards.

The technical specifications highlight its strength and flexibility, with a minimum tensile strength of 276 MPa and yield strength of 124 MPa, allowing it to withstand significant pressure and stress. With a minimum elongation of 20%, it can also endure some bending, making it adaptable in various applications, from valves to pumps.

ASTM B271 C92300

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 85.000 - 89.000
- Lead Pb % 0.300 - 1.000

Technical Specifications:

Tensile Strength in Mpa	248 min.
Yield Strength in Mpa	110 min.
Elongation in %	18 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

The ASTM B271 C92300 is a type of valve made from a special copper-based alloy. This alloy is designed to be

strong and durable, making it suitable for various applications where reliability is crucial. It's typically used in valves, pumps, general engineering, and even in automotive components.

- Key features of the ASTM B271 C92300 include:
- Contains 85% to 89% copper, which ensures excellent conductivity and corrosion resistance.
 - Has a small amount of lead (0.3% to 1%), enhancing its machinability while still maintaining safety standards.

The material exhibits good strength characteristics, with a minimum tensile strength of 248 MPa and a yield strength of 110 MPa, allowing it to withstand significant stress. Additionally, it can stretch up to 18% before breaking, which provides flexibility in various applications.

ASTM B271 C85700

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 58.000 - 64.000
- Lead Pb % 0.800 - 1.500

Technical Specifications:

Tensile Strength in Mpa	276 min.
Yield Strength in Mpa	97 min.
Elongation in %	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C85700 is a type of valve made from a special copper-based alloy that is designed for durability and strength. This material is particularly useful in various industries because it can withstand high stress and has good resistance to wear. The valves made from this alloy are commonly used in applications like pumps and automotive systems, ensuring they perform well under pressure and in diverse engineering scenarios.

- Key benefits of ASTM B271 C85700 include:
- ****Strong and Durable****: With a minimum tensile strength of 276 MPa and yield strength of 97 MPa, these valves are built to last, making them reliable for critical applications.
 - ****Versatile Applications****: Suitable for various uses, including valves, pumps, general engineering, and automotive, this product can meet the needs of many industries.
 - ****Resistant to Wear****: The copper composition ensures that the valves maintain their integrity and functionality over time, reducing the need for frequent replacements.

ASTM B271 C95410

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 83.000 min.
- Iron Fe % 3.000 - 5.000

Technical Specifications:

Tensile Strength in Mpa	515 - 620
Yield Strength in Mpa	205 - 310
Elongation in %	6 - 12
Hardness in HB	150 - 190

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C95410 is a type of valve made from a specialized copper-based alloy that is designed to be strong and durable. This material is particularly useful in applications where reliability is crucial, such as in valves, pumps, and automotive components. The alloy contains at least 83% copper and a small percentage of iron, which helps enhance its strength and resistance to wear.

Key benefits of ASTM B271 C95410 include:

- **High Strength**: It has a tensile strength ranging from 515 to 620 MPa, making it suitable for demanding applications.
- **Durability**: The material's hardness and elongation properties ensure it can withstand stress without breaking.
- **Versatile Applications**: Ideal for various uses in engineering and automotive industries, ensuring it meets diverse operational needs.

ASTM B271 C93500

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 83.000 - 86.000
- Lead Pb % 8.000 - 10.000

Technical Specifications:

Tensile Strength in Mpa	193 min.
Yield Strength in Mpa	83 min.
Elongation in %	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C93500 is a high-quality valve made from a special copper-based alloy designed for durability and reliability. This product is primarily used in various applications, including valves, pumps, general engineering projects, and automotive components. Its composition includes 83% to 86% copper and 8% to 10% lead, which

contributes to its strength and performance.

Key benefits of ASTM B271 C93500 include:

- **High Strength**: With a minimum tensile strength of 193 MPa and yield strength of 83 MPa, it can withstand significant pressure and stress.
- **Good Flexibility**: The material can elongate up to 15%, allowing it to handle bending and stretching without breaking.
- **Versatile Applications**: Ideal for use in a variety of industries, making it a reliable choice for many engineering and automotive needs.

ASTM B271 C93600

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 79.000 - 83.000
- Lead Pb % 11.000 - 13.000

Technical Specifications:

Tensile Strength in Mpa	221 min.
Yield Strength in Mpa	110 min.
Elongation in %	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C93600 is a type of valve made from a specific copper-based alloy that is cast using a centrifugal process. This means the metal is spun to ensure a strong and uniform structure. The alloy contains a high percentage of copper, which gives it good conductivity and resistance to corrosion, making it suitable for various applications in industries like automotive and general engineering.

Key benefits include:

- **Durability**: With a tensile strength of at least 221 MPa, this valve is built to withstand significant pressure and wear.
- **Versatility**: It's suitable for use in valves, pumps, and other engineering applications, making it a valuable component in many systems.
- **Corrosion Resistance**: The copper content helps resist rust and degradation over time, ensuring longevity and reliability in performance.

ASTM B271 C95500

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

Key Features:

- Copper Cu % 78.000 min.
- Iron Fe % 3.000 - 5.000

Technical Specifications:

Tensile Strength in Mpa	620 - 760
Yield Strength in Mpa	275 - 415
Elongation in %	5 - 6
Hardness in HB	190 - 200

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B271 C95500 is a type of valve made from a special copper-based alloy that is designed to be durable and efficient. This product is used in various applications, including valves, pumps, and automotive components. Its strong composition allows it to withstand high pressure and stress, making it suitable for general engineering tasks.

Key benefits include:

- **High Strength**: With a tensile strength of 620 to 760 MPa, this alloy can handle demanding conditions.
- **Durability**: The alloy features a minimum of 78% copper, ensuring longevity and resistance to wear and tear.
- **Versatility**: It is suitable for multiple industries, including automotive and general engineering, providing flexibility in use.

ASTM B275 360.0

Category: Valves

Standard Practice for Codification of Certain Nonferrous Metals and Alloys.

Key Features:

- Copper Cu % 0.600 max.
- Iron Fe % 2.000 max.

Technical Specifications:

Tensile Strength in Mpa	305 min.
Yield Strength in Mpa	170 min.
Elongation in %	2.5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B275 360.0 is a standard specification that outlines the requirements for certain nonferrous metals and alloys, specifically designed for use in creating valves and other engineering components. This product features a maximum copper content of 0.600% and a maximum iron content of 2.000%, ensuring high quality and performance. It is particularly beneficial in industries such as automotive and general engineering where

reliability and durability are crucial.

Key benefits include:

- **High Strength**: With a minimum tensile strength of 305 MPa and a yield strength of 170 MPa, it can withstand significant stress and pressure.
- **Versatility**: Suitable for various applications, including valves and pumps, making it a valuable material for engineers and manufacturers.

ASTM B275 380.0

Category: Valves

Standard Practice for Codification of Certain Nonferrous Metals and Alloys.

Key Features:

- Copper Cu % 3.000 - 4.000
- Iron Fe % 2.000 max.

Technical Specifications:

Tensile Strength in Mpa	315 min.
Yield Strength in Mpa	160 min.
Elongation in %	2.5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B275 380.0 is a standard specification that outlines the properties of certain nonferrous metals and alloys, specifically focusing on their use in valves and related applications. This standard ensures that these materials meet specific performance criteria, making them reliable for various industries such as automotive and general engineering. Essentially, it serves as a guideline to help manufacturers and engineers select the right materials for their products.

Key features of ASTM B275 380.0 include:

- Copper content ranging from 3.000% to 4.000%, which provides excellent conductivity and corrosion resistance.
- A maximum iron content of 2.000%, ensuring the material's integrity and preventing rust.

The technical specifications include a minimum tensile strength of 315 MPa, a yield strength of 160 MPa, and a minimum elongation of 2.5%. These properties indicate that the materials are durable and can withstand significant stress, making them ideal for applications in valves, pumps, and other engineering projects.

ASTM B275 383.0

Category: Valves

Standard Practice for Codification of Certain Nonferrous Metals and Alloys.

Key Features:

- Copper Cu % 2.000 - 3.000
- Iron Fe % 1.300 max.

Technical Specifications:

Tensile Strength in Mpa	310 min.
Yield Strength in Mpa	150 min.
Elongation in %	3.5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B275 383.0 is a standard for a specific type of nonferrous metal and alloy, primarily used in the manufacturing of valves and pumps. This standard outlines the composition and properties of the material, ensuring that it meets specific performance criteria. The key elements in this alloy include copper (Cu) and a maximum iron (Fe) content, which contribute to its strength and durability in various applications.

Key Benefits:

- **Strength and Durability**: With a minimum tensile strength of 310 MPa and a yield strength of 150 MPa, this material is robust enough for demanding environments.
- **Versatile Applications**: It is suitable for use in valves, pumps, general engineering, and automotive industries, making it a reliable choice for various manufacturing needs.

ASTM B275 443.1

Category: Valves

Standard Practice for Codification of Certain Nonferrous Metals and Alloys.

Key Features:

- Copper Cu % 0.600 max.
- Iron Fe % 0.600 max.

Technical Specifications:

Tensile Strength in Mpa	145 min.
Yield Strength in Mpa	49 min.
Elongation in %	2 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B275 443.1 is a standard used in the manufacturing of valves and other components made from certain nonferrous metals and alloys. Essentially, it provides guidelines on how to classify and work with specific metals, ensuring quality and consistency in various applications. This standard is especially important for industries that rely on durable materials, such as automotive and general engineering.

Key features of ASTM B275 443.1 include strict limitations on the amounts of copper and iron that can be

present in the metals used, which helps to enhance their performance and longevity. The materials meet minimum standards for tensile and yield strength, ensuring they can withstand significant stress without breaking.

- Benefits of using ASTM B275 443.1 include:
- Ensures high-quality metal components for better performance in valves and pumps.
 - Provides clarity and consistency in engineering practices.
 - Supports industries that require reliable materials for safety and efficiency.

ASTM B271 C95520

Category: Valves

Standard Specification for Copper - Base Alloy Centrifugal Casting.

- Key Features:
- Copper Cu % 74.500 min.
 - Iron Fe % 4.000 - 5.500

Technical Specifications:

Tensile Strength in Mpa	862 min.
Yield Strength in Mpa	655 min.
Elongation in %	3 min.

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM B271 C95520 is a type of valve made from a special copper-based alloy that is known for its strength and durability. This valve is designed to handle high pressures and is commonly used in various industries, such as automotive and general engineering. Its unique composition ensures that it performs well in demanding environments, making it a reliable choice for many applications.

- Key benefits of ASTM B271 C95520 include:
- **High Strength**: With a minimum tensile strength of 862 MPa, it can withstand significant force without breaking.
 - **Durability**: The alloy's resistance to wear and tear makes it ideal for long-term use in pumps and valves.
 - **Versatility**: Suitable for various applications, including automotive and general engineering projects, ensuring it meets diverse industry needs.

ASTM B275 384.0

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

- Key Features:
- Copper content: 3.000 - 4.500%
 - Iron content: 1.300 max.
 - Magnesium content: 0.100 max.

Technical Specifications:

Tensile Strength (MPa)	330 min.
Yield Strength (MPa)	165 min.
Elongation (%)	2.5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA
- Australia
- Brazil
- Mexico

AI Layman Summary:

ASTM B275 384.0 is a standard guideline that helps identify and classify certain types of nonferrous metals and alloys, particularly those that contain copper, iron, and magnesium. This standard ensures that manufacturers and engineers use materials that meet specific quality and performance criteria, thereby enhancing the reliability of their products.

Key benefits of ASTM B275 384.0 include:

- **Consistent Quality**: By following this standard, businesses can ensure the materials they use have specific chemical compositions and mechanical properties, which leads to better performance in various applications.
- **Wide Application**: This material is ideal for making valves, pumps, and components in automotive and general engineering sectors, making it versatile for many industries.

Additionally, ASTM B275 384.0 is recognized in major markets like the USA, Australia, Brazil, and Mexico, making it a valuable standard for companies looking to produce reliable and high-quality nonferrous metal products.

ASTM B275 518.0

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

Key Features:

- Copper content: 0.250 max.
- Iron content: 1.800 max.
- Magnesium content: 7.500 - 8.500%

Technical Specifications:

Tensile Strength (MPa)	310 min.
Yield Strength (MPa)	190 min.
Elongation (%)	5 min.

Use Cases:

- Valve
- Pump

- General Engineering
- Automotive

Export Markets:

- USA
- Japan

AI Layman Summary:

ASTM B275 518.0 is a standard guideline that helps manufacturers identify and classify certain nonferrous metals and alloys, specifically those that are not based on iron. This ensures that industries use the right materials for their applications, leading to better product quality and performance. Essentially, it's like a reference book for engineers and manufacturers to understand what metals can be used for specific purposes.

****Key Features:****

- Copper content is capped at 0.250%, ensuring the right balance in metal composition.
- Iron content is limited to a maximum of 1.800%, reducing the risk of rust and corrosion.
- Magnesium content ranges from 7.500% to 8.500%, which enhances strength and durability.

The materials classified under ASTM B275 518.0 are essential in various industries, including:

- ****Valves and Pumps:**** Used to manufacture components that require high reliability.
- ****General Engineering:**** Applicable in a wide range of engineering projects.
- ****Automotive:**** Ensures the production of lightweight yet strong parts for vehicles.
- This standard is widely recognized and utilized in markets like the USA and Japan, promoting international trade and quality assurance.

ASTM B275 G8A

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

Key Features:

- Copper content: 0.250 max.
- Iron content: 1.800 max.
- Magnesium content: 7.500 - 8.500%

Technical Specifications:

Tensile Strength (MPa)	310 min.
Yield Strength (MPa)	190 min.
Elongation (%)	5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA
- Japan

AI Layman Summary:

ASTM B275 G8A is a standard guideline for categorizing specific nonferrous metals and alloys, which are

metals that do not contain iron. This standard is particularly useful in industries such as automotive, engineering, and manufacturing, where precise material specifications are crucial. It ensures that the materials used in products like valves and pumps meet certain quality criteria for performance and durability.

- Key features of ASTM B275 G8A include:
- Limited copper content (maximum 0.250%) to enhance material strength.
 - Controlled iron content (maximum 1.800%) to prevent corrosion issues.
 - A specified range of magnesium content (7.500% to 8.500%) to improve the material's resilience.

The standard also outlines important technical specifications, such as a minimum tensile strength of 310 MPa and yield strength of 190 MPa, ensuring that products made from these materials can withstand significant stress without failing. By adhering to this standard, manufacturers can guarantee the reliability and performance of their products in various applications.

ASTM B275 S12A

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

Key Features:

- Copper content: 1.000 max.
- Iron content: 1.300 max.
- Silicon content: 11.000 - 13.000%

Technical Specifications:

Tensile Strength (MPa)	290 min.
Yield Strength (MPa)	130 min.
Elongation (%)	3.5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA
- Brazil
- Mexico

AI Layman Summary:

ASTM B275 S12A is a standard guideline that helps in the classification of specific nonferrous metals and alloys, primarily for industrial use. This means it helps manufacturers and engineers understand the properties and specifications of these materials, ensuring they choose the right type for their applications. The standard includes details about the composition of metals, including limits on copper, iron, and silicon content, making it easier to identify the right material for a project.

- Key benefits of ASTM B275 S12A include:
- **Quality Assurance**: Ensures that the materials meet specific standards for strength and composition, which is crucial for safety and performance in applications like valves and pumps.
 - **Versatility**: Suitable for various industries such as automotive and general engineering, making it a valuable resource for manufacturers across different sectors.
 - **Market Acceptance**: Recognized in key markets like the USA, Brazil, and Mexico, facilitating international

trade and compliance with regulations.

ASTM B275 CG100A

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

Key Features:

- Copper content: 9.200 - 10.700%
- Iron content: 1.500 max.
- Magnesium content: 0.150 - 0.350%

Technical Specifications:

Tensile Strength (MPa) 160 - 275

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA
- UK
- India

AI Layman Summary:

ASTM B275 CG100A is a standardized material used for specific types of nonferrous metals and alloys. This means it provides guidelines on how to categorize and use these materials effectively in various applications. It's particularly useful in industries where the quality and composition of metals are crucial for performance and safety.

Key Benefits:

- **Versatile Usage**: Ideal for manufacturing components in valves, pumps, automotive parts, and general engineering.
- **Quality Assurance**: Ensures that the materials meet strict criteria, including specific percentages of copper, iron, and magnesium, which contribute to their strength and durability.
- **Global Reach**: This standard is recognized and used in major markets like the USA, UK, and India, making it a reliable choice for international projects.

ASTM B275 S12B

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

Key Features:

- Copper content: 1.000 max.
- Iron content: 2.000 max.
- Silicon content: 11.000 - 13.000%

Technical Specifications:

Tensile Strength (MPa) 295 min.

Yield Strength (MPa)	145 min.
Elongation (%)	2.5 min.

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

- Export Markets:
- USA
 - Brazil
 - Mexico

AI Layman Summary:

ASTM B275 S12B is a standard guideline that helps manufacturers categorize and identify specific nonferrous metals and alloys, particularly those involving copper and silicon. This means that industries can ensure they are using the right materials for their products, which is crucial for quality and safety. The guidelines specify the maximum allowable amounts of certain elements, such as copper and iron, and the range for silicon, to maintain consistency and reliability in manufacturing processes.

- Key benefits of using ASTM B275 S12B include:
- **Quality Assurance:** Ensures that the materials used in products like valves, pumps, and automotive components meet strict standards, leading to safer and more durable products.
 - **Versatility:** Applicable in various industries, including general engineering and automotive, making it a valuable resource for many manufacturers.
 - **Global Reach:** Recognized in markets such as the USA, Brazil, and Mexico, facilitating international trade and compliance with safety standards.

ASTM B275 SG100B

Category: Materials

Standard practice for codification of certain nonferrous metals and alloys.

- Key Features:
- Copper content: 0.600 max.
 - Iron content: 2.000 max.
 - Magnesium content: 0.400 - 0.600%

Technical Specifications:

Tensile Strength (MPa)	305 min.
Yield Strength (MPa)	170 min.
Elongation (%)	2.5 min.

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

- Export Markets:
- USA

- Brazil

AI Layman Summary:

ASTM B275 SG100B is a standard guideline used for identifying specific types of nonferrous metals and alloys, which are metals that do not contain significant amounts of iron. This standard helps manufacturers and engineers ensure that they are using the right materials for their products, especially in industries such as automotive and general engineering. Essentially, it provides a way to codify the properties of these metals, allowing for better quality control and consistency in production.

Key Features of ASTM B275 SG100B:

- Maximum copper content of 0.600%, ensuring durability and resistance to corrosion.
- Maximum iron content of 2.000%, which helps maintain the nonferrous properties of the alloy.
- Magnesium content between 0.400% and 0.600%, which enhances the alloy's strength and performance.

Benefits of using ASTM B275 SG100B include:

- Improved material selection for applications like valves and pumps, resulting in better performance.
- Enhanced reliability in engineering projects due to standardized material properties.
- Access to export markets such as the USA and Brazil, promoting broader business opportunities.

ASTM B292 - B

Category: Materials

Standard specification for copper alloy sand casting for general applications.

Key Features:

- Copper content: 84.000 - 89.000%
- Iron content: 0.250 max.
- Lead content: 0.300 - 1.000%

Technical Specifications:

Tensile Strength (MPa)	276 min.
Yield Strength (MPa)	138 min.
Elongation (%)	20 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B292 - B is a standard specification for a type of metal made primarily from copper, designed for various general applications. This metal is particularly useful in industries that require strong and durable materials, such as automotive and engineering sectors. It is made up of about 84% to 89% copper, with small amounts of iron and lead, which contribute to its strength and workability.

Key benefits include:

- ****High Strength****: With a minimum tensile strength of 276 MPa, it can withstand significant stress without breaking.

- **Versatility**: Suitable for a range of applications, including valves and pumps, making it a go-to choice for many engineers.
- **Durability**: The material's properties ensure it performs well in various environments, enhancing the longevity of the products made from it.

ASTM B292 - A

Category: Materials

Standard specification for copper alloy sand casting for general applications.

Key Features:

- Copper content: 85.000 - 90.000%
- Iron content: 0.250 max.
- Lead content: 0.090 max.

Technical Specifications:

Tensile Strength (MPa)	310 - 517
Yield Strength (MPa)	138 - 345
Elongation (%)	5 - 25

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B292 - A is a standard specification that defines the quality and characteristics of copper alloy sand casting materials used in various applications. This type of material is primarily composed of copper, making it a reliable choice for creating durable and sturdy products. With a copper content ranging from 85% to 90%, and minimal amounts of iron and lead, this alloy is designed for safety and performance in demanding environments.

Key Benefits:

- **High Strength**: With tensile strength between 310 to 517 MPa, the material can withstand significant stress, making it suitable for heavy-duty applications.
- **Versatile Use**: Ideal for manufacturing components like valves, pumps, and parts in the automotive and general engineering sectors, ensuring broad applicability across different industries.
- **Quality Assurance**: The specification ensures that the material meets high standards for reliability and safety, making it a trusted choice for manufacturers in the USA.

ASTM B30 2B

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 85.000 - 89.000%
- Iron content: 0.200 max.
- Lead content: 0.300 - 0.900

Technical Specifications:

Tensile Strength (MPa)	248 min.
Yield Strength (MPa)	110 min.
Elongation (%)	18 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 2B is a standard specification for high-quality copper alloys used in various industries. This material primarily consists of 85% to 89% copper, making it highly durable and effective for applications that require strength and resistance to corrosion. The copper alloy can also contain a small amount of iron and lead, which contributes to its performance in specific uses.

Key benefits of ASTM B30 2B include:

- **High Strength**: With a minimum tensile strength of 248 MPa and yield strength of 110 MPa, it can withstand significant stress and pressure.
- **Versatility**: It's suitable for a range of applications such as valves, pumps, and automotive components, making it a go-to choice for engineers and manufacturers.
- **Durability**: The material's composition ensures longevity, reducing the need for frequent replacements or repairs.

ASTM B30 C83600

Category: Materials

Standard specification for copper alloys ingot and casting for general engineering purpose.

Key Features:

- Copper content: 84.000 - 86.000%
- Iron content: 0.250 max.
- Lead content: 4.000 - 5.700

Technical Specifications:

Tensile Strength (MPa)	207 min.
Yield Strength (MPa)	97 min.
Elongation (%)	20 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C83600 is a type of copper alloy specifically designed for various engineering applications. This material is made primarily of copper, with a little bit of iron and lead added to enhance its properties. It is particularly strong and flexible, making it suitable for creating parts that need to withstand pressure and movement, like valves and pumps.

****Key Features:****

- Contains 84% to 86% copper, which gives it excellent conductivity and durability.
- Has a maximum iron content of 0.25%, ensuring it remains lightweight yet strong.
- Includes 4% to 5.7% lead, which improves machinability for easy shaping into complex designs.

****Benefits:****

- Offers high tensile and yield strength, making it reliable for demanding applications.
- Provides good elongation, allowing it to stretch without breaking, which is ideal for parts that experience bending.
- Suitable for a variety of industries, including automotive and general engineering, making it a versatile choice for manufacturers.

ASTM B30 C83800

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 82.000 - 83.500%
- Iron content: 0.250 max.
- Lead content: 5.800 - 6.800

Technical Specifications:

Tensile Strength (MPa)	205 min.
Yield Strength (MPa)	90 - 95
Elongation (%)	16 - 20

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C83800 is a type of copper alloy that is used to create ingots and castings for various industrial applications. This material is primarily made up of copper, with a content ranging from 82% to 83.5%. It also contains small amounts of iron and lead, which help enhance its properties. The alloy is known for its strength and flexibility, making it suitable for manufacturing components like valves and pumps, as well as for general engineering and automotive applications.

Key benefits of ASTM B30 C83800 include:

- High tensile strength (minimum of 205 MPa) ensures durability under pressure.
- Good yield strength (90 - 95 MPa) allows it to maintain shape under stress.

- Elongation of 16 - 20% provides flexibility, making it easier to work with during manufacturing.

Overall, this copper alloy is an excellent choice for companies in the USA looking for reliable materials for their engineering projects.

ASTM B30 C84200

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 78.000 - 82.000%
- Iron content: 0.350 max.
- Lead content: 2.000 - 2.800

Technical Specifications:

Tensile Strength (MPa)	221 min.
Yield Strength (MPa)	110 min.
Elongation (%)	13 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C84200 is a type of copper alloy that is primarily used to make various industrial components such as valves and pumps. It is designed to be strong and durable while maintaining good performance in different engineering applications. This material is made up of about 78% to 82% copper, with small amounts of iron and lead, ensuring it has the right balance of strength and workability.

Key Benefits:

- **High Strength**: With a minimum tensile strength of 221 MPa, it can withstand significant stress without breaking.
- **Versatile Use**: Ideal for general engineering applications, automotive parts, and various machinery, making it a go-to material in manufacturing.
- **Reliability**: Its composition allows it to perform consistently, ensuring long-lasting performance in demanding environments.

ASTM B30 C84400

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 79.000 - 82.000%
- Iron content: 0.350 max.
- Lead content: 6.300 - 7.700

Technical Specifications:

Tensile Strength (MPa)	200 min.
Yield Strength (MPa)	90 min.
Elongation (%)	18 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C84400 is a type of copper alloy that is made primarily from copper and is used to create strong and durable materials. This alloy contains between 79% to 82% copper and includes small amounts of iron and lead, which help to enhance its properties. It is commonly used in various applications such as valves, pumps, and automotive parts, making it a versatile choice in engineering and manufacturing.

Key benefits of ASTM B30 C84400 include:

- High tensile strength (minimum 200 MPa) ensuring it can withstand heavy loads without breaking.
- Good yield strength (minimum 90 MPa) allowing it to maintain its shape under pressure.
- Adequate elongation (minimum 18%) which means it can stretch without snapping, providing flexibility in use.

Overall, this copper alloy is an essential material for industries in the USA, offering reliability and performance for a wide range of engineering needs.

ASTM B30 5A

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 79.000 - 82.000%
- Iron content: 0.350 max.
- Lead content: 6.300 - 7.700

Technical Specifications:

Tensile Strength (MPa)	200 min.
Yield Strength (MPa)	90 min.
Elongation (%)	18 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 5A is a standard specification for a type of metal made from copper alloys, which are mixtures that provide enhanced properties for various applications. This material is primarily used in industries like automotive, engineering, and manufacturing of components such as valves and pumps. The copper content in this alloy is between 79% and 82%, ensuring good conductivity and resistance to corrosion, making it ideal for many mechanical uses.

- Key benefits of ASTM B30 5A include:
- **Strong and Durable**: With a minimum tensile strength of 200 MPa and yield strength of 90 MPa, it can withstand significant stress without breaking.
 - **Flexible**: It can stretch up to 18% before it breaks, allowing for various shapes and forms in manufacturing.
 - **Versatile Use**: Suitable for a wide range of applications including valves, pumps, and other engineering tasks, which makes it a valuable material in the production process.

ASTM B30 C84800

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 75.000 - 76.700%
- Iron content: 0.350 max.
- Lead content: 5.500 - 6.700

Technical Specifications:

Tensile Strength (MPa)	193 min.
Yield Strength (MPa)	83 min.
Elongation (%)	16 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C84800 is a specialized material composed primarily of copper, designed for creating durable and reliable metal components. It is often used in manufacturing parts like valves and pumps due to its strong, corrosion-resistant properties. This alloy contains a specific percentage of copper (between 75% to 76.7%), with small amounts of iron and lead, which enhance its performance in various applications.

- Key benefits of ASTM B30 C84800 include:
- **High Strength**: With a minimum tensile strength of 193 MPa, it can withstand significant stress without breaking.
 - **Versatility**: Suitable for a wide range of uses in general engineering and automotive industries, making it a go-to choice for many manufacturers.
 - **Corrosion Resistance**: Its copper-rich composition helps it resist rust and degradation, ensuring a longer lifespan for the components made from it.

ASTM B30 C85700

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 58.000 - 63.000%
- Iron content: 0.500 max.
- Lead content: 0.800 - 1.500

Technical Specifications:

Tensile Strength (MPa)	276 min.
Yield Strength (MPa)	97 min.
Elongation (%)	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C85700 is a type of copper alloy that is primarily used for making ingots and castings. This material is known for its durability and strength, making it ideal for various applications, including valves, pumps, and general engineering projects. The alloy contains a specific percentage of copper, along with controlled amounts of iron and lead, which contribute to its overall performance and reliability.

Key benefits of ASTM B30 C85700 include:

- **High Strength**: With a minimum tensile strength of 276 MPa and yield strength of 97 MPa, it can withstand significant stress and pressure.
- **Versatility**: Suitable for multiple use cases in industries like automotive and engineering, providing flexibility for manufacturers.
- **Controlled Composition**: The precise ratios of copper, iron, and lead ensure consistent quality and performance in finished products.

ASTM B30 C90300

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 86.000 - 89.000%
- Iron content: 0.150 max.
- Lead content: 0.250 max.

Technical Specifications:

Tensile Strength (MPa)	241 min.
Yield Strength (MPa)	124 min.
Elongation (%)	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C90300 is a type of copper alloy specifically designed for making high-quality ingots and castings. This material is notable for its high copper content, which ranges from 86% to 89%, ensuring excellent conductivity and durability. It also has very low levels of iron and lead, making it safe and reliable for various applications.

The benefits of using ASTM B30 C90300 include:

- **Strength**: It has a minimum tensile strength of 241 MPa and a yield strength of 124 MPa, which means it can withstand significant stress without breaking.
- **Versatility**: This alloy is suitable for a variety of uses, including valves, pumps, general engineering projects, and automotive components, making it a versatile choice for manufacturers.
- **Market Availability**: Primarily exported to the USA, it meets strict quality standards, ensuring that it can be relied upon for important industrial applications.

ASTM B30 C86300

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 60.000 - 66.000%
- Iron content: 2.000 - 4.000
- Lead content: 0.100 max.

Technical Specifications:

Tensile Strength (MPa)	758 min.
Yield Strength (MPa)	414 min.
Elongation (%)	12 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C86300 is a high-quality copper alloy used primarily for making various industrial components. This material is made up of 60% to 66% copper, along with small amounts of iron and lead, which give it desirable properties for different applications. It is specifically designed for casting and ingots, making it versatile for manufacturing parts that require durability and strength.

Some key benefits of ASTM B30 C86300 include:

- **Strong Performance**: It has a minimum tensile strength of 758 MPa and a yield strength of 414 MPa, ensuring that parts made from this material can withstand significant stress and pressure.
- **Versatile Applications**: It can be used in valves, pumps, general engineering projects, and the automotive industry, making it a go-to choice for various manufacturing needs.
- **Reliable Quality**: The controlled composition ensures consistent quality, which is crucial for businesses that rely on reliable materials for their products.

ASTM B30 C89320

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 87.000 - 91.000%
- Iron content: 0.200 max.
- Lead content: 0.090 max.

Technical Specifications:

Tensile Strength (MPa)	241 min.
Yield Strength (MPa)	124 min.
Elongation (%)	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C89320 is a type of copper alloy used primarily for making ingots and castings. This standard specification outlines the key features and properties of this material, ensuring it's suitable for various applications. It has a high copper content (between 87% and 91%) which gives it excellent conductivity and durability, making it ideal for manufacturing components that need to withstand stress and wear.

Key benefits of ASTM B30 C89320 include:

- **Strong and Durable**: With a minimum tensile strength of 241 MPa and yield strength of 124 MPa, it can handle significant force without breaking.
- **Versatile Applications**: It's suitable for use in valves, pumps, automotive components, and general engineering projects, showcasing its adaptability across different industries.
- **Low Impurities**: The low maximum content of iron and lead ensures the alloy maintains its quality and performance in various environments.

ASTM B30 C90700

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 86.000 - 90.000%
- Iron content: 0.150 max.
- Lead content: 0.500 max.

Technical Specifications:

Tensile Strength (MPa)	280 min.
Yield Strength (MPa)	160 min.
Elongation (%)	10 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C90700 is a high-quality copper alloy that is primarily used for making various components in industries like automotive and engineering. This material is crafted under strict specifications to ensure its strength and durability, making it ideal for applications such as valves and pumps. With a copper content ranging between 86% to 90%, it provides excellent conductivity and resistance to corrosion.

Key features of ASTM B30 C90700 include:

- High copper content for superior performance.
- Low levels of iron and lead, enhancing the material's reliability and safety.
- Strong tensile and yield strengths, ensuring it can withstand heavy use without deforming.

This copper alloy not only meets high standards for performance but also offers benefits such as:

- Versatility for use in various engineering applications.
- Enhanced longevity and reduced maintenance costs due to its durable nature.
- Compliance with industry standards for safety and effectiveness, making it a trusted choice for manufacturers.

ASTM B30 C90500

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 86.000 - 89.000%
- Iron content: 0.150 max.
- Lead content: 0.250 max.

Technical Specifications:

Tensile Strength (MPa)	276 min.
Yield Strength (MPa)	124 min.
Elongation (%)	20 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C90500 is a type of copper alloy that is specifically designed for making ingots and castings. This material is primarily composed of about 86% to 89% copper, ensuring it has good strength and durability. It also contains minimal amounts of iron and lead, making it suitable for various industrial applications. This standard specification ensures that manufacturers can rely on consistent quality when using this alloy in their products.

Key Benefits of ASTM B30 C90500:

- **Strong and Durable**: With a minimum tensile strength of 276 MPa, it can withstand significant stress and strain, making it ideal for heavy-duty applications.
- **Versatile Uses**: Commonly used in valves, pumps, and general engineering projects, as well as in the automotive sector.
- **High Quality**: The strict content specifications help ensure that the resulting products maintain high performance and reliability in their applications.

ASTM B30 C91000

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 84.000 - 86.000%
- Iron content: 0.100 max.
- Lead content: 0.200 max.

Technical Specifications:

Tensile Strength (MPa) 207 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C91000 is a standard specification for a type of copper alloy that is primarily used to create ingots and cast parts. This material is composed mostly of copper, making it highly conductive and resistant to corrosion, which are vital properties for many industrial applications. The alloy contains a specific range of copper (between 84% and 86%), with very low amounts of iron and lead, ensuring durability and safety in use.

Key benefits of ASTM B30 C91000 include:

- **High conductivity**: Ideal for electrical components.
- **Corrosion resistance**: Suitable for outdoor and marine applications.

- **Versatility:** Used in valves, pumps, and automotive parts, making it valuable in various engineering fields. This product is particularly exported to the USA, where it serves numerous industries that rely on high-quality metal components.

ASTM B30 C92200

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 86.000 - 89.000%
- Iron content: 0.200 max.
- Lead content: 1.000 - 1.8000

Technical Specifications:

Tensile Strength (MPa)	234 min.
Yield Strength (MPa)	110 min.
Elongation (%)	22 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C92200 is a type of copper alloy used for making strong and durable metal parts. It is primarily an ingot and casting material, which means it is melted down and shaped into various products. This alloy contains a high percentage of copper (between 86% and 89%), along with small amounts of iron and lead, making it suitable for various applications, especially in industries that require reliable materials like valves and pumps.

Benefits of ASTM B30 C92200 include:

- **High Strength:** With a minimum tensile strength of 234 MPa, it can withstand significant stress, making it ideal for heavy-duty applications.
- **Versatile Use:** This alloy is perfect for general engineering projects and automotive parts, ensuring it can meet the needs of different industries.
- **Good Workability:** Its properties allow for easy casting and shaping, which helps manufacturers create complex designs efficiently.

ASTM B30 C92500

Category: Materials

Standard specification for copper alloys ingot and casting.

Key Features:

- Copper content: 85.000 - 88.000%
- Iron content: 0.200 max.
- Lead content: 1.000 - 1.500

Technical Specifications:

Tensile Strength (MPa)	276 min.
Yield Strength (MPa)	165 min.
Elongation (%)	10 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

Export Markets:

- USA

AI Layman Summary:

ASTM B30 C92500 is a high-quality copper alloy used primarily in manufacturing processes. This material contains a significant amount of copper (between 85% and 88%), along with controlled amounts of iron and lead, making it robust and suitable for various applications. It is designed to meet specific standards, ensuring that it has a minimum tensile strength of 276 MPa and a yield strength of 165 MPa, which means it can withstand significant stress without deforming. Additionally, it has a minimum elongation of 10%, indicating that it can be stretched without breaking, which is crucial for many applications.

The benefits of using ASTM B30 C92500 include:

- **Durability**: Its strong composition allows it to withstand harsh conditions, making it ideal for use in valves, pumps, and automotive components.
- **Versatility**: This alloy can be used in a variety of engineering projects, providing flexibility for manufacturers.
- **Quality Assurance**: By adhering to established standards, users can trust the material's performance and longevity.

ASTM B505 C95410

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Aluminium Al %	10.000 - 11.500
Iron Fe %	3.000 - 5.000
Manganese Mn %	0.500 max.
Ni + Co Ni% + Co%	1.500 - 2.500
Copper Cu %	83.000 min.
Tensile Strength in Mpa	586 - 655
Yield Strength in Mpa	221 - 310
Elongation in %	10 - 12

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 C95410 is a type of copper alloy specifically designed for continuous casting, which means it can be formed into shapes without needing to be reshaped later. This alloy is made primarily from copper, with significant amounts of aluminum and iron, making it strong and durable. It is particularly useful in various industries, including automotive, general engineering, and for components like valves and pumps.

- Key benefits of ASTM B505 C95410 include:
- **High Strength**: It has a tensile strength between 586 to 655 MPa, ensuring that it can withstand significant stress and pressure.
 - **Versatile Applications**: Suitable for a wide range of uses, from automotive parts to general engineering applications, making it a flexible material for manufacturers.
 - **Durability**: With a yield strength of 221 to 310 MPa and good elongation properties, this alloy can endure wear and tear, extending the lifespan of the products made from it.

ASTM B505 C95520

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Aluminium Al %	10.500 - 11.500
Cobalt Co %	0.200 max.
Iron Fe %	4.000 - 5.500
Manganese Mn %	1.500 max.
Copper Cu %	74.500 min.
Tensile Strength in Mpa	862 min.
Yield Strength in Mpa	655 min.
Elongation in %	2 min.

- Use Cases:
- Valve
 - Pump
 - General Engineering
 - Automotive

AI Layman Summary:

ASTM B505 C95520 is a type of copper alloy that is designed for continuous casting, which is a manufacturing process that creates metal shapes by pouring molten metal into molds. This specific alloy contains a mix of copper, aluminum, cobalt, iron, and manganese, making it strong and durable for various industrial applications. It is particularly known for its good mechanical properties, including high tensile and yield strength, which means it can withstand significant stress without breaking.

- The benefits of using ASTM B505 C95520 include:
- **Versatile Applications**: Ideal for making valves, pumps, and components in automotive and general engineering sectors.
 - **High Strength**: With a minimum tensile strength of 862 MPa, it offers excellent durability for demanding environments.
 - **Corrosion Resistance**: The alloy's composition provides resistance against corrosion, extending the life of products made from it.

ASTM B505 C96900

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Magnesium Mg %	0.150 max.
Manganese Mn %	0.050 - 0.300
Ni + Co Ni% + Co%	14.500 - 15.500
Copper Cu %	Balance
Tensile Strength in Mpa	758 min.
Yield Strength in Mpa	724 min.
Elongation in %	4 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 C96900 is a type of copper alloy that is made through a continuous casting process. This means that the metal is poured into a mold, allowing it to solidify and form a solid shape without interruption. The composition of this alloy includes a mix of copper, magnesium, manganese, and nickel, which gives it special properties that make it suitable for various industrial applications.

Benefits of ASTM B505 C96900 include:

- **High Strength**: With a minimum tensile strength of 758 MPa, it can withstand significant stress and pressure, making it ideal for demanding applications.
- **Versatility**: This alloy is used in a range of industries, including automotive, valves, pumps, and general engineering, demonstrating its adaptability to different needs.
- **Durability**: The alloy's composition contributes to its longevity and resistance to wear, ensuring that products made from it last longer and perform reliably.

ASTM B505 C96400

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Carbon C %	0.150 max.
Iron Fe %	0.250 - 1.500
Manganese Mn %	1.500 max.
Ni + Co Ni% + Co%	28.000 - 32.000
Copper Cu %	Balance
Tensile Strength in Mpa	448 min.
Yield Strength in Mpa	241 min.
Elongation in %	25 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 C96400 is a type of copper alloy that is designed for continuous casting, which means it can be produced in long, uninterrupted lengths. This material is primarily made of copper, but it also contains small amounts of other elements like iron and manganese. It is known for its strength and durability, making it suitable for a variety of applications, including valves, pumps, and automotive components.

The key benefits of using ASTM B505 C96400 include:

- **High Strength**: It has a minimum tensile strength of 448 MPa, ensuring it can withstand significant stress.
- **Versatile Applications**: This alloy is ideal for various engineering applications, helping manufacturers produce reliable and long-lasting products.
- **Good Workability**: Its composition allows for easy shaping and forming, which is crucial in manufacturing processes.

ASTM B505 High - Leaded Tin Bronze UNS C93200

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	6.300 - 7.500
Lead Pb %	6.000 - 8.000
Copper Cu %	81.000 - 85.000
Tensile Strength in Mpa	241 min.
Yield Strength in Mpa	138 min.
Elongation in %	10 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High-Leaded Tin Bronze UNS C93200 is a special type of metal alloy made primarily from copper, tin, and lead. This alloy is created through a continuous casting process, which means it is produced in a steady stream for consistent quality. It is designed to be strong and durable, making it suitable for various applications like valves, pumps, and parts used in general engineering and automotive industries.

Key benefits of this alloy include:

- **Strength and Durability**: With a minimum tensile strength of 241 MPa, it can withstand significant stress, ideal for mechanical uses.
- **Corrosion Resistance**: The combination of tin and lead provides good resistance against wear and tear, extending the life of components.
- **Versatility**: Its applications range from everyday machinery to specialized automotive parts, making it a valuable material across multiple industries.

ASTM B505 High - Leaded Tin Bronze UNS C93400

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	7.000 - 9.000
Lead Pb %	7.000 - 9.000
Copper Cu %	82.000 - 85.000
Tensile Strength in Mpa	234 min.
Yield Strength in Mpa	138 min.
Elongation in %	8 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C93400 is a specialized type of metal alloy primarily made of copper, tin, and lead. This alloy is known for its excellent durability and strength, making it suitable for various applications, especially in industries such as automotive and engineering. It is designed to be cast continuously, which means it can be produced in long, uniform shapes, making it easier to work with in manufacturing processes.

- **Benefits of ASTM B505 High - Leaded Tin Bronze UNS C93400:****
- ****Strong and Durable:**** With a minimum tensile strength of 234 MPa, this alloy can withstand significant stress and pressure, making it ideal for heavy-duty applications.
 - ****Versatile Applications:**** It is commonly used in valves, pumps, and other engineering components, ensuring reliability in various machinery and automotive parts.
 - ****Easy to Work With:**** The continuous casting process enhances its machinability, allowing manufacturers to create precise components efficiently.

ASTM B505 High - Leaded Tin Bronze UNS C93600

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	6.000 - 8.000
Lead Pb %	11.000 - 13.000
Copper Cu %	79.000 - 83.000
Tensile Strength in Mpa	227 min.
Yield Strength in Mpa	138 min.
Elongation in %	10 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C93600 is a special type of metal alloy made primarily of copper, tin, and lead. It is designed for continuous casting, which means it can be shaped into long, uniform pieces that are easy to work with in manufacturing. This alloy is particularly strong, with a minimum tensile strength of 227 MPa, making it suitable for heavy-duty applications. It contains about 6% to 8% tin and 11% to 13% lead, giving it unique properties that enhance its durability and resistance to wear.

The main benefits of using ASTM B505 High - Leaded Tin Bronze include:

- **Versatility**: It is ideal for a variety of applications such as valves, pumps, and automotive components, making it a great choice for engineers in different industries.
- **Strength and Durability**: Its high tensile and yield strength ensures that products made from this alloy can withstand significant stress and wear, extending their lifespan and reducing maintenance costs.
- **Ease of Machining**: The lead content helps in machining processes, allowing for precise shapes and finishes, which is essential in engineering and manufacturing.

ASTM B505 High - Leaded Tin Bronze UNS C93700

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	9.000 - 11.000
Lead Pb %	8.000 - 11.000
Copper Cu %	78.000 - 82.000
Tensile Strength in Mpa	241 min.
Yield Strength in Mpa	138 min.
Elongation in %	6 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C93700 is a type of metal alloy that combines copper, tin, and lead, making it strong and durable. This alloy is specifically designed for continuous casting, which means it can be formed into solid shapes and used in various industries. It is particularly valuable because of its excellent mechanical properties, including high tensile strength and yield strength, which means it can withstand significant stress without breaking.

Key Benefits:

- **Durable and Strong**: With a minimum tensile strength of 241 MPa, this alloy is built to last, making it ideal for heavy-duty applications.
- **Versatile Use**: It's commonly used in valves, pumps, general engineering, and automotive parts, showcasing its adaptability across different sectors.

- ****High Lead and Tin Content****: The high percentages of lead (8-11%) and tin (9-11%) enhance its machinability and resistance to wear, ensuring that components made from this alloy perform well over time.

ASTM B505 High - Leaded Tin Bronze UNS C93800

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	6.300 - 7.500
Lead Pb %	13.000 - 16.000
Copper Cu %	75.000 - 79.000
Tensile Strength in Mpa	172 min.
Yield Strength in Mpa	110 min.
Elongation in %	5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C93800 is a type of metal alloy primarily made of copper, tin, and lead. This specific alloy is designed for continuous casting, which means it is produced in a way that allows it to be formed into shapes for various industrial uses. Its composition includes 6.3% to 7.5% tin, 13% to 16% lead, and 75% to 79% copper, providing a balance of strength and corrosion resistance.

This alloy is particularly useful in applications such as valves, pumps, and general engineering, making it a versatile choice for many industries, including automotive. The benefits of using ASTM B505 include:

- High tensile strength (minimum 172 MPa) and yield strength (minimum 110 MPa), ensuring durability under stress.
- Good elongation properties (minimum 5%), allowing it to be shaped without breaking.
- Excellent machinability due to the lead content, making it easier to work with in manufacturing processes.

ASTM B505 High - Leaded Tin Bronze UNS C93900

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	5.000 - 7.000
Lead Pb %	14.000 - 18.000
Copper Cu %	76.500 - 79.500
Tensile Strength in Mpa	172 min.
Yield Strength in Mpa	110 min.
Elongation in %	5 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C93900 is a specialized metal alloy primarily composed of copper, tin, and lead. This alloy is designed for continuous casting, which means it can be produced in long lengths and is ideal for various industrial applications. Its unique composition allows it to be durable and resilient under pressure, making it suitable for use in valves, pumps, automotive components, and general engineering tasks.

Key benefits of ASTM B505 include:

- **High Strength**: With a minimum tensile strength of 172 MPa, it can withstand significant stress without breaking.
- **Corrosion Resistance**: The presence of tin and lead enhances its resistance to corrosion, ensuring a longer lifespan in harsh environments.
- **Versatile Applications**: This alloy is perfect for manufacturing critical components in multiple industries, including automotive and engineering, providing reliability and performance.

ASTM B505 High - Leaded Tin Bronze UNS C94000

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	12.000 - 14.000
Lead Pb %	14.000 - 16.000
Copper Cu %	69.000 - 72.000
Tensile Strength in Mpa	-
Yield Strength in Mpa	-
Elongation in %	-

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C94000 is a type of copper alloy that is primarily used in continuous casting processes. It is made up of a blend of tin, lead, and copper, with specific percentages of each metal to enhance its properties. This alloy is known for its durability and ability to withstand wear, making it an ideal choice for various industrial applications.

Here are some key points about this product:

- **Composition**: Contains 12-14% tin, 14-16% lead, and 69-72% copper, which contribute to its strength and corrosion resistance.
- **Applications**: Commonly used in manufacturing valves, pumps, and parts for automotive and general engineering purposes, due to its excellent machinability and structural integrity.
- **Benefits**: Offers high performance and longevity, reducing maintenance needs and costs in industrial settings.

ASTM B505 High - Leaded Tin Bronze UNS C94300

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	4.500 - 6.000
Lead Pb %	23.000 - 27.000
Copper Cu %	67.000 - 72.000
Tensile Strength in Mpa	145 min.
Yield Strength in Mpa	103 min.
Elongation in %	-

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 High - Leaded Tin Bronze UNS C94300 is a specialized type of metal alloy made primarily from copper, tin, and lead. This alloy is used in continuous casting processes, which means it can be shaped into various forms for manufacturing. Its key components include 4.5% to 6% tin, 23% to 27% lead, and 67% to 72% copper. This combination gives the material unique properties that make it ideal for specific applications.

The benefits of using this alloy include:

- **Strength**: It has a minimum tensile strength of 145 MPa, making it durable and reliable in demanding environments.
- **Versatility**: Commonly used in valves, pumps, and general engineering applications, as well as in the automotive industry, showcasing its wide range of uses.
- **Ease of Machining**: The lead content enhances machinability, allowing for easier and more efficient manufacturing processes.

ASTM B505 Leaded Naval Brass UNS C85700

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	0.500 - 1.500
Lead Pb %	0.800 - 1.500
Copper Cu %	58.000 - 64.000
Tensile Strength in Mpa	276 min.
Yield Strength in Mpa	97 min.
Elongation in %	15 min.

Use Cases:

- Valve

- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Naval Brass UNS C85700 is a type of metal alloy primarily made of copper, with added lead and tin. This alloy is designed for continuous casting, which means it can be produced in long, uniform shapes that are ideal for various manufacturing processes. The combination of metals gives it unique properties that make it suitable for a range of applications.

- Here are some key points about its benefits:
- **Strength and Durability**: This brass has a minimum tensile strength of 276 MPa and a yield strength of 97 MPa, making it robust for demanding environments.
 - **Versatile Use**: It's commonly used in valves, pumps, and general engineering, as well as in the automotive industry, showcasing its adaptability.
 - **Good Workability**: The presence of lead in the alloy improves machinability, meaning it can be easily shaped and formed during manufacturing processes.

ASTM B505 Leaded Red Brass UNS C83600

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	4.000 - 6.000
Lead Pb %	4.000 - 6.000
Copper Cu %	84.000 - 86.000
Tensile Strength in Mpa	248 min.
Yield Strength in Mpa	131 min.
Elongation in %	15 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Red Brass UNS C83600 is a type of metal that is primarily made of copper, with added tin and lead to enhance its properties. It is designed for continuous casting, which means it can be produced in long shapes without needing to be melted and reformed repeatedly. This material is strong and durable, making it suitable for various applications in different industries.

- Key benefits of ASTM B505 Leaded Red Brass include:
- **High Strength**: With a minimum tensile strength of 248 MPa, it can withstand significant stress and pressure.
 - **Versatile Applications**: It is used in valves, pumps, automotive parts, and general engineering, making it a reliable choice for many manufacturing needs.
 - **Good Workability**: The alloy's composition allows it to be easily shaped and processed, which is essential for producing complex components efficiently.

ASTM B505 Leaded Nickel - Tin Bronze UNS C92900

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	9.000 - 11.000
Lead Pb %	2.000 - 3.200
Copper Cu %	82.000 - 86.000
Tensile Strength in Mpa	310 min.
Yield Strength in Mpa	172 min.
Elongation in %	8 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Nickel - Tin Bronze UNS C92900 is a special type of metal used primarily in manufacturing various components. It's made from a blend of copper, tin, and lead, which gives it unique properties that make it strong and durable. This metal is commonly used in applications like valves, pumps, and automotive parts due to its excellent strength and resistance to wear and tear.

****Key Benefits:****

- ****Strength and Durability:**** With a minimum tensile strength of 310 MPa, this alloy can withstand significant stress and pressure.
- ****Versatile Applications:**** Perfect for a range of industries including automotive and general engineering, making it a reliable choice for many manufacturers.
- ****Corrosion Resistance:**** The blend of metals helps prevent rust and deterioration, ensuring long-lasting performance in various environments.

ASTM B505 Leaded Semi - Red Brass UNS C84200

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	4.000 - 6.000
Lead Pb %	2.000 - 3.000
Copper Cu %	78.000 - 82.000
Tensile Strength in Mpa	221 min.
Yield Strength in Mpa	110 min.
Elongation in %	13 min.

Use Cases:

- Valve
- Pump
- General Engineering

- Automotive

AI Layman Summary:

ASTM B505 Leaded Semi - Red Brass UNS C84200 is a type of metal alloy primarily made of copper, tin, and lead, designed for use in various industrial applications. It is produced through a continuous casting process, which ensures a uniform and high-quality material. This alloy is particularly valued for its strength and durability, making it a reliable choice for components in machinery and other engineering solutions.

Key benefits of this material include:

- **High Strength**: With a minimum tensile strength of 221 MPa, it can withstand considerable stress, making it suitable for demanding applications such as valves and pumps.
- **Corrosion Resistance**: The composition of copper and tin helps resist corrosion, ensuring longevity in various environments.
- **Versatile Applications**: Ideal for automotive parts, general engineering, and various mechanical components, making it a versatile choice for manufacturers.

ASTM B505 Leaded Semi - Red Brass UNS C84800

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	2.000 - 3.000
Lead Pb %	5.500 - 7.000
Copper Cu %	75.000 - 77.000
Tensile Strength in Mpa	207 min.
Yield Strength in Mpa	103 min.
Elongation in %	16 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Semi - Red Brass UNS C84800 is a type of copper alloy that is commonly used for manufacturing various engineering components. This material is specifically designed for continuous casting, which means it can be produced in long, uniform shapes that are easy to work with. It contains a blend of metals, including copper, lead, and tin, which gives it unique properties such as good strength and corrosion resistance.

Key benefits of using this brass include:

- **Durability**: With a tensile strength of at least 207 MPa, it can withstand significant stress without breaking.
- **Versatility**: It's suitable for a range of applications like valves, pumps, and automotive parts, making it a go-to material for many engineering projects.
- **Workability**: The alloy's composition allows for easy machining and shaping, which is essential in manufacturing processes.

ASTM B505 Leaded Semi - Red Brass UNS C84400

Category: Copper Alloy Continuous Casting

Standard Specification for Copper Alloy Continuous Casting.

Technical Specifications:

Tin Sn %	2.300 - 3.500
Lead Pb %	6.000 - 8.000
Copper Cu %	78.000 - 82.000
Tensile Strength in Mpa	207 min.
Yield Strength in Mpa	103 min.
Elongation in %	16 min.

Use Cases:

- Valve
- Pump
- General Engineering
- Automotive

AI Layman Summary:

ASTM B505 Leaded Semi - Red Brass UNS C84400 is a type of copper alloy that is specifically designed for continuous casting processes. This means it can be made into various shapes and forms, making it ideal for a wide range of applications. It contains a mix of copper, lead, and tin, which gives it unique properties that enhance its performance in different environments.

Key benefits of this material include:

- **Durability**: With a minimum tensile strength of 207 MPa, it can withstand significant stress, making it suitable for demanding applications like valves and pumps.
- **Versatility**: Its composition allows it to be used in various fields, including automotive and general engineering, providing flexibility for manufacturers.
- **Ease of manufacturing**: The continuous casting process allows for efficient production, helping businesses save time and resources.

ASTM B505 Leaded Tin Bronze UNS C92200

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 5.500 - 6.500%, Lead 1.000 - 2.000%, Zinc 3.000 - 5.000%
- Mechanical Properties: Tensile Strength 262 min. MPa, Yield Strength 131 min. MPa, Elongation 18 min.

Technical Specifications:

Iron	0.250 max.
Antimony	0.250 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	86.000 - 90.000%

AI Layman Summary:

ASTM B505 Leaded Tin Bronze UNS C92200 is a type of metal alloy made primarily from copper, tin, and lead. This material is known for its strength and durability, making it suitable for various industrial applications. The alloy contains specific percentages of tin (5.5% to 6.5%), lead (1% to 2%), and zinc (3% to 5%), which together give it unique properties that enhance its performance in different environments.

- Key benefits of this alloy include:
- **High Strength**: Provides strong resistance to wear and tear, ensuring longevity in use.
 - **Excellent Casting Properties**: Can be easily shaped and molded, which is essential for manufacturing complex parts.
 - **Corrosion Resistance**: Offers protection against rust and degradation, making it ideal for use in harsh conditions.

This alloy is often used in applications where durability and reliability are crucial, such as in machinery parts and construction materials.

ASTM B505 Leaded Tin Bronze UNS C92300

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 7.500 - 9.000%, Lead 0.300 - 1.000%, Zinc 2.500 - 5.000%
- Mechanical Properties: Tensile Strength 276 min. MPa, Yield Strength 131 min. MPa, Elongation 16 min.

Technical Specifications:

Iron	0.250 max.
Antimony	0.250 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	85.000 - 89.000%

AI Layman Summary:

ASTM B505 Leaded Tin Bronze UNS C92300 is a specialized metal alloy made primarily of copper, tin, and lead, designed for various applications in industry. It combines the benefits of strength and durability with excellent machinability, making it a popular choice for manufacturing components that require precision and reliability. The alloy consists of around 85-89% copper, with additional elements like tin (7.5-9%), lead (0.3-1%), and zinc, which enhance its properties.

- Key features include:
- High tensile and yield strength, ensuring it can withstand significant stress and pressure.
 - Good elongation, allowing it to be shaped and formed easily without breaking.
 - Low levels of impurities, ensuring high quality and performance in applications.

This material is ideal for industries that need strong, reliable metal parts, as it offers a balance of mechanical strength and ease of machining, making it easier to produce intricate designs and components efficiently.

ASTM B505 Nickel - Aluminium Bronze UNS C95520

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 0.250 max., Lead 0.030 max., Zinc 0.300 max.
- Mechanical Properties: Tensile Strength 862 min. MPa, Yield Strength 655 min. MPa, Elongation 8 min.

Technical Specifications:

Iron	4.000 - 5.500%
Manganese	1.500 max.
Aluminium	10.000 - 11.500%
Copper	74.500 min.

AI Layman Summary:

ASTM B505 Nickel - Aluminium Bronze UNS C95520 is a specialized metal alloy that combines copper with nickel and aluminum, making it particularly strong and resistant to corrosion. This material is often used in various industrial applications, thanks to its impressive mechanical properties and ability to withstand harsh environments.

Key Benefits:

- ****Strong and Durable****: With a minimum tensile strength of 862 MPa, this alloy is designed to handle significant stress without breaking.
- ****Corrosion Resistance****: The unique composition of this alloy helps it resist rust and wear, making it ideal for marine and industrial uses.
- ****Versatile Applications****: While specific use cases aren't listed, this type of alloy is typically used in manufacturing components that require both strength and resistance to environmental factors.

ASTM B505 Leaded Tin Bronze UNS C92800

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 15.000 - 17.000%, Lead 4.000 - 6.000%, Zinc 0.800 max.
- Mechanical Properties: Tensile Strength - , Yield Strength - , Elongation -

Technical Specifications:

Iron	0.200 max.
Antimony	0.250 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	78.000 - 82.000%

AI Layman Summary:

ASTM B505 Leaded Tin Bronze UNS C92800 is a special type of metal made primarily from copper, tin, and lead, designed for various industrial applications. It's a part of a category known as non-ferrous copper alloys,

meaning it doesn't contain iron and is resistant to rust. This metal is notable for its strength and durability, making it ideal for creating parts that need to withstand wear and tear over time.

- **Key Benefits:****
- ****Strong and Durable:**** With a good mix of tin and lead, this bronze alloy offers excellent mechanical strength, which means it can handle heavy loads without breaking.
 - ****Versatile Applications:**** Although specific use cases aren't listed, materials like this are commonly used in manufacturing machinery parts, electrical components, and plumbing fittings due to their reliable performance and resistance to corrosion.
 - ****Easy to Work With:**** The alloy can be cast into different shapes, making it easier for manufacturers to create customized parts for various industrial needs.

ASTM B505 Nickel - Phosphor Bronze UNS C92500

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 10.000 - 12.000%, Lead 1.000 - 1.500%, Zinc 0.500 max.
- Mechanical Properties: Tensile Strength 276 min. MPa, Yield Strength 165 min. MPa, Elongation 10 min.

Technical Specifications:

Iron	0.300 max.
Antimony	0.250 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	85.000 - 88.000%

AI Layman Summary:

ASTM B505 Nickel - Phosphor Bronze UNS C92500 is a type of copper alloy that is specifically designed for various industrial applications. It is known for its excellent mechanical properties and corrosion resistance, making it a reliable choice for manufacturers. This material is primarily made up of copper, with significant amounts of tin and small quantities of lead and zinc, which enhance its durability and strength.

- Key benefits of this alloy include:
- ****High Strength****: With a minimum tensile strength of 276 MPa, it can withstand heavy loads without deforming.
 - ****Corrosion Resistance****: The special composition of the alloy helps prevent rust and degradation over time, ensuring longevity in various environments.
 - ****Manufacturing Versatility****: This material can be easily shaped and molded, making it suitable for numerous applications in the manufacturing sector.

ASTM B505 Nickel - Tin Bronze UNS C94700

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 4.500 - 6.000%, Lead 0.100 max., Zinc 1.000 - 2.500%

- Mechanical Properties: Tensile Strength 310 min. MPa, Yield Strength 138 min. MPa, Elongation 25 min.

Technical Specifications:

Iron	0.250 max.
Antimony	0.150 max.
Sulphur	0.050 max.
Phosphorus	0.050 max.
Aluminium	0.005 max.
Manganese	0.200 max.
Copper	85.000 - 90.000%

AI Layman Summary:

ASTM B505 Nickel - Tin Bronze UNS C94700 is a specialized metal alloy made primarily of copper, tin, and small amounts of other elements. This material is designed for a variety of industrial applications due to its excellent strength and durability. It is particularly useful in situations where resistance to wear and corrosion is crucial.

Key benefits of this alloy include:

- **Strong and Durable**: With a minimum tensile strength of 310 MPa, it can withstand heavy loads and pressures.
- **Versatile Applications**: Suitable for many non-ferrous applications, making it a reliable choice for manufacturers.
- **Controlled Composition**: The specific percentages of tin, lead, and zinc ensure consistent quality and performance, which means fewer failures and longer-lasting products.

ASTM B505 Tin Bronze UNS C90300

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 7.500 - 9.000%, Lead 0.300 max., Zinc 3.000 - 5.000%
- Mechanical Properties: Tensile Strength 303 min. MPa, Yield Strength 152 min. MPa, Elongation 18 min.

Technical Specifications:

Iron	0.200 max.
Antimony	0.200 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	86.000 - 89.000%

AI Layman Summary:

ASTM B505 Tin Bronze UNS C90300 is a specialized type of metal alloy primarily made from copper, tin, and a small amount of lead and zinc. This alloy is known for its excellent strength and durability, making it ideal for various applications in industries that require robust materials. The tin content provides corrosion resistance,

while the lead and zinc enhance its workability, allowing it to be shaped and molded easily into different forms.

Key benefits of ASTM B505 Tin Bronze UNS C90300 include:

- **Strong and durable:** With a minimum tensile strength of 303 MPa, this alloy can withstand significant stress and strain.
- **Corrosion-resistant:** The tin composition helps protect the material from environmental damage, extending its lifespan.
- **Versatile:** This material can be used in various applications, thanks to its favorable mechanical properties and chemical composition.

ASTM B505 Leaded Tin Bronze UNS C92700

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 9.000 - 11.000%, Lead 1.000 - 2.500%, Zinc 0.700 max.
- Mechanical Properties: Tensile Strength 252 min. MPa, Yield Strength 138 min. MPa, Elongation 8 min.

Technical Specifications:

Iron	0.200 max.
Antimony	0.250 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	86.000 - 89.000%

AI Layman Summary:

ASTM B505 Leaded Tin Bronze UNS C92700 is a special type of metal alloy primarily made up of copper, tin, and lead. It is designed for use in various applications where durability and resistance to wear are important. This alloy is created through a continuous casting process, which helps maintain its quality and consistency. The key components include 9-11% tin for strength, 1-2.5% lead for improved machinability, and small amounts of zinc and other elements to enhance performance.

Benefits of using ASTM B505 Leaded Tin Bronze UNS C92700 include:

- ****High Strength**:** With a minimum tensile strength of 252 MPa, this alloy can withstand significant stress.
- ****Good Machinability**:** The lead content allows for easier shaping and cutting, making it ideal for manufacturing.
- ****Corrosion Resistance**:** The copper base provides good resistance to corrosion, extending the life of products made from this alloy.

ASTM B505 Tin Bronze UNS C90500

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 9.000 - 11.000%, Lead 0.300 max., Zinc 1.000 - 3.000%
- Mechanical Properties: Tensile Strength 303 min. MPa, Yield Strength 172 min. MPa, Elongation 10 min.

Technical Specifications:

Iron	0.200 max.
Antimony	0.200 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	86.000 - 89.000%

AI Layman Summary:

ASTM B505 Tin Bronze UNS C90500 is a type of metal alloy primarily made of copper, with tin added to enhance its strength and resistance to corrosion. This material is designed for continuous casting, which means it can be shaped into various forms while maintaining its excellent properties. The alloy is notable for its specific chemical composition, including 9-11% tin and minimal amounts of lead and zinc, making it suitable for a variety of industrial applications.

- Key benefits of ASTM B505 Tin Bronze UNS C90500 include:
- **High Strength and Durability**: With a tensile strength of at least 303 MPa, this alloy can withstand significant stress without breaking, making it ideal for use in demanding environments.
 - **Corrosion Resistance**: The addition of tin helps protect the metal from rust and other forms of degradation, ensuring longevity in applications such as marine or chemical environments.
 - **Versatility**: This material can be used in various applications due to its favorable mechanical properties and chemical makeup.

ASTM B505 Tin Bronze UNS C91000

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 14.000 - 16.000%, Lead 0.200 max., Zinc 1.500 max.
- Mechanical Properties: Tensile Strength 207 min. MPa, Yield Strength - , Elongation -

Technical Specifications:

Iron	0.100 max.
Antimony	0.200 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	84.000 - 86.000%

AI Layman Summary:

ASTM B505 Tin Bronze UNS C91000 is a type of metal made primarily from copper with added tin, making it a strong and durable material. It is part of a group of non-ferrous copper alloys, meaning it does not contain iron, which helps prevent rust and corrosion. This alloy is created through a continuous casting process, ensuring it is uniform and reliable for various applications.

Key benefits of ASTM B505 Tin Bronze include:

- **Strength and Durability**: With a minimum tensile strength of 207 MPa, it can withstand significant stress without breaking.
- **Corrosion Resistance**: The composition helps it resist corrosion, making it suitable for use in challenging environments.
- **Versatile Applications**: While specific use cases are not listed, materials like this are typically used in industries such as construction, automotive, and manufacturing for parts that require both strength and resistance to wear.

ASTM B505 UNS C95700

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Lead 0.030 max., Ni + Cu 1.500 - 3.000%
- Mechanical Properties: Tensile Strength 620 min. MPa, Yield Strength 275 min. MPa, Elongation 15 min.

Technical Specifications:

Iron	2.000 - 4.000%
Manganese	11.000 - 14.000%
Aluminium	7.000 - 8.000%
Copper	71.000 min.

AI Layman Summary:

ASTM B505 UNS C95700 is a type of copper alloy that is specifically designed for continuous casting processes. This means it is created in a way that allows it to be formed into shapes without interruption, making it ideal for manufacturing various components in industries that require durable and reliable materials. The alloy consists mainly of copper, along with small amounts of other elements like lead, iron, manganese, and aluminum, which enhance its performance in different applications.

Key benefits of ASTM B505 UNS C95700 include:

- **High Strength**: With a minimum tensile strength of 620 MPa, this alloy can withstand significant stress, making it suitable for demanding environments.
- **Durability**: Its mechanical properties, including yield strength and elongation, ensure that it can endure wear and tear over time, increasing the lifespan of the products made from it.
- **Versatile Applications**: While specific use cases aren't detailed, its robust nature allows it to be employed in various sectors, such as automotive, aerospace, and electrical components.

ASTM B505 Tin Bronze UNS C91300

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 18.000 - 20.000%, Lead 0.250 max., Zinc 0.250 max.
- Mechanical Properties: Tensile Strength - , Yield Strength - , Elongation -

Technical Specifications:

Iron	0.250 max.
------	------------

Antimony	0.200 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	79.000 - 82.000%

AI Layman Summary:

ASTM B505 Tin Bronze UNS C91300 is a specialized type of bronze made primarily from copper and tin. This alloy is designed for continuous casting and is part of a broader category of non-ferrous metals. It features a chemical composition that includes 18% to 20% tin, along with very minimal amounts of lead and zinc, ensuring it is durable and resistant to corrosion. This makes it ideal for various industrial applications where strength and longevity are key requirements.

- Key benefits of this material include:
- ****High Strength****: Offers excellent tensile and yield strength, making it suitable for demanding environments.
 - ****Corrosion Resistance****: The composition helps in preventing rust and wear over time, extending the lifespan of components made from it.
 - ****Versatility****: Can be used in a range of applications, especially in industries that require reliable metal performance.

ASTM B505 Tin Bronze UNS C90700

Category: Materials

Standard Specification for Copper Alloy Continuous Casting Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 10.000 - 12.000%, Lead 0.500 max., Zinc 0.500 max.
- Mechanical Properties: Tensile Strength 276 min. MPa, Yield Strength 172 min. MPa, Elongation 10 min.

Technical Specifications:

Iron	0.150 max.
Antimony	0.200 max.
Sulphur	0.050 max.
Phosphorus	1.500 max.
Aluminium	0.005 max.
Silicon	0.005 max.
Copper	88.000 - 90.000%

AI Layman Summary:

ASTM B505 Tin Bronze UNS C90700 is a high-quality copper alloy primarily used in various industrial applications. This specific alloy contains a significant amount of tin, which enhances its strength and corrosion resistance, making it ideal for parts that need to withstand wear and tear. The alloy is produced using a continuous casting process, allowing for consistent quality and performance in manufacturing.

- **Key Benefits:****
- ****Strong and Durable:**** With a tensile strength of at least 276 MPa, this material can handle heavy loads without breaking.

- **Corrosion Resistant:** The tin content helps protect it from rust and other forms of corrosion, extending the lifespan of the products made from it.
- **Versatile Use:** Suitable for various applications, particularly where strength and resistance to wear are crucial, although specific use cases are not detailed here.

ASTM B505 UNS C95800

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Lead 0.030 max., Ni + Cu 4.000 - 5.000%
- Mechanical Properties: Tensile Strength 586 min. MPa, Yield Strength 241 min. MPa, Elongation 18 min.

Technical Specifications:

Iron	3.500 - 4.500%
Manganese	0.800 - 1.500%
Aluminium	8.500 - 9.500%
Copper	79.000 min.

AI Layman Summary:

ASTM B505 UNS C95800 is a specific type of copper alloy that is designed for use in various manufacturing processes. This material is categorized under non-ferrous copper alloys and is made through a continuous casting method, which means it is produced in a streamlined, efficient way. It contains a mix of copper and other elements like aluminum, iron, and manganese, ensuring it has strong properties suitable for various applications.

Key benefits of ASTM B505 UNS C95800 include:

- **Strong and Durable:** With a minimum tensile strength of 586 MPa and yield strength of 241 MPa, this alloy can withstand significant stress without breaking.
- **Good Workability:** The alloy has an elongation of 18%, meaning it can be easily shaped and formed for different uses.
- **Controlled Composition:** The precise chemical makeup ensures reliable performance in demanding environments, making it ideal for industries that require high-quality materials.

ASTM B505 UNS C96400

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Lead 0.010 max., Ni + Cu 28.000 - 32.000%
- Mechanical Properties: Tensile Strength 448 min. MPa, Yield Strength 241 min. MPa, Elongation 25 min.

Technical Specifications:

Iron	0.250 - 1.500%
------	----------------

AI Layman Summary:

ASTM B505 UNS C96400 is a type of high-quality copper alloy that is manufactured using a continuous casting process. This means that the metal is melted and poured into shapes that can be easily used in various

applications without the need for extensive further processing. It is specifically designed for use in industries that require strong and durable materials, offering a reliable choice for businesses looking for performance and longevity in their products.

Key benefits of ASTM B505 UNS C96400 include:

- **Strong Mechanical Properties**: It has a minimum tensile strength of 448 MPa and yield strength of 241 MPa, ensuring that it can withstand significant stress and strain without breaking.
- **Controlled Chemical Composition**: With a very low lead content and a specific balance of nickel and copper, this alloy is not only robust but also safe for various applications, making it suitable for industries that prioritize health and safety standards.

ASTM B505 UNS C95900

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Ni + Cu 0.500 max., Iron 3.000 - 5.000%
- Mechanical Properties: Tensile Strength - , Yield Strength - , Elongation -

Technical Specifications:

Manganese	1.500 max.
Aluminium	12.000 - 13.500%
Copper	Balance

AI Layman Summary:

ASTM B505 UNS C95900 is a specific type of copper alloy that is primarily used in creating continuous castings, which are solid metal pieces formed by pouring molten metal into a mold. This alloy is part of a non-ferrous group, meaning it does not contain iron, making it resistant to rust and corrosion. It is designed to offer excellent mechanical properties and is carefully composed of different metals to ensure durability and strength.

Key benefits of ASTM B505 UNS C95900 include:

- **Durability**: Its composition provides resistance to wear and tear, making it suitable for various industrial applications.
- **Corrosion Resistance**: Being a non-ferrous alloy, it is less likely to corrode over time, ensuring longer-lasting products.
- **Versatility**: This alloy can be used in a range of manufacturing processes, enhancing its appeal in different industries.

ASTM B505 UNS C96900

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 7.500 - 8.500%, Lead 0.020 max., Zinc 0.500 max.
- Mechanical Properties: Tensile Strength 758 min. MPa, Yield Strength 724 min. MPa, Elongation 4 min.

Technical Specifications:

Iron	0.500 max.
------	------------

AI Layman Summary:

ASTM B505 UNS C96900 is a type of specialized copper alloy designed for continuous casting processes. This means it can be shaped into different forms while maintaining high quality and strength. It is particularly notable for its composition, which includes a significant amount of tin, a small amount of lead, and a limited amount of zinc, ensuring that it performs well in various applications.

- Some key benefits of this material include:
- **High Strength**: It has a minimum tensile strength of 758 MPa and a yield strength of 724 MPa, making it robust for demanding uses.
 - **Durability**: The specific chemical makeup enhances its ability to withstand stress and wear over time, which is crucial for industrial applications.
 - **Versatility**: While primarily used in non-ferrous applications, its properties allow it to be utilized in a variety of industries, ensuring reliability and performance.

ASTM B505 UNS C97300

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 1.500 - 3.000%, Lead 8.000 - 11.000%, Zinc 17.000 - 25.000%
- Mechanical Properties: Tensile Strength 207 min. MPa, Yield Strength 103 min. MPa, Elongation 8 min.

Technical Specifications:

Iron	1.500 max.
Antimony	0.350 max.
Sulphur	0.080 max.
Phosphorus	0.050 max.
Aluminium	0.005 max.
Manganese	0.500 max.
Copper	53.000 - 58.000%

AI Layman Summary:

ASTM B505 UNS C97300 is a type of copper alloy that is specifically designed for continuous casting processes, meaning it is produced in a way that allows for long, uniform pieces to be made. This alloy is made up of a mix of copper, tin, lead, and zinc, which gives it unique properties that make it suitable for various applications in industries that require durable and reliable materials.

- Key Benefits:**
- **Strong and Reliable**: With a minimum tensile strength of 207 MPa, this material is tough and can withstand significant force without breaking.
 - **Long-lasting Performance**: The specific chemical composition, including a controlled amount of tin and lead, ensures that the alloy is resistant to wear and can last longer in demanding environments.
 - **Versatile Use**: Although specific use cases are not listed, the properties of this alloy make it suitable for numerous applications where strength and durability are essential.

ASTM B505 UNS C97600

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Tin 3.500 - 4.500%, Lead 3.000 - 5.000%, Zinc 3.000 - 9.000%
- Mechanical Properties: Tensile Strength 276 min. MPa, Yield Strength 138 min. MPa, Elongation 10 min.

Technical Specifications:

Iron	1.500 max.
Antimony	0.250 max.
Sulphur	0.080 max.
Phosphorus	0.050 max.
Aluminium	0.005 max.
Manganese	1.000 max.
Copper	63.000 - 67.000%

AI Layman Summary:

ASTM B505 UNS C97600 is a type of copper alloy that is specifically designed for continuous casting, which is a manufacturing process that allows for the creation of metal shapes efficiently. This alloy contains a mixture of tin, lead, and zinc, making it both strong and durable for various applications. It's commonly used in industries where reliable and resilient materials are needed.

Key benefits of ASTM B505 UNS C97600 include:

- ****Strength and Flexibility****: With a minimum tensile strength of 276 MPa, it can withstand significant force while allowing for some stretching without breaking.
- ****Variety of Applications****: Its unique chemical composition makes it ideal for various uses in the manufacturing sector, particularly in products that require a combination of strength and resistance to wear.

ASTM B505 UNS C99500

Category: Materials

Standard Specification for Copper Alloy Continuous Castings Group - Non-Ferrous Copper Alloy Sub Group

Key Features:

- Chemical Composition: Lead 0.250 max., Zinc 0.500 - 2.000%
- Mechanical Properties: Tensile Strength 483 min. MPa, Yield Strength 276 min. MPa, Elongation 12 min.

Technical Specifications:

Iron	3.000 - 5.000%
Manganese	0.500 max.
Aluminium	0.500 - 2.000%
Copper	Balance

AI Layman Summary:

ASTM B505 UNS C99500 is a specialized type of copper alloy that is manufactured using a continuous casting process. This means that the metal is formed into a solid shape in a continuous manner, which helps to ensure consistency and quality in the material. The alloy primarily consists of copper, along with small amounts of lead, zinc, iron, manganese, and aluminum. It's designed for various applications in industries that require durable and reliable metal components.

Key benefits of ASTM B505 UNS C99500 include:

- **Strong Performance**: It has a minimum tensile strength of 483 MPa and a yield strength of 276 MPa, making it a robust choice for demanding applications.
- **Versatile Use**: This alloy's unique chemical composition allows it to be used in a variety of settings where non-ferrous materials are preferred, ensuring longevity and resistance to corrosion.
- **Manufacturing Consistency**: The continuous casting process results in uniform properties throughout the alloy, leading to predictable performance in real-world applications.

ASTM B584 C83800

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium 0.005 max., Iron 0.300 max., Lead 5.000 - 7.000%
- Mechanical Properties: Tensile Strength 207 min. MPa, Yield Strength 90 min. MPa, Elongation 20 min.

Technical Specifications:

Phosphorus	0.030 max.
Sulphur	0.080 max.
Antimony	0.250 max.
Silicon	0.005 max.
Tin	3.300 - 4.200%
Zinc	5.000 - 8.000%
Copper	82.000 - 83.800%

AI Layman Summary:

ASTM B584 C83800 is a type of copper alloy that is specifically designed for sand casting applications. This material is used in a variety of general applications where strength and durability are important. The alloy consists mainly of copper, with additional elements like tin and lead, which enhance its properties and make it suitable for different uses.

Key benefits of ASTM B584 C83800 include:

- **Strong and Durable**: With a minimum tensile strength of 207 MPa, this alloy can withstand significant stress, making it ideal for heavy-duty applications.
- **Versatile Composition**: The alloy's chemical makeup allows it to perform well in various environments, ensuring longevity and reliability in products made from it.
- **Easy to Work With**: As a sand-cast alloy, it is straightforward to mold into various shapes, making it a preferred choice for manufacturers looking for ease of fabrication.

ASTM B584 C84800

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium 0.005 max., Iron 0.400 max., Lead 5.500 - 7.000%
- Mechanical Properties: Tensile Strength 193 min. MPa, Yield Strength 83 min. MPa, Elongation 16 min.

Technical Specifications:

Phosphorus	0.020 max.
------------	------------

Sulphur	0.080 max.
Antimony	0.250 max.
Silicon	0.005 max.
Tin	2.000 - 3.000%
Zinc	13.000 - 17.000%
Copper	75.000 - 77.000%

AI Layman Summary:

ASTM B584 C84800 is a standard type of copper alloy that is commonly used for making sand castings in various applications. Essentially, this material is a blend of copper with other metals, which enhances its strength and durability while maintaining good workability. It is particularly valued for its resistance to wear and corrosion, making it suitable for industrial uses where reliability is crucial.

- Key benefits of ASTM B584 C84800 include:
- **Strong and Durable**: It has a minimum tensile strength of 193 MPa, ensuring that it can withstand significant stress without breaking.
 - **Versatile Applications**: This alloy can be used in various industries, making it a go-to choice for manufacturers looking for reliable materials for their products.
 - **Low Impurity Levels**: With controlled levels of elements like phosphorus and sulphur, it minimizes the risk of defects during casting, leading to higher-quality final products.

ASTM B584 C85700

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium 0.800 max., Iron 0.700 max., Lead 0.800 - 1.500%
- Mechanical Properties: Tensile Strength 276 min. MPa, Yield Strength 97 min. MPa, Elongation 15 min.

Technical Specifications:

Phosphorus	0.050 max.
Sulphur	0.080 max.
Antimony	0.250 max.
Silicon	0.050 max.
Tin	0.500 - 1.500%
Zinc	32.000 - 40.000%
Copper	58.000 - 64.000%

AI Layman Summary:

ASTM B584 C85700 is a type of copper alloy designed specifically for sand casting, which is a process used to create metal parts by pouring molten metal into a sand mold. This material is suitable for general applications where a strong and durable metal is needed. Its unique composition includes a mix of metals that provide both strength and resistance to wear, making it an ideal choice for various industrial uses.

- Key benefits of ASTM B584 C85700 include:
- **Strong Mechanical Properties**: With a minimum tensile strength of 276 MPa, it can withstand significant stress without breaking.

- **Versatile Composition**: The alloy contains a balanced mix of copper, zinc, and other elements that enhance its performance and longevity, ensuring it meets diverse application needs.
- **Reliable Casting Characteristics**: Being designed for sand casting helps manufacturers achieve precision and quality in their metal parts, leading to better overall product reliability.

ASTM B584 C83600

Category: Materials

Standard Specification for Copper Alloy Sand Casting for General Applications

Key Features:

- Chemical Composition: Aluminium 0.005 max., Iron 0.300 max., Lead 4.000 - 6.000%
- Mechanical Properties: Tensile Strength 207 min. MPa, Yield Strength 97 min. MPa, Elongation 20 min.

Technical Specifications:

Phosphorus	0.050 max.
Sulphur	0.080 max.
Antimony	0.250 max.
Silicon	0.005 max.
Tin	4.000 - 6.000%
Zinc	4.000 - 6.000%
Copper	84.000 - 86.000%

AI Layman Summary:

ASTM B584 C83600 is a type of copper alloy specifically designed for sand casting, which is a process used to create metal parts by pouring molten metal into a mold. This alloy is suitable for general applications, making it versatile for various manufacturing needs. Its formulation includes several elements, such as copper, tin, and lead, contributing to its strength and durability.

Key benefits of ASTM B584 C83600 include:

- **Strength and Durability**: With a minimum tensile strength of 207 MPa, this alloy can withstand significant stress, making it ideal for robust applications.
- **Versatile Applications**: Its general specification allows it to be used in various industries, enhancing its utility across different projects.
- **Controlled Composition**: The precise chemical makeup ensures consistent quality and performance, which is crucial for manufacturers looking for reliable materials.

Services Offered

Manufacturing

Comprehensive manufacturing services from prototyping to high-volume production.

Applications:

- Precision engineering
- Custom metal parts

Processes Involved:

- Investment casting
- CNC machining
- Quality assurance testing

AI Layman Summary:

Manufacturing services provide businesses with a complete solution for creating products, from the initial design stage to producing large quantities. This means that if you have an idea for a product, manufacturing services can help turn that idea into reality by making prototypes and then producing the final items in bulk.

Key benefits include:

- ***Precision Engineering***: Ensures that all parts fit and work together perfectly.
- ***Custom Metal Parts***: Allows for the creation of unique components tailored to specific needs.
- ***Advanced Processes***: Utilizes methods like investment casting and CNC machining to create high-quality products.
- ***Quality Assurance***: Testing throughout the manufacturing process guarantees that the end product meets industry standards.

Precision Casting

Manufacturing precision castings that meet global performance and safety standards.

Applications:

- Process instrumentation
- General engineering
- Automotive components

Processes Involved:

- High-Precision Molding
- Controlled Pouring
- Heat Treatment
- CNC Machining
- Final Inspection

AI Layman Summary:

Precision Casting is a manufacturing service that creates highly accurate metal parts using a special casting process. This service ensures that the parts not only meet rigorous safety and performance standards but are also tailored for various industries, including instrumentation, engineering, and automotive. The process involves several key steps to guarantee quality and precision.

Key features of Precision Casting include:

- ***High-Precision Molding***: This technique shapes the metal with great accuracy.

- **Controlled Pouring**: Metal is poured into molds carefully to avoid defects.
- **Heat Treatment**: The parts are heated to enhance their strength and durability.
- **CNC Machining**: Advanced machines refine the parts to exact specifications.
- **Final Inspection**: Each piece undergoes thorough checks to ensure it meets all standards.

Benefits of using Precision Casting:

- Produces components that are reliable and safe for use in critical applications.
- Offers versatility across various industries, ensuring customized solutions.
- Enhances the performance and lifespan of products with high-quality manufacturing techniques.

Sand Casting

Reliable sand casting solutions for diverse industrial needs.

Applications:

- Large and complex castings
- Low to medium-volume production

Processes Involved:

- Pattern Making
- Mold Preparation
- Core Placement
- Mold Assembly
- Metal Pouring
- Cooling
- Mold Breakout
- Fettling & Cleaning
- Machining & Finishing

AI Layman Summary:

Sand casting is a manufacturing process used to create metal parts by pouring molten metal into a sand mold. This method is particularly useful for producing large and complex shapes, making it ideal for various industrial applications. The process involves several steps, including creating patterns, preparing molds, placing cores, pouring metal, and finally cleaning and finishing the cast pieces.

Benefits of Sand Casting:

- **Versatility**: Suitable for a wide range of industrial needs, from small to large parts.
- **Cost-Effective**: Efficient for low to medium-volume production, reducing costs compared to other methods.
- **Complex Shapes**: Capable of producing intricate designs that might be difficult to achieve with other manufacturing techniques.

Investment Casting

Precision casting process using wax patterns for complex geometries.

Applications:

- Automotive components
- Valve and pump parts
- Industrial machinery

Processes Involved:

- Wax pattern creation
- Ceramic shell building
- Metal pouring
- Finishing and inspection

AI Layman Summary:

Investment casting is a precise manufacturing process used to create complex metal parts by forming molds from wax patterns. Here's how it works: first, a wax pattern of the desired part is created. This pattern is then surrounded by a ceramic shell, which hardens to form a mold. Once the mold is ready, the wax is melted away, and molten metal is poured into the cavity to form the final product. After cooling, the mold is broken away, leaving a clean, detailed metal part that requires minimal finishing.

Benefits of investment casting include:

- *Ability to create intricate shapes and designs that are difficult to achieve with other methods.*
- *High precision and a smooth surface finish, reducing the need for further machining.*
- *Versatility in applications, making it suitable for parts used in automotive components, valve and pump parts, and industrial machinery.*

Forging

Manufacturing process that shapes metal using compressive forces.

Applications:

- High-strength components
- Aerospace parts
- Oil and gas equipment

Processes Involved:

- Raw material selection
- Heating and shaping
- Heat treatment
- Machining and finishing

AI Layman Summary:

Forging is a manufacturing process that shapes metal by applying strong compressive forces. This technique is commonly used to create high-strength components that are essential in various industries. For example, forging is crucial in producing parts for aerospace, as well as equipment used in the oil and gas sector.

The forging process involves several key steps:

- ***Raw material selection***: Choosing the right type of metal for the job.
- ***Heating and shaping***: Applying heat to make the metal malleable, then shaping it using powerful presses or hammers.
- ***Heat treatment***: Strengthening the shaped metal by heating and cooling it in a controlled manner.
- ***Machining and finishing***: Refining the forged piece to meet precise specifications.

The benefits of forging include increased strength and durability of the final products, making them ideal for demanding applications. By using this method, manufacturers can produce reliable components that perform well under pressure.

Manufacturing & R&D Processes

Manufacturing Process

The process of producing nickel and nickel-cobalt alloy tubes and components.

Capacity: High-volume production

Workflow / Steps:

1. Pattern & Mold Preparation
2. Metal Melting & Pouring
3. Cooling & Heat Treatment
4. Machining & Finishing
5. Inspection & Testing
6. Material selection
7. Alloy preparation
8. Casting
9. Heat treatment
10. Quality control

Machinery Used:

- CNC Machines
- VMC Machines
- HMC Machines
- Robotic Shelling
- Casting machines
- Heat treatment furnaces
- Quality testing equipment

Manufacturing

Processes involved in the production of nickel alloy forgings.

Capacity: High-volume production

Workflow / Steps:

1. Material selection
2. Forging or rolling
3. Heat treatment
4. Quality control
5. Heating and forging
6. Machining and finishing

Machinery Used:

- Forging presses
- Rolling mills
- Heat treatment furnaces
- CNC machines

Investment Casting

A precision casting process that allows for complex geometries and tight tolerances.

Capacity: High-volume production capabilities

Workflow / Steps:

1. Wax pattern creation
2. Tree assembly
3. Ceramic shell investment
4. Robotic shelling
5. Dewaxing and firing
6. Metal pouring
7. Shell removal and cutoff
8. Heat treatment
9. CNC precision machining
10. Final inspection and dispatch

Machinery Used:

- CNC Machines
- Induction Furnaces
- Robotic Shelling Systems

Continuous Casting

A manufacturing process for producing metal products.

Capacity: High-volume production

Workflow / Steps:

1. Melting of raw materials
2. Pouring into molds
3. Cooling and solidification
4. Cutting to desired lengths

Machinery Used:

- Induction furnaces
- Casting machines
- Cooling systems

Sand Casting Process

A versatile and widely used manufacturing technique for producing large, complex, and durable castings.

Capacity: High-volume production capabilities

Workflow / Steps:

1. Pattern Making
2. Mold Preparation
3. Core Placement
4. Mold Assembly
5. Metal Pouring
6. Cooling
7. Mold Breakout
8. Fettling & Cleaning
9. Machining & Finishing

Machinery Used:

- CNC machines

- Molding machines
- Casting furnaces

Precision Casting Process

A controlled casting process ensuring dimensional accuracy and high strength.

Capacity: High tolerance and repeatable accuracy

Workflow / Steps:

1. High-Precision Molding
2. Controlled Pouring
3. Heat Treatment
4. CNC Machining
5. Final Inspection

Machinery Used:

- CNC machines
- Heat treatment furnaces
- Inspection equipment

Investment Casting Process

A detailed process transforming wax patterns into high-precision metal components.

Capacity: High-volume production capabilities

Workflow / Steps:

1. Wax pattern creation
2. Assembly of wax patterns
3. Ceramic shell building
4. Dewaxing
5. Metal pouring
6. Cooling and shell removal
7. Finishing and inspection

Machinery Used:

- CNC machines
- Robotic shelling systems
- Heat treatment furnaces

Forging Process

A process that shapes metal through compressive forces for enhanced strength.

Capacity: Large production volumes for simple shapes

Workflow / Steps:

1. Raw material selection
2. Heating metals
3. Forging operation
4. Flash removal and trimming
5. Heat treatment
6. Machining and finishing

Machinery Used:

- Hydraulic presses
- Drop hammers
- CNC machining centers

Audience & Market

Target Buyer Personas

- Manufacturing and production managers

Target Industries

- manufacturing
- technology
- construction
- energy

Value Propositions (USPs)

- High-tolerance metal parts with complex geometries
- Integrated manufacturing environment for efficiency
- Rigorous testing and compliance with global standards
- High-quality materials for demanding applications
- Expertise in non-ferrous nickel alloys
- Comprehensive product range for various industries
- High-quality nickel alloys for various industrial applications.
- Customizable solutions for specific engineering needs.
- Expertise in manufacturing and processing nickel alloys.
- High-quality materials for valves and pumps
- Compliance with ASTM standards
- Wide range of applications in various industries
- High precision and repeatability in manufacturing
- Robotic shelling technology for consistent quality
- Integrated machining capabilities for ready-to-assemble components
- Export-ready documentation and compliance support
- High-quality nickel alloy materials
- Customizable solutions for various industries
- Expertise in manufacturing and engineering
- High-quality materials for industrial applications
- Compliance with international standards
- Expertise in nickel and nickel-cobalt alloys
- High-quality nickel alloy products
- Custom solutions for industrial applications
- Expertise in material specifications and standards
- Wide range of applications in engineering and automotive industries
- High-quality materials for industrial applications.
- Compliance with ASTM standards.
- Diverse product range for various engineering needs.
- High-quality materials
- Wide range of applications
- High-quality copper alloys for various applications.
- Expertise in manufacturing and engineering.
- Wide range of products for various industries.
- Custom solutions for various industries

- Expertise in copper alloy manufacturing
- High-quality casting materials
- Wide range of applications in engineering and automotive sectors
- Expertise in non-ferrous alloys
- High-quality, durable products
- Cost-effective solutions
- Expertise in diverse industrial applications
- Commitment to precision and reliability
- Robust quality assurance processes
- Comprehensive documentation and traceability
- Advanced manufacturing technologies including robotic shelling

Brand Tone

Professional and enterprise-oriented

Digital Maturity & SEO Observations

Overall Maturity Score: 71/100

Website Quality Score: 100/100

SEO Readiness Score: 75/100

SEO Gaps & Opportunities:

N/A

R&D Insights & Recommendations

- Increase social proof, content consistency, and visible engagement signals.